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MA1101 Classnotes Calculus

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Syllabus

Functions of two and three variables: Regions in plane, level curves and level surfaces, limit, continuity, partial derivatives, the increment theorem, directional derivatives and gradient, tangent planes and normal lines, Taylor's formula, extreme values, Lagrange multipliers, Volumes by slicing, Volumes of solids of revolution, cylindrical shell method, arc length, area of surface of revolution, double integral and iterated integral, double integral in polar coordinates, triple integral, triple integral in cylindrical coordinates, triple integral in spherical coordinates, change of variables. [T]: Ch 14, §6.1-6.4, Ch. 15 (Exclude Sect. 15.6).

Series Expansion of Functions: Series, Taylor series, Fourier series, odd and even functions, half range Fourier series, Functions of arbitrary period. [P]: §4.1-4.2, §4.16-4.17, §4.19-4.20, §5.1-5.6

Text:
[T] J.R. Hass, C.E. Heil, M.D. Weir and P. Bogacki, Thomas' Calculus: Early Transcendentals, 15th Edition, Pearson, 2022.
[P] N. Piskunov, Differential and Integral Calculus Vol. II, Mir Publishers, Eng. Trans. 1974.

References:

1. E. Kreyszig, Advanced Engineering Mathematics, 10th Ed., John Willey & Sons, 2010.
2. S.T. Tan, Calculus, Cengage Learning, 2010.
3. R.A. Adams and C. Essex, Calculus: A Complete Course, 10th Edition, Pearson Education, 2021.
4. N. Piskunov, Differential and Integral Calculus Vol. 1 & II, Mir Publishers, Eng. Trans.1974.

1

Differential Calculus

1.1 Regions in the plane and surfaces

Most of the results we discuss will not hold for arbitrary subsets of $\mathbb{R}^2$, the plane. We thus distinguish special types of subsets of the plane by requiring that certain properties are satisfied.

The **distance** between any two points (a, b) and (c, d) in $\mathbb{R}^2$ is

$$|(a, b) - (c, d)| := \sqrt{(a - c)^2 + (b - d)^2}.$$

Let D be a subset of $\mathbb{R}^2$.

Let $\epsilon > 0$. An **ϵ-disk** around (a, b) is the set of all points $(x, y) \in \mathbb{R}^2$ whose distance from (a, b) is less than ϵ. Such a disk is also called a disk with center at (a, b) and radius ϵ.

A **disk** around (a, b) is the ϵ-disk around (a, b) for some $\epsilon > 0$.

Let D be a subset of $\mathbb{R}^2$.

D is called **bounded** iff D is contained in a disk around some point of $\mathbb{R}^2$.

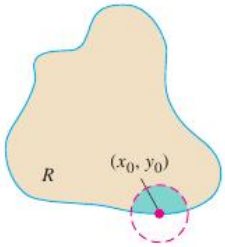
A point $(a, b) \in \mathbb{R}^2$ is called a **a boundary point** of D iff *every* disk around (a, b) contains at least one point from D *and* at least one point from $\mathbb{R} \setminus D$.

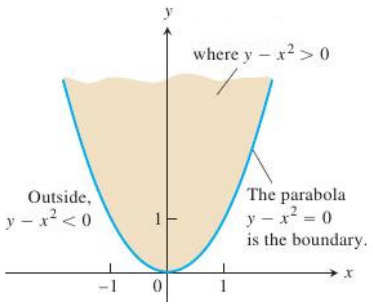
The set of all boundary points of D, written as ∂D, is called the **boundary** of D.

The set $\overline{D} = D \cup \partial D$ is called the **closure** of D.

A point $(a, b) \in D$ is called an **interior point** of D iff some disk around (a, b) is contained in D.

A point $(a, b) \in D$ is called an **isolated point** of D iff (a, b) is the only point of D that is contained in *some* disk around (a, b).





D is called a **region** in the plane iff no point of D is an isolated point, and D has at least one interior point.

Thus, regions are *fat sets* so to say; curves in the plane are not regions.

D is called a **bounded region** iff D is a region and D is a bounded set.

D is called an **open region** iff D is a region and D does not contain any of its boundary points.

D is called a **closed region** iff D is a region and D contains all of its boundary points.

D is called a **connected region** iff D is a region and any two points in D can be joined by a polygonal path entirely lying in D.

Remark 1.1 The following facts can be proved easily:

1. Every disk around a point is an open region.
2. The closure of any disk is a closed region.
3. The boundary of an ϵ-disk around any point (a, b) is the circle with center at (a, b) and radius ϵ.
4. If D is a region, then each disk around each point of D contains infinitely many points of D.
5. A region is open iff every point of it is its interior point.

Let D be a region in the plane. Let $f : D \to \mathbb{R}$ be a function.

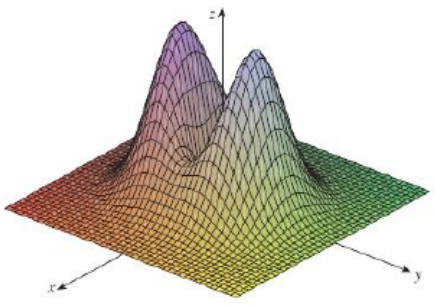
The **domain** of f is D and the **co-domain** of f is $\mathbb{R}$.

The **range** of f is $\{z \in \mathbb{R} : z = f(x, y) \text{ for some } (x, y) \in D\}$.

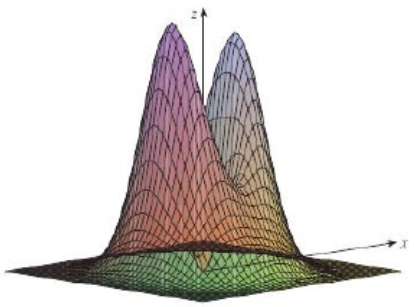
Sometimes, we do not fix the domain and range of a function, but ask you to find it. For instance, The function $f(x, y) = \sqrt{y - x^2}$ has domain $D = \{(x, y) : x^2 \le y\}$. Its range is the set of all non-negative real numbers.

The **graph** of f is $\{(x, y, z) \in \mathbb{R}^3 : z = f(x, y),\ (x, y) \in D\}$. Such a graph is also called the **surface** $z = f(x, y)$.

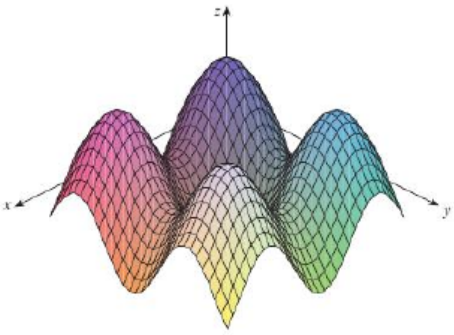
Some examples of surfaces are here:



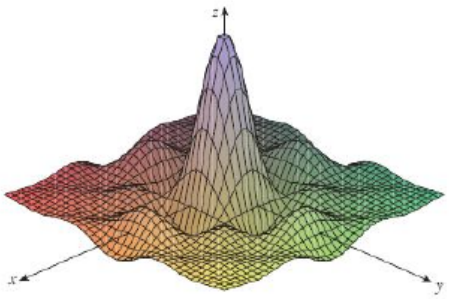
(a) $f(x, y) = (x^2 + 3y^2)e^{-x^2-y^2}$



(b) $f(x, y) = (x^2 + 3y^2)e^{-x^2-y^2}$



(c) $f(x, y) = \sin x + \sin y$



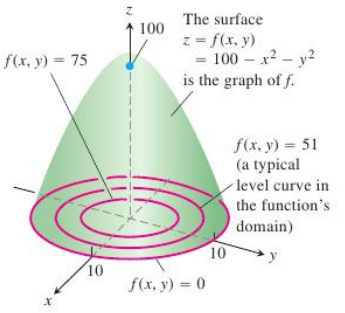
(d) $f(x, y) = \dfrac{\sin x \sin y}{xy}$

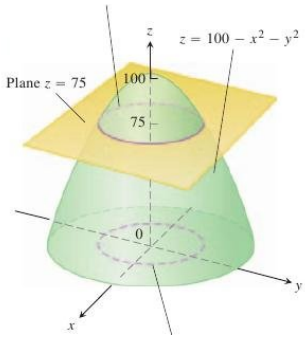
Let $f(x, y)$ be a (real valued) function of two variables. That is, $f : D \to \mathbb{R}$, where D is a region in $\mathbb{R}^2$.

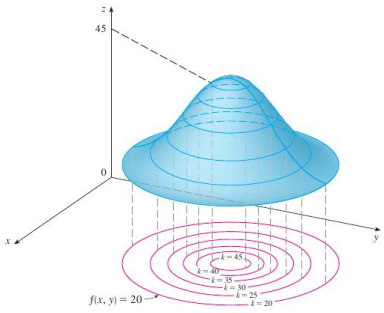
A **contour curve** of f is the curve of intersection of the surface $z = f(x, y)$ and the plane $z = c$ for some constant c in the range of f. It is the curve $f(x, y) = c$ for some constant c in the range of f. Thus, the union of all contour curves is the surface $z = f(x, y)$; it is the graph of f.

A **level curve** of f is the set of points (x, y) in the domain of f for which $f(x, y) = c$ for some constant c in the range of f. Thus, a level curve is the projection of a contour curve on the xy-plane. Thus, the union of all level curves of the surface $z = f(x, y)$ is the the domain of f.

For example, consider the function $f(x, y) = 100 - x^2 - y^2$ depicted below. Its domain is $\mathbb{R}^2$. Its range is the interval $(-\infty, 100]$. The level curve $f(x, y) = 0$ is $\{(x, y) : x^2 + y^2 = 100\}$. The level curve $f(x, y) = 51$ is $\{(x, y) : x^2 + y^2 = 49\}$.







Similarly, for a function $f(x, y, z)$ of three variables, the *level surfaces* are the sets of points (x, y, z) such that $f(x, y, z) = c$ for numbers c in the range of f.

Unless otherwise stated, we assume that the domain D of a function $f : D \to \mathbb{R}$ is a region in the plane. Also, we write "$f(x, y)$ is a real valued function defined on D" to mean that $f : D \to \mathbb{R}$ is a function.

We remark that there are degenerate cases when level curves are not really curves.

For instance, consider the surface $z = f(x, y)$, where the function $f : \mathbb{R}^2 \to \mathbb{R}$ is the constant function $f(x, y) = 5$. The surface is actually the plane $z = 5$ parallel to the xy-plane. The range of the function f is the singleton $\{5\}$. The contour curve with $c = 5$ is the whole plane $z = 5$, and the corresponding level curve is the whole xy-plane. In the sequel, we will not be considering such degenerate cases.

1.2 Limit

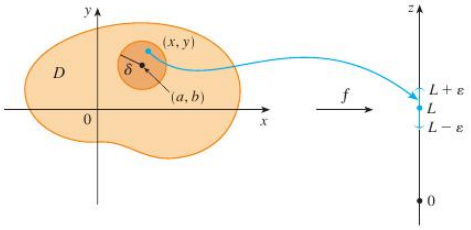
Let $f(x, y)$ be a real valued function defined on a region D in the plane, $(a, b) \in \overline{D}$ and let L be a real number.

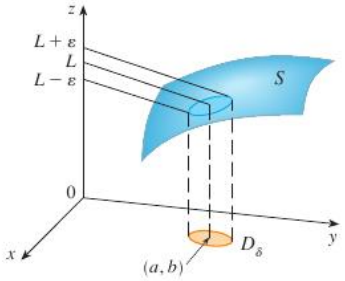
Limit of $f(x, y)$ as (x, y) approaches (a, b) is L iff corresponding to each $\epsilon > 0$, there exists $\delta > 0$ such that for all $(x, y) \in D$ with $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$, we have $|f(x, y) - L| < \epsilon$.

In this case, we also say that **limit of f at (a, b) is L**, and write

$$\lim_{(x,y)\to(a,b)} f(x, y) = L.$$

We say that **limit of f at (a, b) exists** iff there exists a real number L such that limit of (x, y) at (a, b) is L. When for no real number L, limit of (x, y) at (a, b) is L, we say that **limit of f at (a, b) does not exist.**





The intuitive understanding of the notion of limit is as follows:

> The distance between $f(x, y)$ and L can be made arbitrarily small by making the distance between (x, y) and (a, b) sufficiently small but not necessarily zero.

Each of the following tells one and the same thing:

The limit of $f(x, y)$ as (x, y) approaches (a, b) is L.

$f(x, y)$ approaches L as (x, y) approaches (a, b).

$f(x, y) \to L$ as $(x, y) \to (a, b)$.

The limit of f at (a, b) is L.

$$\lim_{(x,y)\to(a,b)} f(x,y) = L.$$

$$\lim_{\substack{x\to a\\ y\to b}} f(x,y) = L.$$

It is often difficult to show that limit of a function does not exist at a point. We will come back to this question soon.

(1.2) ***Example***

Determine if $\lim_{(x,y)\to(0,0)} \dfrac{4xy^2}{x^2+y^2}$ exists.

Observe that the domain D of f is $\mathbb{R}^2 \setminus \{(0,0)\}$. And $f(0,y) = 0$ for $y \neq 0$; $f(x,0) = 0$ for $x \neq 0$. We guess that the limit would be 0. To see that it is the case, we start with any $\epsilon > 0$. We want to choose a $\delta > 0$ such that the following sentence becomes true:

$$\text{If } \ 0 < \sqrt{x^2+y^2} < \delta, \ \text{ then } \ \left|\frac{4xy^2}{x^2+y^2}\right| < \epsilon.$$

Since $|y^2| = y^2 \leq x^2+y^2$ and $|x^2| = x^2 \leq x^2+y^2$, we have

$$\left|\frac{4xy^2}{x^2+y^2}\right| \leq 4|x| \leq 4\sqrt{x^2+y^2}.$$

So, we choose $\delta = \epsilon/4$. Let us verify whether our choice is all right.

Assume that $0 < \sqrt{x^2+y^2} < \delta$. Then

$$\left|\frac{4xy^2}{x^2+y^2} - 0\right| \leq 4\sqrt{x^2+y^2} < 4\delta = \epsilon.$$

Hence $\lim_{(x,y)\to(0,0)} \dfrac{4xy^2}{x^2+y^2} = 0.$ □

(1.3) Observation Suppose we have obtained a δ corresponding to some ϵ. If we take ϵ_1 which is larger than the earlier ϵ, then the same δ will satisfy the requirement in the definition of the limit. Thus while showing that the limit of a function is such and such at a point, we are free to choose a pre-assigned upper bound for our ϵ.

Similarly, suppose for some ϵ, we have already obtained a δ such that the limit requirement is satisfied. If we choose another δ, say δ_1, which is smaller than δ, then the limit requirement is also satisfied. Thus, we are free to choose a pre-assigned upper bound for our δ provided it is convenient to us and it works.

(1.4) ***Example***

Consider $f(x,y) = \sqrt{1-x^2-y^2}$ where $D = \{(x,y) : x^2+y^2 \leq 1\}$.

We guess that limit of $f(x,y)$ is 1 as $(x,y) \to (0,0)$. To show that the guess is right, let $\epsilon > 0$. Notice that $0 \le f(x,y) \le 1$ on D. Using (1.3), assume that $0 < \epsilon < 1$. Choose $\delta = \sqrt{1-(1-\epsilon)^2}$. Let $|(x,y)-(0,0)| < \delta$. Then

$$0 < x^2 + y^2 < 1 - (1-\epsilon)^2 \Rightarrow 1 - x^2 - y^2 > (1-\epsilon)^2 \Rightarrow f(x,y) > 1 - \epsilon.$$

That is, $|f(x,y) - 1| = 1 - f(x,y) < \epsilon$. So, $f(x,y) \to 1$ as $(x,y) \to (0,0)$. □

(1.5) ***Theorem*** (Uniqueness of limit)
Let D be a region in the plane, $(a,b) \in \overline{D}$ and let $f : D \to \mathbb{R}$ be a function. If limit of $f(x,y)$ at (a,b) exists, then it is unique.

Proof. Suppose $f(x,y) \to \ell$ and also $f(x,y) \to m$ as $(x,y) \to (a,b)$. Let $\epsilon > 0$. For $\epsilon/2$, we have $\delta_1 > 0,\ \delta_2 > 0$ such that for all $(x,y) \in D$,

$$0 < (x-a)^2 + (y-b)^2 < \delta_1^2 \Rightarrow |f(x,y) - \ell| < \epsilon/2,$$
$$0 < (x-a)^2 + (y-b)^2 < \delta_2^2 \Rightarrow |f(x,y) - m| < \epsilon/2.$$

Take $\delta = \min\{\delta_1, \delta_2\}$. Since D is a region, it contains no isolated points. So, there exists a point (α,β) in the δ-disk around (a,b) and $(\alpha,\beta) \ne (a,b)$. This means

$$0 < (\alpha-a)^2 + (\beta-b)^2 < \delta^2, \quad \delta^2 \le \delta_1^2,\ \delta^2 \le \delta_2^2.$$

The above implications now guarantee that $|f(\alpha,\beta) - \ell| < \epsilon/2$ and also $|f(\alpha,\beta) - m| < \epsilon/2$. Then,

$$|\ell - m| \le |\ell - f(\alpha,\beta)| + |f(\alpha,\beta) - m| < \epsilon/2 + \epsilon/2 = \epsilon.$$

That is, for each $\epsilon > 0$, we have $|\ell - m| < \epsilon$. Hence $\ell = m$. ∎

For a function of one variable, there are only two directions for approaching a point; from left or from right. Whereas for a function of two variables, the limit refers only to the distance between (x,y) and (a,b). It does not refer to any specific direction of approach to (a,b). There are infinitely many directions, and infinite number of paths on which one can vary a point (x,y) to approach this point (a,b).

To account for this idea, we look at the definition of limit. It urges us to verify the condition $|f(x,y) - L| < \epsilon$ for all points (x,y) whose distance from (a,b) is less than δ. Suppose we restrict these points to lie on a curve C given by $y = g(x)$ for a function $g : I \to \mathbb{R}$, where I is an interval. Then we obtain a restricted version of the notion of limit. We describe this restricted version as follows.

Let D be a region in the plane, $(a,b) \in \overline{D}$, $f : D \to \mathbb{R}$ be a function, $L \in \mathbb{R}$ and let a curve lying in D be given by $y = g(x)$ for a function $g : I \to \mathbb{R}$, where I is an interval.

We say that the **limit of $f(x, y)$ as (x, y) approaches (a, b) along $y = g(x)$ is L** iff corresponding to each $\epsilon > 0$, there exists a $\delta > 0$ such that for all $(x, y) \in D$ with $y = g(x)$, $\lim_{x\to a} g(x) = b$ and $\sqrt{(x-a)^2 + (y-b)^2} < \delta$, we have $|f(x, y) - L| < \epsilon$. In this case, we write

$$\lim_{\substack{(x,y)\to(a,b)\\ \text{along } y=g(x)}} f(x, y) = L.$$

The extra condition $\lim_{x\to a} g(x) = b$ is required since (x, y) approaches (a, b).

We see that

$$\lim_{\substack{(x,y)\to(a,b)\\ \text{along } y=g(x)}} f(x, y) = \lim_{x\to a} f\big(x, g(x)\big), \quad \text{where } \lim_{x\to a} g(x) = b.$$

Obviously, if the limit of $f(x, y)$ as (x, y) approaches (a, b) exists, then this must be equal to the limit of $f(x, y)$ as (x, y) approaches (a, b) along C. We mention it as our next result.

(1.6) *Theorem*

Let D be a region in the plane, $(a, b) \in \overline{D}$, $f : D \to \mathbb{R}$ be a function, $L \in \mathbb{R}$ and let a curve lying in $\overline{D}$ be given by $y = g(x)$, where $\lim_{x\to a} g(x) = b$. If $\lim_{(x,y)\to(a,b)} f(x, y) = L$ then $\lim_{\substack{(x,y)\to(a,b)\\ \text{along } y=g(x)}} f(x, y) = L$.

Analogously, if a curve lying in $\overline{D}$ is given by $x = g(y)$, then

$$\lim_{\substack{(x,y)\to(a,b)\\ \text{along } x=g(y)}} f(x, y) = \lim_{y\to b} f\big(g(y), y\big), \quad \text{where } \lim_{y\to b} g(y) = a.$$

And, if a curve lying in $\overline{D}$ is given in parametric form $x = x(t),\ y = y(t)$, then

$$\lim_{\substack{(x,y)\to(a,b)\\ \text{along } C}} f(x, y) = \lim_{t\to\alpha} f\big(x(t), y(t)\big), \quad \text{where } \lim_{t\to\alpha} x(t) = a \text{ and } \lim_{t\to\alpha} y(t) = b.$$

In general, if a curve C lying in $\overline{D}$ is such that (x, y) approaches to (a, b) along C is meaningful and the limit of $f(x, y)$ exists as (x, y) approaches (a, b), then this limit equals the limit of $f(x, y)$ as (x, y) approaches (a, b) along C.

In most cases if limit of $f(x, y)$ as (x, y) approaches (a, b) does not exist, then it can be shown so by using Theorem 1.6. How?

Suppose C_1 and C_2 are two curves lying in $\overline{D}$ so that

$$\lim_{\substack{(x,y)\to(a,b)\\ \text{along } C_1}} f(x, y) = \ell_1, \qquad \lim_{\substack{(x,y)\to(a,b)\\ \text{along } C_2}} f(x, y) = \ell_2.$$

If $\ell_1 \neq \ell_2$, then $\lim\limits_{(x,y)\to(a,b)} f(x,y)$ does not exist.

(1.7) ***Example***

Consider $f(x,y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ for $(x,y) \neq (0,0)$. What is its limit at $(0,0)$?

When $y = 0$, limit of $f(x,y)$ as $x \to 0$ is $\lim\limits_{x\to 0} \dfrac{x^2}{x^2} = \lim\limits_{x\to 0}(1) = 1$.

That is, $f(x,y) \to 1$ as $(x,y) \to (0,0)$ along the x-axis.

When $x = 0$, limit of $f(x,y)$ as $y \to 0$ is $\lim\limits_{y\to 0} \dfrac{-y^2}{y^2} = -1$.

That is, $f(x,y) \to -1$ as $(x,y) \to (0,0)$ along the y-axis.

Hence $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ does not exist. □

(1.8) ***Example***

Consider $f(x,y) = \dfrac{xy}{x^2 + y^2}$ for $(x,y) \neq (0,0)$. What is its limit at $(0,0)$?

Along the x-axis, $y = 0$; the limit of $f(x,y)$ as $(x,y) \to (0,0)$ is 0.

Along the y-axis, $x = 0$; the limit of $f(x,y)$ as $(x,y) \to (0,0)$ is 0.

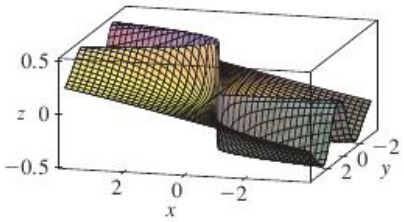
Does it say that limit of $f(x,y)$ as $(x,y) \to (0,0)$ is 0?

Along the line $y = x$, limit of $f(x,y)$ as $(x,y) \to 0$ is $\lim\limits_{x\to 0} \dfrac{x^2}{x^2 + x^2} = 1/2$.

Hence $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ does not exist. □

(1.9) ***Example***

Consider $f(x,y) = \dfrac{xy^2}{x^2 + y^4}$ for $(x,y) \neq (0,0)$. What is its limit at $(0,0)$?



If $y = mx$, for some $m \in \mathbb{R}$, then $f(x,y) = \dfrac{m^2 x}{1 + m^4 x^2}$.

So, the limit of $f(x,y)$ as $(x,y) \to (0,0)$ along all straight lines is 0.

If $x = y^2,\ y \neq 0$, then $f(x,y) = \dfrac{y^4}{y^4 + y^4} = 1/2$.

That is, as $(x,y) \to (0,0)$ along the curve $x = y^2$, $f(x,y) \to 1/2$.

Hence $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ does not exist. □

Are $\lim\limits_{(x,y)\to(a,b)} f(x,y)$, $\lim\limits_{x\to a}\lim\limits_{y\to b} f(x,y)$, $\lim\limits_{y\to b}\lim\limits_{x\to a} f(x,y)$ all equal?

(1.10) ***Example***

Let $f(x,y) = \dfrac{(y-x)(1+x)}{(y+x)(1+y)}$ for $x+y \neq 0, -1 < x,\ y < 1$. Then

$$\lim_{y\to 0}\lim_{x\to 0} f(x,y) = \lim_{y\to 0} \frac{y}{y(1+y)} = 1.$$

$$\lim_{x\to 0}\lim_{y\to 0} f(x,y) = \lim_{x\to 0} \frac{-x(1+x)}{x} = -1.$$

$$\text{Along } y = mx,\ \lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \frac{x(m-1)(1+x)}{x(1+m)(1+mx)} = \frac{m-1}{m+1}.$$

For different values of m, we get the last limit value different. For instance, along the path $y = x$, the limit is 0, whereas along the path $y = 2x$, the limit is $1/3$. So, limit of $f(x,y)$ as $(x,y) \to (0,0)$ does not exist; but the *iterated limits* exist. □

(1.11) ***Example***

Let $f(x,y) = x\sin\dfrac{1}{y} + y\sin\dfrac{1}{x}$ for $x \neq 0,\ y \neq 0$. Then,

$\lim\limits_{y\to 0} f(x,y) = \lim\limits_{y\to 0} x\sin\dfrac{1}{y}$ does not exist. So, $\lim\limits_{x\to 0}\lim\limits_{y\to 0} f(x,y)$ does not exist.

$\lim\limits_{x\to 0} f(x,y) = \lim\limits_{x\to 0} y\sin\dfrac{1}{x}$ does not exist. So, $\lim\limits_{y\to 0}\lim\limits_{x\to 0} f(x,y)$ does not exist.

For the limit of $f(x,y)$ as (x,y) approaches (a,b), let $\epsilon > 0$. Take $\delta = \epsilon/2$. Suppose $0 < \sqrt{x^2+y^2} < \delta$. Then,

$$|f(x,y) - 0| \le |x| + |y| = \sqrt{x^2} + \sqrt{y^2} \le 2\sqrt{x^2+y^2} < 2\delta = \epsilon.$$

Therefore, $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0$.

That is, the two iterated limits do not exist, but the limit exists. □

Hence existence of the limit of $f(x,y)$ as $(x,y) \to (a,b)$ and the existence of two iterated limits have no connection.

Suppose $f(x,y)$ does not depend on y, say $f(x,y) = g(x)$. Does it follow that $\lim\limits_{(x,y)\to(a,b)} f(x,y) = \lim\limits_{x\to a} g(x)$? The answer is 'yes', but with a constraint.

(1.12) ***Theorem***

Let D be a region in the plane, $(a,b) \in \overline{D}$, $g : \{x \in \mathbb{R} : (x,y) \in D\} \to \mathbb{R}$ be a function and let $f : D \to \mathbb{R}$ be a function such that $f(x,y) = g(x)$ for $x \neq a$.

Suppose $\lim_{x\to a} g(x) = \ell$*, and either* $g(x)$ *is not defined for* $x = a$ *or* $g(a) = \ell$*. Then* $\lim_{(x,y)\to(a,b)} f(x,y) = \ell$.

Proof. Let $\epsilon > 0$. We have a $\delta > 0$ such that if $x \in \{x \in \mathbb{R} : (x,y) \in D\}$ and $|x-a| < \delta$, $x \neq a$, then $|g(x) - \ell| < \epsilon$. We plan to use the same δ for proving the conclusion. So, suppose $(x,y) \in D$, $(x,y) \neq (a,b)$ and $\sqrt{(x-a)^2 + (y-b)^2} < \delta$.

Consider two cases:

Case 1: $x \neq a$. Then, $|x-a| \leq \sqrt{(x-a)^2 + (y-b)^2} < \delta$. It implies $|g(x) - \ell| < \epsilon$. That is, $|f(x,y) - \ell| < \epsilon$.

Case 2: $x = a$. Here, $f(x,y) = g(x)$ is defined for $x = a$, and $y \neq b$. We have $|f(x,y) - \ell| = |f(a,y) - \ell| = |g(a) - \ell| = 0 < \epsilon$.

Hence, $\lim_{(x,y)\to(a,b)} f(x,y) = \ell$. ∎

Why do we need an extra condition such as "either $g(x)$ is not defined at $x = a$ or $g(a) = \ell$" in Theorem 1.12? See the following example.

(1.13) *Example*

1. Let $f(x,y) = x$ for $x \neq 0$.

   The domain of $f(x,y)$ is the plane except the y-axis. For $x \neq 0$, $f(x,y) = g(x) = x$. The domain of $g(x)$ is $\{x \in \mathbb{R} : (x,y) \in \mathbb{R}^2, x \neq 0\}$ = the x-axis except the point $x = 0$. The function $g(x)$ is not defined at $x = 0$. By Theorem 1.12, $\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} x = 0$.

   We can verify the limit directly. Let $\epsilon > 0$. Choose $\delta = \epsilon$. Assume that $(x,y) \in \mathbb{R}^2$, $(x,y) \neq (0,0)$ and $\sqrt{(x-0)^2 + (y-0)^2} < \delta$. Then,

   $$|f(x,y) - 0| = |x| \leq \sqrt{x^2 + y^2} < \delta = \epsilon.$$

   Hence, $\lim_{(x,y)\to(0,0)} f(x,y) = 0$.

2. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be such that $f(x,y) = \begin{cases} x & \text{if } x \neq 0 \\ 1 & \text{if } x = 0. \end{cases}$

   For $x \neq 0$, $f(x,y) = g(x) = x$. The domain of $g(x)$ is $\mathbb{R}$. The function $g(x)$ is defined at $x = 0$ with $g(0) = 1$. We have

   $$\lim_{\substack{(x,y)\to(0,0) \\ \text{along } x=0}} f(x,y) = \lim_{y\to 0} f(0,y) = \lim_{y\to 0} 1 = 1.$$

   $$\lim_{\substack{(x,y)\to(0,0) \\ \text{along } y=x}} f(x,y) = \lim_{x\to 0} f(x,x) = \lim_{x\to 0} x = 0.$$

Hence, $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist. However, $\lim_{x\to 0} g(x) = 0$.

The second case does not contradict Theorem 1.12 since $g(x)$ is defined at $x = 0$ but $g(0) \neq 0 = \lim_{x\to 0} g(x)$. □

As the above example shows the condition "either $g(x)$ is not defined at $x = a$ or $g(a) = \lim_{x\to 0} g(x)$" in Theorem 1.12 is essential in concluding that the limit of $f(x,y)$ as (x,y) approaches (a,b) is same as the limit of $g(x)$ as x approaches a.

We will follow an analogous procedure to evaluate the limit of $f(x,y)$ as (x,y) approaches (a,b) when $f(x,y)$ is a function of y alone. In such a case, we must verify the crucial condition that "either $g(y)$ is not defined at $y = b$ or $g(b) = \lim_{y\to b} g(y)$". Then, we would proceed to evaluate the limit as follows:

$$\lim_{(x,y)\to(a,b)} f(x,y) = \lim_{y\to b} g(y).$$

For example, consider $f(x,y) = g(y) = \dfrac{\sin y}{y}$ for $y \neq 0$. The domain of $f(x,y)$ is the plane except the x-axis. The domain of $g(y)$ is $\mathbb{R}\setminus\{0\}$. We see that $\lim_{y\to 0} g(y) = 1$. Since $g(y)$ is not defined at $y = 0$, we have $\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{y\to 0} \dfrac{\sin y}{y} = 1$.

The usual operations of addition, multiplication etc have the expected effects as the following theorem shows. Its proof is analogous to the single variable limits.

(1.14) ***Theorem*** (Algebra of limits)

Let D be a region in the plane, $(a,b) \in \overline{D}$, $f : D \to \mathbb{R}$ be a function and let $L, M, c, r \in \mathbb{R}$. Suppose $\lim_{(x,y)\to(a,b)} f(x,y) = L$ and $\lim_{(x,y)\to(a,b)} g(x,y) = M$. Then the following are true.

(1) **Constant Multiple**: $\lim_{(x,y)\to(a,b)} cf(x,y) = cL$.

(2) **Sum**: $\lim_{(x,y)\to(a,b)} (f(x,y) + g(x,y)) = L + M$.

(3) **Product**: $\lim_{(x,y)\to(a,b)} (f(x,y)\, g(x,y)) = LM$.

(4) **Quotient**: *If $M \neq 0$ and $g(x,y) \neq 0$ in an open disk around the point (a,b), then* $\lim_{(x,y)\to(a,b)} (f(x,y)/g(x,y)) = L/M$

(5) **Power**: *If L^r and $(f(x,y))^r$ are well-defined and $\lim_{(x,y)\to(a,b)} f(x,y) = L$, then* $\lim_{(x,y)\to(a,b)} (f(x,y))^r = L^r$.

1.3 Continuity

Let D be a region in $\mathbb{R}^2$, $(a,b) \in D$ and let $f(x,y)$ be a real valued function defined on D. We say that **$f(x,y)$ is continuous at (a,b)** iff for each $\epsilon > 0$, there exists $\delta > 0$ such that for all points $(x,y) \in D$ with $\sqrt{(x-a)^2+(y-b)^2} < \delta$ we have $|f(x,y) - f(a,b)| < \epsilon$.

We see that $f(x,y)$ is continuous at (a,b) iff $\lim\limits_{(x,y)\to(a,b)} f(x,y) = f(a,b)$.

The function $f(x,y)$ is said to be **continuous on** a subset of D iff $f(x,y)$ is continuous at all points in the subset.

It then follows from the algebra of limits that constant multiples, sum, difference, product, quotient, and rational powers of continuous functions defined on a region are continuous provided they are well defined.

Polynomials in two variables are continuous functions.

Rational functions, i.e., ratios of polynomials, are continuous functions provided they are well defined.

(1.15) ***Example***

Is the function $f(x,y) = \begin{cases} \dfrac{3x^2y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$ continuous on $\mathbb{R}^2$?

At any point other than the origin, $f(x,y)$ is a rational function; therefore, it is continuous. To see that $f(x,y)$ is continuous at the origin, let $\epsilon > 0$ be given. Take $\delta = \epsilon/3$. Assume that $\sqrt{x^2+y^2} < \delta$. Then

$$\left|\frac{3x^2y}{x^2+y^2} - f(0,0)\right| \leq \left|\frac{3(x^2+y^2)y}{x^2+y^2}\right| \leq 3|y| \leq 3\sqrt{x^2+y^2} < 3\delta = \epsilon. \qquad \square$$

(1.16) ***Example***

Is the function $f(x,y) = \begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$ continuous on $\mathbb{R}^2$?

Being a rational function, it is continuous at all nonzero points. For the point $(0,0)$, let $\epsilon > 0$ be given. Choose $\delta = \sqrt{\epsilon}$. Notice that $xy \leq x^2+y^2$ and $x^2 - y^2 \leq x^2+y^2$. For any (x,y) with $\sqrt{x^2+y^2} < \delta$, we have

$$|f(x,y) - 0| = \frac{|x| \cdot |y| \cdot (x^2-y^2)}{x^2+y^2} \leq \frac{(x^2+y^2)(x^2+y^2)}{x^2+y^2} < \delta^2 = \epsilon.$$

Hence $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$.

Therefore, $f(x, y)$ is continuous. □

(1.17) ***Example***

The function $f(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ is continuous on $D = \mathbb{R}^2 \setminus \{(0, 0)\}$.
But it is not continuous at $(0, 0)$ since $(0, 0) \notin D$. Well, we redefine.

Consider the function $g(x, y)$, where

$$g(x, y) = \begin{cases} \dfrac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

By (1.7), $\lim\limits_{(x,y)\to(0,0)} g(x, y)$ does not exist.
Hence $g(x, y)$ is not continuous at $(0, 0)$. □

As in the single variable case, composition of continuous functions is continuous. Try to prove the following result.

(1.18) ***Theorem***

Let D be region in the plane, $f : D \to \mathbb{R}$ be continuous at $(a, b) \in D$ with $f(a, b) = c$ and let $g : I \to \mathbb{R}$ be continuous at $c \in I$ for some interval I in $\mathbb{R}$. Then the function $g(f(x, y))$ from D to $\mathbb{R}$ is continuous at (a, b).

For example,

e^{x-y} is continuous at all points in the plane.

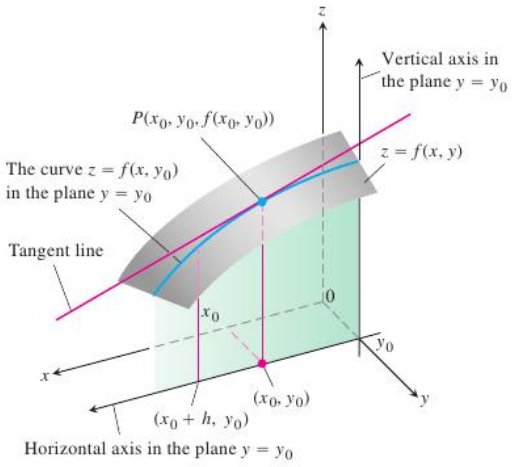
$\cos \dfrac{xy}{1 + x^2}$ and $\ln(1 + x^2 + y^2)$ are continuous on $\mathbb{R}^2$.

$\tan^{-1}(y/x)$ is continuous everywhere except on the y-axis, where it is not defined.

$(x^2 + y^2 + z^2 - 1)^{-1}$ is not continuous on the sphere $x^2 + y^2 + z^2 = 1$, and is continuous everywhere else.

1.4 Partial derivatives

Let $f(x, y)$ be a real valued function defined on a region $D \subseteq \mathbb{R}^2$. Let $(a, b) \in D$. If C is the curve of intersection of the surface $z = f(x, y)$ with the plane $y = b$, then the slope of the tangent line to C at $(a, b, f(a, b))$ is the partial derivative of $f(x, y)$ with respect to x at (a, b). In the figure below, take $x_0 = a, y_0 = b$.



A formal definition of the partial derivative follows.

The **partial derivative of $f(x, y)$ with respect to x at (a, b)** is

$$f_x(a,b) = \frac{\partial f}{\partial x}(a,b) = \left.\frac{df(x,b)}{dx}\right|_{x=a} = \lim_{h\to 0}\frac{f(a+h,b)-f(a,b)}{h},$$

provided this limit exists.
Notice that if $f_x(a, b)$ exists, then $f(x, b)$, as a function of x, is continuous at $x = a$.

The **partial derivative of $f(x, y)$ with respect to y at (a, b)** is

$$f_y(a,b) = \frac{\partial f}{\partial y}(a,b) = \left.\frac{df(a,y)}{dy}\right|_{y=b} = \lim_{k\to 0}\frac{f(a,b+k)-f(a,b)}{k},$$

provided this limit exists. If $f_y(a, b)$ exists, then $f(a, y)$ is continuous at $y = b$.

Remark 1.19 In the definition of $f_x(a, b)$, the expression $\left.\frac{df(x,b)}{dx}\right|_{x=a}$ is to be interpreted as the value of the limit $\lim_{h\to 0}\frac{f(a+h,b)-f(a,b)}{h}$. It is not necessarily equal to "the derivative of $f(x, b)$ for $x \neq a$, taken with respect to x, and then evaluated at $x = a$".

For example, consider $f(x,y) = \begin{cases} 1 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

We have $f_x(0,0) = \lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0}\frac{1}{h}$, which does not exist.

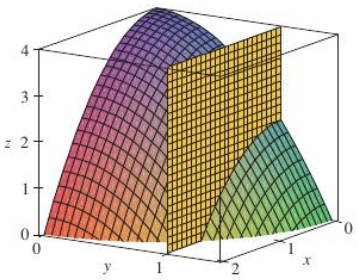
Whereas if wrongly interpreted, then one would find $f'(x, 0) = \frac{d(1)}{dx} = 0$ for $x \neq 0$, and then evaluated at $x = 0$ would give the spurious result 0. Similar comments go for the expression in determining $f_y(a, b)$.

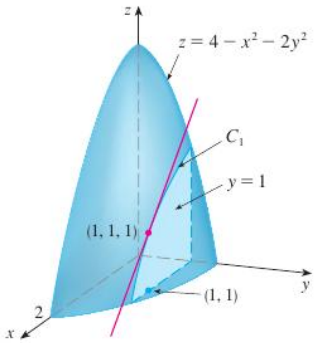
Notice that in the example given above, $f(x, 0)$ as a function of x is not differentiable at $x = 0$.

It can be shown that if the function $f(x,b)$ as a function of x is differentiable at $x = a$, then $f_x(a,b) = \left.\dfrac{df(x,b)}{dx}\right|_{x=a}$. In fact, in most cases we use this to obtain partial derivatives relatives easily without going to the limit definition. It means that we treat y as a constant and differentiate $f(x,y)$ with respect to x, and then evaluate the result at $(x,y) = (a,b)$. This method gives the correct result provided that $f(x,b)$ as a function of x is differentiable at $x = a$.

(1.20) ***Example***

Find $f_x(1,1)$ where $f(x,y) = 4 - x^2 - 2y^2$.





$$f_x(1,1) = \lim_{h\to 0} \frac{(4-(1+h)^2-2)-(4-1-2)}{h} = \lim_{h\to 0} \frac{-2h-h^2}{h} = -2.$$

That is, treat y as a constant and differentiate with respect to x.

$$f_x(1,1) = f_x(x,y)\big|_{(1,1)} = -2x\big|_{(1,1)} = -2.$$

The vertical plane $y = 1$ crosses the paraboloid in the curve $C_1 : z = 2 - x^2, y = 1$. This curve is a parabola. The slope of the tangent line to this parabola at the point $(1,1,1)$ is $f_x(1,1) = -2$. It corresponds to $(x,y) = (1,1)$.

Notice that treating y as a constant, say c, we have $f(x,c) = 4 - x^2 - 2c^2$ is differentiable as a function of x at $x = 1$. □

(1.21) ***Example***

The plane $x = 1$ intersects the surface $z = x^2 + y^2$ in a parabola. Find the slope of the tangent to the parabola at the point $(1,2,5)$.

The asked slope is $\partial z/\partial y$ at $(1,2)$. It is

$$\frac{\partial(x^2+y^2)}{\partial y}(1,2) = (2y)(1,2) = 4.$$

Alternatively, the parabola is $z = x^2 + y^2, x = 1$ OR, $z = 1 + y^2$. So, the slope at $(1,2,5)$ is

$$\left.\frac{dz}{dy}\right|_{y=2} = \left.\frac{d(1+y^2)}{dy}\right|_{y=2} = (2y)|_{y=2} = 4. \qquad \square$$

In general, $f_x(a,b)$ and $\lim\limits_{(x,y)\to(a,b)} f_x(x,y)$ are different. See the following two examples.

(1.22) ***Example***

Let $f(x,y) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \le 0. \end{cases}$

Then $f_x(x,y) = 0$ for all $x > 0$; and $f_x(x,y) = 0$ for all $x < 0$.
So, $\lim\limits_{(x,y)\to(0,0)} f_x(x,y) = 0$. Notice that

$$\lim_{h\to 0-} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{0}{h} = 0, \quad \lim_{h\to 0+} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{1}{h}.$$

The latter limit does not exist. Hence, $f_x(0,0)$ does not exist. $\square$

(1.23) ***Example***

Let $f(x) = \begin{cases} \dfrac{x^3}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$

Then, $f_x(0,0) = \lim\limits_{h\to 0} \dfrac{f(0+h,0) - f(0,0)}{h} = \lim\limits_{h\to 0} \dfrac{h^3}{h^2 \cdot h} = 1.$

For $(x,y) \neq (0,0)$, $f_x(x,y) = \dfrac{(x^2+y^2)\cdot 3x^2 - x^3 \cdot 2x^2}{(x^2+y^2)^2}$. Thus,

$$\lim_{\substack{(x,y)\to(0,0) \\ \text{along } x=0}} f_x(x,y) = \lim_{y\to 0} f_x(0,y) = \lim_{y\to 0} 0 = 0.$$

$$\lim_{\substack{(x,y)\to(0,0) \\ \text{along } y=0}} f_x(x,y) = \lim_{x\to 0} f_x(x,0) = \lim_{x\to 0} \frac{3x^4 - 6x^5}{x^4} = 3.$$

Hence, $\lim\limits_{(x,y)\to(0,0)} f_x(x,y)$ does not exist. $\square$

Remark 1.24 Observe that $f_x(a,b)$ is not the same as $\lim\limits_{(x,y)\to(a,b)} f_x(x,y)$. In fact, $f_x(a,b)$ can exist even if $\lim\limits_{(x,y)\to(a,b)} f_x(x,y)$ does not exist.

Of course, if $f_x(x,y)$ is continuous at (a,b), then $f_x(a,b) = \lim\limits_{(x,y)\to(a,b)} f_x(x,y)$.

For a function of one variable, $f'(t)$ exists at $t = a$ implies that $f(t)$ is continuous at $t = a$. Thus, if $f_x(a,b) = df(x,b)/dx$ at $x = a$ exists, then $f(x,b)$ is continuous at $x = a$. Similarly, if $f_y(a,b) = df(a,y)/dy$ at $y = b$ exists, then $f(a,y)$ is continuous

at $y = b$. Is it true that if both $f_x(x, y)$ and $f_y(x, y)$ exist at (a, b), then $f(x, y)$ is continuous at (a, b)?

(1.25) ***Example***

Consider the function $f(x, y) = \begin{cases} 1 & \text{if } xy \neq 0 \\ 0 & \text{otherwise.} \end{cases}$

We compute its partial derivatives at the origin.

$$f_x(0, 0) = \lim_{h \to 0} \frac{f(0 + h, 0) - f(0, 0)}{h} = \lim_{h \to 0} \frac{0 - 0}{h} = 0,$$

$$f_y(0, 0) = \lim_{k \to 0} \frac{f(0, 0 + k) - f(0, 0)}{h} = \lim_{h \to 0} \frac{0 - 0}{h} = 0.$$

So, both the partial derivatives at $(0, 0)$, that is, $f_x(0, 0)$ and $f_y(0, 0)$ exist.

Let us check whether $f(x, y)$ is continuous at (a, b).

$$\lim_{\substack{(x,y) \to (0,0) \\ \text{along } x=0}} f(x, y) = \lim_{y \to 0} f(0, y) = \lim_{y \to 0} 0 = 0.$$

$$\lim_{\substack{(x,y) \to (0,0) \\ \text{along } y=x}} f(x, y) = \lim_{x \to 0} f(x, x) = \lim_{x \to 0} 1 = 1.$$

Thus, $\lim_{(x,y) \to (0,0)} f(x, y)$ does not exist; $f(x, y)$ is not continuous at $(0, 0)$. □

Thus, even if both $f_x(a, b)$ and $f_y(a, b)$ exist, $f(x, y)$ need not be continuous at (a, b). For another example, look at (1.29).

Remark 1.26 It can be shown that if both $f_x(x, y)$ and $f_y(x, y)$ exist in a disk around (a, b) and at least one of these is continuous at (a, b) (or, bounded in a disk around (a, b)), then $f(x, y)$ is continuous at (a, b).

1.5 The increment theorem

The partial derivative $f_x(a, b)$ is defined as the limit of the ratio of $f(a+h, b) - f(a, b)$ to h as $h \to 0$. Conventionally, this increment h in the independent variable x is written as Δx. The expression $f(a + \Delta x, b) - f(a, b)$ measures the increment in $f(x, y)$ at (a, b) in the direction of x. Similarly, write the increment k in the variable y as Δy. Then, the expression $f(a, b + \Delta y) - f(a, b)$ measures the increment of $f(x, y)$ at (a, b) in the direction of y. The **total increment** in $f(x, y)$ at (a, b) is

then measured by the expression $f(a+\Delta x, b+\Delta y) - f(a,b)$ and it is denoted by $(\Delta f)(a,b)$, or sometimes, by Δf with the understanding that it is taken at the point (a,b). That is,

$$\Delta f = f(a+\Delta x, b+\Delta y) - f(a,b) \quad \text{at the point } (a,b).$$

If the partial derivatives of $f(x,y)$ are continuous, then the total increment Δf can be written in a more suggestive form.

(1.27) ***Theorem*** (Increment Theorem)
Let D be an open region in the plane. Let the function $f : D \to \mathbb{R}$ have continuous partial derivatives. Then f is continuous and the total increment Δf at $(a,b) \in D$ can be written as

$$\Delta f = f_x(a,b)\Delta x + f_y(a,b)\Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y,$$

where $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0,0)$.

Proof. We prove the formula for Δf, from which continuity of $f(x,y)$ follows. Write h instead of Δx and k in place of Δy for better readability. Now,

$$\Delta f := f(a+h, b+k) - f(a+h,b) + f(a+h,b) - f(a,b).$$

By MVT, there exist $c \in [a, a+h]$ and $d \in [b, b+k]$ such that

$$f(a+h,b) - f(a,b) = h[f_x(c,b) - f_x(a,b)] + hf_x(a,b),$$
$$f(a+h,b+k) - f(a+h,b) = k[f_y(a+h,d) - f_y(a,b)] + kf_y(a,b).$$

Write $\epsilon_1 = f_x(c,b) - f_x(a,b)$ and $\epsilon_2 = f_y(a+h,d) - f_y(a,b)$. When $h \to 0$ and $k \to 0$, we see that $c \to a$ and $d \to b$. Since f_x and f_y are assumed to be continuous, we have $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$. Then the total increment can be written as

$$\Delta f = f(a+h,b+k) - f(a,b) = hf_x(a,b) + kf_y(a,b) + \epsilon_1 h + \epsilon_2 k,$$

where $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$ as $(h,k) \to (0,0)$. ∎

Remark 1.28 Let (a,b) be an interior point of a region D. Let $f : D \to \mathbb{R}$ be a function. When the total increment Δf at (a,b) can be written as

$$\Delta f = f_x(a,b)\Delta x + f_y(a,b)\Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y,$$

where $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0,0)$, then we say that f is **differentiable** at (a,b). The following statements give some connections between differentiability, continuity and the partial derivatives.

- Let D be a region in $\mathbb{R}^2$. Let $f : D \to R$ be such that both f_x and f_y exist on D and at least one of them is continuous at an interior point (a, b) of D. Then f is differentiable at (a, b).

- Let D be a region in $\mathbb{R}^2$. Let $f : D \to R$ be differentiable at an interior point (a, b) of D. Then f is continuous at (a, b).

The first statement strengthens the increment theorem. In most of the results that follow, we will assume that both f_x and f_y are continuous. All those results hold true if this assumption is replaced by the weaker assumption that f is differentiable. However, to use differentiability will increase load on terminology, and it is also difficult to verify the condition of differentiability. Instead, we will work with the stronger assumption that f_x and f_y are continuous.

For a function $f(x, y)$, partial derivatives of second order are defined as follows:

$$f_{xx} = (f_x)_x = \frac{\partial}{\partial x}\frac{\partial f}{\partial x} = \frac{\partial^2 f}{\partial x^2}.$$
$$f_{xy} = (f_x)_y = \frac{\partial f_x}{\partial y} = \frac{\partial}{\partial y}\frac{\partial f}{\partial x} = \frac{\partial^2 f}{\partial y \partial x}.$$
$$f_{yx} = (f_y)_x = \frac{\partial f_y}{\partial x} = \frac{\partial}{\partial x}\frac{\partial f}{\partial y} = \frac{\partial^2 f}{\partial x \partial y}.$$
$$f_{yy} = (f_y)_y = \frac{\partial}{\partial y}\frac{\partial f}{\partial y} = \frac{\partial^2 f}{\partial y^2}.$$

Similarly, higher order partial derivatives are defined. For instance,

$$f_{xxy} = \frac{\partial}{\partial y}\frac{\partial}{\partial x}\frac{\partial f}{\partial x} = \frac{\partial^3 f}{\partial y \partial x \partial x}.$$

The following example shows that some of the higher order partial derivatives at a point may exist but others do not.

(1.29) ***Example***

Let $f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$

We have $f(x, 0) = 0 = f(0, y)$. Then,

$$\lim_{\substack{(x,y)\to(0,0) \\ \text{along } x=0}} f(x, y) = \lim_{y\to 0} f(0, y) = 0, \qquad \lim_{\substack{(x,y)\to(0,0) \\ \text{along } y=x}} f(x, y) = \lim_{x\to 0} \frac{x^2}{x^2 + x^2} = \frac{1}{2}.$$

So, limit of $f(x,y)$ as $(x,y) \to (0,0)$ does not exist. Consequently, $f(x,y)$ is not continuous at $(0,0)$. For any x, we have

$$f_x(x,0) = \lim_{h\to 0} \frac{f(x+h,0) - f(x,0)}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0.$$
$$f_{xx}(x,0) = \lim_{h\to 0} \frac{f_x(h,0) - f_x(0,0)}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0.$$

Similarly, it follows that $f_y(0,y) = 0$ and $f_{yy}(0,y) = 0$ for any y. In particular, we get $f_{xx}(0,0) = 0 = f_{yy}(0,0)$. What about $f_{xy}(0,0)$ and $f_{yx}(0,0)$?

$$f_x(0,y) = \lim_{h\to 0} \frac{f(h,y) - f(0,y)}{h} = \lim_{h\to 0} \frac{y}{h^2+y^2} = \frac{1}{y}.$$
$$f_y(x,0) = \lim_{k\to 0} \frac{f(x,k) - f(x,0)}{k} = \lim_{k\to 0} \frac{x}{x^2+k^2} = \frac{1}{x}.$$
$$f_{xy}(0,0) = \lim_{k\to 0} \frac{f_x(0,k) - f_x(0,0)}{k} = \lim_{k\to 0} \frac{1/k - 0}{k} \text{ does not exist.}$$
$$f_{yx}(0,0) = \lim_{h\to 0} \frac{f_y(h,0) - f_y(0,0)}{h} = \lim_{h\to 0} \frac{1/h - 0}{h} \text{ does not exist.}$$ □

(1.30) ***Example***

Consider $f(x,y) = \begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$

$$f(x,0) = f(0,y) = f(0,0) = 0.$$
$$f_x(x,0) = f_y(0,y) = f_{xx}(0,0) = f_{yy}(0,0) = 0.$$
$$f_x(0,y) = \lim_{h\to 0} \frac{f(h,y) - f(0,y)}{h} = -y.$$
$$f_y(x,0) = \lim_{k\to 0} \frac{f(x,k) - f(x,0)}{k} = x.$$
$$f_{xy}(0,0) = \lim_{k\to 0} \frac{f_x(0,k) - f_x(0,0)}{k} = \lim_{k\to 0} \frac{-k}{k} = -1.$$
$$f_{yx}(0,0) = \lim_{h\to 0} \frac{f_y(h,0) - f_y(0,0)}{h} = \lim_{h\to 0} \frac{h}{h} = 1.$$

That is, $f_{xy}(0,0) \neq f_{yx}(0,0)$. □

Thus, even if both f_{xy} and f_{yx} exist at a point, they need not be equal. However, if f_x, f_y, f_{xy} and f_{yx} are continuous, then they are equal.

(1.31) ***Theorem*** (Clairaut)

Let D be an open region in the plane. Let $f : D \to \mathbb{R}$ be a function, where f_x, f_y, f_{xy} and f_{yx} are continuous. Then $f_{xy} = f_{yx}$.

Proof. Assume that f_{xy} and f_{yx} are continuous. Let $(a, b) \in D$. Let $h \neq 0$. Write $g(x) = f(x, b + h) - f(x, b)$ and $\tilde{g}(y) = f(a + h, y) - f(a, y)$. Now,

$$\begin{aligned}\phi(h) := g(a + h) - g(a) &= [f(a + h, b + h) - f(a + h, b)] - [f(a, b + h) - f(a, b)] \\ &= [f(a + h, b + h) - f(a, b + h)] - [f(a + h, b) - f(a, b)] \\ &= \tilde{g}(b + h) - \tilde{g}(b).\end{aligned}$$

Notice that $\phi(h)$ is a function of h. Consider the equality $\phi(h) = g(a + h) - g(a)$. Since f_x is continuous, $g'(x)$ is continuous. By the Mean Value Theorem (MVT), we have c between a and $a + h$ such that

$$\phi(h) = g'(c)h = h[f_x(c, b + h) - f_x(c, b)].$$

Now, $f_x(c, y)$ is a function of y. Since f_{xy} is continuous, the function $f_x(c, y)$ as a function of y, is continuously differentiable. Again, applying MVT on $f_x(c, y)$, we get d between b and $b + h$ such that

$$\phi(h) = h \cdot h \cdot f_{xy}(c, d) = h^2 f_{xy}(c, d).$$

Due to continuity of f_{xy}, we have

$$\lim_{h \to 0} \frac{\phi(h)}{h^2} = \lim_{(c,d) \to (a,b)} f_{xy}(c, d) = f_{xy}(a, b).$$

Similarly, considering the equality $\phi(h) = \tilde{g}(b + h) - \tilde{g}(b)$, we obtain

$$\lim_{h \to 0} \frac{\phi(h)}{h^2} = f_{yx}(a, b).$$

Hence, $f_{xy}(a, b) = f_{yx}(a, b)$. ∎

1.6 Chain rules

Th chain rules are concerned with taking partial derivatives of composite functions. Before attempting this for functions of many variables, let us look at derivatives of composite functions in one variable. Suppose $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are two functions. Informally, we write f as $f(x)$ and g as $g(x)$. Recall that the chain rule for derivative of the composite function $g \circ f : \mathbb{R} \to \mathbb{R}$ is as follows:

$$\frac{dg(f(x))}{dx} = \frac{dg(f(x))}{df(x)} \frac{df(x)}{dx}.$$

Writing $y = f(x)$, we also informally write the chain rule as

$$\frac{dg}{dx} = \frac{dg}{dy}\,\frac{dy}{dx}.$$

In this notation, the g on the LHS is interpreted as a function of x, namely, the composite function $g(f(x))$ while on the RHS, the g is interpreted as a function of y, namely, $g(y)$. For example, suppose f is given by $f(x) = 2x$ and g is given by $g(x) = x^2$. Then, $y = f(x) = 2x$, $g(y) = y^2$, and $g(f(x)) = (2x)^2 = 4x^2$. The above equality reads as follows:

$$\frac{d(4x^2)}{dx} = \frac{dy^2}{dy}\,\frac{dy}{dx} = (2y)(2x) = 2(2x)(2x) = 8x^2.$$

we must remember the above correct interpretation of g, which happens to be different on LHS from that on the RHS.

In case of taking partial derivatives of functions of two or more variables similar interpretation is required when composition of functions is involved. Consider the functions $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = x + 2y$ and $g : \mathbb{R}^3 \to \mathbb{R}$ given by $g(x, y, z) = 3x + 4y + 5z$, where $z = f(x, y) = x + 2y$. As such, we see that $g_x = 3$. However, with $z = x + 2y$, we have

$$g(x, y, z) = 3x + 4y + 5z = 3x + 4y + 5(x + 2y) = 8x + 14y \Rightarrow g_x = 8.$$

Look at the meaning underlying these computations. In the first case, we are considering g as a function of three variables whereas in the second case, we are considering g as the function given by $g(x, y, f(x, y))$, which is a function of two variables. Of course, $g : \mathbb{R}^3 \to \mathbb{R}$ is a function of three variables. Whereas the function given by $g(x, y, f(x, y))$ is a function from $\mathbb{R}^2$ to $\mathbb{R}$; it is not same as g. It is some other function, the composition of g with f taken in this particular way. It is a misuse of notation to write the function from $\mathbb{R}^2$ to $\mathbb{R}$ given by $g(x, y, f(x, y))$ as g. However, we continue misusing the notation for our convenience, but be aware of the correct interpretation.

With this in the background we now discuss partial differentiation of composite functions. Our main tool is the increment theorem. Further, we go ahead with informal formulations of our results without talking about the domains and co-domains of functions involved. You are supposed to augment the results with appropriate background information so that the compositions make sense.

Moreover, we formulate and discuss our results for a function $f(x, y)$ of two variables. Analogously, all these notions and results can be formulated and proved for a function $f(x_1, \ldots, x_n)$ of n variables for $n > 2$.

(1.32) ***Theorem*** (Chain Rule 1)
Let $f(x,y)$ have continuous partial derivatives. If $x = x(t)$ and $y = y(t)$ are differentiable functions of t, then

$$\frac{df\big(x(t),y(t)\big)}{dt} = \frac{\partial f(x,y)}{\partial x}\frac{dx}{dt} + \frac{\partial f(x,y)}{\partial y}\frac{dy}{dt}.$$

Proof. Using the increment theorem (1.27) at a point P we obtain

$$\frac{\Delta f}{\Delta t} = \frac{\partial f}{\partial x}\frac{\Delta x}{\Delta t} + \frac{\partial f}{\partial y}\frac{\Delta y}{\Delta t} + \epsilon_1\frac{\Delta x}{\Delta t} + \epsilon_2\frac{\Delta y}{\Delta t}.$$

As $\Delta t \to 0$, we have $(\Delta x, \Delta y) \to (0,0)$ so that $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$. Then the result follows. ∎

You may remember the above formula as

$$\frac{df}{dt} = f_x\frac{dx}{dt} + f_y\frac{dy}{dt}.$$

However, you must correctly interpret it. The f on the LHS is the composite function $f\big(x(t),y(t)\big)$, which is a function of t alone, and the f on the RHS is $f(x,y)$, which is a function of x and y.

For example, if $z = xy$ and $x = \sin t,\ y = \cos t$, then

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}x'(t) + \frac{\partial z}{\partial y}y'(t) = yx'(t) + xy'(t) = \cos^2 t - \sin^2 t.$$

Check: $z(t) = \sin t\cos t = \frac{1}{2}\sin 2t$. So, $\dfrac{dz}{dt} = \cos 2t = \cos^2 t - \sin^2 t$.

(1.33) ***Theorem*** (Chain Rule 2)
Let $f(x,y)$, $x(s,t)$ and $y(s,t)$ have continuous partial derivatives. Then

$$\frac{\partial f\big(x(s,t),y(s,t)\big)}{\partial s} = \frac{\partial f(x,y)}{\partial x}\,\frac{\partial x(s,t)}{\partial s} + \frac{\partial f(x,y)}{\partial y}\,\frac{\partial y(s,t)}{\partial s},$$
$$\frac{\partial f\big(x(s,t),y(s,t)\big)}{\partial t} = \frac{\partial f(x,y)}{\partial x}\,\frac{\partial x(s,t)}{\partial t} + \frac{\partial f(x,y)}{\partial y}\,\frac{\partial y(s,t)}{\partial t}.$$

Its proof is similar to that of (1.33). In subscript notation, it looks like

$$f_s = f_x x_s + f_y y_s, \quad f_t = f_x x_t + f_y y_t.$$

But you have been warned! The f on the LHS is $f\big(x(s,t),y(s,t)\big)$, which is a function of s,t, and the f on the RHS is $f(x,y)$, which is a function of x,y.

(1.34) ***Example***

Let $z = e^x \sin y,\ x = st^2,\ y = s^2t$. Then

$$\frac{\partial z}{\partial s} = (e^x \sin y)t^2 + (e^x \cos y)2st = te^{st^2}(t\sin(s^2t) + 2s\cos(s^2t)).$$

$$\frac{\partial z}{\partial t} = (e^x \sin y)2st + (e^x \cos y)s^2 = se^{st^2}(2t\sin(s^2t) + s\cos(s^2t)).$$

Notice that the above partial derivatives are obtained for the function $z = z(s,t)$ and not for the function $z = z(x,y)$. Of course, on the RHS, we had taken the partial derivatives of $z = z(x,y)$ as the chain rule requires. Substitute expressions for x and y to get $z = z(s,t)$ and then check that the results are correct. □

Sometimes we write $z = z(x,y)$ to mean that $z = f(x,y)$, where f is a function from a suitable region in $\mathbb{R}^2$ to $\mathbb{R}$.

(1.35) ***Example***

Suppose $z = z(x,y)$ has continuous second order partial derivatives and that $x = r^2 + s^2$ and $y = 2rs$. Find z_{rr}. Again, we are required to find the second order partial derivative of the function $z = z(r,s)$ with respect to r.

$$\begin{aligned}
z_r &= 2rz_x + 2sz_y.\\
z_{xr} &= z_{xx}x_r + z_{xy}y_r = 2rz_{xx} + 2sz_{xy}.\\
z_{yr} &= z_{yx}x_r + z_{yy}y_r = 2rz_{yx} + 2sz_{yy}.\\
z_{rr} &= \frac{\partial z_r}{\partial r} = \frac{\partial}{\partial r}(2rz_x + 2sz_y) = 2z_x + 2rz_{xr} + 2sz_{yr}\\
&= 2z_x + 2r(2rz_{xx} + 2sz_{xy}) + 2s(2rz_{yx} + 2sz_{yy})\\
&= 2z_x + 4r^2z_{xx} + 8rsz_{xy} + 4s^2z_{yy}.
\end{aligned}$$

□

Sometimes we will require differentiating a function given implicitly. Suppose a function $F(x,y,z)$ is defined on an open region D in $\mathbb{R}^3$ containing a point (a,b,c), where $F(a,b,c) = 0,\ F_z(a,b,c) \neq 0$, and F_x, F_y, F_z are continuous, then the equation $F(x,y,z) = 0$ defines a function $z = f(x,y)$ in an open disk around (a,b) in the plane. Moreover, the function $z = f(x,y)$ can now be differentiated partially with

$$z_x = -F_x/F_z,\ z_y = -F_y/F_z.$$

It is easier to differentiate implicitly than remembering the formula.

(1.36) ***Example***

Find z_x and z_y if $z = z(x,y)$ is given by $x^3 + y^3 + z^3 + 6xyz = 1$.

We differentiate 'the equation' with respect to x and y as follows:

$$3x^2 + 3z^2 z_x + 6y(z + xz_x) = 0 \Rightarrow z_x = -\frac{(x^2 + 2yz)}{z^2 + 2xy}.$$
$$3y^2 + 3z^2 z_y + 6x(z + xz_y) = 0 \Rightarrow z_y = -\frac{(y^2 + 2xz)}{z^2 + 2xy}.$$

Of course, this is similar to implicit differentiation of functions of one variable. □

(1.37) ***Example***

Find w_x if $w = x^2 + y^2 + z^2$ and $x^2 + y^2 - z = 0$.

Here, the question is open to at least two interpretations depending on whether to consider the second equation or not.

Considering $w = w(x, y, z)$, a function of the variables x, y, z, we have $w_x = 2x$.

If we consider the other equation $x^2 + y^2 - z = 0$, then one of the variables x, y, z is possibly a function of the other two in an appropriate domain. In that case, we are asked to find the partial derivative of the composite function w with respect to x. Again, this is open to various interpretations depending on which one among x, y, z depends on the other two.

Suppose the equation $x^2 + y^2 - z = 0$ defines a function $z = z(x, y)$ in some domain. That is, z is the dependent variable and x, y are independent variables. The composite function w is a function of x, y; it is $w = w\big(x, y, z(x, y)\big)$. Here, the equation that defines $z = z(x, y)$ gives $z = x^2 + y^2$ so that $w = w(x, y) = x^2 + y^2 + (x^2 + y^2)^2$. Thus,

$$w_x = w_x(x, y) = 2x + 4x^3 + 4xy^2.$$

Also, by the chain rule,

$$\begin{aligned} w_x = w_x\big(x, y, z(x, y)\big) &= w_x(x, y, z) + w_z(x, y, z)z_x \\ &= 2x + 2z\,2x = 2x + (x^2 + y^2)(2x) = 2x + 4x^3 + 4xy^2. \end{aligned}$$

We also consider other interpretations.

If y is the dependent variable and x, z are independent variables, then $y^2 = z - x^2$ so that $w = w(x, z) = x^2 + (z - x^2) + z^2 = z + z^2$. Then $w_x = 0$.

If x is the dependent variable and y, z are independent variables, then $w = w(y, z)$. We see that w_x has no meaning. □

The correct formulation of the question in (1.37) needs to specify two things. Firstly, whether w_x is the partial derivative of the function $w(x, y, z)$ disregarding the other given equation, or whether w_x is the partial derivative of $w(x, y, z)$ where the other equation might define one or many variables as function(s) of other variables.

Notice that the latter case shows up w as a composite function. Secondly, in this latter case, it must be specified which of the variables is dependent and which ones are independent so that the other equation is correctly interpreted to give rise to a function.

If nothing is specified, then the question can be interpreted in many ways, and the answer may depend on the particular interpretation.

As a default rule, when a condition such as the second equation in (1.37) is given, one is supposed to differentiate (partially) the composite function provided that taking such a derivative is meaningful.

(1.38) ***Example***

Find z_x, where $z = 2x^2 - 3y^3$, $x = s^2t$ and $y = st^2$.

Here, it is not clear whether the function $z = z(x, y)$ or $z = z\big(x(s,t), y(s,t)\big)$ is to be differentiated partially with respect to x. However, partial derivative of the composite function $z = z\big(x(s,t), y(s,t)\big)$ with respect to x is not meaningful since x is no more in z. Hence, following the default rule, we ignore the second condition and get $z_x = \partial(2x^2 - 3y^3)/\partial x = 4x$. □

(1.39) ***Example***

Given that $w = x^2 + y^2 + z^2$ and $z = z(x, y)$ satisfies $z^3 - xy + yz + y^3 = 1$, evaluate w_x at $(2, -1, 1)$.

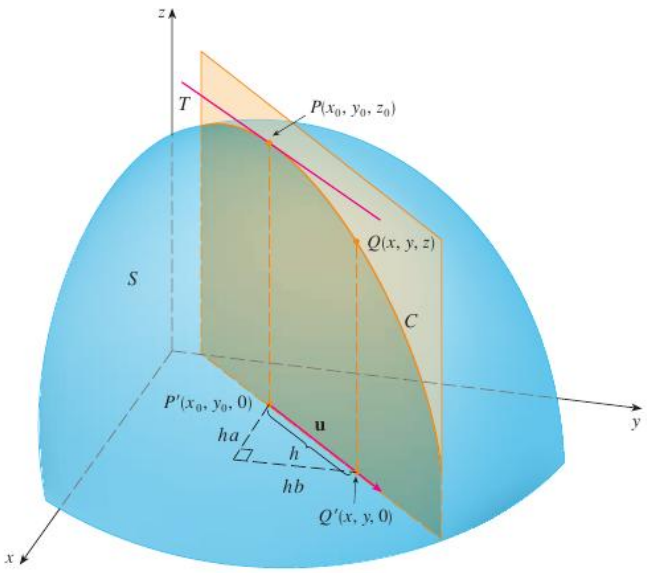
Here, z is given to be a dependent variable and x, y are independent variables. Then, w as a function of x, y is $w = w(x, y, z(x, y))$. This is the intended meaning. We use the chain rule to differentiate the given equations partially with respect to x to obtain

$$w_x = 2x + 2zz_x, \quad 3z^2z_x - y + yz_x = 0.$$

These two together imply $w_x = 2x + \dfrac{2yz}{y + 3z^2}$.
Evaluating it at $(2, -1, 1)$ gives $w_x(2, -1, 1) = 3$. □

1.7 Directional derivative

Recall that if $f(x, y)$ is a function, then $f_x(x_0, y_0)$ is the rate of change in f with respect to change in x, at (x_0, y_0), that is, in the direction $\hat{\imath}$. Similarly, $f_y(x_0, y_0)$ is the rate of change at (x_0, y_0) in the direction $\hat{\jmath}$. How do we find the rate of change of $f(x, y)$ at (x_0, y_0) in the direction of any unit vector $\hat{u}$ in the xy-plane?



Consider a smooth surface S given by the equation $z = f(x, y)$. Let $z_0 = f(x_0, y_0)$. The point $P(x_0, y_0, z_0)$ lies on S. Let $\hat{u}$ be a unit vector in the xy-plane. Translating $\hat{u}$ suitably, we may assume that $\hat{u}$ contains the point P. The vertical plane that passes through P and contains the vector $\hat{u}$ intersects S in a curve C. The slope of the tangent line T to the curve C at the point P is the rate of change of z in the direction of $\hat{u}$.

Convention: A vector is written as $\vec{u}$ whereas a unit vector is written as $\hat{u}$. In the context of the directional derivative $D_u z$, the vector $\hat{u}$ is taken as a unit vector.

Let $f(x, y)$ be a function defined in an open region D. Let $(x_0, y_0) \in D$. The **directional derivative** of $f(x, y)$ in the direction of a unit vector $\hat{u} = a\hat{\imath} + b\hat{\jmath}$ at (x_0, y_0) is given by

$$(D_u f)(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}.$$

(1.40) *Example*

Find the derivative of $z = x^2 + y^2$ at $(1, 2)$ in the direction $\hat{\imath} + \hat{\jmath}$.

The unit vector in the given direction is $\hat{u} = \frac{\hat{\imath}}{\sqrt{2}} + \frac{\hat{\jmath}}{\sqrt{2}}$. Then,

$$(D_u z)(1, 2) = \lim_{h \to 0} \frac{f\left(1 + \frac{h}{\sqrt{2}}, 2 + \frac{h}{\sqrt{2}}\right) - f(1, 2)}{h} = \lim_{h \to 0} \frac{\frac{2h}{\sqrt{2}} + 2 \cdot \frac{2h}{\sqrt{2}}}{h} = 3\sqrt{2}.$$

Notice that $f_x(1, 2) \cdot \frac{1}{\sqrt{2}} + f_y(1, 2) \cdot \frac{1}{\sqrt{2}} = (2 + 2 \cdot 2) \cdot \frac{1}{\sqrt{2}} = 3\sqrt{2}$. □

The following result shows how to compute the directional derivative from the partial derivatives in the direction of axes.

(1.41) ***Theorem***

Let D be an open region in the plane. Let $f : D \to \mathbb{R}$ have continuous partial derivatives. Then f has a directional derivative at each $(x, y) \in D$ in any direction $\hat{u} = a\hat{\imath} + b\hat{\jmath}$, and it is given by

$$(D_u f)(x, y) = f_x(x, y)a + f_y(x, y)b.$$

Proof. Due to our convention, it is assumed that $\hat{u}$ is a unit vector.

Let $(x_0, y_0) \in D$. Define the function $g : \mathbb{R} \to \mathbb{R}$ by $g(h) = f(x_0 + ah, y_0 + bh)$. Then $g(h)$ is a continuously differentiable function of h. Now,

$$(D_u f)(x_0, y_0) = \lim_{h\to 0} \frac{f(x_0 + ah, y_0 + bh) - f(x_0, y_0)}{h} = \lim_{h\to 0} \frac{g(h) - g(0)}{h} = g'(0).$$

Using the Chain rule and continuity of $f_x,\ f_y$, we have

$$g'(h) = f_x \frac{dx}{dh} + f_y \frac{dy}{dh} = f_x\, a + f_y\, b \Rightarrow g'(0) = f_x(x_0, y_0)\, a + f_y(x_0, y_0)\, b.$$

Therefore, $(D_u f)(x_0, y_0) = f_x(x_0, y_0)a + f_y(x_0, y_0)b$. ∎

(1.42) ***Example***

Find the directional derivative of $f(x, y) = x^3 - 3xy + 4y^2$ in the direction of the line that makes an angle of $\pi/6$ with the x-axis.

The direction is given by the unit vector $\hat{u} = \cos\left(\frac{\pi}{6}\right)\hat{\imath} + \sin\left(\frac{\pi}{6}\right)\hat{\jmath} = \frac{\sqrt{3}}{2}\hat{\imath} + \frac{1}{2}\hat{\jmath}$. Thus

$$\begin{aligned}(D_u f)(x, y) &= \frac{\sqrt{3}}{2} f_x + \frac{1}{2} f_y = \frac{\sqrt{3}}{2}\big(3x^2 - 3y\big) + \frac{1}{2}\big(-3x + 8y\big) \\ &= \frac{1}{2}\Big[3\sqrt{3}x^2 - 3x + (8 - 3\sqrt{3})y\Big].\end{aligned}$$

□

The formula for the directional derivative in the direction of the unit vector $\hat{u} = a\hat{\imath} + b\hat{\jmath}$ can be written as

$$D_u f = f_x a + f_y b = (f_x\hat{\imath} + f_y\hat{\jmath}) \cdot (a\hat{\imath} + b\hat{\jmath}).$$

The vector operator $\nabla := \dfrac{\partial}{\partial x}\hat{\imath} + \dfrac{\partial}{\partial y}\hat{\jmath}$ is called the **gradient** and the **gradient of** $f(x, y)$ is

$$\nabla f := \operatorname{grad} f := \frac{\partial f}{\partial x}\hat{\imath} + \frac{\partial f}{\partial y}\hat{\jmath}.$$

Therefore, $D_u f = \nabla f \cdot \hat{u}$. That is, at (x_0, y_0), the directional derivative is given by

$$(D_u f)(x_0, y_0) = (\nabla f)(x_0, y_0) \cdot \hat{u}.$$

Caution: This formula has been derived under the assumptions that f_x and f_y are continuous at (x_0, y_0), and $\hat{u}$ is a unit vector.

(1.43) *Example*

Find the directional derivative of $f(x, y) = xe^y + \cos(xy)$ in the direction of the vector $3\hat{\imath} - 4\hat{\jmath}$ at $(2, 0)$.

Here $\hat{u} = \frac{3}{5}\,\hat{\imath} - \frac{4}{5}\,\hat{\jmath}$. The partial derivatives of f are continuous. We have

$$\nabla f = f_x\hat{\imath} + f_y\hat{\jmath} = \big(e^y - y\sin(xy)\big)\hat{\imath} + \big(xe^y - x\sin(xy)\big)\hat{\jmath}.$$
$$(\nabla f)(2, 0) = 1 \cdot \hat{\imath} + 2 \cdot \hat{\jmath}.$$

Thus, the directional derivative of f in the direction of $3\hat{\imath} - 4\hat{\jmath}$ at $(2, 0)$ is

$$(D_u f)(2, 0) = (\nabla f)(2, 0) \cdot \left(\tfrac{3}{5}\,\hat{\imath} - \tfrac{4}{5}\,\hat{\jmath}\right) = (\hat{\imath} + 2\hat{\jmath}) \cdot \left(\tfrac{3}{5}\,\hat{\imath} - \tfrac{4}{5}\,\hat{\jmath}\right) = \tfrac{3}{5} - \tfrac{8}{5} = -1. \qquad \square$$

The directional derivative can be used to approximate a function at a neighboring point. Assume that h is (intuitively) small. Let $\hat{u} = a\hat{\imath} + b\hat{\jmath}$ be a unit vector. Then, $(x_0 + ah, y_0 + bh)$ is taken as a point close to (x_0, y_0) in the direction of $\hat{u}$. The rate of change in $f(x, y)$ at (x_0, y_0) in the direction of $\hat{u}$ is given by the directional derivative $(D_u f)(x_0, y_0)$. Thus the change is approximately equal to $h\,(D_u f)(x_0, y_0)$. We get the approximation

$$f(x_0 + ah, y_0 + bh) \approx f(x_0, y_0) + h\,(D_u f)(x_0, y_0).$$

(1.44) *Example*

How much the value of $y\sin x + 2yz$ change if the point (x, y, z) moves 0.1 units from $(0, 1, 0)$ toward $(2, 2, -2)$?

Let $f(x, y, z) = y\sin x + 2yz$. $P(0, 1, 0)$, $Q(2, 2, -2)$. $\vec{u} = \overrightarrow{PQ} = 2\hat{\imath} + \hat{\jmath} - 2\hat{k}$. The unit vector in the direction of $\vec{u}$ is $\hat{u} = \frac{1}{3}\,\vec{v}$. We find $D_u f$ at P which requires ∇f.

$$\nabla f = (y\cos x)\hat{\imath} + (\sin x + 2z)\hat{\jmath} + 2y\,\hat{k}.$$

Then $(D_u f)(P) = (\nabla f)(0, 1, 0) \cdot \vec{u} = (\hat{\imath} + 2\hat{k}) \cdot \left(\frac{2}{3}\hat{\imath} + \frac{1}{3}\hat{\jmath} - \frac{2}{3}\hat{k}\right) = -\frac{2}{3}$.

The change of $f(x, y, z)$ in the direction of $\vec{u}$ in moving 0.1 units is approximately

$$\Delta f \approx D_u(P) \times 0.1 = -\tfrac{2}{3}\,(0.1) = -0.067 \text{ units.} \qquad \square$$

The formula $D_u f = \nabla f \cdot \hat{u}$ means that the directional derivative is the length of the projection of the gradient in the direction of $\hat{u}$.

(1.45) *Theorem*

Let D be an open region in the plane. Let the function $f : D \to \mathbb{R}$ have continuous partial derivatives. Then the maximum value of the directional derivative $(D_u f)(x, y)$ is $|\nabla f|$, and it is achieved in the direction of ∇f.

Proof. Let $(x, y) \in D$. At the point (x, y) we have

$$D_u f \leq |D_u f| = |\nabla f \cdot \hat{u}| = |\nabla f|\,|\hat{u}|\,\cos\theta = |\nabla f|\cos\theta,$$

where θ is the angle between ∇f and $\hat{u}$. Since maximum of $\cos\theta$ is 1, maximum of $D_u f$ is $|\nabla f|$. The maximum is achieved when $\theta = 0$, that is, when the directions of ∇f and $\hat{u}$ coincide. ∎

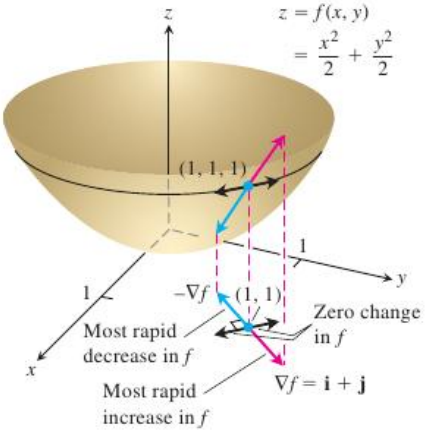
From (1.45) we observe the following:

$f(x, y)$ increases most rapidly in the direction of its gradient.

$f(x, y)$ decreases most rapidly in the opposite direction of its gradient.

(1.46) ***Example***

Find the directions in which the function $f(x, y) = \frac{1}{2}(x^2 + y^2)$ changes most, least, and not at all, at the point $(1, 1)$.



Note that when we ask for a direction, we mean a unit vector. Now,

$$\nabla f = f_x \hat{\imath} + f_y \hat{\jmath} = x\hat{\imath} + y\hat{\jmath}. \quad (\nabla f)(1, 1) = \hat{\imath} + \hat{\jmath}.$$

Thus the function $f(x, y)$ increases most at $(1, 1)$ in the direction $\frac{\hat{\imath}}{\sqrt{2}} + \frac{\hat{\jmath}}{\sqrt{2}}$. It decreases most at $(1, 1)$ in the direction $-\frac{\hat{\imath}}{\sqrt{2}} - \frac{\hat{\jmath}}{\sqrt{2}}$. □

In higher dimensions, if $f(x_1, \ldots, x_n)$ is a function of n independent variables defined on $D \subseteq \mathbb{R}^n$, then its gradient at any point is

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right).$$

The directional derivative at any point $\vec{x}$ in the direction of a unit vector $\hat{u} = (u_1, \ldots, u_n)$ is

$$D_u f = \lim_{h \to 0} \frac{f(\vec{x} + h\hat{u}) - f(\vec{x})}{h} = \nabla f \cdot \hat{u} = f_{x_1} u_1 + \cdots + f_{x_n} u_n$$

provided the partial derivatives $f_{x_1}, \ldots, f_{x_n}$ are continuous.

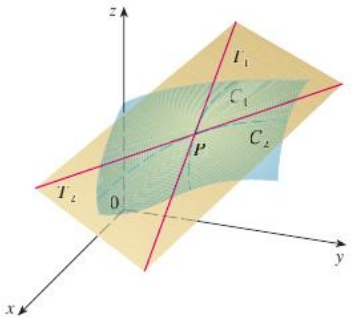
The algebraic rules for the gradient are as follows:

1. Constant multiple: $\nabla(kf) = k\,\nabla f$ for $k \in \mathbb{R}$.
2. Sum: $\nabla(f+g) = \nabla f + \nabla g$.
3. Difference: $\nabla(f-g) = \nabla f - \nabla g$.
4. Product: $\nabla(fg) = f\,\nabla g + g\,\nabla f$.
5. Quotient: $\nabla\left(\dfrac{f}{g}\right) = \dfrac{g\,\nabla f - f\,\nabla g}{g^2}$.

1.8 Tangent planes and normal lines

Partial derivatives can be used to derive an equation of the tangent plane to a surface at a point. Let S be a *smooth* surface given by the equation $z = f(x,y)$ for $(x,y) \in D$, an open region in the plane. Here, S is smooth means that both f_x and f_y are continuous on D.

Let $(a,b) \in D$. Write $c = f(a,b)$. The point $P = (a,b,c)$ lies on S. The **tangent plane** to S at P consists of the tangent lines at P to all possible curves that lie on S and pass through P. This plane approximates S at P most closely.



Let C_1 be the curve of intersection of the plane $y = b$ with S and let T_1 be the tangent line to the curve C_1 at the point $P = (a,b,c)$. Similarly, let T_2 be the tangent line at P to the curve C_2 obtained by the intersection of the plane $x = a$ with S. Then, the tangent plane to S at P is the plane that contains the lines T_1 and T_2.

For the equation of the tangent plane, notice that any plane passing through $P = (a,b,c)$ is given by $z - c = A(x-a) + B(y-b)$. When $y = b$, this tangent plane gives us the tangent line T_1 with slope as A. That is, $A = f_x(a,b)$. Similarly, $B = f_y(a,b)$, the slope of the tangent line T_2. Hence the equation of the tangent plane to the smooth surface S given by $z = f(x,y)$ at the point (a,b,c) is

$$z - c = f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

Recall that the **normal line** to the surface $z = f(x, y)$ at (a, b, c) is the line that passes through the point (a, b, c) and is orthogonal to the tangent plane at the same point. Then the normal line has direction ratios $f_x(a, b),\ f_y(a, b),\ -1$. Thus the equation of the normal line to $z = f(x, y)$ at $(a, b, f(a, b))$ in parametric form is

$$x = a + f_x(a, b)\, t,\ y = b + f_y(a, b)\, t,\ z = f(a, b) - t.$$

(1.47) ***Example***

Find the equation of the tangent plane and the normal line to the elliptic paraboloid $z = 2x^2 + y^2$ at $(1, 1, 3)$.

Here, $z_x = 4x, z_y = 2y$. So, $z_x(1, 1) = 4, z_y(1, 1) = 2$. Then the equation of the tangent plane is $z - 3 = 4(x - 1) + 2(y - 1)$. It simplifies to $z = 4x + 2y - 3$.

The equation of the normal line is $x = 1 + 4t,\ y = 1 + 2t,\ z = 3 - t$. □

An alternative derivation of the equations of tangent planes and normal lines uses the geometric meaning of the gradient. Let $z = f(x, y)$ be a given surface. Assume that f_x and f_y are continuous. Recall that a level curve to this surface is a curve in the plane where $f(x, y)$ is a constant. Fix some constant c in the range of f. On the corresponding level curve, $f(x, y)$ takes the constant value c. Suppose $\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath}$ is a parametrization of this level curve. For each point on this level curve, $f(x, y) = c$, that is, $f\big(x(t), y(t)\big) = c$ for all t. Differentiating, we have $\frac{d}{dt} f\big(x(t), y(t)\big) = 0$. Or,

$$f_x \frac{dx}{dt} + f_y \frac{dy}{dt} = \nabla f \cdot \frac{d\,\vec{r}(t)}{dt} = 0.$$

We also write $d\,\vec{r}/dt$ as $\vec{r}\,'(t)$. Since $\vec{r}\,'(t)$ is the tangent to the curve, ∇f is the normal to the level curve, at any point. We thus obtain the following:

> Let D be an open region in the plane, $(a, b) \in D$ and let $f : D \to \mathbb{R}$ have continuous partial derivatives at (a, b). If $(\nabla f)(a, b) \neq 0$, then $(\nabla f)(a, b)$ is the direction of the normal to the level curve $f(x, y) = f(a, b)$ at (a, b).

To determine the tangent planes and normal lines to a surface at a point $P = (a, b, c)$ defined implicitly, say, by $f(x, y, z) = 0$, we consider the function $w = f(x, y, z)$, whose graph is in $\mathbb{R}^4$. Suppose f_x, f_y, f_z are continuous in a **ball** around P. A ball around P is a set of the form $\{(x, y, z) \in \mathbb{R}^3 : (x-a)^2+(y-b)^2+(z-c)^2 < \epsilon^2\}$ for some $\epsilon > 0$. The surface $f(x, y, z) = 0$ is a level surface of the function $w = f(x, y, z)$. Let $\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k}$ be a parametrization of a smooth curve on the level surface $f(x, y, z) = 0$ passing through P. Then $f\big(x(t), y(t), z(t)\big) = 0$ for all t. Differentiating this we get $df/dt = 0$. By the Chain rule, it gives

$$\nabla f \cdot \vec{r}\,'(t) = 0.$$

Look at all such smooth curves that pass through P on the level surface. The above equation asserts that the velocity vectors $\vec{r}\,'(t)$ at P to all these smooth curves are orthogonal to the gradient at P. That is, the direction of the normal line to the surface $f(x, y, z) = 0$ at P is $(\nabla f)(P)$. Then, the equation of the **normal line** to the surface $f(x, y, z) = 0$ at $P = (a, b, c)$ in parametric form is

$$x = a + f_x(a, b, c)\, t,\ y = b + f_y(a, b, c)\, t,\ z = c + f_z(a, b, c)\, t.$$

The **tangent plane** at $P = (a, b, c)$ on the level surface $f(x, y, z) = 0$ is the plane through P which is orthogonal to ∇f at P. Then, its equation is

$$f_x(a, b, c)(x - a) + f_y(a, b, c)(y - b) + f_z(a, b, c)(z - c) = 0.$$

If the surface is given in the form $z = f(x, y)$, then write $F(x, y, z) = f(x, y) - z$; the surface is given by $F(x, y, z) = 0$. Now, $F_x = f_x$, $F_y = f_y$, $F_z = -1$. Using the equations derived in case of a surface given in implicit form, we see that the equation of the normal line is

$$x = a + f_x(a, b)\, t,\ y = b + f_y(a, b)\, t,\ z = c - t.$$

Similarly, the equation of the tangent plane is

$$f_x(a, b)(x - a) + f_y(a, b)(y - b) - (z - f(a, b)) = 0$$

as obtained earlier.

(1.48) *Example*

Find the tangent plane and the normal line of the surface $x^2 + y^2 + z = 9$ at the point $(1, 2, 4)$.

Here, $f(x, y, z) = x^2 + y^2 + z - 9$. First, check that the point $(1, 2, 4)$ lies on the surface. Next,

$$f_x(1, 2, 4) = 2, \quad f_y(1, 2, 4) = 4, \quad f_z(1, 2, 4) = 1.$$

The tangent plane at $(1, 2, 4)$ is given by

$$2(x - 1) + 4(y - 2) + (z - 4) = 0.$$

The normal line at $(1, 2, 4)$ is given by

$$x = 1 + 2t,\ y = 2 + 4t,\ z = 4 + t.$$ □

(1.49) *Example*

Find the tangent line to the curve of intersection of the surfaces $f(x, y, z) = 0$ and $g(x, y, z) = 0$ at the point $(1, 1, 3)$, where

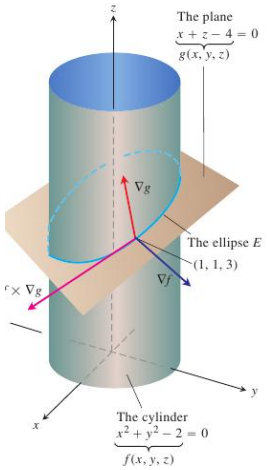
$$f(x, y, z) = x^2 + y^2 - 2 = 0, \quad g(x, y, z) = x + z - 4 = 0.$$

The tangent line lies in the tangent plane to $f(x,y,z) = 0$ and also it lies in the tangent plane to $g(x,y,z) = 0$, at $(1,1,3)$. So, the tangent line is orthogonal to both ∇f and ∇g at $(1,1,3)$. Thus it is parallel to

$$\nabla f \times \nabla g = (2\hat{\imath} + 2\hat{\jmath}) \times (\hat{\imath} + \hat{k}) = 2\hat{\imath} - 2\hat{\jmath} - 2\hat{k}.$$

Then the equation of the tangent line is

$$x = 1 + 2t,\ y = 1 - 2t,\ z = 3 - 2t.$$



□

1.9 Taylor's formulas

For a function of one variable, a polynomial approximation is given by Taylor's formula. It has two forms: integral form and differential form. The differential form is a generalization of the Mean Value Theorem for differentiable functions.

(1.50) ***Theorem*** (Taylor's Formula)
Let $n \in \mathbb{N} \cup \{0\}$ and let $f : (a,b) \to \mathbb{R}$ be a function such that $f^{(n+1)}(x)$ is continuous. Let $x_0, x \in (a,b)$. Then,

$$f(x) = f(x_0) + \sum_{m=1}^{n} \frac{f^{(m)}(x_0)}{m!}(x - x_0)^m + R_n(x), \quad R_n(x) = \int_{x_0}^{x} \frac{(x-t)^n}{n!} f^{(n+1)}(t)\, dt.$$

Further, there exists c between x and x_0 such that $R_n(x) = \dfrac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$.

Proof. First, we prove the *integral form* by induction on n. Due to the Fundamental theorem of calculus,

$$f(x) = f(x_0) + R_0(x) = f(x_0) + \int_{x_0}^{x} f'(t)\, dt.$$

This proves the formula for $n = 0$. Now, suppose that Taylor's formula holds for $n = k$. That is, we have

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + R_k(x),$$

where $R_k(x) = \displaystyle\int_{x_0}^{x} \frac{(x-t)^k}{k!} f^{(k+1)}(t)\, dt$. We evaluate $R_k(x)$ using integration by parts with the first function as $f^{(k+1)}(t)$ and the second function as $(x-t)^k/k!$. Remember that the variable of integration is t and x is a fixed number. Then

$$\begin{aligned} R_k(x) &= \Big[-f^{(k+1)}(t)\frac{(x-t)^{k+1}}{(k+1)!}\Big]_{x_0}^{x} + \int_{x_0}^{x} f^{(k+2)}(t)\frac{(x-t)^{k+1}}{(k+1)!}\, dt \\ &= f^{(k+1)}(x_0)\frac{(x-x_0)^{k+1}}{(k+1)!} + \int_{x_0}^{x} f^{(k+2)}(t)\frac{(x-t)^{k+1}}{(k+1)!}\, dt \\ &= \frac{f^{(k+1)}(x_0)}{(k+1)!}(x-x_0)^{k+1} + R_{k+1}(x). \end{aligned}$$

This completes the proof of Taylor's formula in integral form.

For the differential form of the remainder term $R_n(x)$ we proceed as follows. Observe that for $x = x_0$, the formula holds. Let $x \in (x_0, b)$. For any $t \in [x_0, x]$, let

$$p(t) = f(x_0) + f'(x_0)(t-x_0) + \frac{f''(x_0)}{2!}(t-x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(t-x_0)^n.$$

Here, we treat x as a certain point, not a variable; and t as a variable. Write

$$g(t) = f(t) - p(t) - \frac{f(x)-p(x)}{(x-x_0)^{n+1}}\,(t-x_0)^{n+1}.$$

We see that $g(x_0) = 0,\ g'(x_0) = 0,\ g''(x_0) = 0,\ \ldots, g^{(n)}(x_0) = 0,$ and $g(x) = 0$.

By Rolle's theorem, there exists $c_1 \in (x_0, x)$ such that $g'(c_1) = 0$. Since $g(x_0) = 0$, apply Rolle's theorem once more to get a $c_2 \in (x_0, c_1)$ such that $g''(c_2) = 0$.

Continuing this way, we get a $c_{n+1} \in (x_0, c_n)$ such that $g^{(n+1)}(c_{n+1}) = 0$.

Since $p(t)$ is a polynomial of degree at most n, $p^{(n+1)}(t) = 0$. Then

$$g^{(n+1)}(t) = f^{(n+1)}(t) - \frac{f(x)-p(x)}{(x-x_0)^{n+1}}\,(n+1)!.$$

Evaluating at $t = c_{n+1}$ we have $f^{(n+1)}(c_{n+1}) - \dfrac{f(x)-p(x)}{(x-x_0)^{n+1}}\,(n+1)! = 0$. That is,

$$\frac{f(x)-p(x)}{(x-x_0)^{n+1}} = \frac{f^{(n+1)}(c_{n+1})}{(n+1)!}.$$

Consequently, $g(t) = f(t) - p(t) - \dfrac{f^{(n+1)}(c_{n+1})}{(n+1)!}\,(t-x_0)^{n+1}$.

Evaluating it at $t = x$ and using the fact that $g(x) = 0$, we get

$$f(x) = p(x) + \frac{f^{(n+1)}(c_{n+1})}{(n+1)!}\,(x - x_0)^{n+1}.$$

Since x is an arbitrary point in (x_0, b), this completes the proof. ∎

The polynomial

$$p_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

in Taylor's formula is called as **Taylor's polynomial of order** n. Notice that the degree of the Taylor's polynomial may be less than or equal to n. The expression that $f(x) = p_n(x) + R_n(x)$ is called **Taylor's formula** for $f(x)$. Taylor's polynomial is an approximation to $f(x)$ with **the error**

$$R_n(x) = \int_{x_0}^{x} \frac{(x-t)^n}{n!} f^{(n+1)}(t)\,dt = \frac{f^{(n+1)}(c_{n+1})}{(n+1)!}\,(x - x_0)^{n+1}.$$

How good $f(x)$ is approximated by $p_n(x)$ depends on the smallness of the error $R_n(x)$. For example, if we use $p_5(x)$, Taylor's polynomial of order 5, for approximating $\sin x$ at $x = 0$, then we get

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + R_5(x), \text{ where } R_5(x) = \frac{\sin\theta}{6!}x^6.$$

Here, θ lies between 0 and x. The absolute error is bounded above by $|x|^6/6!$. However, if we take the Taylor's polynomial of order 6, then $p(x)$ is the same as in the above, but the absolute error is now $|x|^7/7!$. If x is near 0, this is smaller than the earlier bound.

Notice that if $f(x)$ is a polynomial of degree n, then Taylor's polynomial of order n is equal to the original polynomial.

We have a similar result for functions of several variables.

(1.51) ***Theorem*** (Taylor)

Let D be a region in the plane containing a disk D_0 around a point (a, b). Let the function $f : D \to \mathbb{R}$ have continuous partial derivatives of orders up to $n+1$ in D_0 for $n \in \mathbb{N} \cup \{0\}$. Then, corresponding to each point $(a+h, b+k) \in D_0$ there exists $\theta \in [0, 1]$ such that

$$f(a+h, b+k) = f(a,b) + \sum_{m=1}^{n} \frac{1}{m!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^m f(a,b) + \frac{1}{(n+1)!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^{n+1} f(a+\theta h, b+\theta k).$$

For example, $\frac{1}{2!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(a,b) = \frac{1}{2}\left(h^2 f_{xx}+2hkf_{xy}+k^2 f_{yy}\right)\Big|_{(a,b)}$.

Proof. Let I be an open interval containing $[0,1]$. Define the function $\phi : I \to \mathbb{R}$ by $\phi(t) = f(a+th, b+tk)$. Due to the hypotheses, $\phi^{(n)}(t)$ is continuous. Then,

$$\phi'(t) = f_x(a+th,b+tk)h + f_y(a+th,b+tk)k = \left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(a+th,b+tk).$$

$$\phi^{(2)}(t) = (f_{xx}h+f_{xy}k)h + (f_{yx}h+f_{yy}k)k = \left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(a+th,b+tk).$$

By induction, we get $\phi^{(m)}(t) = \left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^m f(a+th,b+tk)$.

Using (1.50) on the single variable function $\phi(t)$, we have

$$\phi(1) = \phi(0) + \sum_{m=1}^{n}\frac{\phi^{(m)}(0)}{m!} + \frac{\phi^{(n+1)}(\theta)}{(n+1)!} \quad \text{for some } \theta \in [0,1].$$

Substituting the expressions for $\phi(1)$, $\phi(0)$, $\phi^{(m)}(0)$ and $\phi^{(n+1)}(\theta)$, we get the required result. ∎

Taylor's theorem can be used to approximate $f(a+h,b+k)$ in terms of $f(a,b)$, the increments h,k and the values of the partial derivatives at (a,b). The 0th order approximation is simply

$$f(a+h,b+k) \approx f(a,b).$$

Here, we neglect all powers of the increments h and k. However,

$$f(a+h,b+k) = f(a,b) + \Delta f.$$

The increment Δf, which is the error term in the 0th order approximation, can again be approximated as follows:

$$\Delta f \approx f_x(a,b)\Delta x + f_y(a,b)\Delta y.$$

Analogous to functions of one variable, we define the RHS as the **total differential** of f, also called the **differential**, and write it as df at (a,b). Conventionally, Δx is written as dx and Δy is written as dy also. That is, for any point (x,y) in the domain of $f(x,y)$ we define the differential as

$$df = f_x(x,y)dx + f_y(x,y)dy.$$

The above approximation formula states that the differential df is an approximation to the total increment Δf. As Taylor's formula shows, the actual value of Δf is df

evaluated at $(a+\theta h, b+\theta k)$ for some $\theta \in [0,1]$. Thus, the total differential evaluated at (a,b) gives a rough estimate of the error we might commit in approximating $f(a+h, b+k)$ with $f(a,b)$.

(1.52) ***Example***

The dimensions of a rectangular box are measured to be 75cm, 60cm, and 40 cm, and each measurement is correct to within 0.2cm. Use differentials to estimate the largest possible error when the volume of the box is calculated from these measurements.

The volume of the box is $V = xyz$. So,

$$dV = V_x dx + V_y dy + V_z dz \quad \text{at } (75, 60, 40).$$

Now, $|\Delta x|, |\Delta y|, |\Delta z| \leq 0.2$cm, $V_x(75,60,40) = (yz)$ at $y = 60, z = 40$, which is equal to 60×40. Similarly, $V_y(75,60,40) = 75 \times 40$ and $V_z(75,60,40) = 75 \times 60$. So, the largest error in cubic cm is

$$|\Delta V| \approx |dV| = 60 \times 40 \times 0.2 + 40 \times 75 \times 0.2 + 75 \times 60 \times 0.2 = 1980.$$

Notice that the relative error is $1980/(75 \times 60 \times 40)$, which is about 1%.

Here, the maximum value of ΔV is $(75+0.2)(60+0.2)(40+0.2) - 75 \times 60 \times 40$. Thus, the difference between the maximum possible value of ΔV and dV is

$$\begin{aligned}&(75+0.2)(60+0.2)(40+0.2) - 75 \times 60 \times 40\\ &-(60 \times 40 \times 0.2 + 40 \times 75 \times 0.2 + 75 \times 60 \times 0.2) = 0.2 \times 0.2 \times 0.2.\end{aligned}$$

Of course the actual error in approximating V using the said measurements is between dV and the maximum possible value of ΔV. □

In Taylor's formula, when the terms involving h^2, k^2 and higher powers of h, k are neglected, we obtain a linear approximation such as $f(a,b) + f_x(a,b)h + f_y(a,b)k$ to $f(a+h, b+k)$. Taking $h = x-a$ and $k = y-b$, we define the **standard linearization** of $f(x,y)$ at (a,b) as

$$L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

We also call $L(x,y)$ as the **linear approximation of** $f(x,y)$ **at** (a,b). Here, $L(x,y)$ is taken as an approximation of $f(x,y)$, assuming that the point (x,y) is (intuitively) close to the point (a,b).

Notice that the notation is slightly ambiguous. Instead of $L(x,y)$, it should have been written as $L(f(x,y),(a,b))$ keeping track of all things involved. However, we will use the customary notation $L(x,y)$ and remember to be careful.

The standard linearization is closely related to the tangent plane. Recall that the equation of the tangent plane to the surface $z = f(x, y)$ at $(a, b, f(a, b))$ is given by

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

This equation is satisfied for all points (x, y, z) on the tangent plane at $(a, b, f(a, b))$. To approximate $f(x, y)$ for a point (x, y) close to (a, b), we may take

$$f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

The LHS is the z-component of the point on the surface $z = f(x, y)$ corresponding to the point (x, y). The RHS is the the z-component of the point on the tangent plane to the surface $z = f(x, y)$ at $(a, b, f(a, b))$ corresponding to the point (x, y). Notice that the RHS is same as $L(x, y)$.

The **linearization error** in the standard linearization at (a, b) can be written as

$$\begin{aligned} E(x, y) &= f(x, y) - L(x, y) \\ &= \frac{1}{2!}\left[(x - a)^2 f_{xx}(c, d) + 2(x - a)(y - b) f_{xy}(c, d) + (y - b)^2 f_{yy}(c, d)\right] \end{aligned}$$

where $c = a + \theta(x - a),\ d = b + \theta(y - b)$ for some $\theta \in [0, 1]$.

We can have an estimate of the linearization error, which may be easier to apply. Suppose the point (a, b) is contained in a rectangle Rwhich is contained in an open region D, $f : D \to \mathbb{R}$ has continuous second order partial derivatives, and there exists a real number M such that $|f_{xx}| \leq M$, $|f_{xy}| \leq M$, and $|f_{yy}| \leq M$ for all points in R. Then

$$|E(x, y)| \leq \frac{M}{2}\left|(x - a)^2 + 2(x - a)(y - b) + (y - b)^2\right| \leq \frac{M}{2}\Big(|(x - a| + |y - b|\Big)^2.$$

(1.53) *Example*

Find the standard linearization of $f(x, y) = x^2 - xy + y^2/2 + 3$ at $(3, 2)$. Also find an upper bound of the linearization error in the rectangle $|x - 3| \leq 0.1,\ |y - 2| \leq 0.1$.

The standard linearization (linear approximation) of $f(x, y)$ at (a, b) is

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Now, $f(3, 2) = 8,\ f_x(3, 2) = (2x - y)|_{(3,2)} = 4$ and $f_y(3, 2) = (-x + y)|_{(3,2)} = -1$. Thus

$$L(x, y) = 8 + 4(x - 3) - (y - 2) = 4x - y - 2.$$

The error in this linearization is

$$E(x, y) = f(x, y) - L(x, y) = x^2 - xy + y^2/2 + 3 - 4x + y + 2.$$

The rectangle is $R:\ |x-3| \le 0.1, |y-2| \le 0.1$. Here, $f_{xx} = 2,\ f_{xy} = -1,\ f_{yy} = 1$.

So, we take $M = 2$ as an upper bound for their absolute values. Then

$$|E(x,y)| \le \big(|x-3| + |y-2|\big)^2 \le (0.1+0.1)^2 = 0.04. \qquad \square$$

Similar error estimates hold for linearization of functions of three or more variables at a point.

(1.54) ***Example***

For the function $f(x,y,z) = x^2 - xy + 2\sin z$, find the linearization and the maximum error incurred at $P(2,1,0)$ in the cuboid $|x-2| \le 0.01$, $|y-1| \le 0.02$, $|z| \le 0.01$.

$f(P) = f(2,1,10) = 4-2 = 2,\ f_x(P) = (2x-y)|_{(2,1,0)} = 3,\ f_y(P) = (-x)|_{(2,1,0)} = -2$ and $f_z(P) = (2\cos z)|_{(2,1,0)} = 2$. Thus

$$L(x,y,z) = f(P) + f_x(P)(x-2) + f_y(P)(y-1) + f_z(P)z = 3x - 2y + 2z - 2.$$

All double derivatives are bounded above by 2. So, the error

$$E(x,y,z) \le \frac{2}{2}\Big(|x-2| + |y-1| + |z|\Big)^2 \le 0.0016. \qquad \square$$

Similar to linearization, the **quadratic approximation** $Q(x,y)$ to $f(x,y)$ at (a,b) is given by

$$\begin{aligned} Q(x,y) = {} & f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) \\ & + \frac{1}{2}\Big((x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\Big). \end{aligned}$$

The **cubic approximation** $C(x,y)$ to $f(x,y)$ at (a,b) is given by

$$\begin{aligned} C(x,y) = {} & f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) \\ & + \frac{1}{2}\Big((x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\Big) \\ & + \frac{1}{3!}\Big((x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\Big)^3 f(a,b). \end{aligned}$$

Notice that $Q(x,y)$ and $C(x,y)$ are polynomials of degree at most 2 and 3, respectively. Further, these approximations to $f(x,y)$ at (a,b) satisfy

$$L(a,b) = Q(a,b) = C(a,b) = f(a,b).$$

In the quadratic approximation $Q(x,y)$ of $f(x,y)$ at (a,b), the error $E(x,y)$ is the term that has been left out in Taylor's theorem for $n = 2$. That is, the error in the quadratic approximation is

$$E(x,y) = \frac{1}{6}\Big((x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\Big)^3 f\big(a+\theta(x-a), b+\theta(y-b)\big)$$

for some $\theta \in [0,1]$. This error can be estimated as earlier provided we know some upper bound for the third order partial derivatives of f in a rectangle containing the point (a,b). Similarly, the error in quadratic approximation can be estimated.

(1.55) ***Example***

Let $f(x,y) = x^2 + xy - y^2,\ a = 1,\ b = -2$.

Here, $f(1,-2) = -5,\ f_x(1,-2) = 0,\ f_y(1,-2) = 5,\ f_{xx} = 2,\ f_{xy} = 1,\ f_{yy} = -2$.

The quadratic approximation to $f(x,y)$ at $(1,-2)$ is given by

$$Q(x,y) = -5 + 5(y+2) + \frac{1}{2}\Big(2(x-1)^2 + 2(x-1)(y+2) - 2(y+2)^2\Big).$$

Upon simplification, we see that $Q(x,y) = f(x,y)$. That is, this approximation becomes exact. The reason is that the third order derivatives are 0. □

Higher order approximations to $f(x,y)$ at (a,b) are analogously obtained from Taylor's theorem. The polynomial

$$p_n(x,y) = f(a,b) + \sum_{m=1}^{n} \frac{1}{m!}\Big((x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\Big)^m f(a,b)$$

in Taylor's theorem is called the **Taylor's polynomial** of order n. We also say that $p_0(x,y) = f(a,b)$ is the Taylor's polynomial of order 0. In general, the degree of $p_n(x,y)$ is at most n. We have $L(x,y) = p_1(x,y)$, $Q(x,y) = p_2(x,y)$ and $C(x,y) = p_3(x,y)$. Notice that if $p_n(x,y)$ is an approximation to $f(x,y)$ of any order n at (a,b), then $p_n(x,y) = f(a,b)+$ terms involving powers of $x-a$ and/or powers of $y-b$; consequently, $p_n(a,b) = f(a,b)$.

In general, if $f(x,y)$ is a polynomial of degree n, (that is, when the maximum sum of degrees of x and y in any term of $f(x,y)$ is n), then any approximation of $f(x,y)$ of order n (or larger) at any point (a,b) coincides with $f(x,y)$. That is, if $f(x,y)$ is a polynomial of degree n in the variables x and y, then $p_n(x,y) = f(x,y)$.

The error term in approximating $f(x,y)$ by the Taylor's polynomial $p_n(x,y)$ is given by

$$f(x,y) - p_n(x,y) = \frac{1}{(n+1)!}\Big((x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\Big)^{n+1} f\big(a+\theta(x-a), b+\theta(y-b)\big).$$

In this nth order approximation of $f(x,y)$ at (a,b), the error is 0 if $f(x,y)$ is a polynomial of degree n, in which case the partial derivatives of order $n+1$ are zero. Again, an estimate of this error term can be obtained if we know the bounds for these derivatives in a disk (or rectangle) around the point (a,b).

1.10 Extreme values

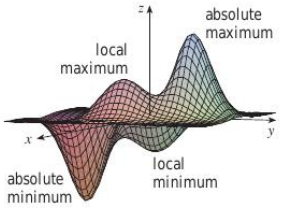
Let D be a region in the plane, $(a, b) \in D$, and let $f : D \to \mathbb{R}$ be a function.

We say that $f(x, y)$ has a **local maximum** at (a, b) iff (a, b) is an interior point of D and $f(x, y) \leq f(a, b)$ for all (x, y) in some open disk around (a, b) that is contained in D. Informally, the condition says that $f(x, y) \leq f(a, b)$ holds for all $(x, y) \in D$ near (a, b). In such a case, the point (a, b) is called a **point of local maximum** of $f(x, y)$, and the number $f(a, b)$ is called a **local maximum value** of $f(x, y)$.

Replace all $\leq$ by $\geq$ in the above definitions; and call all those **minimum** instead of *maximum*. The function $f(x, y)$ has a **local extremum** at (a, b) iff $f(x, y)$ has a local maximum or a local minimum at (a, b).

We say that f has the **absolute maximum** at $(a, b) \in D$ iff for all $(x, y) \in D$, $f(x, y) \leq f(a, b)$. In this case, the point (a, b) is called a **point of absolute maximum** of $f(x, y)$, and the number $f(a, b)$ is called the **(absolute) maximum value** of f. The *points of absolute minimum* and the *(absolute) minimum value* are defined by replacing $\leq$ with $\geq$. Both 'absolute maximum' and 'absolute minimum' are commonly referred to as 'absolute extremum'.

Notice that a local extremum point must be an interior point whereas an absolute extremum point need not be an interior point; it is allowed to be any point from D.



An interior point (a, b) of D is called a **critical point** of $f(x, y)$ iff either at least one of $f_x(a, b)$, $f_y(a, b)$ does not exist, or both exist and $f_x(a, b) = 0 = f_y(a, b)$.

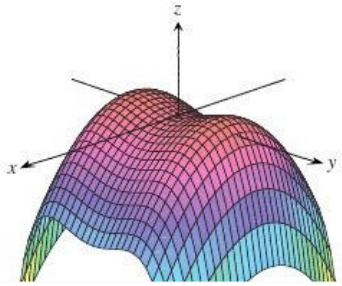
(1.56) ***Theorem***

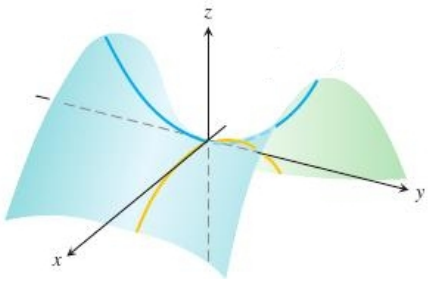
Let D be a region in the plane and let $f : D \to \mathbb{R}$ be a function. If (a, b) is a point of local extremum of f, then it is a critical point of f.

Proof. Suppose (a, b) is a point of local maximum of f. Then the function $g(x) = f(x, b)$ has a local maximum at $x = a$. We have $g'(a) = 0$ if it exists. So, either $f_x(a, b)$ does not exist or $f_x(a, b) = 0$. Similarly, consider $h(y) = f(a, y)$ and conclude that either $f_y(a, b)$ does not exist or $f_y(a, b) = 0$. Give similar argument if f has a local minimum at (a, b). ∎

Geometrically, it says that if at an interior point (a, b), there exists a tangent plane to the surface $z = f(x, y)$, and if this point (a, b) happens to be an extremum point, then there exists a horizontal tangent plane to the surface at (a, b).

Let D be a region in the plane, $f : D \to \mathbb{R}$ be a function, and let (a, b) be a critical point of $f(x, y)$. The point $(a, b, f(a, b))$ on the surface is called a **saddle point** of $f(x, y)$ if in every disk around (a, b) and contained in D, there are points (x_1, y_1), (x_2, y_2) such that $f(x_1, y_1) < f(a, b) < f(x_2, y_2)$. In such a case, we also say that $f(x, y)$ *has a saddle point at* (a, b). At a saddle point, the function has neither a local maximum nor a local minimum.





In order to devise a test for determining whether a certain critical point is a point of local maximum, local minimum or a saddle point we need the notion of a Hessian.

The **Hessian** of a function $f(x, y)$ is defined by $H(f) := \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - f_{xy}^2$.

(1.57) ***Theorem***

Let D be a region in the plane, $(a, b) \in D$ and let the function $f : D \to \mathbb{R}$ have continuous second order partial derivatives in a disk D_0 around (a, b), where $D_0 \subseteq D$. Assume that (a, b) is a critical point of f.

(1) *If $H(f)(a, b) > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum at (a, b).*

(2) *If $H(f)(a, b) > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum at (a, b).*

(3) *If $H(f)(a, b) < 0$ then $f(x, y)$ has a saddle point at (a, b).*

(4) *If $H(f)(a, b) = 0$, then nothing can be said about extrema at (a, b), in general.*

Proof. Let $(a + h, b + k) \in D_0$. By Taylor's formula,

$$f(a+h, b+k) = \left(f + hf_x + kf_y\right)\Big|_{(a,b)} + \frac{1}{2}\left(h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\right)\Big|_{(a+\theta h, b+\theta k)}.$$

Since (a, b) is a critical point of f, $f_x(a, b) = 0 = f_y(a, b)$. Then

$$f(a+h, b+k) - f(a, b) = \frac{1}{2}\left(h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\right)\Big|_{(a+\theta h, b+\theta k)} \tag{1.10.1}$$

(1) Suppose $H(f)(a, b) > 0$ and $f_{xx}(a, b) < 0$. Then, multiply both sides of (1.10.1) by $2f_{xx}$, add and subtract $(f_{xy})^2 k^2$, and rearrange to get (All of f_{xx}, f_{xy}, f_{yy} are evaluated at the point $(a + \theta h, b + \theta k)$.)

$$2f_{xx}[f(a+h, b+k) - f(a, b)] = (hf_{xx} + kf_{xy})^2 + k^2 H(f)(a + \theta h, b + \theta k).$$

By continuity of f_{xx}, f_xy, f_{yy}, we have $f_{xx} < 0$ and $H(f) > 0$ at $(a+\theta h, b+\theta k)$ in a disk D_1 around (a,b), where $D_1 \subseteq D_0$ and $\theta \in [0,1]$. The RHS is positive. Therefore, $f(a+h,b+k) - f(a,b) < 0$. That is, (a,b) is a local maximum point.

(2) Let $H(f)(a,b) > 0$ and $f_{xx}(a,b) > 0$. As in (1), we have $f_{xx}(a+\theta h, b+\theta k) > 0$ in a disk D_2 around (a,b), where $D_2 \subseteq D_0$. So, $f(a+h,b+k) - f(a,b) > 0$. That is, (a,b) is a local minimum point.

(3) Let $H(f)(a,b) < 0$. We want to show that $f(a+h,b+k) - f(a,b)$ has opposite signs at different points in any (small) disk around (a,b). We break this case into three sub-cases:

(3A) $f_{xx}(a,b) \neq 0$. (3B) $f_{yy}(a,b) \neq 0$, (3C) $f_{xx}(a,b) = f_{yy}(a,b) = 0$.

(3A) Let $H(f)(a,b) < 0$ and $f_{xx}(a,b) \neq 0$. First, set $h = t, k = 0$ in (1.10.1) and evaluate the following limit:

$$\lim_{t\to 0} \frac{f(a+h,b+k) - f(a,b)}{t^2} = \lim_{t\to 0} \frac{t^2 f_{xx}(a+t,b)}{2t^2} = \frac{f_{xx}(a,b)}{2}.$$

Next, set $h = -tf_{xy}(a,b),\ k = tf_{xx}(a,b)$. Use (1.10.1) to obtain

$$\lim_{t\to 0} \frac{f(a+h,b+k) - f(a,b)}{t^2} = \lim_{t\to 0} \frac{1}{2}(f_{xy}^2 f_{xx} - 2f_{xx}f_{xy}^2 + f_{xx}^2 f_{yy}) = \frac{f_{xx}(a,b)}{2} H(f)(a,b).$$

Since $H(f)(a,b) < 0$, these two limits have opposite signs. Due to continuity, $f(a+h,b+k) - f(a,b)$ has opposite signs in a disk D_3 around (a,b), where $D_3 \subseteq D_0$.

(3B) Let $H(f)(a,b) < 0$ and $f_{yy}(a,b) \neq 0$. This is similar to (3A).

(3C) Let $H(f)(a,b) < 0$ and $f_{xx}(a,b) = f_{yy}(a,b) = 0$. First, set $h = k = t$. Use (1.10.1) to get

$$\lim_{t\to 0} \frac{f(a+h,b+k) - f(a,b)}{t^2} = \lim_{t\to 0} \frac{1}{2}\big(f_{xx} + 2f_{xy} + f_{yy}\big)\Big|_{(a+t,b+t)} = f_{xy}(a,b).$$

Next, set $h = t,\ k = -t$. Using (1.10.1) again, we have

$$\lim_{t\to 0} \frac{f(a+h,b+k) - f(a,b)}{t^2} = \lim_{t\to 0} \frac{1}{2}\big(f_{xx} - 2f_{xy} + f_{yy}\big)\Big|_{(a+t,b+t)} = -f_{xy}(a,b).$$

As in (3A), $f(a+h,b+k) - f(a,b)$ has opposite signs in any disk $D_4 \subseteq D_0$ around (a,b).

(4) Consider $f(x,y) = x^2y^2$. At $(0,0)$, all of $f_x, f_y, f_{xx}, f_{xy}, f_{yy}$ are 0. So, $(0,0)$ is a critical point and $H(f)(0,0) = 0$. If $(x,y) \neq (0,0)$, then $f(x,y) \geq 0 = f(0,0)$. That is, $H(f)(0,0) = 0$ and $f(x,y)$ has a point of local minimum at $(0,0)$.

For the function $g(x,y) = x^3y^3$, we see that $(0,0)$ is a critical point with $g(0,0) = 0$ and $H(g)(0,0) = 0$. However, in any disk around $(0,0)$, there are points where

$g(x, y)$ is positive and also there are points where g is negative. So, $(0, 0)$ is a saddle point of $g(x, y)$. ∎

Notice that the case $H(f)(a, b) > 0$ and $f_{xx}(a, b) = 0$ is not possible. Moreover, Under the condition that $H(f)(a, b) > 0$, both $f_{xx}(a, b)$ and $f_{yy}(a, b)$ have the same sign. Thus, in (1.57-1 & 2), the sign condition on $f_{xx}(a, b)$ can be replaced by the corresponding sign condition on $f_{yy}(a, b)$. It also says that if $f_{xx}(a, b)$ and $f_{yy}(a, b)$ have opposite signs, then the critical point (a, b) is not a point of extremum of $f(x, y)$. Geometrically, the condition $H(f)(a, b) > 0$ implies that the surface $z = f(x, y)$ curves the same way (either upward or downward) in all directions near (a, b); and if $H(f)(a, b) < 0$, then the surface curves in different directions near (a, b).

Caution: The point (a, b) in (1.57) is assumed to be an interior point of the domain of $f(x, y)$. If the domain has points which are not interior points, then those can also be possible absolute extremum points.

(1.58) *Example*

Find the extreme values of $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$.

Domain of f is the whole plane. Its extreme values are all local extrema. The second order partial derivatives of f are continuous. Now,

$$f_x = y - 2x - 2, \quad f_y = x - 2y - 2 \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

The critical points satisfy $f_x = 0 = f_y$. That is, $x = y = -2$.

We get $f_{xx}(-2, -2) = -2,\ f_{xy}(-2, -2) = 1,\ f_{yy}(-2, -2) = -2$.

Then $H(f)(-2, -2) = 3 > 0$.

Since $f_{xx} < 0$, the function f has a local maximum at $(-2, -2)$.

Also f has absolute maximum at $(-2, 2)$; the maximum value is $f(-2, -2) = 8$. □

(1.59) *Example*

Investigate $f(x, y) = x^4 + y^4 - 4xy + 1$ for extreme values.

The function has continuous first and second order partial derivatives everywhere.

The critical points are at (x, y) where $f_x = 4x^3 - 4y = 0 = f_y = 4y^3 - 4x$.

That is, when $x^3 = y$ and $y^3 = x$. This gives $x^9 = x$, which has solutions $x = 0, 1, -1$ in $\mathbb{R}$. The corresponding y values are $0, 1, -1$.

The possible points of extremum are $(0, 0)$, $(1, 1)$ and $(-1, -1)$.

Now, $f_{xx} = 12x^2,\ f_{xy} = -4,\ f_{yy} = 12y^2$. Thus $H(f) = 144x^2y^2 - 16$.

$H(f)(0, 0) = -16 < 0$. Thus $f(x, y)$ has a saddle point at $(0, 0)$.

$H(f)(1, 1) = 128 > 0,\ f_{xx}(1, 1) = 12 > 0$. So, f has a local minimum at $(1, 1)$.

$H(f)(-1, -1) = 128,\ f_{xx}(-1, -1) = 12$. Thus f has a local minimum at $(-1, -1)$.

The local minimum values are $f(1,1)$ and $f(-1,-1)$; here both are equal to -1. Also, the absolute minimum value of $f(x,y)$ is -1, which is achieved at $(1,1)$ and $(-1,-1)$. □

(1.60) ***Example***

Find absolute extrema of $f(x,y) = 2 + 2x + 2y - x^2 - y^2$ defined on the triangular region bounded by the straight lines $x = 0,\ y = 0$, and $x + y = 9$.

The critical points are solutions of $f_x = 2 - 2x = 0 = f_y = 2 - 2y$. That is, $x = 1, y = 1$. This accounts for the interior points of the region.

Draw the picture. We should consider the boundary in detail. First, the vertices of the triangle such as $A(0,0),\ B(0,9),\ C(9,0)$ are possible extremum points.

Next, on the line segment $AB,\ x = 0$; thus $f(x,y)$ is given by:

$$g(y) = f(0,y) = 2 + 2y - y^2 \quad \text{for } 0 \le y \le 9.$$

Then $g'(y) = 0 \Rightarrow y = 1$. Thus, a possible extremum point is $(0,1)$.

Similarly, on the line segment $AC,\ y = 0$; thus $f(x,y)$ is given by:

$$h(x) = f(x,0) = 2 + 2x - x^2 \quad \text{for } 0 \le x \le 9.$$

Now, $g'(x) = 0 \Rightarrow x = 1$. Thus $(1,0)$ is a possible extremum point.

On the line segment $BC,\ x + y = 9$; thus $f(x,y)$ is given by:

$$\phi(x) = f(x, 9-x) = 2+2x+2(9-x)-x^2-(9-x)^2 = -61+18x-2x^2 \quad \text{for } 0 \le x \le 9.$$

Now, $\phi'(x) = 0 \Rightarrow 18 - 4x = 0 \Rightarrow x = 9/2,\ y = 9 - x = 9/2$. Thus $(9/2, 9/2)$ is a possible extremum point.

Then, all possible points at which $f(x,y)$ may have an extremum are

$$(1,1),\ (0,0).\ (0,9),\ (9,0),\ (1.0),\ (0,1),\ (9/2, 9/2).$$

The values of $f(x,y)$ at these points are

$$f(1,1) = 4, f(0,0) = 2, f(0,9) = -61, f(9,0) = -61,$$
$$f(1,0) = 3, f(0,1) = 3, f(9/2, 9/2) = -41/2.$$

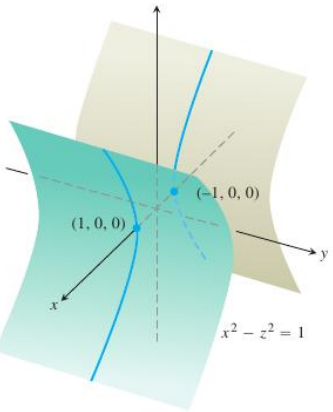
Comparing these values, we see that $f(x,y)$ has absolute minimum at $(0,9)$ and $(9,0)$ with the minimum value as -61. It has absolute maximum at $(1,1)$ with the maximum value as 4. □

1.11 Lagrange multipliers

In some problems we are required to maximize or minimize a function subject to a given constraint. See the following example.

(1.61) ***Example***

Find the points closest to the origin on the hyperbolic cylinder $x^2 - z^2 = 1$.



We seek a point (x, y, z) that minimizes $f(x, y, z) = x^2+y^2+z^2$ subject to $x^2-z^2 = 1$. Taking $z^2 = x^2 - 1$, we seek (x, y) that minimizes

$$g(x, y) := f\left(x, y, \pm\sqrt{x^2 - 1}\right) = x^2 + y^2 + x^2 - 1 = 2x^2 + y^2 - 1.$$

Now, $g_x = 4x,\ g_y = 2y$. Equating them to zero gives $x = 0$ and $y = 0$. But $x = 0$ does not correspond to any point on the surface $x^2 - z^2 = 1$. So, the method fails!

Instead of eliminating z, suppose we eliminate x. In that case, we seek to minimize

$$h(y, z) := f\left(\pm \sqrt{1 + z_2}, y, z\right) = 1 + z^2 + y^2 + z^2 = y^2 + 2z^2 + 1.$$

Then $h_y = 0 = h_z$ implies that $2y = 0 = 4z$. The point so obtained is $y = 0,\ z = 0$. This corresponds to the points $(\pm 1, 0, 0)$ on the surface.

Now, of course, we can proceed as earlier for minimizing $h(y, z)$.

Here, $h_{yy} = 2,\ h_{yz} = 0,\ h_{zz} = 4$.

At $y = 0, z = 0$, we have $H(h)(0, 0) = h_{yy}h_{zz} - h_{yz}^2 = 8 > 0$.

Since $h_{yy}(0, 0) > 0$, we conclude that $h(y, z)$ has a local minimum at $(0, 0)$.

These points $(\pm 1, 0, 0)$ of local minima give the minimum value of the distance $f(x, y, z)$ as 1. □

It raises the question: "How do we know which variable is to be eliminated so that a solution may be reached?" We would rather look for alternative ways of solving

such *constrained optimization problems*.

(1.62) *Theorem*

Let D be an open region in the plane. Let $f, g : D \to \mathbb{R}$ have continuous partial derivatives. Let f have an extreme value at a point (a, b) on the curve $g(x, y) = 0$, where $\nabla g(a, b) \neq 0$. Then a, b and λ satisfy the equations

$$f_x(a, b) + \lambda g_x(a, b) = 0, \quad f_y(a, b) + \lambda g_y(a, b) = 0, \quad g(a, b) = 0.$$

Proof. The constraint $g(x, y) = 0$ is a level curve of the surface $z = g(x, y)$. Let $f(x, y)$ have an extreme value at (a, b) on this curve. Let $\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath}$ be a prametrization of this curve, where $(a, b) = \vec{r}(t_0)$.

Now, $h(t) = g\big(x(t), y(t)\big) = 0$ for all t. Differentiating with respect to t and evaluating at $t = t_0$, we have $h'(t_0) = 0$. That is,

$$0 = h'(t_0) = g_x(a, b)x'(t_0) + g_y(a, b)y'(t_0) = \nabla g(a, b) \cdot \vec{r}\,'(t_0).$$

On the other hand, the composite function $\phi(t) = f\big(x(t), y(t)\big)$ represents the values that f takes on C. Since f has an extreme value at $(a, b) = \vec{r}(t_0)$, the function $\phi(t)$ has an extreme value at $t = t_0$. Then $\phi'(t_0) = 0$. That is,

$$0 = \phi'(t_0) = f_x(a, b)x'(t_0) + f_y(a, b)y'(t_0) = (\nabla f)(a, b) \cdot \vec{r}\,'(t_0).$$

So, both $(\nabla f)(a, b)$ and $(\nabla g)(a, b)$ are orthogonal to $\vec{r}\,'(t_0)$, and $(\nabla g)(a, b) \neq 0$. Thus, $(\nabla f)(a, b)$ and $(\nabla g)(a, b)$ are parallel. That is, there exists $\lambda \in \mathbb{R}$ such that $(\nabla f + \lambda\, \nabla g)(a, b) = 0$. Breaking into components, we obtain $f_x(a, b) + \lambda g_x(a, b) = 0$ and $f_y(a, b) + \lambda g_y(a, b) = 0$. Further, (a, b) lies on the curve $g(x, y) = 0$; so, $g(a, b) = 0$. Therefore, a, b and λ satisfy the required equations. ∎

Notice that if we set $F(x, y, \lambda) := f(x, y) + \lambda g(x, y)$, then

$$F_x = f_x + \lambda g_x,\ F_y = f_y + \lambda g_y,\ F_\lambda = g(x, y).$$

These are the equations mentioned in (1.62) that a, b and λ satisfy. A result similar to (1.62) holds when f is a function of n variables and there are m number of constraints. We thus reach at the following method of solving a constrained optimization problem, called the method of **Lgarange multipliers**.

Requirement: Find extrema of the function $f(x_1, \ldots, x_n)$ subject to the conditions

$$g_1(x_1, \ldots, x_n) = 0,\ \cdots,\ g_m(x_1, \ldots, x_n) = 0.$$

Method: Set the **auxiliary function**

$$F(x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_m) := f(x_1, \ldots, x_n) + \lambda_1 g_1(x_1, \ldots, x_n) + \cdots \lambda_m g_m(x_1, \ldots, x_n).$$

Equate to zero the partial derivatives of F with respect to $x_1, \ldots, x_n,\ \lambda_1, \ldots, \lambda_m$.
It results in $m + n$ equations in the unknowns $x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_m$.
Determine the values of $x_1, \ldots, x_n\ \lambda_1, \ldots, \lambda_m$ from these equations.
The points of extremum are among these values of $x_1, \ldots, x_n$.

Caution: Lagrange multipliers method succeeds under the assumption that such extreme values exist where $\nabla g_j \neq 0$ for any j. Further, the points thus obtained are only possible points of extremum. All of those need not be points of extremum. Other considerations may be required to determine whether any such point is an actual maximum or minimum of f.

(1.63) *Example*

Consider the problem in (1.61). We need to minimize $f(x, y, z) = x^2+y^2+z^2$ subject to $g(x, y, z) = x^2 - z^2 - 1 = 0$. The auxiliary function is

$$F(x, y, z, \lambda) = f + \lambda g = x^2 + y^2 + z^2 + \lambda(x^2 - z^2 - 1).$$

Now, $F_x = 2x + 2\lambda x$, $F_y = 2y$, $F_z = 2z - 2\lambda z$ and $F_\lambda = x^2 - z^2 - 1$. We then equate the partial derivatives of F at (a, b, c) to 0. This gives

$$2a + 2\lambda a = 0,\ 2b = 0,\ 2c - 2\lambda c = 0,\ a^2 - c^2 - 1 = 0.$$

We then have $\big(a = 0 \text{ or } \lambda = -1\big)$, $b = 0$, $\big(c = 0 \text{ or } \lambda = 1\big)$, and $a^2 - c^2 - 1 = 0$.

From these options, $a = 0$ is not possible since for any c, $a^2 - c^2 - 1 = -c^2 - 1 \neq 0$. And, $\lambda = 1$ leads to $a = 0$, which is not possible. We are left with $\lambda = -1, b = 0$ and $c = 0$. Now, $a^2 - c^2 - 1 = 0$ gives $a = \pm 1$.

The corresponding points are $(\pm 1, 0, 0)$. Notice that $\nabla g(\pm 1, 0, 0) = \pm 2\hat{\imath} \neq 0$. At these points $f(x, y, z)$ has value 1. Since $f(x, y, z) = x^2+y^2+z^2$ can be made as large as possible by choosing (x, y, z) satisfying $x^2 = z^2 + 1$ suitably, $f(x, y, z)$ does not have a maximum. So, $f(x, y, z)$ at these points is minimum. Therefore, the points closest to the origin on the hyperbolic cylinder are $(\pm 1, 0, 0)$. □

(1.64) *Example*

Find the maximum volume of a box of length x, width y and height z given that $x + 2y + 2z = 108$.

We maximize $V(x, y, z) = xyz$ subject to the constraint $x + 2y + 2z - 108 = 0$. using Lagrange multipliers method, we set up the auxiliary function

$$F(x, y, z) = xyz + \lambda(x + 2y + 2z - 108).$$

Now, $F_x = yz + \lambda,\ F_y = xz + 2\lambda,\ F_z = xy + 2\lambda$ and $F_\lambda = x + 2y + 2z - 108$.
A possible point of extremum (a, b, c) satisfies

$$bc + \lambda = 0, \quad ac + 2\lambda = 0, \quad ab + 2\lambda = 0, \quad a + 2b + 2c - 108 = 0.$$

If $a = 0$ or $b = 0$ or $c = 0$, then $\lambda = 0$ and $V(a, b, c) = abc = 0$.
Otherwise, assume that $a \neq 0,\ b \neq 0$ and $c \neq 0$. Then, $\lambda = -bc$, $a = 108 - 2b - 2c$, $ac - 2bc = 0$ and $ab - 2bc = 0$. The latter two equations give $a = 2b = 2c$. Thus $b = c$ and the second equation gives $2b = 108 - 2b - 2b \Rightarrow b = 108/6 = 18$. Then, $c = 18$, $a = 36$ and $\lambda = -364$. Then, $V = abc = 11664$.

The two possible extreme values of V are 0 and 11664, the maximum is 11664. Therefore, the maximum volume is 11644 cubic units, when the box has length 36 units, width as 18 units and height as 18 units.

Notice that $\nabla g(36, 18, 18) = \hat{\imath} + 2\hat{\jmath} + 2\hat{k} \neq 0$. □

(1.65) ***Example***

Find the maximum value of $f(x, y, z) = x + 2y + 3z$ on the curve of intersection of the plane $x - y + z = 1$ and the cylinder $x^2 + y^2 = 1$.

With $g(x, y, z) = x - y + z - 1$ and $h(x, y, z) = x^2 + y^2 - 1$, the auxiliary function is

$$F(x, y, z, \lambda, \mu) := f + \lambda g + \mu h = x + 2y + 3z + \lambda(x - y + z - 1) + \mu(x^2 + y^2 - 1).$$

Setting $F_x = F_y = F_z = F_\lambda = F_\mu = 0$, for (a, b, c), we obtain

$$1 + \lambda + 2a\mu = 0,\ 2 - \lambda + 2b\mu = 0,\ 3 + \lambda = 0,\ a - b + c - 1 = 0,\ a^2 + b^2 - 1 = 0.$$

If $\mu = 0$, then $1 + \lambda = 0 = 2 - \lambda$; and it leads to inconsistency. So, let $\mu \neq 0$. We obtain: $\lambda = -3,\ a = 1/\mu,\ b = -5/(2\mu),\ 1/\mu^2 + 25/(4\mu^2) = 1$. That is, $\mu^2 = 29/4$. Then possible extreme points are

$$a = \pm 2/\sqrt{29},\ b = \mp 5/\sqrt{29},\ c = 1 \mp 7/\sqrt{29}.$$

Notice that at these points (a, b, c), $\nabla g(a, b, c) \neq 0$ and $\nabla(h)(a, b, c) \neq 0$. Evaluating $f(a, b, c)$, we see that the maximum of $f(a, b, c) = a + 2b + 3c$ is obtained for $a = -2/\sqrt{29}$, $b = 5/\sqrt{29}$ and $c = 1 + 7/\sqrt{29}$, and it is $3 + \sqrt{29}$. □

1.12 Review problems

Problem 1.1 Where is the function $f(x, y) = \dfrac{2xy}{x^2 + y^2}$ continuous? What are the limits of f at the points of discontinuity?

The function $f(x, y)$ is defined everywhere in the plane except at the origin. When $(x, y) \neq (0, 0)$, the functions $g(x) = 2xy$ and $h(x, y) = x^2 + y^2$ are continuous. Hence

$f(x,y)$ is continuous everywhere except at $(0,0)$. And, at $(0,0)$,

$$\lim_{\substack{x\to 0\\ \text{along } y=x}} f(x,y) = \lim_{x\to 0}\frac{2x^2}{2x^2} = 1, \qquad \lim_{\substack{x\to 0\\ \text{along } y=-x}} f(x,y) = \lim_{x\to 0}\frac{-2x^2}{2x^2} = -1.$$

Then, $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ does not exist so that $f(x,y)$ is discontinuous at $(0,0)$.

Problem 1.2 Determine the directions in which the function $f(x,y)$ has directional derivatives at $(0,0)$, where

$$f(x,y) = \begin{cases} \dfrac{xy^2}{x^2+y^4} & \text{if } (x,y)\neq(0,0) \\ 0 & \text{if } (x,y)=(0,0). \end{cases}$$

Consider a unit vector $\hat{u} = a\hat{\imath} + b\hat{\jmath}$. The directional derivative of $f(x,y)$ at $(0,0)$ in the direction of $\hat{u}$ is

$$(D_u f)(0,0) = \lim_{h\to 0}\frac{f(ah,bh)-f(0,0)}{h} = \lim_{h\to 0}\frac{ab^2}{a^2+bh^2} = \begin{cases} b^2/a & \text{for } a\neq 0 \\ 0 & \text{for } a=0. \end{cases}$$

Hence the directional derivative of $f(x,y)$ at $(0,0)$ exists in each direction.

Notice that $(\nabla f)(0,0) = 0\hat{\imath}+0\hat{\jmath}$. If you use the formula for the directional derivative at $(0,0)$ blindly, then it would give the wrong result that in every direction, the directional derivative of $f(x,y)$ is 0. What is the reason for this anomaly?

Problem 1.3 Let $f(x,y,z) = x^2+y^2+z^2$. Find the directional derivative of $f(x,y,z)$ in the direction of the vector $\vec{v} = 2\hat{\imath}+\hat{\jmath}+3\hat{k}$ at $(1,1,1)$.

The unit vector in the direction of $\vec{v}$ is $\hat{u} = \frac{2}{\sqrt{14}}\hat{\imath}+\frac{1}{\sqrt{14}}\hat{\jmath}+\frac{3}{\sqrt{14}}\hat{k}$. The gradient of f at $(1,1,1)$ is $\nabla f(1,1,1) = (f_x\hat{\imath}+f_y\hat{\jmath}+f_z\hat{k})(1,1,1) = 2\hat{\imath}+2\hat{\jmath}+2\hat{k}$. Then

$$(D_u f)(1,1,1) = (\nabla f\cdot\hat{u})(1,1,1) = \frac{12}{\sqrt{14}}.$$

Problem 1.4 Find the total increment Δz and the total differential dz of the function $z = xy$ at $(2,3)$ for $\Delta x = 0.1,\ \Delta y = 0.2$.

At $(2,3)$ with $\Delta x = 0.1,\ \Delta y = 0.2$, we have

$$\Delta z = (x+\Delta x)(y+\Delta y)-xy = y\Delta x+x\Delta y+\Delta x\Delta y = 3\times 0.1+2\times 0.2+0.1\times 0.2 = 0.72.$$

$$dz = z_x dx + z_y dy = ydx+xdy = y\Delta x + x\Delta y. = 3\times 0.1+2\times 0.2 = 0.7.$$

Problem 1.5 It is known that in computing the coordinates of a point (x,y,z,t) certain (small) errors such as $\Delta x, \Delta y, \Delta z, \Delta t$ might have been committed. Find the

maximum absolute error so committed when we evaluate a function $f(x, y, z, t)$ at that point.

Let $\Delta u = f(x + \Delta x, y + \Delta y, z + \Delta z, t + \Delta t) - f(x, y, z, t)$. We want to find $\max \Delta u$. By Taylor's formula,

$$\Delta u = (f_x \Delta x + f_y \Delta y + f_z \Delta z + f_t \Delta t)(a, b, c, d)$$

where (a, b, c, d) lies on the line segment joining (x, y, z, t) to $(x + \Delta x, y + \Delta y, z + \Delta z, t + \Delta t)$. Therefore,

$$|\Delta u| \leq |f_x|\,|\Delta x| + |f_y|\,|\Delta y| + |f_z|\,|\Delta z| + |f_t|\,|\Delta t|.$$

Problem 1.6 The hypotenuse c and the side a of a right angled triangle ABC determined with maximum absolute errors $|\Delta c| = 0.2,\ |\Delta a| = 0.1$ are, respectively, $c = 75,\ a = 32$. Determine the angle A and determine the maximum absolute error ΔA in the calculation of the angle A.

$$A(a, c) = \sin^{-1}\frac{a}{c} \Rightarrow \frac{\partial A}{\partial a} = \frac{1}{\sqrt{c^2 - a^2}},\ \frac{\partial A}{\partial c} = \frac{-a}{c\sqrt{c^2 - a^2}}.$$

Then $|\Delta A| \leq \dfrac{1}{\sqrt{(75)^2 - (32)^2}} \times 0.1 + \dfrac{32}{75\sqrt{(75)^2 - (32)^2}} \times 0.2 = 0.00273.$

Therefore $\sin^{-1}\frac{32}{75} - 0.00273 \leq A \leq \sin^{-1}\frac{32}{75} + 0.00273.$

Problem 1.7 Find a point in the plane where the function $f(x, y) = \frac{1}{2} - \sin(x^2 + y^2)$ has a local maximum.

We see that at $(0, 0)$, the function has a maximum value of $\frac{1}{2}$. To prove this, consider the neighborhood $B = \{(x, y) : x^2 + y^2 \leq \pi/9\}$ of $(0, 0)$. Now, for any point $(a, b) \in B$ other than $(0, 0)$, we have

$$f(a, b) = \frac{1}{2} - \sin(a^2 + b^2) \leq \frac{1}{2} = f(0, 0).$$

Problem 1.8 Decompose a given positive number a into three parts to make their product maximum.

Let $a = x + y + (a - x - y)$, for $0 \leq x, y, a - x - y \leq a$. Then x and y can take values from the region D bounded by the straight lines $x = 0, y = 0$ and $x + y = a$. The function to be maximized is

$$f(x, y) = xy(a - x - y) \quad \text{on } D.$$

On the line $x = 0$, $f(x, y)$ is 0; on the line $y = 0$, $f(x, y) = 0$; and on the line $x + y = a$, $f(x, y) = 0$.

In the interior of D, the partial derivatives of f are continuous. They are

$$f_x = y(a - 2x - y),\ f_y = x(a - x - 2y).$$

The critical points satisfy $y(a - 2x - y) = 0$ and $x(a - x - 2y) = 0$, where $0 < x, y, a - x - y < a$. The solutions of these equations give:

$$P_1 = (0, 0),\ P_2 = (0, a),\ P_3 = (a, 0),\ P_4 = \big(a/3, a/3\big).$$

Of these, the points P_1, P_2, P_3 are on the boundary of D; the only interior point is P_4. At P_4, the value of $f(x, y) = \frac{a^3}{27}$. Comparing $f(P_4)$ and the values of $f(x, y)$ on the boundary of D, we see that $f(P_4)$ is the maximum. The point P_4 corresponds to the decomposition of a as $a = \frac{a}{3} + \frac{a}{3} + \frac{a}{3}$.

Problem 1.9 Find extreme points of the function $z = x^3 + y^3 - 3xy$.

Here, z_x and z_y are continuous. Thus the critical points are obtained by solving

$$z_x = 3x^2 - 3y = 0,\ z_y = 3y^2 - 3x = 0.$$

These are $P_1 = (1, 1)$ and $P_2 = (0, 0)$.

The second derivatives are $z_{xx} = 6x,\ z_{xy} = -3,\ z_{yy} = 6y$.

$H(P_1) = (z_{xx}z_{yy} - z_{xy}^2)(P_1) = 36 - 9 = 27 > 0,\ z_{xx}(P_1) = 6 > 0$. Thus, P_1 is a minimum point and the minimum value of z is $z(1, 1) = -1$.

$H(P_2) = (z_{xx}z_{yy} - z_{xy}^2)(P_2) = -9 < 0$. Hence P_2 is a saddle point.

Problem 1.10 Find the maximum of $w = xyz$ given that $xy + zx + yz = a$ for a given positive number a, and $x > 0,\ y > 0,\ z > 0$.

The auxiliary function is

$$F(x, y, z, \lambda) = xyz + \lambda(xy + zx + yz - a).$$

Equating its partial derivatives to zero, we have

$$yz + \lambda(y + z) = 0,\ xz + \lambda(x + z) = 0,\ xy + \lambda(x + y) = 0.$$

Multiply the first by x, the second by y, and subtract to obtain:

$$\lambda x(y + z) - \lambda y(x + z) = 0 \Rightarrow \lambda z(x - y) = 0.$$

If $\lambda = 0$, then $xy + \lambda(x + y) = 0$ would imply $x = 0$ or $y = 0$. But $x > 0$ and $y > 0$. So, $\lambda \neq 0$. Also, $z > 0$. Therefore, $x = y$. Similarly, using the second and third equations, we get $y = z$. Therefore, $x = y = z$. Then

$$xy + zx + yz = a \text{ gives } x = y = z = \sqrt{a/3}.$$

The corresponding value of w is $(a/3)^{3/2}$. This cannot be the minimum value of w; reason: by reducing x, y close to 0, and taking a large value of z so that $xy+zx+yz = a$ is satisfied, w can be made as small as possible. Hence w has a maximum at $(\sqrt{a/3}, \sqrt{a/3}, \sqrt{a/3})$ and the maximum value is $(a/3)^{3/2}$.

Problem 1.11 Determine the maximum value of $z = (x_1 \cdots x_n)^{1/n}$ provided that $x_1 + \cdots + x_n = a$, where a is a given positive number.

Maximizing z is equivalent to maximizing $f(x_1, \ldots, x_n) = z^n = x_1 x_2 \cdots x_n$. Set up the auxiliary function

$$F(x_1, \ldots, x_n, \lambda) = x_1 x_2 \cdots x_n + \lambda(x_1 + \cdots x_n - a).$$

Equate the partial derivatives F_{x_i} to zero to obtain

$$x_1 \cdots x_{i-1} x_{i+1} \cdots x_n + \lambda = 0 \text{ for } i = 1, 2, \ldots, n.$$

Notice that $\lambda \neq 0$. Then multiplying by x_i, we see that $-\lambda x_i = x_1 x_2 \cdots x_n$ for each i. Therefore, $x_1 = x_2 = \cdots = x_n = a/n$. In that case, $f = (a/n)^n$ and $z = a/n$. This value is not a minimum value of z since z can be made arbitrarily small by choosing x_1 close to 0. Thus, the maximum of z is a/n.

This gives an alternative proof that the geometric mean of n positive numbers is no more than the arithmetic mean of those numbers.

Problem 1.12 A function $f(x, y)$ is called **homogeneous** of degree n in an open region $D \subseteq \mathbb{R}^2$ iff for all $(x, y) \in D$, and for each positive λ, $f(\lambda x, \lambda y) = \lambda^n f(x, y)$. For example,

$f(x, y) = \left(x^2 + y^2\right)^{3/2}$ is homogeneous of degree 3 in the whole plane;
$f(x, y) = x^{1/3} y^{-4/3} \tan^{-1}(y/x)$ is homogeneous of degree -1 in the region D, which is any quadrant without the axes.

Prove the following theorem by Euler:
Let D be an open region in the plane. Let $f : D \to \mathbb{R}$ have continuous partial derivatives. Then f is a homogeneous function of degree n iff $xf_x + yf_y = nf$.

Proof of Euler's theorem: Let f be a homogeneous function of degree n. Then $f(\lambda x, \lambda y) - \lambda^n f(x, y) = 0$. Differentiate it partially with respect to λ to obtain:

$$x f_x(\lambda x, \lambda y) + y f_y(\lambda x, \lambda y) = n\lambda^{n-1} f(x, y).$$

Here, f_x means the partial derivative of f with respect to the first variable, and f_y means the partial derivative of f with respect to the second variable.

Then set $\lambda = 1$ to get $x f_x(x, y) + y f_y(x, y) = n f(x, y)$.

Conversely, let $(a, b) \in D$. Define $\phi(\lambda) = f(\lambda a, \lambda b)$. Differentiate with respect to λ to get

$$\phi'(\lambda) = af_x(\lambda a, \lambda b) + bf_y(\lambda a, \lambda b).$$

However, $nf(\lambda a, \lambda b) = \lambda a f_x(\lambda a, \lambda b) + \lambda b f_y(\lambda a, \lambda b)$. That is, $\lambda\phi'(\lambda) = n\phi(\lambda)$.
Now, differentiate $\lambda^{-n}\phi(\lambda)$ with respect to λ to obtain

$$\big(\phi(\lambda)\lambda^{-n}\big)' = \phi'(\lambda)\lambda^{-n} - n\phi(\lambda)\lambda^{-n-1} = 0.$$

Therefore, $\phi(\lambda)\lambda^{-n} = c$ for some constant c. Set $\lambda = 1$ to get $c = f(a, b)$. Then

$$f(\lambda a, \lambda b) = \lambda^n f(a, b).$$

Since (a, b) is any arbitrary point in D, we have $f(\lambda x, \lambda y) = \lambda^n f(x, y)$. This completes the proof.

For our earlier examples, you can check that

$$x\,\frac{\partial}{\partial x}\Big(x^{1/3}y^{-4/3}\tan^{-1}(y/x)\Big) + y\frac{\partial}{\partial y}\Big(x^{1/3}y^{-4/3}\tan^{-1}(y/x)\Big) + x^{1/3}y^{-4/3}\tan^{-1}\left(\frac{y}{x}\right) = 0.$$

$$x\frac{\partial}{\partial x}\Big(x^2+y^2\Big)^{3/2} + y\frac{\partial}{\partial y}\Big(x^2+y^2\Big)^{3/2} = 3\big(x^2+y^2\big)^{3/2}.$$

1.13 Exercises for Chapter 1

1. Find the limit, if exists:
   (a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{e^y \sin x}{x}$ (b) $\lim\limits_{\substack{(x,y)\to(0,0)\\ x\neq y}} \dfrac{x - y + 2\sqrt{x} - 2\sqrt{y}}{\sqrt{x} - \sqrt{y}}$
   (c) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{|xy|}$ (d) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2}{x^2 - y}$
   (e) $\lim\limits_{(x,y)\to(0,0)} \cos\left(\dfrac{x^3 - y^3}{x^2 + y^2}\right)$ (f) $\lim\limits_{(x,y)\to(0,0)} \dfrac{2x}{x^2 + x + y^2}$
2. At what points (x, y) or (x, y, z) are the functions below continuous?
   (a) $g(x, y) = \dfrac{x^2 + y^2}{x^2 - 3x + 2}$ (b) $h(x, y, z) = \dfrac{1}{x^2 + z^2 - 1}$
3. Let $f(x, y) = \ln\left(\dfrac{3x^2 - x^2y^2 + 3y^2}{x^2 + y^2}\right)$ for $(x, y) \neq (0, 0)$. Define $f(0, 0)$ in a way that extends f to be continuous at the origin.
4. Let $f(x, y) = \dfrac{y}{x^2 + 1}$. Given $\epsilon = 0.05$, find a $\delta > 0$ such that for all (x, y) with $\sqrt{x^2 + y^2} < \delta$, we have $|f(x, y) - f(0, 0)| < \epsilon$.

5. Find f_x and f_y, where
   (a) $f(x,y) = \dfrac{x}{x^2+y^2}$ (b) $f(x,y) = e^{xy}\ln y$
6. Find f_x, f_y and f_z, where
   (a) $f(x,y,z) = \sin^{-1}(xyz)$ (b) $f(x,y,z) = yz\ln(xy)$
7. Does a function f with continuous first partial derivatives throughout an open region R have to be continuous on R?
8. Find all the second order partial derivatives:
   (a) $g(x,y) = x^2y + \cos y + y\sin x$ (b) $g(x,y) = \tan^{-1}(y/x)$
9. Express $\partial z/\partial u$ and $\partial z/\partial v$ in terms of u and v, where $z = 4e^x \ln y,\ x = \ln(u\cos v),\ y = u\sin v$.
10. Express $\partial w/\partial u$ and $\partial w/\partial v$ in terms of u and v, where $w = xy + yz + zx,\ x = u+v,\ y = u-v,\ z = uv$.
11. Find dy/dx if $xe^y + \sin xy + y - \ln 2 = 0$.
12. If $z = x^2+y^2+z^2$ and $z = x^2+y^2$, then find
    (a) w_y, with y and z as independent variables.
    (b) w_z, with z and x as independent variables.
    (c) w_z, with y and z as independent variables.
13. Suppose that $f(x,y,z,w) = 0$ and $g(x,y,z,w) = 0$ determine z and w as differentiable functions of the independent variables x and y. Suppose also that $f_z g_w - f_w g_z \neq 0$. Then show the following:
    (a) $\dfrac{\partial z}{\partial x} = -\dfrac{f_x g_w - f_w g_x}{f_z g_w - f_w g_z}$ (b) $\dfrac{\partial w}{\partial y} = -\dfrac{f_z g_y - f_y g_z}{f_z g_w - f_w g_z}$
14. Find the directional derivative of f at P_0 in the direction of $\vec{A}$:
    (a) $f(x,y) = \tan^{-1}(y/x) + \sqrt{3}\sin^{-1}(xy/2),\ P_0(1,1),\ \vec{A} = 3\hat{i} - 2\hat{j}$
    (b) $f(x,y,z) = x^2 + 2y^2 - 3z^2,\ P_0(1,1,1),\ \vec{A} = \hat{i} + \hat{j} + \hat{k}$
15. By about how much will $g(x,y,z) = x + x\cos z - y\sin z + y$ change if the point P moves from $(2,-1,0)$ a distance $ds = 0.2$ units toward the point $(0,1,2)$?
16. Find the directions in which the functions increase and decrease most rapidly at P_0. Then find the derivative of the function in these directions:
    (a) $f(x,y) = x^2+xy+y^2,\ P_0(1,1)$ (b) $f(x,y,z) = xe^y+z^2, P_0(1,\ln 2, 1/2)$
17. Find the equation of the tangent plane and normal line at P_0 on the given surface.
    (a) $x^2+y^2-z^2 = 18,\ P_0(3,5,-4)$ (b) $x^2 - xy - y^2 - z = 0,\ P_0(1,1,-1)$
18. Use Taylor's formula for $f(x,y) = (1-x-y+xy)^{-1}$ at the origin to find quadratic and cubic approximations of f near the origin.

19. Use Taylor's formula to find a quadratic approximation of $e^x \sin y$ at the origin. Estimate the error in the approximation if $|x| \leq 0.1$ and $|y| \leq 0.1$.
20. Find all the local maxima, local minima, and saddle points of the following functions:
    (a) $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2 - 8$ (b) $f(x, y) = y \sin x$
    (c) $f(x, y) = 2x^3 + 2y^3 + -9x^2 + 3y^2 - 12y$ (d) $f(x, y) = e^{2x} \cos y$
21. Find the absolute maxima and minima of the functions on the given domains:
    (a) $f(x, y) = 2x^2 - 4x + y^2 - 4y + 1$ on the closed triangular plate bounded by the lines $x = 0,\ y = 2,\ y = 2x$ in the first quadrant.
    (b) $f(x, y) = (4x - x^2) \cos y$ on the rectangular plate $1 \leq x \leq 3, -\pi/4 \leq y \leq \pi/4$.
22. Find $a, b \in \mathbb{R}$ with $a \leq b$ such that I has its largest value:
    (a) $I = \int_a^b (6 - x - x^2)\, dx$ (b) $I = \int_a^b (24 - 2x - x^2)^{3/2}\, dx$
23. Find the maxima, minima, and saddle points of (x, y), if any, given that $f_x = 9x^2 - 9$ and $f_y = 2y + 4$.
24. For what values of the constant k does the Second Derivative Test guarantee that the function $f(x, y) = x^2 + kxy + y^2$ will have a saddle point at $(0, 0)$? A local minimum at $(0, 0)$? For what values of k is the Second Derivative Test inconclusive? Give reasons for your answers.
25. Find the absolute maximum and minimum values of $f(x, y) = 2x^2 + y^2$ on the quarter circle $x^2 + y^2 = 4,\ x \geq 0,\ y \geq 0$.
26. Find the points on the curve $x^2 + xy + y^2 = 1$ in the xy-plane that are nearest to and farthest from the origin.
27. The temperature at a point (x, y) on a metal plate is $T(x, y) = 4x^2 - 4xy + y^2$. An ant on the plate walks around the circle of radius 5 centered at the origin. What are the highest and lowest temperatures encountered by the ant?
28. A space probe in the shape of the ellipsoid $4x^2 + y^2 + 4z^2 = 16$ enters Earth's atmosphere and its surface begins to heat. After 1 hour, the temperature at the point (x, y, z) on the probe's surface is $T(x, y, z) = 8x^2 + 4yz - 16z + 600$. Find the hottest point on the probe's surface.
29. Find the extreme values of the function $f(x, y, z) = xy + z^2$ on the circle in which the plane $y - x = 0$ intersects the sphere $x^2 + y^2 + z^2 = 4$.

Answers:
1a. 1 1b. 2 1c. -1 1d. doesnt exist 1e. 1 1f. doesnt exist
2a. $x \neq 1, x \neq 2$ 2b. $x^2 + z^2 \neq 1$ 3. $f(0, 0) = \ln 3$ 4. $\delta = 0.05$
5a. $-\frac{2xy}{(x^2+y^2)^2}$ 5b. $xe^{xy} \ln y + \frac{e^{xy}}{y}$ 7. Yes 14a. $-3/(2\sqrt{13})$ 14b. 0

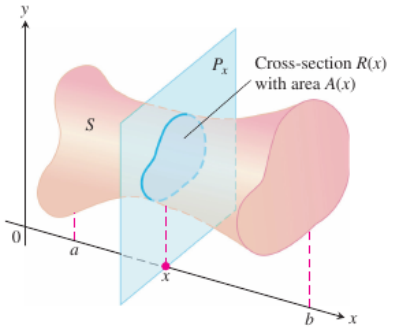
15. 0 16. Increases most in direction (a) $-\hat{\imath}/\sqrt{2} + \hat{\jmath}/\sqrt{2}$ (b) $(2\hat{\imath} + 2\hat{\jmath} + k)/3$
17. Tangents: (a) $3x + 5y + 4z = 18$ (b) $x - 3y - z = -1$
18. Cubic Approximation: $1 + x + y + x^2 + xy + y^2 + x^3 + x^2y + xy^2 + y^3$
19. Quad Approx: $y + xy$, Error: ≤ 0.000814 20a & c. one loc min, one loc max, two sadl pts. 20c. only sadl pts 20d. no extrema, no sadl pts
21a. Max: 1, Min: -5 21b. Max: 4, Min: $3/\sqrt{2}$
22a. $a = -3,\ b = 2$ 22b. $a = -6,\ b = 4$ 23. Loc Min at $(1, -2)$, no loc. max, sadl pt at $(-1, -2)$ 24. Loc min for $|k| < 2$, Sadl pt for $|k| > 2$, Inconclusive for $|k| = 2$ 25. Max. 2, Min. 0 26. Nearest: $\pm(1/\sqrt{3}, 1/\sqrt{3})$, Farthest: $\pm(1, -1)$
27. Highest: $125°$, Lowest: $0°$ 28. $(\pm 4/3, -4/3, -4/3)$ 29. Max: 4, Min: 2

2

Multiple Integrals

2.1 Volumes by slicing

We know that the integral of a continuous function $f(x)$ from a to b is the area of the region enclosed between the graph of $y = f(x)$, the x-axis, the lines $x = a$ and $x = b$. Analogously, an integral can be used to compute the volume of a solid in certain cases. Given a solid, let us choose our axes in such a way that the solid is bounded between and touching the planes $x = a$ and $x = b$, where $a < b$. Suppose the the plane at any point $x \in [a, b]$ parallel to the yz-plane crosses the solid at a region $R(x)$ whose area is $A(x)$. Assume that $A(x)$ is a continuous function of x.



We divide the interval $[a, b]$ into n sub-intervals. Let the partition be

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

On the ith subinterval we approximate the slice of the solid by $A(x_i^*)$ for a point $x_i^* \in [x_{i-1}, x_i]$. Write $\Delta x_i = x_{i+1} - x_i$. Then the volume of the solid is approximated by the sum

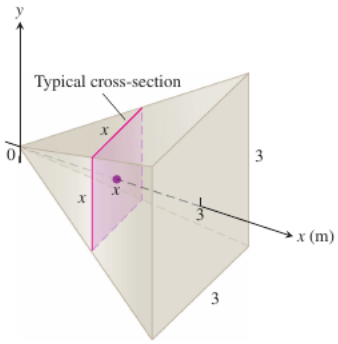
$$\sum_{i=1}^{n} A(x_i^*)\Delta x_i.$$

The volume of the solid is the limit of the above sum where each Δx_i approaches 0 (necessarily $n \to \infty$). Since $A(x)$ is a continuous function of x, the volume is,

$$V = \int_a^b A(x)\, dx.$$

(2.1) ***Example***

A pyramid 3 m high has a square base that is 3 m on a side. The cross-section of the pyramid perpendicular to the altitude x m down from the vertex is a square with each side x m. Find the volume of the pyramid.

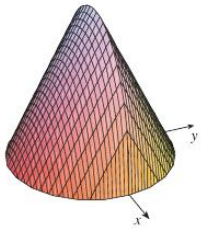


The cross section of the pyramid made by the plane at $x \in [0, 3]$ parallel to the yz-plane has area $A(x) = x^2$.

So, the volume $V = \int_0^3 A(x)\,dx = \int_0^3 x^2\,dx = \dfrac{x^3}{3}\Big|_0^3 = 9\,\text{m}^3$. □

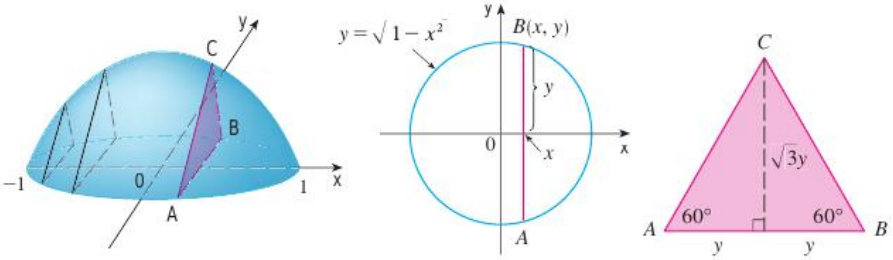
(2.2) ***Example***

In the figure is shown a solid with a circular base of radius 1. Parallel cross sections perpendicular to the base are equilateral triangles. Find the volume of the solid.



Take the base of the solid as the disk $x^2 + y^2 \le 1$.

The solid, its base, and a typical triangle at a distance x from the origin are shown in the figure below.



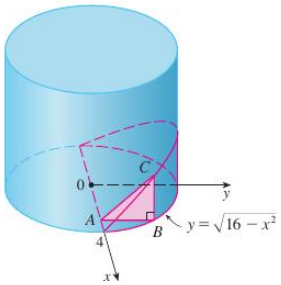
The point B lies on the circle $y = \sqrt{1 - x^2}$. So, the length of AB is $2\sqrt{1 - x^2}$. Since the triangle is equilateral, its height is $\sqrt{3}\sqrt{1 - x^2}$. The cross sectional area is

$$A(x) = \frac{1}{2}\, 2\, \sqrt{1 - x^2}\, \sqrt{3}\, \sqrt{1 - x^2} = \sqrt{3}\,(1 - x^2).$$

The volume of the solid is $V = \int_{-1}^{1} A(x)\,dx = \int_{-1}^{1} \sqrt{3}\,(1 - x^2)\,dx = \dfrac{4}{\sqrt{3}}$. □

(2.3) ***Example***

A wedge is cut out of a circular cylinder of radius 4 by two planes. One plane is perpendicular to the axis of the cylinder. The other intersects the first at an angle of 30° along a diameter of the cylinder. Find the volume of the wedge.



If we place the x-axis along the diameter where the planes meet, then the base of the solid is the semicircle with equation

$$y = \sqrt{16 - x^2}, \quad -4 \le x \le 4.$$

A cross-section perpendicular to the x-axis at a distance x from the origin is the triangle ABC, whose base is $y = \sqrt{16 - x^2}$; its height is $|BC| = y \tan 30° = \sqrt{16 - x^2}/\sqrt{3}$. Thus the cross sectional area is

$$A(x) = \frac{1}{2}\sqrt{16 - x^2}\,\frac{\sqrt{16 - x^2}}{\sqrt{3}} = \frac{16 - x^2}{2\sqrt{3}}.$$

Then the required volume of the wedge is

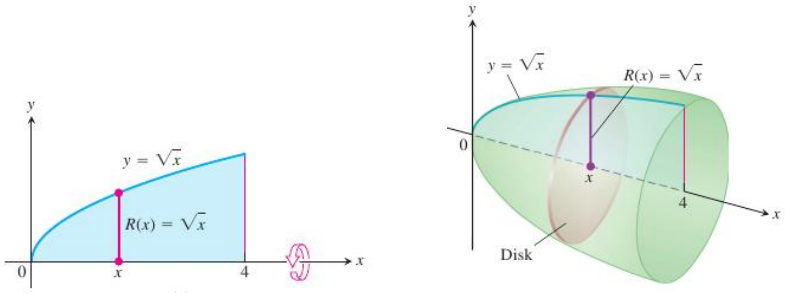
$$V = \int_{-4}^{4} A(x)\,dx = \int_{-4}^{4} \frac{16 - x^2}{2\sqrt{3}}\,dx = \frac{128}{3\sqrt{3}}.$$

Aliter: Consider the cross section of the solid made by the plane at a point y on the y-axis parallel to the zx-plane. The cross section is a rectangle $R(y)$. Now, the length of $R(y)$ is the length of the chord of the circle $x^2 + y^2 = 4$ where the perpendicular distance from the origin to the chord is y. Thus, it is $2\sqrt{16 - y^2}$. The width of $R(y)$ is $y \tan 30° = y/\sqrt{3}$. Then, the volume is

$$V = \int_0^4 2\sqrt{16 - y^2} \cdot \frac{y}{\sqrt{3}}\,dy = -\frac{1}{\sqrt{3}}\int_0^4 \sqrt{16 - y^2}\,d(16 - y^2) = \frac{128}{3\sqrt{3}}. \qquad \square$$

2.2 Volume of a solid of revolution

We can use the method of slicing for computing the volume of a solid of revolution. The solid obtained by rotating a plane region about a straight line in the same plane is called a **solid of revolution**. The line is called the **axis of revolution**.



Suppose the region is bounded above by the curve $y = f(x)$ and below by the x-axis, where $a \le x \le b$. Suppose the axis of revolution is the x-axis. The cross section of the solid made by the plane at the point $x \in [a, b]$ parallel to the yz-plane is a disk of radius $f(x)$. Thus, $A(x) = \pi[f(x)]^2$. The volume of the solid of revolution is

$$V = \int_a^b A(x)\,dx = \int_a^b \pi[f(x)]^2\,dx.$$

This method of computing the volume of a solid of revolution is sometimes called the **disk method**.

If the axis of revolution is a straight line other than the x-axis, similar formulas can be obtained for the volume.

(2.4) *Example*

The region between the curves $y = \sqrt{x},\ 0 \le x \le 4$ and $y = 0$ is revolved around the x-axis. Find the volume of the solid of revolution.

As shown in the above figure, the required volume is

$$V = \int_0^4 \pi\left(\sqrt{x}\right)^2 dx = \int_0^4 \pi\, x\, dx = \pi\left[\frac{x^2}{2}\right]_0^4 = 8\pi. \qquad \square$$

(2.5) *Example*

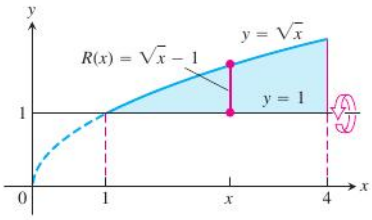
Find the volume of the sphere of radius $a > 0$.

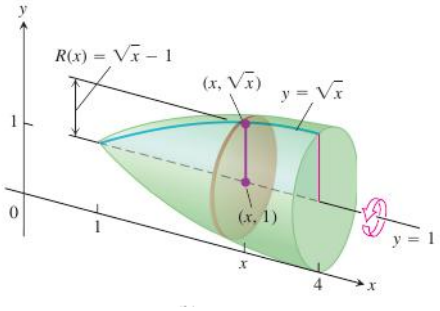
We think of the sphere as the solid of revolution of the region bounded by the upper semi-circle $x^2 + y^2 = a^2,\ y \ge 0$ with $-a \le x \le a$. The curve is thus $y = \sqrt{a^2 - x^2}$. Then the volume of the sphere is

$$V = \int_{-a}^{a} \pi\left(\sqrt{a^2 - x^2}\right)^2 dx = \int_{-a}^{a} \pi\left(a^2 - x^2\right) dx = \pi\left[a^2 x - \frac{x^3}{3}\right]_{-a}^{a} = \frac{4}{3}\pi a^3. \qquad \square$$

(2.6) *Example*

Find the volume of the solid obtained by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 1,\ x = 4$ about the line $y = 1$.

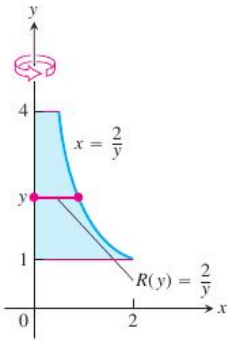


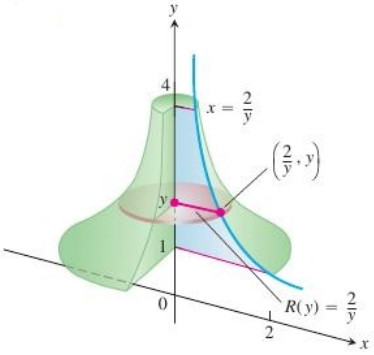


$$V = \int_1^4 \pi[R(x)]^2\,dx = \int_1^4 \pi\left(\sqrt{x}-1\right)^2 dx = \int_1^4 \pi\left(x - 2\sqrt{x} + 1\right) dx = \frac{7\pi}{6}. \qquad \square$$

(2.7) *Example*

Find the volume of the solid generated by revolving the region between the y-axis and the curve $xy = 2,\ 1 \le y \le 4$, about the y-axis.

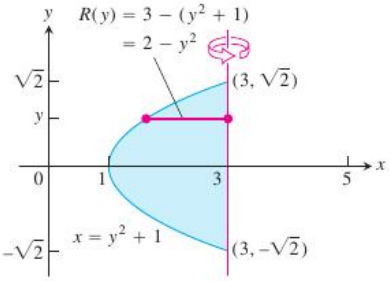


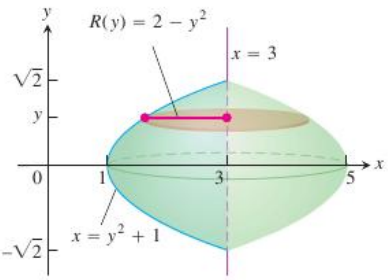


The volume is $V = \int_1^4 \pi[R(y)]^2\,dy = \pi\int_1^4 \frac{4}{y^2}\,dy = 3\pi.$ $\square$

(2.8) *Example*

Find the volume of the solid generated by revolving the region between the parabola $x = y^2 + 1$ and the line $x = 3$ about the line $x = 3$.





Notice that the cross sections are perpendicular to the axis of revolution: $x = 3$.

The volume is $V = \int_{-\sqrt{2}}^{\sqrt{2}} \pi [R(y)]^2 dy = \int_{-\sqrt{2}}^{\sqrt{2}} \pi \left(2 - y^2\right)^2 dy = \frac{64\pi\sqrt{2}}{15}$. □

(2.9) ***Example***

Find the volume of the solid generated by revolving the region bounded by the lines $x = 0,\ x = 2,\ y = 0$ and the curve $y = 4/(x^2 + 4)$, about the x-axis.

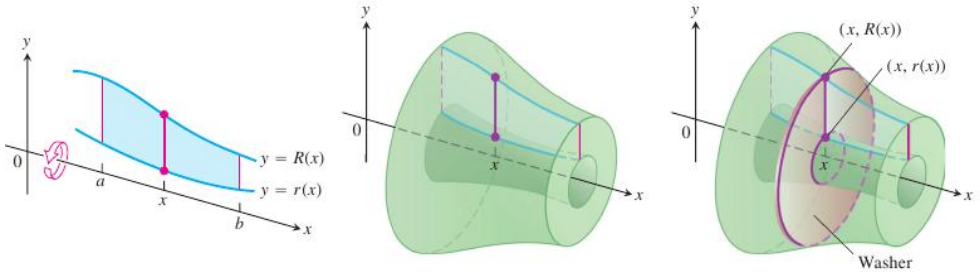
The volume is $V = \int_0^2 \pi \frac{16}{(x^2+4)^2}\, dx$.

Substitute $x = 2\tan t$. $dx = 2\sec^2 t\, dt$, $(x^2 + 4)^2 = 16\sec^4 t$ for $0 \le t \le \pi/4$. So,

$$V = \int_0^{\pi/4} 16\pi \frac{2\sec^2 t}{16\sec^4 t}\, dt = \int_0^{\pi/4} 2\pi \cos^2 t\, dt = \pi\left(\frac{\pi}{4} + \frac{1}{2}\right).$$ □

2.3 Washer method

If the region which revolves does not border the axis of revolution, then there are **holes** in the solid. In this case, we subtract the volume of the hole to obtain the volume of the solid of revolution. This is sometimes called the **washer method**.



Thus, the volume of the the solid of revolution is given by

$$V = \int_a^b A(x)\, dx = \int_a^b \pi\left[(R(x))^2 - (r(x))^2\right] dx.$$

(2.10) ***Example***

The region bounded by the curve $y = x^2 + 1$ and the line $x + y = 3$ is revolved about the x-axis to generate a solid. Find the volume of the solid.

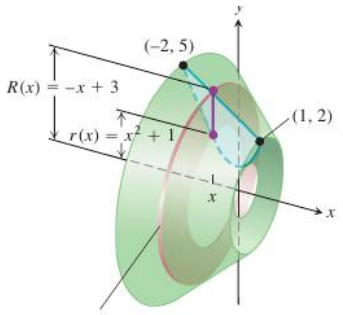
The outer radius of the washer is $R(x) = -x+3$ and the inner radius is $r(x) = x^2+1$.

The limits of integration are obtained by finding the points of intersection of the given curves:

$$x^2 + 1 = -x + 3 \Rightarrow x = -2,\ 1.$$

The required volume is

$$V = \int_{-2}^{1} \pi\left[(-x+3)^2 - (x^2+1)^2\right] dx = \frac{117\pi}{5}.$$

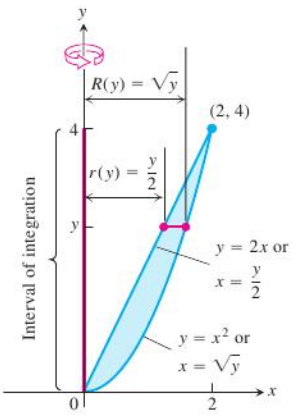


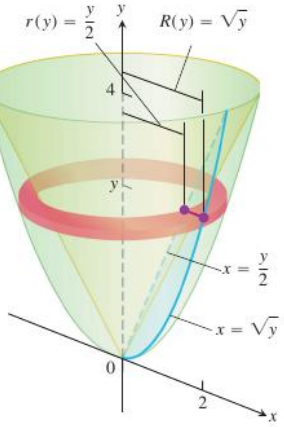
□

(2.11) *Example*

Find the volume of the solid obtained by revolving the region bounded by the curves $y = x^2$ and $y = 2x$, about the y-axis.

The given curves intersect at $y = 0$ and $y = 4$. The required volume is



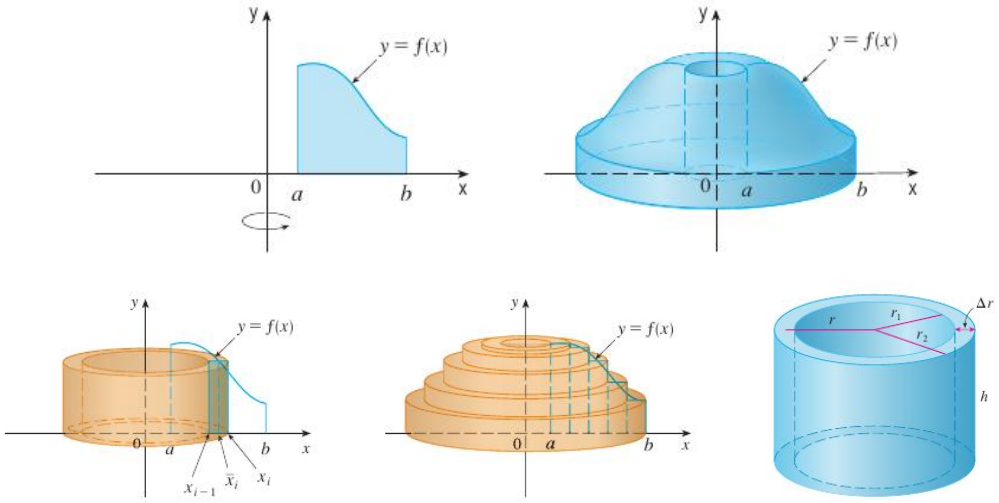


$$V = \int_0^4 \pi\left[(R(y))^2 - (r(y))^2\right] dy = \int_0^4 \pi\left[(\sqrt{y})^2 - (y/2)^2\right] dy = \frac{8\pi}{3}.$$

□

2.4 Cylindrical shell method

There is another way we can compute the volumes of solids of revolution with holes. Let $a \geq 0$ and let $f : [a, b] \to \mathbb{R}$ be a continuous function with $f(x) \geq 0$. Let S be the solid obtained by revolving the region bounded by the curve $y = f(x)$ and the lines $y = 0$, $x = a$ and $x = b$, about the y-axis.



In **Cylindrical Shell method**, we slice the solid into cylindrical shells as shown in the figures. The sum of these cylindrical shells is taken as an approximation to the volume of the solid. When the width of each cylindrical shell approaches zero, the limit of this sum gives the volume.

To carry out this idea, we divide the interval $[a, b]$ into n subintervals $[x_{i-1}, x_i]$ of equal width Δx and take $\overline{x}_i$ as the mid-point of the subinterval. The ith cylindrical shell is obtained by revolving the rectangle with base $[x_{i-1}, x_i]$ and average height $f(\overline{x}_i)$ about the y-axis. The base of this cylindrical shell has inner radius x_i, outer radius $x_{i+1} = x_i + \Delta x$, and height $f(\overline{x}_i)$. So, its volume is

$$V_i = \pi\left[x_{i+1}^2 - x_i^2\right] f(\overline{x}_i) = \pi\left[x_{i+1} + x_i\right]\Delta x f(\overline{x}_i) = 2\pi\overline{x}_i\, f(\overline{x}_i)\, \Delta x.$$

Then an approximation to the volume V is given by

$$\sum_{i=1}^{n} V_i = \sum_{i=1}^{n} 2\pi\overline{x}_i\, f(\overline{x}_i)\, \Delta x.$$

When n approach ∞, we get the volume as

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} V_i = \int_a^b 2\pi x f(x)\, dx.$$

In the integrand of this formula, the term x is called the **shell radius** and the term $f(x)$ is called the **shell height**.

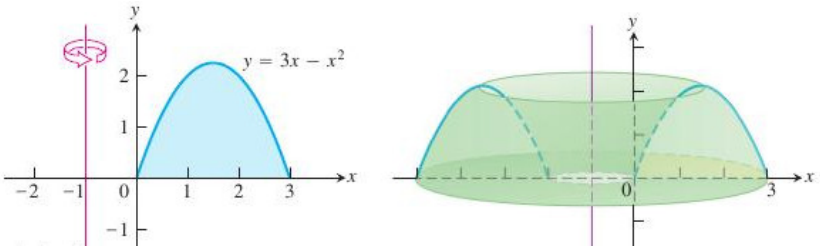
Instead of taking the axis of revolution as the y-axis, we may take the vertical line $x = \ell$. In that case, the shell radius will be $x - \ell$ instead of $x = x - 0$. With the assumptions used, we may state the result as follows:

The volume of the solid generated by revolving the region between the x-axis and the curve $y = f(x)$, $a \le x \le b$ with $f(x) \ge 0$ about a vertical line $x = \ell$, where $\ell \le a$, is

$$V = \int_a^b 2\pi(x - \ell)f(x)\,dx = \int_a^b 2\pi(\text{shell radius})(\text{shell height})\,dx.$$

(2.12) ***Example***

Find the volume of the solid generated by revolving the region bounded by the parabola $y = 3x - x^2$ and the x-axis, about the line $x = -1$.

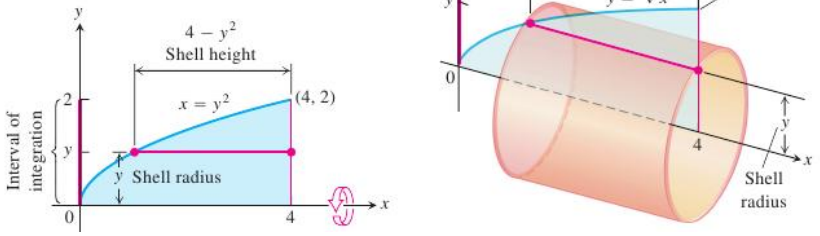


The parabola intersects the x-axis at $x = 0$ and $x = 3$. The required volume is

$$V = \int_0^3 2\pi(x + 1)(3x - x^2)\,dx = 2\pi \int_0^3 \left(2x^2 + 3x - x^3\right) dx = \frac{45\pi}{2}. \qquad \square$$

(2.13) ***Example***

The region bounded by the x-axis, the line $x = 4$ and the curve $y = \sqrt{x}$ is revolved about the x-axis. Find the volume of the solid of revolution.



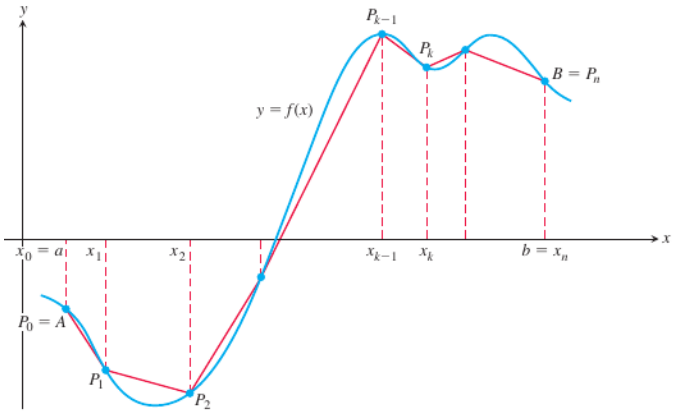
Since the revolution is about the x-axis, that is, the line $y = 0$, the variable of integration is y. The limits of integration are $y = 0$ and $y = 2$. The shell radius is y and the shell height is $4 - y^2$. Thus the volume of the solid of revolution is

$$V = \int_0^2 2\pi\, y(4 - y^2)\,dy = 2\pi\left[2y^2 - \frac{y^4}{4}\right]_0^2 = 8\pi. \qquad \square$$

2.5 Arc length

Suppose a curve in the plane is given by a function $y = f(x)$, $a \le x \le b$, where $f(x)$ is continuously differentiable on $[a, b]$. Such a curve is said to be a **smooth curve**. We can approximate the length of the curve by the length of the line segment joining the end-points $(a, f(a))$ and $(b, f(b))$. This approximation may be very crude. However, if a and b are very close to each other, then our approximation may be a good one. Intuitively, we may thus approximate the length of the curve by choosing many close-by points on the curve and then take the sum of the lengths of line segments joining those successive points. This leads to the following way to approximate the length of the curve:

Partition the interval $[a, b]$ into n number of sub-intervals $[x_0, x_1]$, $[x_1, x_2]$, ..., $[x_{n-1}, x_n]$ with $x_0 = a$, $x_n = b$ for some large natural number n. On each sub-interval $[x_k, x_{k+1}]$ we approximate the length of the curve by the length of the line segment joining the points $(x_i, f(x_i))$ and $(x_{i+1}, f(x_{i+1}))$. Writing $\Delta x_k = x_{k+1} - x_k$ and $\Delta y_k = f(x_{k+1}) - f(x_k))$, this length is $\sqrt{\Delta x_k^2 + \Delta y_k^2}$. Then, the approximate length of the curve, using this partition, is $\sum_{k=0}^{n-1} \sqrt{\Delta x_k^2 + \Delta y_k^2}$. Of course when the sub-intervals become smaller, the approximation of the length of the curve becomes better.



Since $f(x)$ is assumed to be differentiable, by the mean value theorem on the sub-interval $[x_k, x_{k+1}]$, $\Delta f(x_k) = f'(c_k)\Delta x_k$ for some c_k lying in the sub-interval $[x_k, x_{k+1}]$. Then, the length of the curve is approximated by

$$\sum_{k=0}^{n-1} \sqrt{\Delta x_k^2 + \Delta y_k^2} = \sum_{k=0}^{n-1} \sqrt{\Delta x_k^2 + [f'(c_k)]^2 (\Delta x_x)^2} = \sum_{k=0}^{n-1} \sqrt{1 + [f'(c_k)]^2}\, \Delta x_k.$$

Since the approximations are good when the lengths of th sub-intervals are small,

we obtain the length of the curve by taking the limit of he above expression when all lengths of the sub-intervals approach 0, that is, when the maximum of the lengths of the subintervals $\max_k(x_{k+1} - x_k)$ approaches 0. Since the above expression is the Riemann sum of the function $\sqrt{1 + [f'(x)]^2}$, the limit is simply the integral of this function taken over the interval $[a, b]$. Therefore, we obtain the following:

Let $f'(x)$ be continuous on the interval $[a, b]$. Then the length of the curve $y = f(x),\ a \le x \le b$, that is, the **arc length** of $y = f(x)$ joining the points $(a, f(a))$ and $(b, f(b))$, is given by

$$\text{Arc length} = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int_a^b \sqrt{1 + (y')^2}\, dx.$$

If a curve is given as $x = g(y)$, $c \le y \le d$, with $g'(x)$ being continuous, then the arc length of the curve joining the points $(g(c), c)$ and $(g(d, d))$ is given by

$$\text{Arc length} = \int_c^d \sqrt{1 + [g'(y)]^2}\, dy = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy = \int_c^d \sqrt{1 + (x')^2}\, dy.$$

If a curve is given parametrically by $x = x(t),\ y = y(t),\ \alpha \le t \le \beta$, with dx/dt and dy/dt continuous, then the length of the curve, that is the arc length of the curve joining the points $\big(x(\alpha), y(\alpha)\big)$ and $\big(x(\beta), y(\beta)\big)$, is given by

$$\text{Arc length} = \int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = \sqrt{(\dot{x})^2 + (\dot{y})^2}\, dt.$$

As a particular case, consider a curve given in polar coordinates:

$$r = f(\theta),\ \alpha \le \theta \le \beta.$$

In polar coordinates, we have $x = r\cos\theta$, $y = r\sin\theta$. Thus, the curve is

$$x = f(\theta)\cos\theta,\ y = f(\theta)\sin\theta, \quad \alpha \le \theta \le \beta.$$

Then, $dx/d\theta = f(\theta)(-\sin\theta) + f'(\theta)\cos\theta,\ dy/d\theta = f(\theta)(\cos\theta) + f'(\theta)\sin\theta$ so that $(dx/d\theta)^2 + (dy/d\theta)^2 = \big(f(\theta)\big)^2 + \big(f'(\theta)\big)^2$. Hence,

$$\text{Arc length} = \int_\alpha^\beta \sqrt{\big(f(\theta)\big)^2 + \big(f'(\theta)\big)^2}\, d\theta = \int_a^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta.$$

(2.14) ***Example***

Find the length of the following curves:

1. $y = \frac{4\sqrt{2}}{3}x^{3/2} - 1,\ 0 \le x \le 1.$

2. $y = (e^x + e^{-x})/2,\ 0 \le x \le 2.$

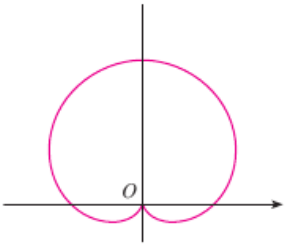
3. $y = (x/2)^{2/3},\ 0 \le x \le 2.$

1. $\dfrac{dy}{dx} = \dfrac{4\sqrt{2}}{3}\dfrac{3}{2}x^{1/2} \Rightarrow \left(\dfrac{dy}{dx}\right)^2 = 8x$. So,
   Arc length= $\displaystyle\int_0^1 \sqrt{1+(y')^2}\,dx = \int_0^1 \sqrt{1+8x}\,dx = \frac{2}{3}\frac{1}{8}(1+8x)^{3/2}\Big|_0^1 = \frac{13}{6}.$
2. $y' = (e^x - e^{-x})/2 \Rightarrow 1 + (y')^2 = \big(e^{2x} + e^{-2x} + 2\big)/4$. So,
   Arc length $= \displaystyle\int_0^2 \sqrt{1+(y')^2}\,dx = \int_0^2 \frac{1}{2}(e^x + e^{-x})\,dx = \frac{1}{2}\big(e^2 - e^{-2}\big).$
3. The derivative of y is not defined at $x = 0$.
   So, we write the curve as $x = 2y^{3/2}$, $0 \le y \le 1$. Then,
   $x' = \dfrac{dx}{dy} = 3y^{1/2} \Rightarrow (1 + \big(x'\big)^2 = 1 + 9y.$
   Arch length $= \displaystyle\int_0^1 \sqrt{1+9y}\,dy = \frac{1}{9}\frac{2}{3}(1+9y)^{3/2}\Big|_0^1 = \frac{2}{27}\big(10\sqrt{10} - 1\big).$ □

(2.15) ***Example***

Compute the length of the cardioid $r = 1 + \sin\theta$.

The cardioid $r = 1 + \sin\theta$ is traced fully once when $0 \le \theta \le 2\pi$.



$$r^2 + \left(\frac{dr}{d\theta}\right)^2 = (1+\sin\theta)^2 + \cos^2\theta = 2 + 2\sin\theta = 2\big(\cos(\theta/2) + \sin(\theta/2)\big)^2.$$

Hence, the length of the cardioid is given by

$$L = \int_0^{2\pi} \sqrt{r^2 + (dr/d\theta)^2}\,d\theta = \int_0^{2\pi} \sqrt{2}\,\big|\cos(\theta/2) + \sin(\theta/2)\big|\,d\theta.$$

Notice that $g(x) = \cos(x/2) + \sin(x/2)$ changes sign from + to − at $x = 3\pi/2$.

Hence,

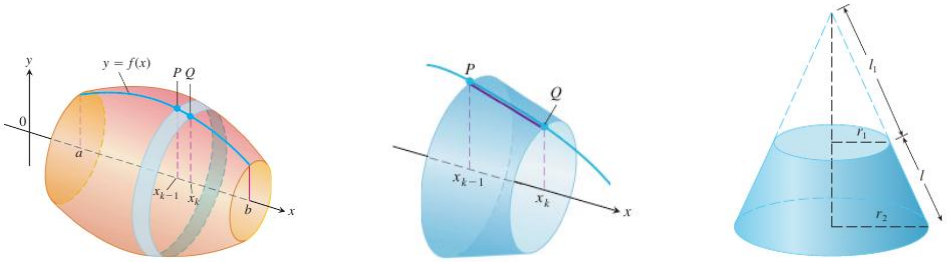
$$L = \int_0^{3\pi/2} \sqrt{2}\big(\cos(\theta/2) + \sin(\theta/2)\big)\,d\theta - \int_{3\pi/2}^{2\pi} \sqrt{2}\big(\cos(\theta/2) + \sin(\theta/2)\big)\,d\theta$$

$$= 2\sqrt{2}\big(\sin(\theta/2) - \cos(\theta/2)\big)\Big|_0^{3\pi/2} - 2\sqrt{2}\big(\sin(\theta/2) - \cos(\theta/2)\big)\Big|_{3\pi/2}^{2\pi} = 8.$$ □

2.6 Area of surface of revolution

Suppose a smooth curve is given by $y = f(x)$, where $f(x) \geq 0$. Its arch when $a \leq x \leq b$ is revolved about the x-axis to generate a solid. How do we compute the area of the surface of this solid?

We follow a strategy similar to computing the volume of revolution. Partition $[a, b]$ into n subintervals $[x_{k-1}, x_k]$. When each Δx_k is small, the surface area corresponding to this subinterval is approximately same as the area on the frustum of a right circular cone.



If a right circular cone has base radius R and slant height ℓ, then its surface area is given by $\pi R\ell$. Now, for the frustum, we subtract the smaller cone surface area from the larger. Look at the figure. The area of the frustum is

$$A = \pi r_2(\ell_1 + \ell) - \pi r_1\ell_1 = \pi[(r_2 - r_1)\ell_1 + r_2\ell].$$

Using similarity of triangles, we have $\dfrac{\ell_1}{r_1} = \dfrac{\ell_1 + \ell}{r_2}$.

This gives $r_2\ell_1 = r_1\ell_1 + r_1\ell \Rightarrow (r_2 - r_1)\ell_1 = r_1\ell$. Therefore,

$$A = \pi(r_1\ell + r_2\ell) = 2\pi\, r\, \ell, \text{ where } r = \frac{r_1 + r_2}{2}.$$

To use this this formula on the frustum obtained on the subinterval $[x_{k-1}, x_k]$, we notice that the slant height ℓ is approximated by $\sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$, where $\Delta x_k = x_k - x_{k-1}$ and $\Delta y_k = f(x_k) - f(x_{k-1})$. Next, the average radius

$$r = \frac{r_1 + r_2}{2} = \frac{f(x_{k-1}) + f(x_k)}{2}.$$

Thus the area of the frustum is

$$A_k = 2\pi\frac{f(x_{k-1}) + f(x_k)}{2}\sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}.$$

Due to MVT, we have $c_k \in [x_{k-1}, x_k]$ such that

$$\Delta y_k = f(x_k) - f(x_{k-1}) = f'(c_k)(x_k - x_{k-1}) = f'(c_k)\Delta x_k.$$

So, $\sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} = \sqrt{1 + (f'(c_k))^2}\,\Delta x_k$. The surface of revolution is approximated by

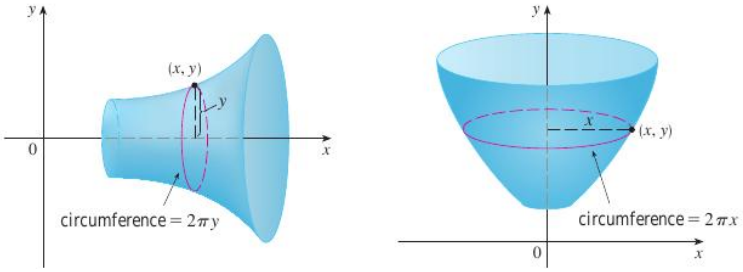
$$\sum_{k=1}^{n} A_k = 2\pi \frac{f(x_{k-1}) + f(x_k)}{2} \sqrt{1 + (f'(c_k))^2}\,\Delta x_k.$$

Its limit as $\max_k(x_k - x_{k-1}) \to 0$ is the Riemann sum of an integral, which is the required area:

$$S = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2}\,dx = \int_a^b 2\pi y \sqrt{1 + (f'(x))^2} dx.$$

If the arch is given by $y = f(x),\ a \le x \le b$, and the axis of revolution is the y-axis, then the surface area of revolution is given by

$$S = \int_a^b 2\pi\, x \sqrt{1 + (f'(x))^2}\,dx.$$



Similar formulas hold when the curve is given by $x = g(y),\ c \le y \le d$.

If the curve is given by $y = f(x),\ a \le x \le b$, we write $ds = \sqrt{1 + (f'(x))^2}\,dx$.

If the curve is given by $x = g(y),\ c \le y \le d$, we write $ds = \sqrt{1 + (g'(y))^2}\,dy$.

For parameterized curves, suppose the smooth curve is given by $x = x(t),\ y = y(t)$ for $\alpha \le t \le \beta$, where the curve is traversed exactly once while t increases from α to β. In such a case, we write $ds = \sqrt{\left(\dot{x}(t)\right)^2 + \left(\dot{y}(t)\right)^2}\,dt$.

With this notation, we see that the surface area of the solid generated by revolving the curve about an axis is given as follows.

1. Revolution about the x-axis $(y \ge 0)$: $S = \int 2\pi\, y\, ds$.
2. Revolution about the y-axis $(x \ge 0)$: $S = \int 2\pi\, x\, ds$.

We should remember that ds is simply an abbreviation here as explained earlier. The integration is with respect to the variables $x,\ y$ or t not with respect to s. If the curve is given by $y = f(x),\ a \le x \le b$, then the integration is with respect to the variable x with the limits of integration: from a to b. If $x = g(y),\ c \le y \le d$, then the integration is with respect to the variable y and the limits of integration: from c

to d. And if $x = x(t),\ y = y(t),\ \alpha \le t \le \beta$, then the integration is with respect to t and the limits of integration: from α to β.

Further caution is to be taken when the curve is given in parametric form. Firstly, the curve is traced exactly once while t varies from α to β. Secondly, the axis of revolution must not break the curve into parts. In case, it is a closed curve this condition says that the region bounded by the curve must not be divided into smaller regions. Though the curve may have its end-points on the axis of revolution.

(2.16) *Example*

Find the surface area of the solid obtained by revolving the curve $y = 2\sqrt{x}, 1 \le x \le 2$ about the x-axis.

Since $y = 2\sqrt{x},\ y' = 1/\sqrt{x},\ \sqrt{1 + (y')^2} = \sqrt{1 + 1/x}$. So, $ds = \sqrt{1 + 1/x}\,dx$. Then

$$S = \int_1^2 2\pi\, y\, ds = \int_1^2 2\pi\, 2\sqrt{x}\,\sqrt{1 + \frac{1}{x}}\,dx = \frac{8\pi}{3}(3\sqrt{3} - 2\sqrt{2}). \qquad \square$$

(2.17) *Example*

The arch of the parabola $y = x^2,\ 1 \le x \le 2$ is revolved about the y-axis. Find the surface area of revolution.

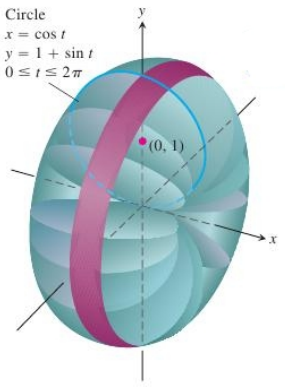
Since $y' = 2x$, the surface area is

$$S = \int_1^2 2\pi\, x\, ds = \int_1^2 2\pi\, x\,\sqrt{1 + (y')^2}\,dx = 2\pi \int_1^2 x\,\sqrt{1 + 4x^2}\,dx$$

$$= \pi \cdot \frac{2}{3} \cdot \frac{1}{4} \int_1^2 \sqrt{1 + 4x^2}\,d(1 + 4x^2) = \frac{\pi}{6}\left[1 + 4x^2\right]_1^2 = \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5}). \qquad \square$$

(2.18) *Example*

The circle of radius 1 centered at $(0, 1)$ is revolved about the x-axis. Find the surface area of the solid so generated.



The circle can be parameterized as

$$x = \cos t,\ y = 1 + \sin t,\ 0 \le t \le 2\pi.$$

Then $(x'(t))^2 + (y'(t))^2 = 1$. Thus the area is

$$S = \int_0^{2\pi} 2\pi\, y\, ds = \int_0^{2\pi} 2\pi\,(1 + \sin t)\,dt = 4\pi^2.$$

Notice that the curve here touches the axis of revolution but is not broken into parts by the axis. $\square$

2.7 Double Integrals

We now consider solids which are not necessarily solids of revolution. First, we consider a simple case, when the given solid has all plane faces except one, which is a portion of a surface given by $z = f(x, y)$.

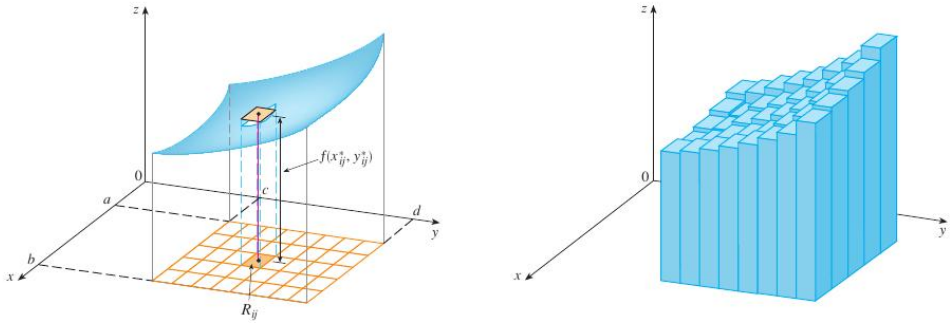
Let $f(x, y)$ be defined on the rectangle $R:\ a \le x \le b,\ c \le y \le d$. For simplicity, let $f(x, y) \ge 0$. The graph of f is the surface $z = f(x, y)$. Consider the solid

$$S:\ \{(x, y, z) \in \mathbb{R}^3 : (x, y) \in R,\ 0 \le z \le f(x, y)\}.$$

We approximate the volume of S by partitioning R and then adding up the volumes of the solid rods. So, consider a **partition** of R as in the following:

$$P:\ R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j],\ 1 \le i \le m,\ 1 \le j \le n,\ a = x_0,\ b = x_m,\ c = y_0,\ d = y_n.$$

Let $A(R_{ij})$ denote the area of the rectangle R_{ij}. Write $\|P\| = \max A(R_{ij})$ and call it the **norm** of P.



Choose **sample points** $(x_i^*, y_j^*) \in R_{ij}$. An approximation to the volume of S is the **Riemann sum**

$$S_{mn} = \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i^*, y_j^*) A(R_{ij}).$$

If limit of S_{mn} exists as $\|P\| \to 0$, then this limit is called the **double integral** of $f(x, y)$. It is denoted by $\iint_R f(x, y)dxdy$. That is, the volume V raised over the region R and bounded above by the surface $z = f(x, y)$ is given by

$$V = \iint_R f(x, y)dxdy.$$

Whenever the integral exists, it is also enough to consider **uniform partitions**, that is, $x_i - x_{i-1} = (b-a)/m = \Delta x$ and $y_j - y_{j-1} = (d-c)/n = \Delta y$. In this case, we write $A(R_{ij}) = \Delta A = \Delta x \Delta y$. Then

$$\iint_R f(x,y)dxdy = \lim_{\|P\|\to 0} S_{mn} = \lim_{m\to\infty}\lim_{n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta x\Delta y.$$

Since $f(x,y) \geq 0$, the value of this integral is the volume of the solid S bounded by the rectangle R on the bottom, the surface $z = f(x,y)$, $a \leq x \leq b$, $c \leq y \leq d$ on the top, and the planes $x = a$, $x = b$, $y = c$, $y = d$ on the sides.

When the integral of $f(x,y)$ exists, we say that f is **Riemann integrable** or just **integrable**.

Riemann sum is well defined even if $f(x,y) \geq 0$ does not hold for each $(x,y) \in R$. However, the double integral computes the signed volume. Analogous to the single variable case, each real valued continuous function defined on a closed bounded rectangle is integrable. Moreover, such an integral can be computed by integrating with respect to the independent variables successively. We mention this result as follows, and omit its proof.

(2.19) ***Theorem*** (Fubuni)

Let R be the rectangle $[a,b]\times[c,d]$. Let $f : R \to \mathbb{R}$ be a continuous function. Then $\iint_R f(x,y)dxdy$ exists and

$$\iint_R f(x,y)dxdy = \int_a^b\int_c^d f(x,y)dydx = \int_c^d\int_a^b f(x,y)dxdy.$$

Expressions such as $\int_c^d\int_a^b f(x,y)dxdy$ and $\int_a^b\int_c^d f(x,y)dydx$ are called the **iterated integrals**. These are to be interpreted as follows:

$$\int_c^d\int_a^b f(x,y)dxdy = \int_c^d\left(\int_a^b f(x,y)dx\right)dy.$$

$$\int_a^b\int_c^d f(x,y)\,dydx = \int_a^b\left(\int_c^d f(x,y)\,dy\right)dx.$$

Caution: Be careful in interpreting the expression $dxdy$ in Fubini's theorem. The $dxdy$ on the LHS is a single entity that goes with the double integration sign; sometimes it is also written as dA and called the *area element*. Whereas on the RHS, $dxdy$ has a different meaning. The dx here goes with the inner integral and the dy goes with the outer integral as shown via the parentheses.

(2.20) ***Example***

Evaluate $\iint_R(1-6x^2y)dxdy$, where $R = [0,2]\times[-1,1]$.

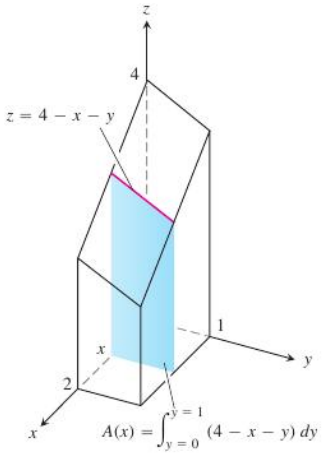
$$\iint_R(1-6x^2y)dxdy = \int_{-1}^{1}\int_0^2(1-6x^2y)dxdy = \int_{-1}^{1}(2-16y)dy = 4.$$

Also, reversing the order of integration, we have

$$\iint_R (1-6x^2y)dxdy = \int_0^2 \int_{-1}^1 (1-6x^2y)dydx = \int_0^2 2dx = 4. \qquad \square$$

(2.21) ***Example***

Find the volume V of the solid raised over the rectangle $R: [0,2]\times[0,1]$ and bounded above by the plane $z = 4-x-y$.



To use the method of slicing, suppose $A(x)$ is the cross sectional area at x. Then $V = \int_0^2 A(x)dx$.

Now, $A(x) = \int_0^1 (4-x-y)dy$.

Thus, $V = \int_0^2 \int_0^1 (4-x-y)dydx$.

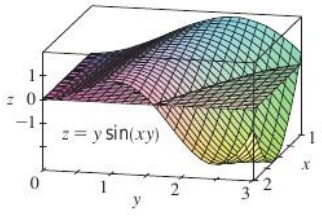
Alternatively, as a double integral,

$$\iint_R (4-x-y)dxdy = \int_0^2 \int_0^1 (4-x-y)dydx = \int_0^2 (4-x-1/2)\,dx = 5. \qquad \square$$

(2.22) ***Example***

Evaluate $\iint_R y\sin(xy)dxdy$, where $R = [1,2]\times[0,\pi]$.

$$\iint_R y\sin(xy)dxdy = \int_0^\pi \int_1^2 y\sin(xy)\,dxdy$$
$$= \int_0^\pi (-\cos 2y + \cos y)dy = 0.$$

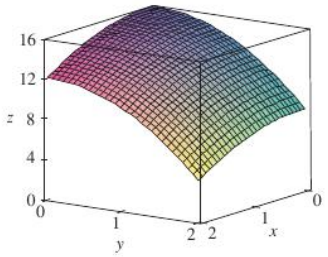


The volume of the solid above R and below the surface $z = y\sin(xy)$ is the same as the volume below R and above the surface. Since they have opposite signs, the net volume is zero. $\square$

(2.23) ***Example***

Find the volume of the solid bounded by the elliptic paraboloid $x^2 + 2y^2 + z = 16$, planes $x = 2$ and $y = 2$, and the three coordinate planes.

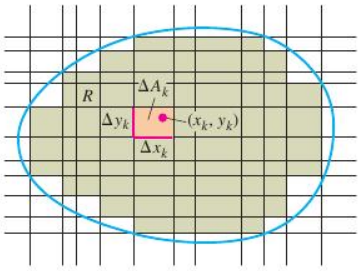
Let R be the rectangle $[0,2] \times [0,2]$. The solid is above R and below the surface defined by $z = f(x,y) = 16 - x^2 - 2y^2$, where f is defined on R. Then,



$$V = \iint_R (16 - x^2 - 2y^2)dxdy$$

$$= \int_0^2 \int_0^2 \left(16 - x^2 - 2y^2\right)dxdy = \int_0^2 \left(16x - \tfrac{1}{3}x^3 - 2xy^2\right) dy = 48. \qquad \square$$

The double integrals can be extended to functions defined on non-rectangular regions. Essentially, the approach is the same as earlier. We partition the region into smaller rectangles, form the Riemann sum, take its limit as the norm of the partition goes to zero.



It can be shown that when $f(x,y)$ is continuous on a closed bounded region R, its double integral exists. The double integral can be computed using iterated integrals in two special cases.

(2.24) ***Theorem*** (Fubini)

Let R be a region in the plane and let $f : R \to \mathbb{R}$ be a continuous function.

(1) *If R is given by $a \le x \le b$, $g_1(x) \le y \le g_2(x)$, where $g_1, g_2 : [a,b] \to \mathbb{R}$ are continuous, then*

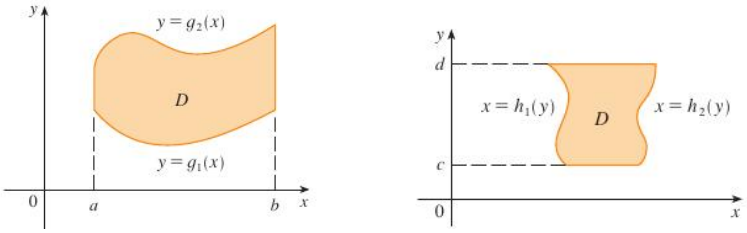
$$\iint_R f(x,y)dxdy = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\, dydx.$$

(2) *If R is given by $c \le y \le d$, $h_1(y) \le x \le h_2(y)$, where $h_1, h_2 : [c,d] \to \mathbb{R}$ are continuous, then*

$$\iint_R f(x,y)dxdy = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\, dxdy.$$

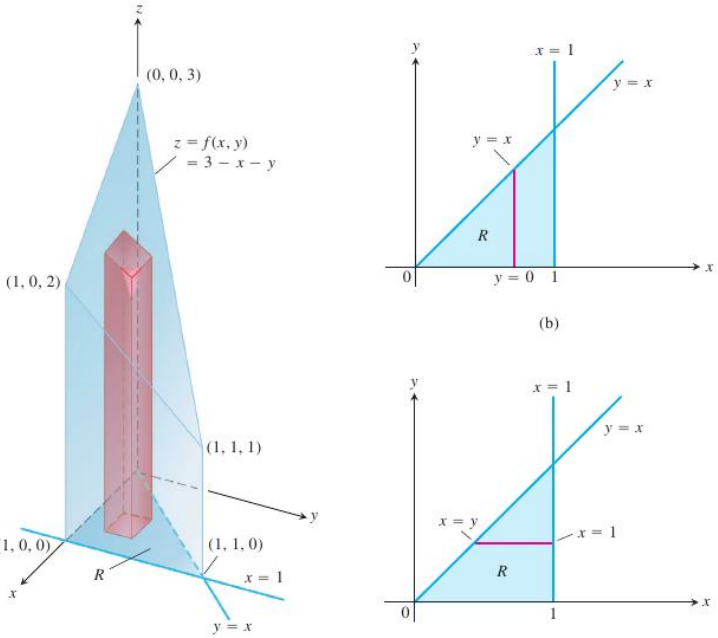
The two special cases in Fubuni's theorem are as follows:

1. R is bounded by the lines $x = a$, $x = b$ and the curves $y = g_1(x)$, $y = g_2(x)$.
2. R is bounded by the lines $y = c$, $y = d$ and the curves $x = h_1(y)$, $x = h_2(y)$.



(2.25) ***Example***

Find the volume of the prism whose base is the triangle in the xy-plane bounded by the lines $y = 0$, $x = 1$ and $y = x$, and whose top lies in the plane $z = 3 - x - y$.



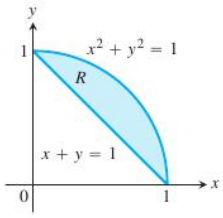
$$V = \int_0^1 \int_0^x (3 - x - y)dydx = \int_0^1 (3x - 3x^2/2)dx = 1.$$

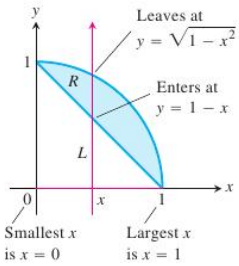
Also, $V = \int_0^1 \int_y^1 (3 - x - y)dxdy = \int_0^1 (5/2 - 4y + 3y^2/2)dy = 1.$ □

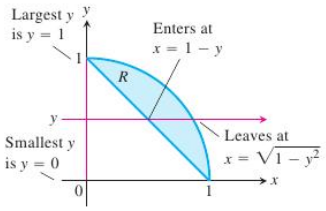
(2.26) ***Example***

Let R be the region bounded by the line $x + y = 1$ and the portion of the circle $x^2 + y^2 = 1$ in the first quadrant. Express $\iint_R f(x, y)\, dxdy$ as iterated integrals.

We sketch the region, find the limits, and then write the appropriate integrals.





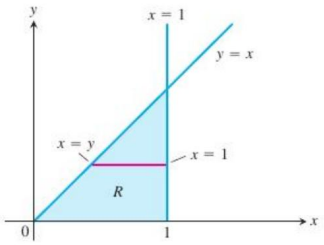


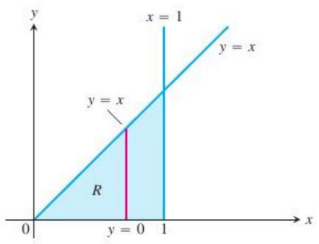
$$\iint_R f(x,y)dxdy = \int_0^1 \int_{1-x}^{\sqrt{1-x^2}} f(x,y)dydx = \int_0^1 \int_{1-y}^{\sqrt{1-y^2}} f(x,y)dxdy. \qquad \square$$

For evaluating a double integral as an iterated integral, choose some order: first x, next y. If it does not work, or if it is complicated, you may have to choose the reverse order.

(2.27) *Example*

Evaluate $\iint_R (\sin x)/x\, dxdy$, where R is the triangle in the xy-plane bounded by the lines $y = 0,\ x = 1$, and $y = x$.





The triangular region R can be expressed as $\{(x,y) : 0 \le y \le 1,\ y \le x \le 1\}$. So,

$$\iint_R \frac{\sin x}{x} dxdy = \int_0^1 \int_y^1 \frac{\sin x}{x} dxdy.$$

We do not see any way to proceed further. Instead, we choose a different order to express the same region. $R = \{(x,y) : 0 \le y \le x,\ 0 \le x \le 1\}$. Then

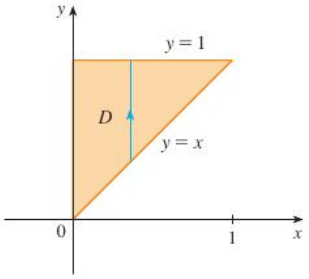
$$\begin{aligned}\iint_R \frac{\sin x}{x} dxdy &= \int_0^1 \int_0^x \frac{\sin x}{x} dydx = \int_0^1 \left(\frac{\sin x}{x} \int_0^x dy\right) dx \\ &= \int_0^1 \sin x dx = -\cos(1) + 1.\end{aligned}$$

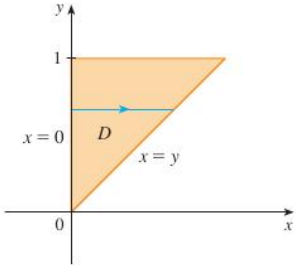
$\square$

(2.28) *Example*

Evaluate the iterated integral $\int_0^1 \int_x^1 \sin(y^2) dxdy$.

Write $D = \{(x,y) \in \mathbb{R}^2 : 0 \le x \le 1,\ x \le y \le 1\}$. We plan to change the order of integration. we have $D = \{(x,y) \in \mathbb{R}^2 : 0 \le x \le y,\ 0 \le y \le 1\}$. Then,





$$\int_0^1 \int_x^1 \sin(y^2)dydx = \iint_D \sin(y^2)dxdy = \int_0^1 \int_0^y \sin(y^2)dxdy$$
$$= \int_0^1 y\sin(y^2)dy = \frac{1}{2}(1-\cos(1).$$

□

Properties of double integrals with respect to addition, multiplication etc. are similar to integrals with a single variable. They are as follows.

(2.29) ***Theorem*** (Algebra of Integrals)
Let D be a region in the plane and let $f, g : D \to \mathbb{R}$ be integrable functions. Then the following are true:

(1) Constant Multiple*:* $\iint_D cf(x,y)dxdy = c\iint_D f(x,y)dxdy$ *for any* $c \in \mathbb{R}$.

(2) Sum-Difference*:*

$$\iint_D [f(x,y) \pm g(x,y)]dxdy = \iint_D f(x,y)dxdy \pm \iint_D g(x,y)dxdy.$$

(3) Additivity*: If* $D = D_1 \cup D_2$ *and* $D_1 \cap D_2 = \emptyset$*, then*

$$\iint_D f(x,y)dxdy = \iint_{D_1} f(x,y)dxdy + \iint_{D_2} f(x,y)dxdy.$$

(4) Domination*:*

If $f(x,y) \le g(x,y)$ *in* D*, then* $\iint_D f(x,y)dxdy \le \iint_D g(x,y)dxdy$.

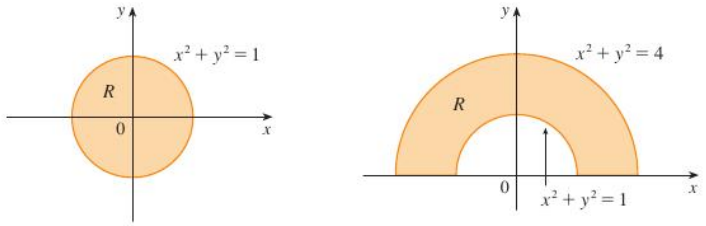
(5) Area*:* $\iint_D 1\,dxdy =$ Area of D.

(6) Boundedness*: If* $m \le f(x,y) \le M$ *in* D *for some* $m, M \in \mathbb{R}$*, then*

$$m\,(\text{Area of } D) \le \iint_D f(x,y)dxdy \le M\,(\text{Area of } D).$$

2.8 Double Integral in polar coordinates

Suppose R is a disk or a semi-annular region in the plane as shown below:

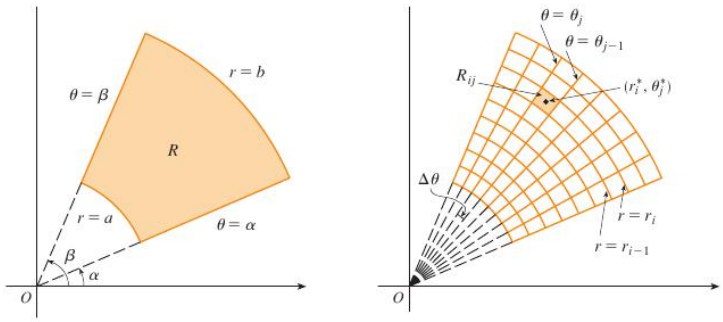


(a) $R = \{(r, \theta) \mid 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$ (b) $R = \{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$

It is easy to describe such regions in polar coordinates. Using polar coordinates, we define a **polar rectangle** as a region given in the form:

$$R = \{(r, \theta) : a \leq r \leq b,\ \alpha \leq \theta \leq \beta,\ 0 \leq \beta - \alpha < 2\pi\}.$$

To compute the volume of the solid raised over such a region and bounded above by the surface $z = f(r, \theta)$, we proceed as follows. We divide a polar rectangle into polar sub-rectangles as in the following:



$$R_{ij} = \{(r, \theta) : r_{i-1} \leq r \leq r_i,\ \theta_{j-1} \leq \theta \leq \theta_j\}.$$

Suppose f is a real valued function defined on a polar rectangle R. Let P be a partition of R into smaller polar rectangles R_{ij}. The area of R_{ij} is

$$A(R_{ij}) = \frac{1}{2}(r_i^2 - r_{i-1}^2)(\theta_j - \theta_{j-1}) = \frac{1}{2}(r_i + r_{i-1})(r_i - r_{i-1})(\theta_j - \theta_{j-1}).$$

Take a uniform grid dividing $r = a$ into m equal parts and $\theta = \beta - \alpha$ into n equal parts. Write $r_i - r_{i-1} = \Delta r$ and $\theta_j - \theta_{j-1} = \Delta\theta$. Also write the mid-point of r_{i-1} and r_i as $r_i^* = \frac{1}{2}(r_i + r_{i-1})$, similarly, $\theta_j^* = \frac{1}{2}(\theta_{j-1} + \theta_j)$. Then the Riemann sum for $f(r, \theta)$ can be written as

$$S = \sum_{i=1}^{m}\sum_{j=1}^{n} f(r_i^*, \theta_j^*)A(R_{ij}) = \sum_{i=1}^{m}\sum_{j=1}^{n} f(r_i^*, \theta_j^*)r_i^*\Delta r\Delta\theta.$$

Therefore, if $f(r,\theta)$ is continuous on the polar rectangle R, then the volume V of the solid raised over the region R and bounded above by the surface $z = f(r,\theta)$ is given by

$$V = \iint_R f(r,\theta)\, r\, dr d\theta.$$

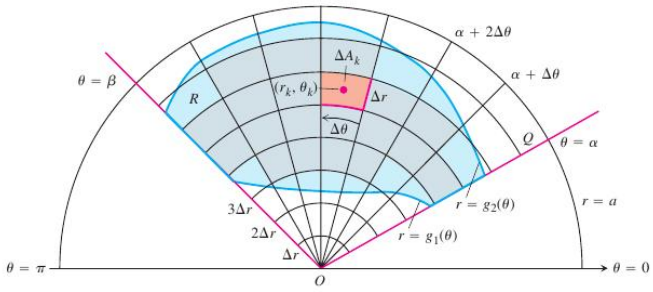
Informally, we say that the area element $dxdy$ in polar coordinates is equal to $r\, dr d\theta$.

It means that if $f(x,y)$ is given in Cartesian coordinates, which is continuous on the polar rectangle $R = \{(r,\theta) : a \le r \le b,\ \alpha \le \theta \le \beta,\ 0 \le \beta - \alpha < 2\pi\}$, then converting its double integral into polar form, we have

$$\iint_R f(x,y)dxdy = \iint_R f(r\cos\theta, r\sin\theta) r\, dr d\theta = \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta.$$

Here, in the rightmost expression, we have an iterated integral where dr goes with the inner integral sign and $d\theta$ goes with the outer integral sign.

The double integral in polar form can be generalized to functions defined on regions other than polar rectangles.



Let $f(r,\theta)$ be a continuous function defined over the region bounded by the rays $\theta = \alpha,\ \theta = \beta$ and the continuous curves $r = g_1(\theta),\ r = g_2(\theta)$. Then the volume V of the solid raised over R and bounded above by the surface $z = f(r,\theta)$ is given by

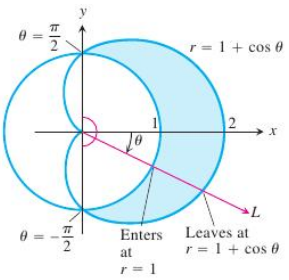
$$V = \iint_R f(r,\theta) r\, dr d\theta = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} f(r,\theta)\, r\, dr\, d\theta.$$

Caution: Do not forget the r on the right hand side.

(2.30) *Example*

Find the limits of integration for computing the volume V os the solid raised over the region R that lies inside the cardioid $r = 1 + \cos\theta$ and outside the circle $x^2 + y^2 = 1$ and bounded above by the surface $z = f(r,\theta)$.

Better write the circle as $r = 1$. Now, R is the region shown in blue:

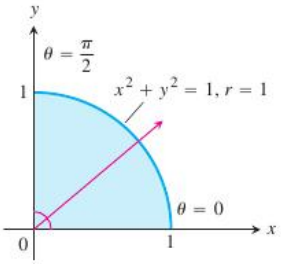


Then, $V = \int_{-\pi/2}^{\pi/2} \int_{1}^{1+\cos\theta} f(r,\theta) r \, dr \, d\theta.$ □

(2.31) ***Example***

Evaluate $I = \int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx.$

The limits of integration say that the region is the quarter of the unit disk in the first quadrant.
The region in polar coordinates is
$R: \ 0 \le \theta \le \pi/2, \ 0 \le r \le 1.$
Changing to polar coordinates, we have
$x = r\cos\theta, y = r\sin\theta$ and then



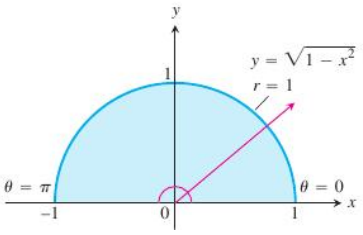
$$I = \int_0^{\pi/2} \int_0^1 r^2 \, r \, dr \, d\theta = \int_0^{\pi/2} \frac{1}{4} \, d\theta = \frac{\pi}{8}.$$ □

(2.32) ***Example***

Evaluate $I = \int_{-1}^{1} \int_0^{\sqrt{1-x^2}} e^{x^2+y^2} dy dx.$

The region is the upper semi-unit-disk, whose polar description is

$$R = \{(r,\theta) : 0 \le \theta \le \pi, \ 0 \le r \le 1\}.$$



Then $I = \iint_R e^{x^2+y^2} dx dy.$
Using integration in polar form,

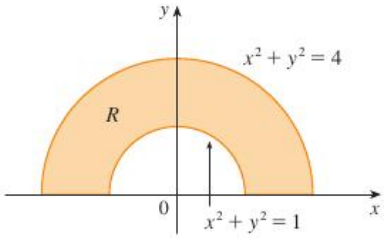
$$I = \int_0^{\pi} \int_0^1 e^{r^2} r \, dr \, d\theta = \int_0^{\pi} \left[\frac{1}{2} e^{r^2}\right]_0^1 d\theta = \int_0^{\pi} \frac{e-1}{2} \, d\theta = \frac{\pi}{2}(e-1).$$ □

(2.33) ***Example***

Evaluate $\iint_R (3x + 4y^2) dx dy$, where R is the region in the upper half plane bounded

by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

$R = \{(r, \theta) : 1 \le r \le 2,\ 0 \le \theta \le \pi\}$. Therefore,



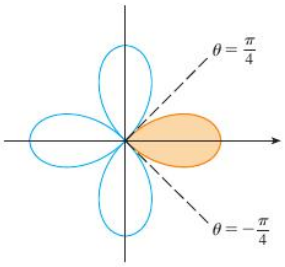
$$\iint_R (3x + 4y^2)\,dx\,dy$$
$$= \int_0^{\pi} \int_1^2 (3r\cos\theta + 4r^2\sin^2\theta) r\,dr\,d\theta$$
$$= \int_0^{\pi} \left[r^3\cos\theta + r^4\sin^2\theta\right]_1^2 d\theta = \int_0^{\pi} (7\cos\theta + 15\sin^2\theta)\,d\theta = \frac{15\pi}{2}.$$

□

(2.34) ***Example***

Find the area enclosed by one of the four leaves of the curve $r = |\cos(2\theta)|$.

The region is $R = \{(r, \theta) : -\pi/4 \le \theta \le \pi/4,\ 0 \le r \le \cos(2\theta)\}$. Then the required area is



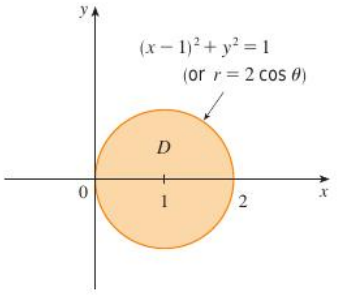
$$\iint_R 1\,dx\,dy = \int_{-\pi/4}^{\pi/4} \int_0^{\cos(2\theta)} r\,dr\,d\theta$$
$$= \int_{-\pi/4}^{\pi/4} \frac{\cos^2(2\theta) - 1}{2}\,d\theta = \frac{\pi}{8}.$$

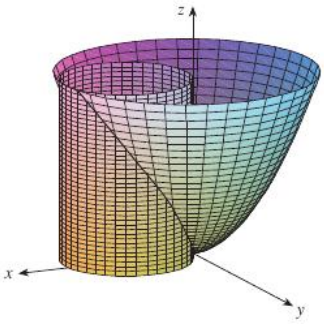
□

(2.35) ***Example***

Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy-plane, and inside the cylinder $x^2 + y^2 = 2x$.

The solid lies above the disk D whose boundary has equation $x^2 + y^2 = 2x$, or in polar coordinates, $r^2 = 2r\cos\theta$, or $r = 2\cos\theta$.





The disk $D = \{(r, \theta) : -\pi/2 \le \theta \le \pi/2,\ 0 \le r \le 2\cos\theta\}$.

Then the required volume V is given by

$$V = \iint_D (x^2 + y^2)dxdy = \int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r^2\, r\, dr\, d\theta = \int_{-\pi/2}^{\pi/2} 4\cos^4\theta\, d\theta$$
$$= \int_{-\pi/2}^{\pi/2} (3 + \cos 4\theta + 4\cos 2\theta)\, d\theta = \frac{3\pi}{2}.$$

□

2.9 Triple integral

Similar to the double integrals, we introduce the triple integral. Let $f(x, y, z)$ be a real valued function defined on a bounded region D in $\mathbb{R}^3$. As earlier we divide the region into smaller cubes enclosed by planes parallel to the coordinate planes. The set of these smaller cubes is called a partition P. The norm of the partition is the maximum volume enclosed by any smaller cube. Then form the Riemann sum S and take its limit as the cubes become smaller and smaller. If the limit exists, we say that the limit is the triple integral of the function over the region D.

$$\iiint_D f(x, y, z)\, dxdydz = \lim_{\|P\|\to 0} \sum f(x_i^*, y_j^*, z_k^*)(x_i - x_{i-1})(y_j - y_{j-1})(z_k - z_{k-1}),$$

where (x_i^*, y_j^*, z_k^*) is a point in the (i, j, k)-th cube in the partition.

As earlier, Fubuni's theorem says that for continuous functions, if the region D can be written as

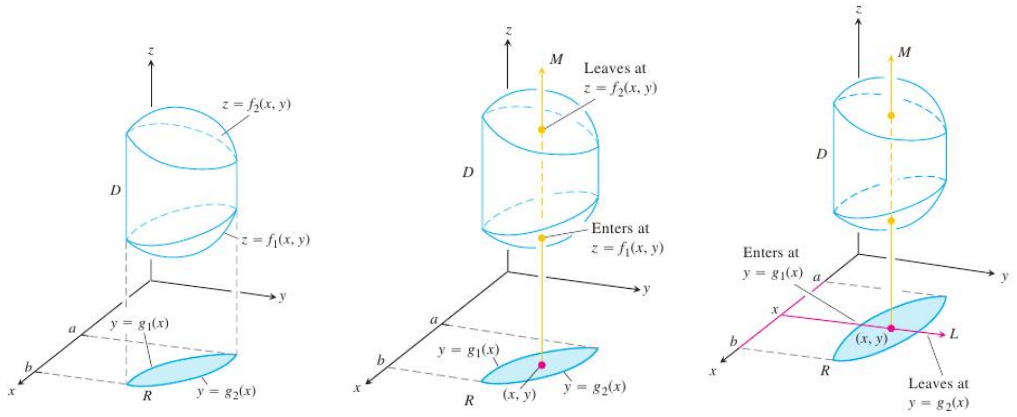
$$D = \{(x, y, z) : a \le x \le b,\ g_1(x) \le y \le g_2(x),\ h_1(x, y) \le z \le h_2(x, y)\},$$

then the triple integral can be written as an iterated integral:

$$\iiint_D f(x, y, z)\, dxdydz = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{h_1(x,y)}^{h_2(x,y)} f(x, y, z)dz\, dy\, dx.$$

Caution: The $dxdydz$ on the LHS is a single entity, sometimes written as dV and called the *volume element*, whereas the dz on the RHS goes with the innermost integral, the dy goes with the middle integral and the dx goes with the outermost integral.

To find the limits of integration, we first sketch the region D along with its shadow on the xy-plane. Next, we find the z-limits, then y-limits and then x-limits.



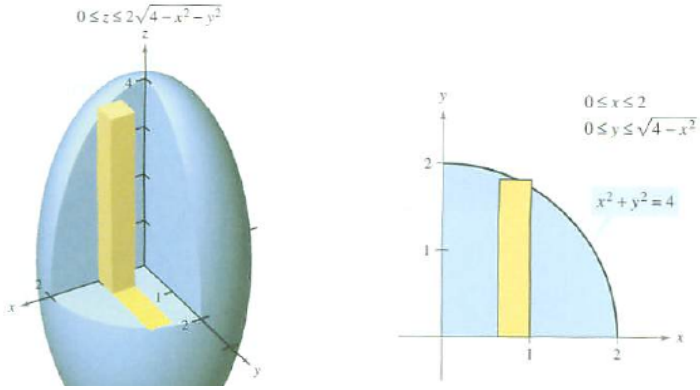
Observe that the volume of D is $\iiint_D 1\, dxdydz$.

All properties for double integrals hold analogously for triple integrals.

(2.36) *Example*

Find the volume of the ellipsoid given by $4x^2 + 4y^2 + z^2 = 16$.

Due to symmetry, the volume is 8 times the volume of D, where D is the portion of the ellipsoid in the first octant. Let us choose the order of integration as $dzdydx$.



Now, $D = \{(x, y, z) : 0 \le x \le 2,\ 0 \le y \le \sqrt{4 - x^2},\ 0 \le z \le 2\sqrt{4 - x^2 - y^2}\}$.
Therefore, Volume of the ellipsoid

$$= 8 \iiint_D dxdydz = 8 \int_0^2 \int_0^{\sqrt{4-x^2}} \int_0^{2\sqrt{4-x^2-y^2}} dzdydx$$

$$= 16 \int_0^2 \int_0^{\sqrt{4-x^2}} \sqrt{(4 - x^2) - y^2}\, dydx = 16 \int_0^2 \frac{\pi}{4}(4 - x^2)\, dx = \frac{64\pi}{3}.$$

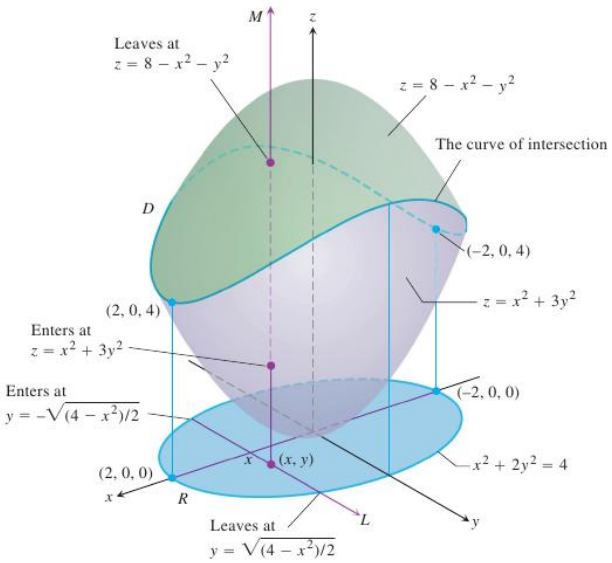
Notice that $\int_0^{\sqrt{4-x^2}} \sqrt{(4 - x^2) - y^2}\, dy$ is the area of the quarter circle of radius $\sqrt{4 - x^2}$; so it is equal to $\frac{\pi}{4}(4 - x^2)$. You can also get the same answer by using the

formula $\int \sqrt{a^2 - t^2}\, dt = \frac{t}{2}\sqrt{a^2 - t^2} + \frac{a^2}{2} \sin^{-1} \frac{t}{a} + C.$ □

(2.37) ***Example***

Find the volume of the solid enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

Eliminating z from the two equations, we get the projection of the curve of intersection of the two surfaces on the xy-plane, which is $x^2 + 2y^2 = 4$. Here, the projection of the solid on the xy-plane is the region $\{(x, y) \in \mathbb{R}^2 : x^2 + 2y^2 \leq 4\}$. The curve of intersection of the two surfaces is given by the function $z = x^2 + 3y^2$ where (x, y) varies over this region. Since we have equated the z-values on the two surfaces, the same curve of intersection is also given by $z = 8 - x^2 - y^2$ where (x, y) varies over this region. Now, the projected region gives the limits of integration for y as $\pm\sqrt{(4 - x^2)/2}$. Clearly, $-2 \leq x \leq 2$. Therefore,



$$V = \iiint_D dx\,dy\,dz = \int_{-2}^{2} \int_{-\sqrt{(4-x^2)/2}}^{\sqrt{(4-x^2)/2}} \int_{x^2+3y^2}^{8-x^2-y^2} dz\, dy\, dx$$

$$= \int_{-2}^{2} \int_{-\sqrt{(4-x^2)/2}}^{\sqrt{(4-x^2)/2}} (8 - 2x^2 - 4y^2)\, dy\, dx$$

$$= \int_{-2}^{2} \left[8\left(\frac{4 - x^2}{2}\right)^{3/2} - \frac{8}{3}\left(\frac{4 - x^2}{2}\right)^{3/2} \right] dx$$

$$= \frac{4\sqrt{2}}{3} \int_{-2}^{2} (4 - x^2)^{3/2}\, dx = 8\pi\sqrt{2}.$$ □

Notice that changing the order of integration involves expressing the region by choosing different order of the limits of values in the axes.

(2.38) *Example*

Evaluate $\int_0^1 \int_0^z \int_0^y e^{(1-x)^3}\, dx\, dy\, dz$ by changing the order of integration.

Here, the region is $D = \{(x, y, z) : 0 \le z \le 1, 0 \le y \le z, 0 \le x \le y\}$. Sketch the region. Notice that for any point $(x, y, z) \in D$, we have $0 \le x, y, z \le 1$.

We plan to change the order of integration from $dxdydz$ to $dzdydx$. Towards this, we project D on the xy-plane. The projection is the triangle R bounded by the lines $x = 0,\ y = 1$ and $y = x$ in the xy-plane. So, $R = \{(x, y, 0) : 0 \le x \le 1,\ x \le y \le 1\}$.

Next, for any point $(x, y, 0) \in R$, we draw a line parallel to z-axis. It enters D at $z = y$ and leaves at $z = 1$. So, $D = \{(x, y, z) : 0 \le x \le 1,\ x \le y \le 1,\ y \le z \le 1\}$. Therefore,

$$\int_0^1 \int_0^z \int_0^y e^{(1-x)^3}\, dx\, dy\, dz = \int_0^1 \int_x^1 \int_y^1 e^{(1-x)^3}\, dz\, dy\, dx$$

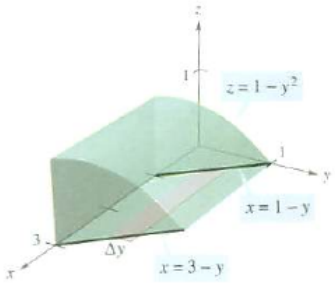
$$= \int_0^1 \int_x^1 (1-y) e^{(1-x)^3}\, dy\, dx = \int_0^1 \frac{(1-x)^2}{2} e^{(1-x)^3}\, dx$$

$$= -\int_{(1-0)^3}^0 \frac{e^t}{6}\, dt = \frac{e-1}{6}. \qquad \text{with } t = (1-x)^3. \qquad \square$$

(2.39) *Example*

Write the volume of the solid in the first octant bounded above by the cylinder $z = 1 - y^2$ and lying between the planes $x + y = 1$ and $x + y = 3$ as an iterated integral.

Projection of the solid D to the plane $z = 0$ gives a parallelogram with sides on the lines $y = 0,\ y = 1$ and $x = 1 - y,\ x = 3 - y$, in the xy-plane. Further, D is bounded below by the plane $z = 0$ and above by the cylinder $z = 1 - y^2$. So, $0 \le z \le 1 - y^2$. Then, $D = \{(x, y, z) : 0 \le y \le 1,\ 1 - y \le x \le 3 - y,\ 0 \le z \le 1 - y^2\}$. Then, the volume of D is



$$V = \iiint_D dxdydz = \int_0^1 \int_{1-y}^{3-y} \int_0^{1-y^2} dzdxdy. \qquad \square$$

(2.40) ***Example***

Evaluate $I = \int_0^{\sqrt{\pi/2}} \int_x^{\sqrt{\pi/2}} \int_1^3 \sin(y^2)\, dzdydx$.
We change the order of integration to $dzdxdy$ so that y is the outer variable of integration. Now,

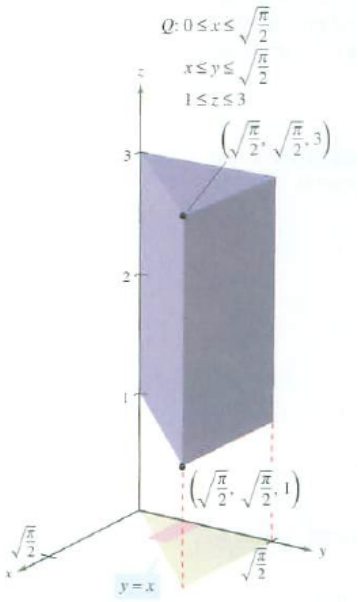
$$D = \{(x, y, z) : 0 \le x \le \sqrt{\pi/2}, \\ 0 \le x \le y \le \sqrt{\pi/2},\ 1 \le z \le 3\}.$$

Projection of D on the xy-plane is the region

$$R = \{(x, y, 0) : 0 \le y \le \sqrt{\pi/2},\ 0 \le x \le y\}.$$

The limits of z are still 1 and 3. Therefore,

$$I = \int_0^{\sqrt{\pi/2}} \int_0^{y} \int_1^3 \sin(y^2)\, dzdxdy$$
$$= \int_0^{\sqrt{\pi/2}} \int_0^{y} \sin(y^2)\, dxdy$$
$$= 2\int_0^{\sqrt{\pi/2}} y \sin(y^2)\, dy = 1.$$

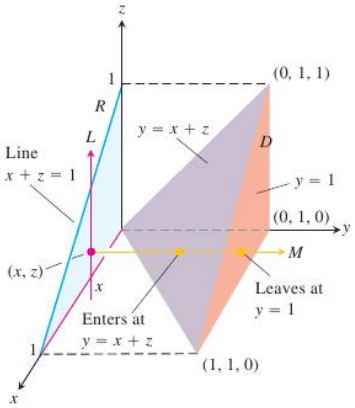


□

(2.41) ***Example***

Write the triple integral of $f(x, y, z)$ over the tetrahedron with vertices at $(0, 0, 0)$, $(1, 1, 0)$, $(0, 1, 0)$, and $(0, 1, 1)$ as an iterated integral.

First, sketch the region D to see the limits geometrically. The right hand side bounding surface of D lies in the plane $y = 1$. The projection of D on the zx-plane is the triangle R with vertices at $(0, 0, 0)$, $(1, 0, 0)$ and $(0, 0, 1)$ in the zx-plane. The upper boundary of R is the line $z = 1 - x$ and the lower boundary of R is the line $z = 0$, in the zx-plane. Thus, $R = \{(x, 0, z) : 0 \le x \le 1,\ 0 \le z \le 1 - x\}$.



To find the y-limits for D, we consider a typical point $(x, 0, z)$ in R and a line through this point parallel to y-axis. It enters D at the left side plane bounding D, whose equation is $y = x + z$, and leaves at the right hand side plane with equation as $y = 1$. Thus, $D = \{(x, y, z) : 0 \le x \le 1,\ 0 \le z \le 1 - x,\ x + z \le y \le 1\}$.

Then the triple integral of a function $f(x, y, z)$ over D is given by

$$\iiint_D f(x, y, z)\, dxdydz = \int_0^1 \int_0^{1-x} \int_{x+z}^1 f(x, y, z)\, dy\, dz\, dx.$$

If we interchange the orders of y and z, then first we consider limits for z and then of y. In this case, we project D on the xy-plane. A line parallel to z-axis through (x, y) in the xy-plane enters D at $z = 0$ and leaves D through the upper plane $z = y - x$. So, $0 \le z \le y - x$.

For the y-limits, on the xy-plane, where $z = 0$, the sloped side of D crosses the plane along the line $y = x$. A line through (x, y) parallel to y-axis enters the xy-plane at $y = x$ and leaves at $y = 1$. So, $x \le y \le 1$. The x-limits are as earlier, that is, $0 \le x \le 1$. So, $D = \{(x, y, z) : 0 \le x \le 1,\ x \le y \le 1,\ 0 \le z \le y - x\}$.

Then, the same triple integral is rewritten as the following iterated integral:

$$\iiint_D f(x, y, z)\, dxdydz = \int_0^1 \int_x^1 \int_0^{y-x} f(x, y, z)\, dz\, dy\, dx. \qquad \square$$

Analogous to a function of a single variable, we can define the average value over a planar region. It is defined as follows.

The **average value** of a function $f(x, y)$ over a planar region D is

$$\frac{\iint_D f(x, y)\, dxdy}{\text{Area of } D} = \frac{\iint_D f(x, y)\, dxdy}{\iint_D dxdy}.$$

Similarly, the average value of $f(x, y, z)$ over a region D in $\mathbb{R}^3$ is

$$\frac{\iiint_D f(x, y, z)\, dxdydz}{\iiint_D dxdydz}.$$

Average value lies within the minimum and maximum values of f.

(2.42) *Example*

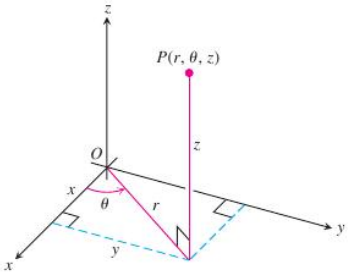
Find the average value of $f(x, y, z) = xyz$ over the cube bounded by the planes $x = 0, x = 2, y = 0, y = 2, z = 0$, and $z = 2$.

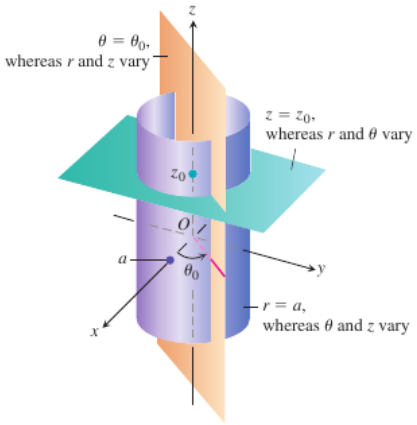
The required average value is

$$\frac{1}{V} \int_0^2 \int_0^2 \int_0^2 xyz\, dz\, dy\, dx = \frac{1}{8} \int_0^2 \int_0^2 2xy\, dy\, dx = \frac{1}{8} \int_0^2 4x\, dx = 1. \qquad \square$$

2.10 Triple integral in cylindrical coordinates

Cylindrical coordinates express a point P in space as a triple (r, θ, z), where (r, θ) is the polar representation of the projection of P on the xy-plane.





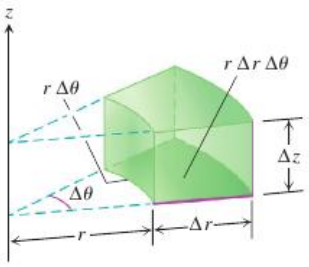
If P has Cartesian representation (x, y, z) and cylindrical representation (r, θ, z), then

$$x = r\cos\theta,\ y = r\sin\theta,\ z = z,\ r^2 = x^2 + y^2,\ \tan\theta = y/x.$$

In cylindrical coordinates,

$r = a$ describes a cylinder with axis as the z-axis.
$\theta = \alpha$ describes a plane containing the z-axis.
$z = b$ describes a plane perpendicular to the z-axis.

The Riemann sum of $f(r, \theta, z)$ uses a partition of D into cylindrical wedges:



The volume element $dxdydz = r\,dr\,d\theta\,dz$. Thus the triple integral of the function $f(r, \theta, z)$ in cylindrical coordinates is given by

$$\iiint_D f(r, \theta, z) r\,dr d\theta dz.$$

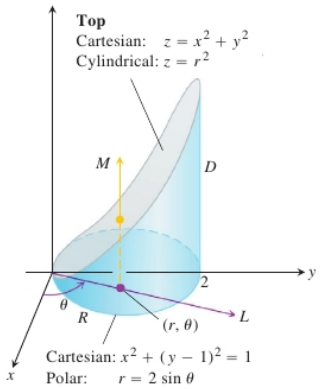
It means that if a function $f(x, y, z)$ is given in Cartesian coordinates, then

$$\iiint_D f(x, y, z)\, dxdydz = \iiint_D f(r\cos\theta, r\sin\theta, z)\, r\, dr\, d\theta\, dz.$$

A triple integral in cylindrical coordinates can be computed by converting it to iterated integrals using a similar technique of determining the limits of integration.

(2.43) ***Example***

Find the limits of integration in cylindrical coordinates for integrating a function $f(r, \theta, z)$ over the region D bounded below by the plane $z = 0$, laterally by the circular cylinder $x^2 + (y-1)^2 = 1$, and above by the paraboloid $z = x^2 + y^2$.



The projection of D onto the xy-plane gives the disk R enclosed by the circle $x^2 + (y-1)^2 = 1$. It simplifies to $x^2 + y^2 = 2y$. Its polar form is $r^2 = 2r\sin\theta$ or, $r = 2\sin\theta$. Here, $0 \le \theta \le \pi$ and $0 \le r \le 2\sin\theta$.

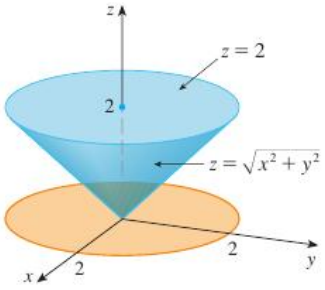
A line through a point $(r, \theta) \in R$ enters D at $z = 0$ and leaves D at $z = x^2 + y^2 = r^2$. Hence

$$\iiint_D f(r, \theta, z) r dr d\theta dz = \int_0^{\pi} \int_0^{2\sin\theta} \int_0^{r^2} f(r, \theta, z) r\, dz\, dr\, d\theta. \qquad \square$$

(2.44) ***Example***

Evaluate $I = \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^{2} (x^2 + y^2)\, dz\, dy\, dx$ using cylindrical coordinates.

The z-limits show that the solid is bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the plane $z = 2$. Its projection on the xy-plane is the disk $x^2 + y^2 = 4$. The limits for y also confirm this. A sketch of the solid looks as follows:



The disk $x^2 + y^2 = 4$ (projected region in the xy-plane) is described in the cylindrical coordinates by $0 \le \theta \le 2\pi,\ 0 \le r \le 2$. For any point (r, θ) in this region, we

have $\sqrt{x^2+y^2} = r \le z \le 2$ for the solid. Thus

$$I = \iiint_D (x^2+y^2)dxdydz = \int_0^{2\pi}\int_0^2\int_r^2 r^2\, r\, dz\, dr\, d\theta$$
$$= \int_0^{2\pi}\int_0^2 (r^3(2-r)\, dr\, d\theta = \int_0^{2\pi}\left(2\frac{2^4}{4} - \frac{2^5}{5}\right)d\theta = \int_0^{2\pi}\frac{8}{5}\,d\theta = \frac{16}{5}\pi. \qquad \square$$

(2.45) ***Example***

Let D be the solid bounded below by the cone $z = \sqrt{x^2+y^2}$ and above by the paraboloid $z = 2 - x^2 - y^2$. Set up the iterated integrals in cylindrical coordinates for computing $\iiint_D f(r,\theta,z)\, dxdydz$ using the following orders of integration:

(a) $dz\, dr\, d\theta$ (b) $dr\, dz\, d\theta$ (c) $d\theta\, dz\, dr$.

In cylindrical coordinates, the equation of the cone is $r = z$ and the equation of the paraboloid is $z = 2 - r^2$ or $r = \sqrt{2-z}$. (Since $r \ge 0$.) Further, eliminating z from these equations, we have $r = 2 - r^2 \Rightarrow (r+2)(r-1) = 0 \Rightarrow r = 1$. The projection of D onto $r\theta$-plane is the unit disk: $0 \le \theta \le 2\pi$, $0 \le r \le 1$. For any $(r,\theta,z) \in D$, neither z nor r depends on θ; also θ does not depend on z and r.

(a) For any $(r,\theta,z) \in D$, $0 \le r \le 1$. For any such r, we have $r \le z \le 2 - r^2$. Hence,

$$\iiint_D f(r,\theta,z)\, rdrd\theta dz = \int_0^{2\pi}\int_0^1\int_r^{2-r^2} f(r,\theta,z)\, r\, dz\, dr\, d\theta.$$

(b) For any $(r,\theta,z) \in D$, $0 \le z \le 2$. To get the dependence of r on z, we see that $D = D_1 \cup D_2$, where D_1 is the portion of D inside the cone bounded by the plane $z = 1$, and D_2 is the rest of D. For any $(r,\theta,z) \in D_1$, $0 \le z \le 1$ and $0 \le r \le z$. For any $(r,\theta,z) \in D_2$, $1 \le z \le 2$ and $0 \le r \le \sqrt{2-z}$. Then

$$\iiint_D f(r,\theta,z)\, rdrdzd\theta = \int_0^{2\pi}\int_0^1\int_0^z f(r,\theta,z)\, r\, dr\, dz\, d\theta$$
$$+ \int_0^{2\pi}\int_1^2\int_0^{\sqrt{2-z}} f(r,\theta,z)\, r\, dr\, dz\, d\theta.$$

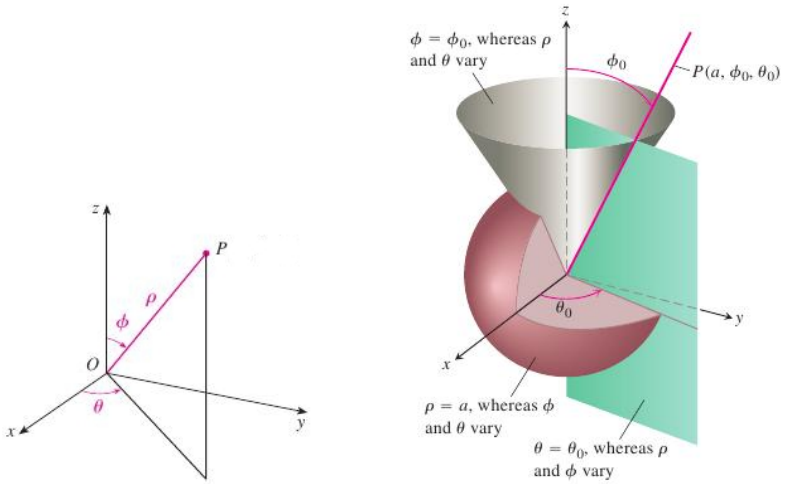
(c) Since θ is independent of r and z, from (a) we get the volume as

$$\iiint_D f(r,\theta,z)\, dxdydz = \int_0^1\int_r^{2-r^2}\int_0^{2\pi} f(r,\theta,z)\, r\, d\theta\, dz\, dr.$$

By taking $f(r,\theta,z) = 1$, you can verify that all the three iterated integrals in (a)-(c) give the volume of D as $5\pi/6$. $\square$

2.11 Triple integral in spherical coordinates

Spherical coordinates express a point P in space as a triple (ρ, ϕ, θ), where ρ is the distance of P from the origin O, ϕ is the angle between +ve z-axis and the line OP,



and θ is the angle between the projected line of OP on the xy-plane and the +ve x-axis. This θ is the same as the 'cylindrical' θ. Moreover, $\rho \geq 0,\ 0 \leq \phi \leq \pi$, and $0 \leq \theta \leq 2\pi$.

If $P(x, y, z)$ has spherical representation (ρ, ϕ, θ), then

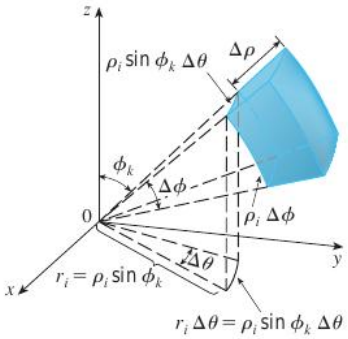
$$x = \rho \sin\phi \cos\theta,\ y = \rho \sin\phi \sin\theta,\ z = \rho \cos\phi,\ \rho = \sqrt{x^2 + y^2 + z^2}.$$

The 'cylindrical' r is equal to $\rho \sin\phi$. In spherical coordinates,

$\rho = a$ describes the sphere of radius a centered at the origin.

$\phi = \phi_0$ describes a cone with its axis as the z-axis.

$\theta = \theta_0$ describes a plane containing the z-axis.



When computing a triple integral over a region D in spherical coordinates, we partition the region into n spherical wedges. The size of the kth spherical wedge, which contains a point $(\rho_k, \phi_k, \theta_k)$, is given by the changes $\Delta\rho_k, \Delta\phi_k, \Delta\theta_k$ in ρ, ϕ, θ. Such a spherical wedge has one edge a circular arc of a circle with radius ρ_k and angle $\Delta\phi_k$, whose length is $\rho_k\Delta\phi_k$; another edge a circular arc of a circle with radius $\rho_k \sin\phi_k$ and angle $\Delta\theta_k$, whose length is $\rho_k \sin\phi_k\Delta\theta_k$; and thickness $\Delta\rho_k$. The volume of such a spherical wedge is approximately that of a rectangular box with dimensions ρ_k, $\rho_k\Delta\phi_k$ and $\rho_k \sin\phi_k\Delta\theta_k$. Thus

$$\Delta V_k \approx \rho_k^2 \sin\phi_k\Delta\rho_k\Delta\phi_k\Delta\theta_k.$$

The corresponding Riemann sum is $S = \sum_{k=1}^{n} f(\rho_k, \phi_k, \theta_k)\rho_k^2 \sin\phi_k\Delta\rho_k\Delta\phi_k\Delta\theta_k$. Accordingly, the triple integral of a function $f(\rho, \phi, \theta)$ given in spherical coordinates is given by

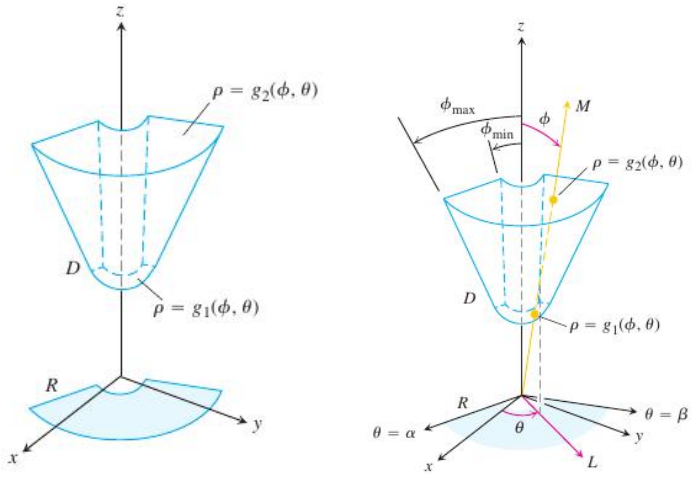
$$\iiint_D f(\rho, \phi, \theta)\, \rho^2 \sin\phi\, d\rho d\phi d\theta.$$

It means that if function $f(x, y, z)$ in given in Cartesian coordinates, then

$$\iiint_D f(x, y, z)\, dxdydz = \iiint_D f(\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi)\, \rho^2 \sin\phi\, d\rho d\phi d\theta.$$

The procedure in computing a triple integral in spherical coordinates is similar to that in cylindrical coordinates:

Sketch the region D and its projection on the xy-plane. Then find the ρ limit, ϕ limit and θ limit.



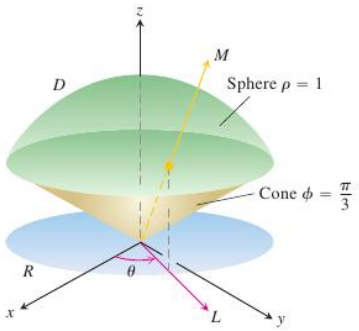
$$\iiint_D f(\rho, \phi, \theta)\rho^2 \sin\phi\, d\rho\, d\phi\, d\theta = \int_{\alpha}^{\beta} \int_{\phi-min}^{\phi-max} \int_{g_1(\phi,\theta)}^{g_2(\phi,\theta)} f(\rho, \phi, \theta)\rho^2 \sin\phi\, d\rho\, d\phi\, d\theta.$$

As earlier the iterated integral on the right hand side is to be interpreted correctly.

(2.46) ***Example***

Find the volume of the solid D cut from the ball $\rho \le 1$ by the cone $\phi = \pi/3$.

Draw a ray M through D from the origin making an angle ϕ with the z-axis. Draw also its projection L on the xy-plane. The line L makes an angle θ with the x-axis. Let R be the projected region of D in the xy-plane.



M enters D at $\rho = 0$ and leaves D at $\rho = 1$.

Angle ϕ runs through 0 to $\pi/3$, since D is bounded by the cone $\phi = \pi/3$.

L sweeps through R as θ varies from 0 to 2π. Thus

$$V = \iiint_D \rho^2 \sin\phi \, d\rho d\phi d\theta = \int_0^{2\pi} \int_0^{\pi/3} \int_0^1 \rho^2 \sin\phi \; d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/3} \frac{1}{3} \sin\phi \, d\phi \, d\theta = \int_0^{2\pi} \left[\frac{-\cos\phi}{3}\right]_0^{\pi/3} = \frac{1}{6} 2\pi = \frac{\pi}{3}. \qquad \square$$

(2.47) ***Example***

Evaluate $I = \displaystyle\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} e^{(x^2+y^2+z^2)^{3/2}} dz \, dy \, dx.$

Notice that $I = \displaystyle\iiint_D e^{(x^2+y^2+z^2)^{3/2}} dxdydz$, where D is the unit ball. In spherical coordinates, $I = \displaystyle\iiint_D e^{\rho^3} \rho^2 \sin\phi \, d\rho d\phi d\theta$. Converting I to an iterated integral, we have

$$I = \int_0^{2\pi} \int_0^{\pi} \int_0^1 e^{\rho^3} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta.$$

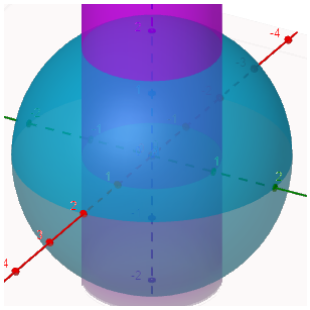
The integrand is a product of a function of ρ, a function of ϕ, and a function of θ. Thus,

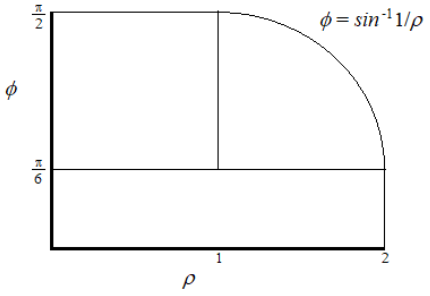
$$I = \int_0^1 e^{\rho^3} \rho^2 \, d\rho \int_0^{\pi} \sin\phi \, d\phi \int_0^{2\pi} d\theta = \left[\frac{e^{\rho^3}}{3}\right]_0^1 \Big[-\cos\phi \Big]_0^{\pi} (2\pi) = \frac{4\pi}{3}(e-1). \qquad \square$$

(2.48) ***Example***

A solid is bounded below by the xy-plane, above by the sphere $x^2+y^2+z^2=4$, and on the sides by the cylinder $x^2+y^2=1$. Set up the iterated integrals in spherical coordinates for computing the volume of the solid by using the following order of integration: (a) $d\rho\,d\phi\,d\theta$ (b) $d\phi\,d\rho\,d\theta$.

The projection of the solid in the xy-plane is the unit disk. So, for any point (ρ,ϕ,θ) in the solid, $0\le\theta\le 2\pi$; the variables ρ and ϕ do not depend on θ, and θ does not depend on ρ and ϕ. The equation of the sphere in spherical coordinates is $\rho=2$, and the equation of the cylinder is $\rho^2\sin^2\phi=1$, or $\rho\sin\phi=1$.





Notice that $0\le\phi\le\pi/2$. Eliminating ρ, we have $2\sin\phi=1\Rightarrow\phi=\pi/6$. This gives the curve of intersection of the surfaces, which is a circle.

(a) We think of the solid as the disjoint union of D_1 and D_2, where D_1 is the spherical sector with the curve of intersection as the bounding circle, and D_2 is the rest of the solid. The top of D_1 is the portion of the sphere cut by the bounding circle, and the bounding wall of D_2 is the cylinder. If the point $(\rho,\phi,\theta)\in D_1$, then $0\le\phi\le\pi/6$ and $0\le\rho\le 2$. If the point $(\rho,\phi,\theta)\in D_2$, then $\pi/6\le\phi\le\pi/2$ and $0\le\rho\le\operatorname{cosec}\phi$. Thus, the volume of the solid is

$$\int_0^{2\pi}\int_0^{\pi/6}\int_0^{2}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta+\int_0^{2\pi}\int_{\pi/6}^{\pi/2}\int_0^{\operatorname{cosec}\phi}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta.$$

(b) We think of the solid as the disjoint union of two solids D_3 and D_4, where D_3 is the portion of the solid cut-out by the unit ball, and D_4 is the rest. If the point $(\rho,\phi,\theta)\in D_3$, then $0\le\rho\le 1$ and $0\le\phi\le\pi/2$. If the point $(\rho,\phi,\theta)\in D_4$, then $1\le\rho\le 2$ and $0\le\phi\le\sin^{-1}(1/\rho)$. Notice that in this case, $0\le\phi\le\pi/6$ describes the top portion of D_4 up to the circle of intersection, and $\pi/6\le\phi\le\sin^{-1}(1/\rho)$ describes the side portion of D_4 from the circle of intersection to the xy-plane along the cylindrical wall. (This relationship of ρ and ϕ is depicted schematically in the second figure above.) Hence, the volume of the solid is

$$\int_0^{2\pi}\int_0^{1}\int_0^{\pi/2}\rho^2\sin\phi\,d\phi\,d\rho\,d\theta+\int_0^{2\pi}\int_1^{2}\int_0^{\sin^{-1}(1/\rho)}\rho^2\sin\phi\,d\phi\,d\rho\,d\theta.$$ ∎

2.12 Change of variables

The change of coordinate system from Cartesian to Cylindrical or to Spherical are examples of change of variables. Let us consider what happens when a different type of change of variables occurs.

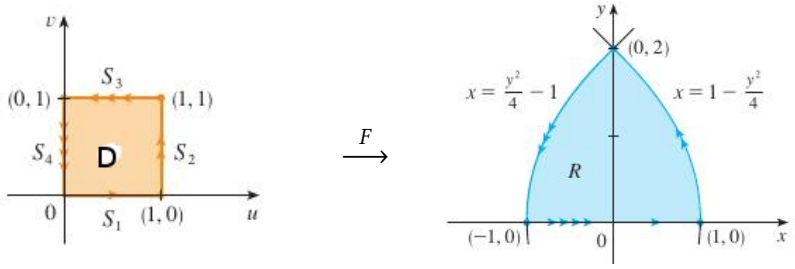
Suppose F maps a region D in $\mathbb{R}^2$ onto a region R in $\mathbb{R}^2$ in a one-one manner. For convenience, we say that D is a region in the uv-plane, R is a region in the xy-plane, and F maps (u, v) to (x, y). The function F is a pair of maps: (F_1, F_2), where $x = F_1(u, v)$ and $y = F_2(u, v)$. We often show this dependence implicitly by writing

$$x = x(u, v), \quad y = y(u, v).$$

(2.49) ***Example***

What is the image of $D = \{(u, v) : 0 \le u \le 1, 0 \le v \le 1\}$ under the map given by $x = u^2 - v^2,\ y = 2uv$?

The region D in the uv-plane is a square of side 1 with vertices at $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$.



The lower boundary of D is the line segment joining the point $(0, 0)$ to $(1, 0)$ in the uv-plane. Here, u varies from 0 to 1 while $v = 0$. It is transformed to the line segment $x = u^2, y = 0$. Then x varies from 0 to 1 while $y = 0$. In the xy-plane it is the line segment joining $(0, 0)$ to $(1, 0)$.

The right side boundary of D is the line segment joining $(1, 0)$ to $(1, 1)$ in the uv-plane. Here, $u = 1$ and v varies from 0 to 1. This is transformed to $x = 1 - v^2, y = 2v$. Eliminating v from these equations we have the arch of the curve $x = 1 - \frac{y^2}{4}$ joining the points $(1, 0)$ to $(0, 2)$ in the xy-plane.

The upper boundary of D is the line segment joining $(1, 1)$ to $(1, 0)$ in the uv-plane. Here, u varies from 1 to 0 while $v = 1$. This is transformed to $x = u^2 - 1, y = 2u$. Eliminating u from these equations, we get the arch of the curve $x = \frac{y^2}{4} - 1$ joining the points $(0, 2)$ to $(-1, 0)$ in the xy-plane.

The left side boundary of D is the line segment joining $(0, 1)$ to $(0, 0)$ in the uv-plane. Here, $u = 0$ and v varies from 1 to 0. This is transformed to $x = -v^2, y = 0$.

Here, x varies from -1 to 0 while $y = 0$. In the xy-plane, it is the line segment joining $(-1, 0)$ to $(0, 0)$.

The interior of D is mapped onto the interior of the region R in the xy-plane whose boundary consists of the above line segments and the archs. ∎

We would like to see how a double integral over a region D in the uv-plane and the double integral over the corresponding region R in the xy-plane are related when $(u, v) \mapsto (x, y)$. For this, let us first find out the change in a small rectangle in the uv-plane under the same map.

Let the region D in the uv-plane be in one-one correspondence with the region R in the xy-plane by the map $(u, v) \mapsto (x, y)$. Assume that $x = x(u, v)$ and $y = y(u, v)$ have continuous partial derivatives with respect to u and v in D. Let a typical small rectangle D' contained in D have vertices at

$$A_1 = (a, b),\ A_2 = (a + \Delta u, b),\ A_3 = (a + \Delta u, b + \Delta v),\ A_4 = (a, b + \Delta v).$$

The rectangle D' has sides as Δu and Δv; so its area is $\Delta u \Delta v$. Suppose the image of D' under the map $(u, v) \mapsto (x, y)$ is the region R' in the xy-plane. We wish to approximate the area of R'. For this purpose, we write the images of the points A_k under the map $(u, v) \mapsto (x, y)$ as $B_k = (a_k, b_k)$ for $k = 1, 2, 3, 4$. We then approximate the numbers a_k, b_k using the partial derivatives evaluated at (a, b), and call those numbers as c_k, d_k, respectively. That is, $c_k \approx a_k$ and $d_k \approx b_k$. Then, we take the area of the parallelogram P' with vertices at (c_k, d_k) as an approximation to the area of R'. Now, the (linear) approximations of the numbers a_k are as follows:

$$\begin{aligned}
a_1 &= x(a, b) = c_1,\\
a_2 &= x(a + \Delta u, b) \approx x(a, b) + x_u \Delta u = c_2,\\
a_3 &= x(a + \Delta u, b + \Delta v) \approx x(a, b) + x_u \Delta u + x_v \Delta v = c_3\\
a_4 &= x(a, b + \Delta v) \approx x(a, b) + x_v \Delta v = c_4.
\end{aligned}$$

Here, $x_u = x_u(a, b)$ and $x_v = x_v(a, b)$. Similar approximations d_k for b_k are given by

$$\begin{aligned}
b_1 &= d_1 = y(a, b), & b_2 &\approx d_2 = y(a, b) + y_u \Delta u,\\
b_3 &\approx d_3 = y(a, b) + y_u \Delta u + y_v \Delta v, & b_4 &\approx d_4 = y(a, b) + y_v \Delta v.
\end{aligned}$$

Here again, $y_u = y_u(a, b)$ and $y_v = y_v(a, b)$.

The area of the parallelogram P' with vertices at (c_1, d_1), (c_2, d_2), (c_3, d_3) and (c_4, d_4) is twice the area of the triangle T with vertices at (c_1, d_1), (c_2, d_2) and (c_4, d_4). By shifting the origin to the point (c_1, d_1), we see that the area of T is equal to the area of the triangle T' with vertices at $(0, 0)$, $(c_2 - c_1, d_2 - d_1)$ and $(c_4 - c_1, d_4 - d_1)$. Recall that the area of a triangle with vertices at $(0, 0)$, (x_1, y_1) and (x_2, y_2) is $\frac{1}{2}|x_1 y_2 - x_2 y_1|$.

Using this, we obtain

$$\begin{aligned}\text{Area of } R' \approx \text{Area of } P' &= 2 \times \text{Area of } T' \\ &= |(c_2 - c_1)(d_4 - d_1) - (c_4 - c_1)(d_2 - d_1)| \\ &= |x_u \Delta u \cdot y_v \Delta v - x_v \Delta v \cdot y_u \Delta u| \\ &= \left| \det \begin{bmatrix} x_u & x_v \\ y_u & y_v \end{bmatrix} \text{ at } (a,b) \right| \Delta u \Delta v.\end{aligned}$$

This determinant is called the **Jacobian** of the map $(u, v) \mapsto (x, y)$; and we write

$$J = J\big(x(u,v), y(u,v)\big) = \frac{\partial(x,y)}{\partial(u,v)} = \det \begin{bmatrix} x_u & x_v \\ y_u & y_v \end{bmatrix}.$$

Using this notation, we get

$$\text{Area of } R' \approx |J(a,b)|\, \Delta u \Delta v = \left| \frac{\partial(x,y)}{\partial(u,v)}(a,b) \right| \Delta u \Delta v. \tag{2.12.1}$$

That is, the area of the image of a rectangle contained in D with one vertex at (a, b) and sides of length Δu and Δv is approximately equal to $|J(a, b)|\Delta u \Delta v$.

We now look at the double integrals. Let $f(x, y)$ be a real valued continuous function defined on the region R. Then we have the map $\tilde{f}(u, v) = f\big(x(u, v), y(u, v)\big)$, which is a continuous real valued function defined on D.

Divide D in the uv-plane into smaller rectangles D_i, and call their images in the xy-plane as R_i. By using (2.12.1), forming the Riemann sum, and taking the limit as $\Delta u \to 0, \Delta v \to 0$, we obtain:

$$\iint_R f(x,y)dxdy = \iint_D \tilde{f}(u,v) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| dudv = \iint_D f\big(x(u,v), y(u,v)\big) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| dudv.$$

Similarly, when $x = x(u, v, w),\ y = y(u, v, w),\ z = z(u, v, w)$, we write the Jacobian as

$$J = J\big(x(u,v,w), y(u,v,w), z(u,v,w)\big) = \frac{\partial(x,y,z)}{\partial(u,v,w)} = \det \begin{bmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{bmatrix}.$$

If R is the region in $\mathbb{R}^3$ on which f has been defined, D is the region in the uvw-space so that the functions $x(u, v, w), y(u, v, w), z(u, v, w)$ map D onto R in a one-one manner, and x, y, z have continuous partial derivatives with respect to u, v, w, then

$$\iiint_R f(x,y,z)dxdydz = \iiint_D f\big(x(u,v,w), y(u,v,w), z(u,v,w)\big) \left| \frac{\partial(x,y,z)}{\partial(u,v,w)} \right| dudvdw.$$

(2.50) ***Example***

(1) In case of polar coordinates, $x = x(r,\theta) = r\cos\theta, y = y(r,\theta) = r\sin\theta$.
Thus, the Jacobian is

$$\frac{\partial(x,y)}{\partial(r,\theta)} = x_r y_\theta - x_\theta y_r = \cos\theta(r\cos\theta) - (-r\sin\theta)\sin\theta = r.$$

Therefore, the double integral in polar coordinates for a function $f(x,y)$ takes the form

$$\iint_R f(x,y)dxdy = \iint_D f(r\cos\theta, r\sin\theta)\, r\, drd\theta.$$

(2) In case of cylindrical coordinates, $x = r\cos\theta,\ y = r\sin\theta,\ z = z$. Then

$$\left|\frac{\partial(x,y,z)}{\partial(r,\theta,z)}\right| = \left|\det\begin{bmatrix} x_r & x_\theta & x_z \\ y_r & y_\theta & y_z \\ z_r & z_\theta & z_z \end{bmatrix}\right| = r.$$

Therefore, $\iiint_R f(x,y,z)dxdydz = \iiint_D f(r\cos\theta, r\sin\theta, z)\, r\, drd\theta dz$.

(3) For the spherical coordinates, we have

$$x = \rho\sin\phi\cos\theta,\ y = \rho\sin\phi\sin\theta,\ z = \rho\cos\phi.$$

Then, $\left|\frac{\partial(x,y,z)}{\partial(\rho,\phi,\theta)}\right| = \left|\det\begin{bmatrix} x_\rho & x_\phi & x_\theta \\ y_\rho & y_\phi & y_\theta \\ z_\rho & z_\phi & z_\theta \end{bmatrix}\right| = \rho^2\sin\phi$. And, the triple integral is given by

$$\iiint_R f(x,y,z)dxdydz = \iiint_D f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2\sin\phi\, d\rho d\phi d\theta.$$

We had already derived these results independently. □

The formula relating the integrals over D and R helps us in evaluating a multiple integral over R by using a suitable transformation from D to R.

(2.51) ***Example***

Evaluate $\iint_R \left(x - \frac{y}{2}\right) dxdy$ by using the transformation $x = u + v,\ y = 2v$, where R is the region in $\mathbb{R}^2$ enclosed by the lines $y = 0$, $y = 2x - 2$, $y = 4$ and $y = 2x$.

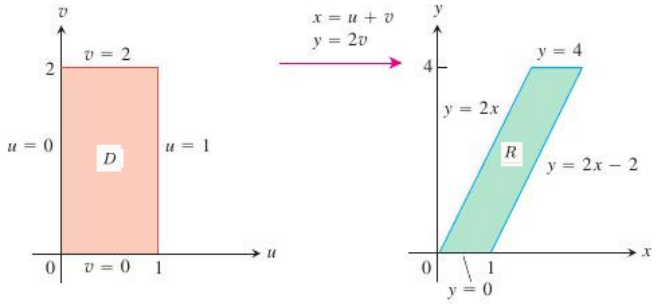
The region R in the xy-plane is (the paralleogram)

$$R = \{(x,y) : 0 \le y \le 4,\ y/2 \le x \le 1 + y/2\}.$$

The transformation has the inverse given by $u = x - \frac{y}{2},\ v = \frac{y}{2}$. Since $0 \le y \le 4$ and $v = y/2$, we have $0 \le v \le 2$. For limits of u, we see that $y/2 \le x \Rightarrow 0 \le x - y/2 = u$,

and $x \le 1 + y/2 \Rightarrow u = x - y/2 \le 1$. That is, $0 \le u \le 1$. So, the region D in the uv-plane is

$$D = \{(u, v) : 0 \le u \le 1, 0 \le v \le 2\}.$$



We have $f(x, y) = x - \dfrac{y}{2} = u$, $\dfrac{\partial(x, y)}{\partial(u, v)} = x_u y_v - x_v y_u = (1)(2) - (0)(1) = 2$. So,

$$\iint_R \left(x - \frac{y}{2}\right) dxdy = \iint_D |2|\, u\, dudv = \int_0^2 \int_0^1 2u\, du\, dv = \int_0^2 1^2\, dv = 2.$$

Also, looking at R from the y-axis, direct integration gives

$$\iint_R \left(x - \frac{y}{2}\right) dxdy = \int_0^4 \int_{y/2}^{1+y/2} \left(x - \frac{y}{2}\right) dx\, dy = \int_0^4 \left(\frac{x^2}{2}\Big|_{y/2}^{1+y/2} - \frac{y}{2} \cdot x \Big|_{y/2}^{1+y/2} \right) dy$$

$$= \int_0^4 \left(\frac{1}{2}\left[\left(1 + \frac{y}{2}\right)^2 - \left(\frac{y}{2}\right)^2 \right] - \frac{y}{2} \right) dy$$

$$= \int_0^4 \left(\frac{1}{2}\left[1 \cdot (1 + y)\right] - \frac{y}{2} \right) dy = \int_0^4 \frac{1}{2}\, dy = 2. \qquad \square$$

(2.52) ***Example***

Evaluate $\iint_R (y - x) dxdy$, where R is the region in the xy-plane enclosed by the lines $y - x = 1$, $y - x = -3$, $3y + x = 7$ and $3y + x = 15$.

Take $u = y - x$, $v = 3y + x$. That is, $x = \frac{1}{4}(v - 3u)$, $y = \frac{1}{4}(u + v)$. Then

$$D = \{(u, v) : -3 \le u \le 1,\ 7 \le v \le 15\}.$$

The Jacobian is $\dfrac{\partial(x, y)}{\partial(u, v)} = x_u y_v - x_v y_u = -\dfrac{3}{4} \cdot \dfrac{1}{4} - \dfrac{1}{4} \cdot \dfrac{1}{4} = -\dfrac{1}{4}$. Therefore,

$$\iint_R (y - x) dxdy = \iint_D u \left| \frac{\partial(x, y)}{\partial(u, v)} \right| dudv = \iint_D \frac{u}{4}\, dudv$$

$$= \int_{-3}^1 \int_7^{15} \frac{u}{4}\, dv\, du = \int_{-3}^1 2u\, du = -8.$$

For a direct evaluation of the double integral, notice that R is a parallelogram which is the union of regions R_1, R_2 and R_3, where R_1 is bounded by the lines $x = 1$, $x = 3$ on the sides, $y = (7 - x)/3$ on the bottom and $y = 1 + x$ on the top; R_2 is bounded by the lines $x = 3$, $x = 4$ on the sides, $y = (7 - x)/3$ on the bottom and $y = 5 - x/3$ on the top; and R_3 is bounded by $x = 4$, $x = 6$ on the sides, $y = x - 3$ on the bottom and $y = 5 - x/3$ on the top. Therefore,

$$\iint_R (y - x)dxdy = \int_1^3 \int_{(7-x)/3}^{1+x} (y - x)\, dy\, dx + \int_3^4 \int_{(7-x)/3}^{5-x/3} (y - x)\, dy\, dx + \int_4^6 \int_{x-3}^{5-x/3} (y - x)\, dy\, dx.$$

Verify that it gives the same value of -8 as earlier. □

The change of variables formula turns an xy-integral into a uv-integral. But the map that changes the variables goes from uv-region onto xy-region. This map must be one-one on the interior of the uv-region. Sometimes it is easier to get such a map from xy-region to uv-region. Then we will be tackling with the inverse of such an easy map. However, for computing the integral it is enough to compute the Jacobian of the inverse map instead of the map itself. The following fact helps us:

Jacobian of the inverse map is the reciprocal of Jacobian of the original map.

This may be expressed as

$$\frac{\partial(x, y)}{\partial(u, v)} = \left(\frac{\partial(u, v)}{\partial(x, y)}\right)^{-1}.$$

For instance, in (2.51), $u = x - y/2$, $v = y/2$ implies that

$$\frac{\partial(u, v)}{\partial(x, y)} = u_x v_y - u_y v_x = 1 \cdot \frac{1}{2} - \left(-\frac{1}{2}\right) \cdot 1 = 1 = \left(\frac{\partial(x, y)}{\partial(u, v)}\right)^{-1}.$$

And in (2.52), $u = y - x$, $v = 3y + x$ implies that

$$\frac{\partial(u, v)}{\partial(x, y)} = u_x v_y - u_y v_x = (-1) \cdot 3 - 1 \cdot 1 = -4 = \left(\frac{\partial(x, y)}{\partial(u, v)}\right)^{-1}.$$

(2.53) ***Example***

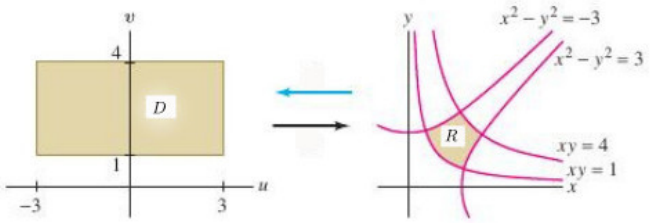
Integrate $f(x, y) = xy(x^2 + y^2)$ over the region

$$R = \{(x, y) \in \mathbb{R}^2 : -3 \le x^2 - y^2 \le 3,\ 1 \le xy \le 4\}.$$

There is a simple map that goes in the *wrong direction*: $u = x^2 - y^2,\ v = xy$. This map $(x, y) \mapsto (u, v)$ maps R in the xy-plane to

$$D = \{(u, v) : -3 \le u \le 3,\ 1 \le v \le 4\},$$

which is a rectangle in the uv-plane.



To compute the integral over R, we need the Jacobian of the map $(u, v) \mapsto (x, y)$. It is given by

$$\frac{\partial(x,y)}{\partial(u,v)} = \left(\frac{\partial(u,v)}{\partial(x,y)}\right)^{-1} = (u_x v_y - u_y v_x)^{-1} = \big(2x \cdot x - (-2y) \cdot y\big)^{-1} = \frac{1}{2(x^2+y^2)}.$$

Then

$$I = \iint_R xy(x^2+y^2)\,dxdy = \iint_D \left[xy(x^2+y^2) \times \left|\frac{1}{2(x^2+y^2)}\right|\right] dudv.$$

The integral on the right side is in the uv-plane and the term inside $[\,\cdot\,]$ is a function of (u, v), which simplifies to $xy/2 = v/2$. Thus,

$$I = \iint_D \frac{v}{2}\,dudv = \frac{1}{2}\int_{-3}^{3}\int_{1}^{4} v\,dv\,du = \frac{1}{2}\int_{-3}^{3}\frac{4^2-1^2}{2}\,du = \frac{15}{4}\big[3-(-3)\big] = \frac{45}{2}. \quad \square$$

Similarly, triple integrals undergo change of variables by using the inverse of the Jacobian of the map $u = u(x, y, z)$, $v = v(x, y, z)$ and $w = w(x, y, z)$.

2.13 Review problems

Problem 2.1 Find the area of the region bounded by the curves $y = x$ and $y = 2 - x^2$.

The points of intersection of the curves satisfy $y = x$ and $x = 2 - x^2$. The last equation is same as $(x + 2)(x - 1) = 0$. Thus the points of intersection are $(-2, -2)$ and $(1, 1)$. Hence the area is

$$\left|\int_{-2}^{1}\int_{x}^{2-x^2} dydx\right| = \left|\int_{-2}^{1}(2 - x^2 - x)dx\right| = \left|\left[2x - \frac{x^3}{3} - \frac{x^2}{2}\right]_{-2}^{1}\right| = \frac{9}{2}.$$

Since the significant portion of the curve $y = 2 - x^2$ lies above the portion of the line $y = x$, there is no need to take the absolute value. The calculation also confirms this.

Problem 2.2 Evaluate $I = \iint_D (4 - x^2 - y^2)\, dxdy$ if D is the region bounded by the straight lines $x = 0,\ x = 1,\ y = 0$ and $y = 3/2$.

$$I = \int_0^{3/2} \int_0^1 (4 - x^2 - y^2) dxdy = \int_0^{3/2} \left[4x - x^3/3 - y^2 x\right]_0^1 dy = \int_0^{3/2} \left(\frac{11}{3} - y^2\right) dy = \frac{35}{8}.$$

Problem 2.3 Evaluate the double integral of $f(x, y) = 1 + x + y$ over the region bounded by the lines $y = -x,\ y = 2$ and the parabola $x = \sqrt{y}$.

Draw the region. The integral is equal to

$$\int_0^2 \int_{-y}^{\sqrt{y}} (1 + x + y) dx\, dy = \int_0^2 \left(\sqrt{y} + \frac{y}{2} + \sqrt{y}\, y\right) - \left(-y + \frac{y^2}{2} - y^2\right) dy$$

$$= \int_0^2 \left(\sqrt{y} + \frac{3y}{2} + y\sqrt{y} + \frac{y^2}{2}\right) dy = \left[\frac{2}{3}y^{3/2} + \frac{3}{4}y^2 + \frac{5}{2}y^{5/2} + \frac{1}{6}y^3\right]_0^2 = \frac{1}{3}(13 + 44\sqrt{2}).$$

Problem 2.4 Change the order of integration in $\int_0^1 \int_x^{\sqrt{x}} f(x, y) dydx$.

The region D of integration is bounded by the straight line $y = x$ and the parabola $y = \sqrt{x}$. Every straight line parallel to x-axis cuts the boundary of D in no more than two points, and it remains in between y^2 to y. Also, y lies between 0 and 1. Hence

$$\int_0^1 \int_x^{\sqrt{x}} f(x, y) dydx = \int_0^1 \int_{y^2}^{y} f(x, y) dx\, dy.$$

Problem 2.5 Evaluate $\iint_D e^{y/x}\, dxdy$, where D is a triangle bounded by the straight lines $y = x$, $y = 0$, and $x = 1$.

In D, the variable x remains in between 0 and 1, and y lies between 0 and x. Hence

$$\iint_D e^{y/x}\, dxdy = \int_0^1 \int_0^x e^{y/x} dy\, dx = \int_0^1 x(e - 1)\, dx = \frac{e - 1}{2}.$$

Problem 2.6 Find $I = \iint_D e^{x+y}\, dxdy$, where D is the annular region bounded by two squares of sides 2 and 4, each having center at $(0, 0)$ and sides parallel to the axes.

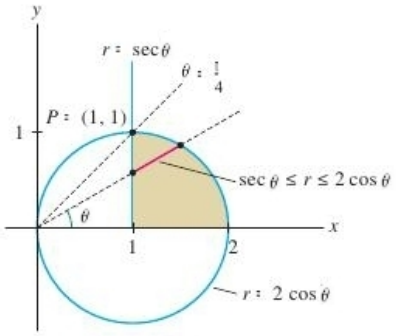
Draw the picture. D is not a simply connected region. Divide D into four simply connected regions by drawing lines $x = -1$ and $x = 1$. Let D_1 be the rectangle to the left of the inner square; D_2 be the square on top of the inner square; D_3 be the square below the inner square; and D_4 be the rectangle to the right of the inner square; so that D is the disjoint union of D_1, D_2, D_3, D_4. Then

$$I = \iint_{D_1} e^{x+y}\, dxdy + \iint_{D_2} e^{x+y}\, dxdy + \iint_{D_3} e^{x+y}\, dxdy + \iint_{D_4} e^{x+y}\, dxdy.$$

Converting each integral to an iterated integral, we have

$$I = \int_{-2}^{-1}\int_{-2}^{2} e^{x+y}dy\,dx + \int_{-1}^{1}\int_{1}^{2} e^{x+y}dy\,dx + \int_{-1}^{1}\int_{-2}^{-1} e^{x+y}dy\,dx + \int_{1}^{2}\int_{-2}^{2} e^{x+y}dy\,dx = e^4 - e^2 - e^{-2} + e^{-4}.$$

Problem 2.7 Evaluate $\iint_D (x^2+y^2)^{-2}\,dxdy$, where D is the shaded region in the figure below:



The integrand in polar coordinates is $f(r,\theta) = r^{-4}$. The region D is given by $0 \le \theta \le \pi/4$, $\sec\theta \le r \le 2\cos\theta$. Thus

$$\iint_D (x^2+y^2)^{-2}\,dxdy = \int_0^{\pi/4}\int_{\sec\theta}^{2\cos\theta} r^{-4}\,r\,dr\,d\theta = \frac{1}{8}\int_0^{\pi/4}(4\cos^2\theta - \sec^2\theta)\,d\theta = \frac{\pi}{16}.$$

Problem 2.8 Calculate the volume of the solid bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$.

The volume $V = \iint_D (1-x-y)dxdy$, where D is the base of the solid on the xy-plane. We see that D is the triangular region bounded by the straight lines $x = 0$, $y = 0$ and $x + y = 1$. Thus,

$$V = \int_0^1\int_0^{1-x}(1-x-y)dydx = \int_0^1 \frac{1}{2}(1-x)^2dx = \frac{1}{6}.$$

Problem 2.9 Compute the volume V of the solid bounded by the spherical surface $x^2+y^2+z^2 = 4a^2$, the cylinder $x^2+y^2 = 2ay$, where $a > 0$, and which is inside the cylinder.

The region of integration is the base of the cylinder. It is the circle $x^2+y^2-2ay = 0$, whose centre is $(0,a)$ and radius a. We calculate $V/4$, the volume of the portion of the solid in the first octant. Now, the region of integration D is the semicircular disk whose boundaries are given by

$$x = g_1(y) = 0,\ x = g_2(y) = \sqrt{2ay - y^2},\ y = 0,\ y = 2a.$$

The integrand is $z = f(x, y) = \sqrt{4a^2 - x^2 - y^2}$. Then

$$\frac{V}{4} = \int_0^{2a} \int_0^{\sqrt{2ay-y^2}} \sqrt{4a^2 - x^2 - y^2}\, dx\, dy.$$

To evaluate this, use polar coordinates: $x = r\cos\theta,\ y = r\sin\theta$. For the limits of integration, use $x^2 + y^2 = r^2,\ y = r\sin\theta$ to get:

$$x^2 + y^2 - 2ay = 0 \Rightarrow r^2 - 2ar\sin\theta = 0 \Rightarrow r = 2a\sin\theta.$$

That is, in polar coordinates, the boundaries of D are given by

$$r = g_1(\theta) = 0,\ r = g_2(\theta) = 2a\sin\theta,\ 0 \le \theta \le \pi/2.$$

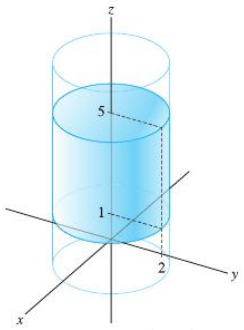
The integrand is $f(r, \theta) = \sqrt{4a^2 - r^2}$. Hence,

$$\begin{aligned} V &= 4\int_0^{\pi/2} \int_0^{2a\sin\theta} \sqrt{4a^2 - r^2}\, r\, dr\, d\theta \\ &= \frac{-4}{3}\int_0^{\pi/2} \left[(4a^2 - 4a^2\sin^2\theta)^{3/2} - (4a^2)^{3/2}\right] d\theta = \frac{16}{9}a^3(3\pi - 4). \end{aligned}$$

Problem 2.10 Integrate $f(x, y, z) = z\sqrt{x^2 + y^2}$ over the solid cylinder $x^2 + y^2 \le 4$ for $1 \le z \le 5$.

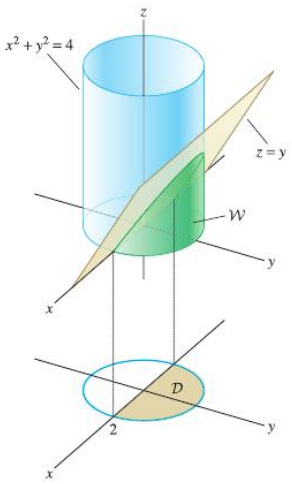
The region of integration D in cylindrical coordinates is given by $0 \le \theta \le 2\pi$, $0 \le r \le 2$, $1 \le z \le 5$. The integrand is zr. Thus

$$\begin{aligned} &\iiint_D z\sqrt{x^2 + y^2}\, dxdydz \\ &= \int_0^{2\pi} \int_0^2 \int_1^5 (zr)\, r\, dz\, dr\, d\theta = 64\pi. \end{aligned}$$



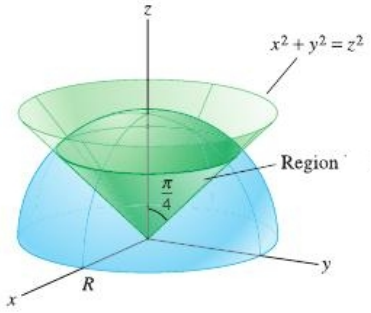
Problem 2.11 Integrate $f(x, y, z) = z$ over the part of the solid cylinder $x^2 + y^2 \le 4$ for $0 \le z \le y$.

The region W has the projection D on the xy-plane as the semicircle depicted in the figure. The z-coordinate varies from 0 to y and $y = r\sin\theta$. Thus W is given by $0 \le \theta \le \pi$, $0 \le r \le 2$, $0 \le z \le r\sin\theta$. In cylindrical coordinates,



$$\iiint_W z\,dxdydz = \int_0^{\pi}\int_0^{2}\int_0^{r\sin\theta} z\,r\,d\theta\,dr\,dz$$
$$= \int_0^{\pi}\int_0^{2}\frac{1}{2}(r\sin\theta)^2\,r\,d\theta\,dr = \pi.$$

Problem 2.12 Compute $\iiint_D z\,dxdydz$, where D is the solid lying above the cone $x^2 + y^2 = z^2$ and below the unit sphere.



The upper branch of the cone, which is relevant to D, has the equation $\phi = \pi/4$ in spherical coordinates. The sphere has the equation $\rho = 1$. Thus D is given by

$$D:\ 0 \le \theta \le 2\pi,\ 0 \le \phi \le \pi/4,\ 0 \le \rho \le 1.$$

Since $z = \rho\cos\phi$, the required integral is

$$\iiint_D z\,dxdydz = \int_0^{2\pi}\int_0^{\pi/4}\int_0^{1}(\rho\cos\phi)\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$$
$$= 2\pi\int_0^{\pi/4}\int_0^{1}\rho^3\cos\phi\sin\phi\,d\rho\,d\phi = \frac{\pi}{2}\int_0^{\pi/4}\cos\phi\sin\phi\,d\phi = \frac{\pi}{8}.$$

Problem 2.13 Compute the volume of the ellipsoid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$.

Projection of this solid on the xy-plane is the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$. Therefore, the

required volume is

$$V = \int_{-a}^{a} \int_{-b\sqrt{1-\frac{x^2}{a^2}}}^{b\sqrt{1-\frac{x^2}{a^2}}} \int_{-c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}}^{c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}} dz\,dy\,dx = 2c \int_{-a}^{a} \int_{-b\sqrt{1-\frac{x^2}{a^2}}}^{b\sqrt{1-\frac{x^2}{a^2}}} \sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}\,dy\,dx.$$

Substitute $y = b(1 - x^2/a^2)^{1/2} \sin t$. Then $dy = b(1 - x^2/a^2) \cos t\, dt$ and $-\pi/2 \le t \le \pi/2$. Therefore,

$$\begin{aligned} V &= 2c \int_{-a}^{a} \int_{-\pi/2}^{\pi/2} \left[\left(1-\frac{x^2}{a^2}\right) - \left(1-\frac{x^2}{a^2}\right)\sin^2 t\right]^{1/2} b\left(1-\frac{x^2}{a^2}\right)\cos t\, dt\, dx \\ &= \frac{bc\pi}{a^2} \int_{-a}^{a} \left(a^2 - x^2\right) dx = \frac{4\pi abc}{3}. \end{aligned}$$

Problem 2.14 Find a continuous bijection from the region R in the xy-plane to to a rectangle D in the uv-plane, where

$$R = \{(x, y) \in \mathbb{R}^2 : x \ge 0,\ y \ge 0,\ 1 \le 2(x + y) \le 2\}.$$

Notice that R is a trapezoidal region bounded by the lines $y = 0$, $x + y = 1$, $x = 0$ and $x + y = 1/2$. The trapezium has vertices at $(1/2, 0)$, $(1, 0)$, $(0, 1)$ and $(0, 1/2)$.

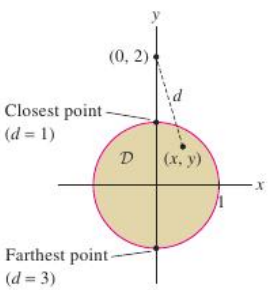
Take $u = y/(x + y)$, $v = 2(x + y)$; so that $x = v(1 - u)/2$, $y = uv/2$; and the rectangle

$$D = \{(u, v) : 0 \le u \le 1,\ 1 \le v \le 2\}.$$

Here is how the boundary of D transforms to the boundary of R:

$u = 0,\ 1 \le v \le 2$: $x = v/2,\ y = 0,\ 1/2 \le x \le 1$;
$0 \le u \le 1,\ v = 2$: $x = 1 - u,\ y = u,\ x + y = 1,\ 0 \le x, y \le 1$;
$u = 1,\ 1 \le v \le 2$: $x = 0,\ y = v/2,\ 1/2 \le y \le 1$;
$0 \le u \le 1,\ v = 1$: $x = (1 - u)/2,\ y = u/2,\ x + y = 1/2,\ 0 \le x, y \le 1/2$.

Problem 2.15 Show that $\dfrac{\pi}{3} \le \displaystyle\iint_D \frac{dxdy}{\sqrt{x^2 + (y-2)^2}} \le \pi$, where D is the unit disc.



The quantity $f(x,y) = \sqrt{x^2+(y-2)^2}$ is the distance of any point (x,y) from $(0,2)$. For $(x,y) \in D$, maximum of $f(x,y)$ is thus 3 and minimum is 1. Therefore,

$$\frac{1}{3} \le \frac{1}{\sqrt{x^2+(y-2)^2}} \le 1.$$

Integrating over D, we have

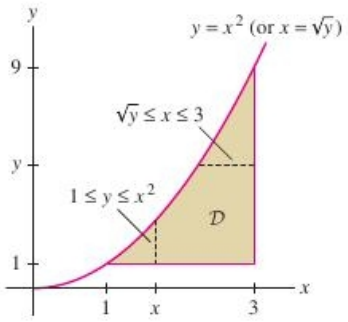
$$\iint_D \frac{1}{3}\,dxdy \le \iint_D \frac{1}{\sqrt{x^2+(y-2)^2}}\,dxdy \le \iint_D 1\,dxdy.$$

Since $\iint_D dxdy$ = area of D, we obtain

$$\frac{\pi}{3} \le \iint_D \frac{dxdy}{\sqrt{x^2+(y-2)^2}} \le \pi.$$

Problem 2.16 Evaluate $\int_1^9 \int_{\sqrt{y}}^3 xe^y\,dx\,dy$.

The region of integration is given by $1 \le y \le 9,\ \sqrt{y} \le x \le 3$. The same is expressed as $1 \le x \le 3,\ 1 \le y \le x^2$. Changing the order of integration, we have



$$\int_1^9 \int_{\sqrt{y}}^3 xe^y\,dx\,dy = \int_1^3 \int_1^{x^2} xe^y\,dx\,dy = \int_1^3 (xe^{x^2} - e\,x)dx = \frac{1}{2}(e^9 - 9e).$$

Problem 2.17 Evaluate $\lim_{a\to\infty} \int_{-a}^{a} e^{-x^2}\,dx$.

$$\left(\int_{-a}^{a} e^{-x^2}\,dx\right)^2 = \left(\int_{-a}^{a} e^{-x^2}\,dx\right)\left(\int_{-a}^{a} e^{-y^2}\,dy\right)$$
$$= \int_{-a}^{a}\int_{-a}^{a} e^{-x^2-y^2}\,dx\,dy = \iint_R e^{-x^2-y^2}\,dxdy$$

where R is the square $[-a,a] \times [-a,a]$ for $a > 0$.

Let $D = B(0, a)$ and $S = B(0, \sqrt{2}\,a)$, the balls centred at 0 and with radii a and $\sqrt{2}\,a$, respectively. Then $D \subseteq R \subseteq S$. Since $e^{-x^2-y^2} > 0$ for all $(x, y) \in \mathbb{R}^2$, we have

$$\iint_D e^{-x^2-y^2}\,dxdy \le \iint_R e^{-x^2-y^2}\,dxdy \le \iint_S e^{-x^2-y^2}\,dxdy.$$

Now,

$$\iint_D e^{-x^2-y^2}\,dxdy = \int_0^{2\pi}\int_0^a e^{-r^2}\,r\,dr\,d\theta = -\frac{1}{2}\int_0^{2\pi}(e^{-a^2}-1)\,d\theta = \pi(1-e^{-a^2}).$$

Similarly, $\iint_S e^{-x^2-y^2}\,dxdy = \pi(1-e^{-2a^2})$. We see that

$$\lim_{a\to\infty}\iint_D e^{-x^2-y^2}\,dxdy = \pi, \quad \lim_{a\to\infty}\iint_S e^{-x^2-y^2}\,dxdy = \pi.$$

By Sandwich theorem, $\lim_{a\to\infty}\iint_R e^{-x^2-y^2}\,dxdy = \pi$. Therefore,

$$\lim_{a\to\infty}\int_{-a}^{a} e^{-x^2}\,dx = \sqrt{\pi}.$$

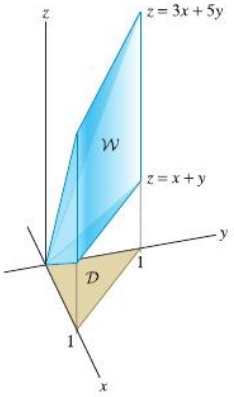
Problem 2.18 Evaluate $\iiint_W z\,dxdydz$, where W is the solid bounded by the planes $x = 0$, $y = 0$, $x + y = 1$, $z = x + y$, and $z = 3x + 5y$ in the first octant.

The shadow of W in the xy-plane is the triangle

$$D = \{(x, y) \in \mathbb{R}^2 : 0 \le x \le 1,\ 0 \le y \le 1 - x\}.$$

Hence

$$\begin{aligned}\iiint_D z\,dxdydz &= \int_0^1\int_0^{1-x}\int_{x+y}^{3x+5y} z\,dz\,dy\,dx \\ &= \int_0^1\int_0^{1-x}(4x^2+14xy+12y^2)\,dy\,dx \\ &= \int_0^1 (4-5x+2x^2-x^3)\,dx = \frac{23}{12}.\end{aligned}$$



Problem 2.19 (Fun Problem) The n-dimensional cube with side a has volume a^n. What is the volume of an n-dimensional ball?

Denote by $V_n(r)$ the volume of the n-dimensional ball with radius r. Also, write $A_n = V_n(1)$. For $n = 1$, we have the interval $[-1, 1]$, whose volume we take as its length, that is, $A_1 = 2,\ V_1(r) = 2r$. For $n = 2$, we have the unit disk, whose volume is its area; that is, $A_2 = \pi,\ V_2 = \pi r^2$. For $n = 3$, we know that $A_3 = 4\pi/3$ and $V_3(r) = 4\pi r^3/3$.

Exercise 1: Show by induction that $V_n(r) = A_n\, r^n$.

Suppose $V_{n-1}(r) = A_{n-1} r^{n-1}$. The slice of the n-dimensional ball $x_1^2 + \cdots x_{n-1}^2 + x_n^2 = r^2$ at the height $x_n = c$, has the equation

$$x_1^2 + \cdots + x_{n-1}^2 + c^2 = r^2.$$

This slice has the radius $\sqrt{r^2 - c^2}$. Thus

$$V_n(r) = \int_{-r}^{r} V_{n-1}\sqrt{r^2 - x_n^2}\, dx_n = A_{n-1}\int_{-r}^{r} \left(r^2 - x_n^2\right)^{(n-1)/2} dx_n.$$

Substitute $x_n = r\sin\theta$. So, $dx_n = r\cos\theta$ and $-\pi/2 \le \theta \le \pi/2$. Then

$$V_n(r) = A_{n-1} r^n \int_{-\pi/2}^{\pi/2} \cos^n\theta\, d\theta = A_{n-1} C_n r^n,$$

where $C_n = \int_{-\pi/2}^{\pi/2} \cos^n\theta\, d\theta$. This says that $A_n = A_{n-1}C_n$.

Exercise 2: Prove that $C_3 = 4/5, C_4 = 3\pi/8$ and $C_n = \frac{n-1}{n} C_{n-2}$.

Exercise 3: Prove that $A_{2m} = \dfrac{\pi^m}{m!}$ and $A_{2m+1} = \dfrac{2^{m+1}\pi^m}{1\cdot 3\cdots(2m+1)}$.

This sequence of numbers has a curious property: A_n increases up to $n = 5$ and then it decreases to 0 as $n \to \infty$.

2.14 Exercises for Chapter 2

1. Find the volume of the solid generated by revolving the region bounded by the curve $y = \sqrt{x}$ and the lines $y = 2$ and $x = 0$ about
   (a) the x-axis. (b) the y-axis. (c) the line $y = 2$. (d) the line $x = 4$.
2. The arch $y = \sin x,\ 0 \le x \le \pi$, is revolved about the line $y = c,\ 0 \le c \le 1$, to generate a solid.
   (a) Find the value of c that minimizes the volume of the solid. What is the minimum volume?

(b) What value of c in $[0, 1]$ maximizes the volume of the solid?

3. Find the length of one arch of the cycloid $x(t) = 2(t - \sin t)$, $y = 2(1 - \cos t)$ for $0 \le t \le 2\pi$.
4. Find the areas of the surfaces by revolving the curves about indicated axes: (a) $y = \sqrt{1 + x}$, $1 \le x \le 5$; x-axis. (b) $y = (3x)^{1/3}$, $0 \le x \le 1/3$; y-axis.
5. Sketch the region of integration and evaluate the integral:
   (a) $\int_1^2 \int_y^{y^2} dx\, dy$ (b) $\int_1^{\ln 8} \int_0^{\ln y} e^{x+y}\, dx\, dy$
6. Integrate $f(x, y) = y - \sqrt{x}$ over the region bounded by the axes and the line $x + y = 1$.
7. Integrate $f(x, y) = e^x \ln y$ over the region bounded by the straight lines $x = 0,\ y = 1, y = 2$ and the curve $x = \ln y$.
8. Sketch the region and reverse the order of integration:
   (a) $\int_0^1 \int_{1-x}^{1-x^2} dy\, dx$ (b) $\int_0^{3/2} \int_0^{9-4x^2} 16x\, dydx$
9. Evaluate the following integrals:
   (a) $\int_0^2 \int_0^{4-x^2} \frac{xe^{2y}}{4 - y}\, dy\, dx$ (b) $\int_0^{1/16} \int_{y^1/4}^{1/2} \cos(16\pi x^5)\, dxdy$
   (c) $\int_{-1}^{1} \int_{-1/\sqrt{1-x^2}}^{1/\sqrt{1-x^2}} (2y + 1)\, dy\, dx$
   (d) $\iint_R (y - 2x^2)\, dxdy$, where R is the region whose boundary is $|x| + |y| = 1$.
   (e) $\iint_R xy\, dxdy$, where R is the region bounded by the straight lines $y = x$, $y = 2x$, and $x + y = 2$.
10. Find the volume of the region bounded by the paraboloid $z = x^2 + y^2$ and below by the triangle enclosed by the lines $y = x,\ x = 0$, and $x + y = 2$ in the xy-plane.
11. Find the volume of the solid that is bounded on the front and back by the planes $x = 2$ and $x = 1$, on the sides by the cylinders $y = \pm 1/x$, and above and below by the planes $z = x + 1$ and $z = 0$.
12. Express the area of the region bounded by the curves $y = \ln x$, $y = 2 \ln x$ and the line $x = e$ in the first quadrant and then find the area.
13. Determine which one is the larger: the average value of xy over the rectangle $0 \le x \le 1$, $0 \le y \le 1$ and the average value of xy over the quarter disc $x^2 + y^2 \le 1$ in the first quadrant?
14. Use Polar co-ordinates to evaluate the integrals:
   (a) $\int_{-1}^{0} \int_{-\sqrt{1-x^2}}^{0} \frac{2\, dy\, dx}{1 + \sqrt{x^2 + y^2}}$ (b) $\int_0^2 \int_0^{\sqrt{1-(x-1)^2}} \frac{x + y}{x^2 + y^2}\, dy\, dx$

15. Find the area of the region cut from the first quadrant by the curve $r = 2\sqrt{2 - \sin 2\theta}$.
16. Find the area of the region common to the interiors of the curves $r = 1 + \cos\theta$ and $r = 1 - \cos\theta$.
17. Find the volume of the solid bounded in back by the plane $x = 0$, on the front and sides by he parabolic cylinder $x = 1 - y^2$, on the top by the paraboloid $z = x^2 + y^2$, and on the bottom by the xy-plane.
18. Evaluate the triple integral $\int_0^2 \int_0^{4-x^2} \int_0^x \frac{\sin 2z}{4-z}\, dy\, dz\, dx$.
19. Evaluate $\int_{-1}^1 \int_0^{\sqrt{1-y^2}} \int_0^x (x^2 + y^2)\, dz\, dx\, dy$ using cylindrical co-ordinates.
20. Using spherical co-ordinates, find the volume of the solid enclosed by the cardioid of revolution $\rho = 1 - \cos\phi$.
21. Use $u = x - y,\ v = 2x + y$ to evaluate $\iint_R (2x^2 - xy - y^2) dxdy$, where R is the region in the first quadrant bounded by the lines $y = -2x + 4$, $y = -2x + 7$, $y = x - 2$, and $y = x + 1$.
22. Using a suitable transformation evaluate $\iint_R (3x^2 + 14xy + 8y^2)\, dxdy$, where R is the region in the first quadrant bounded by the lines $2y = -3x + 2$, $2y = -3x + 6$, $4y = -x$, and $4y = -x + 4$.
23. Let R be the region in the first quadrant of the xy-plane bounded by the hyperbolas $xy = 1,\ xy = 9$ and the lines $y = x,\ y = 4x$. Using the transformation $x = u/v,\ y = uv$ with $u > 0,\ v > 0$ evaluate $\iint_R (\sqrt{y/x} + \sqrt{xy}) dxdy$.
24. Let D be the region given by $1 \le x \le 2$, $0 \le xy \le 2$, $0 \le z \le 1$. Using $u = x$, $v = xy$, $w = 3z$, evaluate $\iiint_D (x^2 y + 3xyz) dxdydz$.

Answers:
1a. 8π 1b. $32\pi/5$ 1c. $8\pi/3$ 1d. $224\pi/15$
2a. $c = 2/\pi,\ V - min = \pi^2/2 - 4$ 2b. $c = 0$ 3. 16. 4a. $49\pi/3$.
4b. $(\pi/9)(\sqrt{8} - 1)$. 5a. 5/6 5b. $8\ln 8 - 16 + e$ 6. $-1/10$ 7. 1/4
9a. $(e^8 - 1)/4$ 9b. $1/80\pi$ 9c. 2π 9d. $-2/3$ 9e. 13/81 10. 4/3
11. $2 + 2\ln 2$ 12. 1 13. Average over square 14a. $\pi - \pi\ln 2$ 14b. $1 + \pi/2$
15. $2\pi - 2$ 16. $3\pi/2 - 4$ 17. 4/7 18. $\frac{1}{2}\sin^2 4$ 19. 2/5 20. $8\pi/3$
21. 33/4 22. 64/5 23. $8 + (52\ln 2)/3$ 24. $2 + \ln 8$

3

Series Expansion of Functions

3.1 Series

A series is an infinite sum of numbers. As it is, two numbers can be added; so by induction, a finite number of them can also be added. However, it does not say how to add an infinite number of them. That is, an infinite sum needs to be defined. For an infinite sum to be meaningful, we look at finite sums starting from the beginning. The infinite sum $x_1+x_2+\cdots$ is meaningful when two sums $x_1+\cdots+x_m$ and $x_1+\cdots+x_n$ do not differ much for large m and n. Further, this infinite sum would result in a real number s provided any finite sum $x_1 + \cdots + x_n$, for large n, remains near s. We will call an infinite sum a **series** and formalize this idea by its *convergence*.

We say that the series $\sum x_n$ **converges to** $s \in \mathbb{R}$ iff for each $\epsilon > 0$, there exists an $m \in \mathbb{N}$ such that for each natural number $n > m$, we have $|\sum_{k=1}^{n} x_k - s| < \epsilon$. When $\sum x_n$ converges to $s \in \mathbb{R}$, we also say that $\sum x_n$ **sums to** s.

It can be shown that if a series sums to s, then it cannot sum to t where $t \neq s$ for $s, t \in \mathbb{R} \cup \{\infty, -\infty\}$. Thus, when a series $\sum x_n$ sums to s, we write $\sum x_n = s$.

Further, we say that a series **converges** iff it converges to some $s \in \mathbb{R}$. A series which converges, is called a **convergent series**.

A series is said to **diverge** iff it does not converge. A series which diverges, is called a **divergent series**.

There are two special types of divergent series. We say that the series $\sum x_n$ **diverges to** ∞ iff for each $r > 0$, there exists $m \in \mathbb{N}$ such that for each natural number $n > m$, $\sum_{k=1}^{n} x_k > r$. When $\sum x_n$ diverges to ∞, we also say that $\sum x_n$ **sums to** ∞.

We say that the series $\sum x_n$ **diverges to** $-\infty$ iff for each $r > 0$, there exists $m \in \mathbb{N}$ such that for each natural number $n > m$, $\sum_{k=1}^{n} x_k < -r$. When $\sum x_n$ diverges to $-\infty$, we also say that $\sum x_n$ **sums to** $-\infty$.

For instance, the series $1 + 2 + 3 + 4 + \cdots$ diverges to (sums to) ∞, and the series $-1 - 2 - 3 - 4 - \cdots$ diverges to (sums to) $-\infty$. The series $1 - 1 + 1 - 1 + 1 - 1 + \cdots$ diverges but it neither diverges to ∞ nor diverges to $-\infty$.

It can be shown that if a series sums to s, then it cannot sum to t where $t \neq s$ for $s, t \in \mathbb{R} \cup \{\infty, -\infty\}$. Thus, when a series $\sum x_n$ sums to s, we write $\sum x_n = s$.

(3.1) ***Example***

Let $a, r \in \mathbb{R}$. Consider the **geometric series**

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \cdots .$$

When $a = 0$, the series sums to 0. So, suppose $a \neq 0$. The sum of the first n terms of the geometric series is

$$s_n = a + ar + ar^2 + ar^3 + \cdots ar^{n-1} = \frac{a(1-r^n)}{1-r} = \frac{a}{1-r} + \frac{a}{1-r} \cdot (-r^n).$$

(a) If $r = 1$, then the geometric series is $a+a+\cdots$; and it diverges to ∞ when $a > 0$; and it diverges to $-\infty$ when $a < 0$.

(b) If $r = -1$, then the series is $a - a + a - a + \cdots$, which diverges.

(c) Let $r > 1$. If $a > 0$, then $s_n > na$. So, the series diverges to ∞. If $a < 0$, write $a = -b$. Now, $b > 0$. And, $s_n < -nb$. So, the series diverges to $-\infty$.

(d) If $r < -1$, write $t = -r$. Then, $t > 1$ and

$$s_n = a\big(1 - t + t^2 - t^3 + \cdots + (-1)^{n+1}t^n\big) = \frac{a\big(1 + (-1)^{n+1}t^{n+1}\big)}{1+t}.$$

For n odd, it has positive sign, and for n even, it has negative sign. Moreover, t^n can be made as large as we please by making n large. So, the series diverges.

(e) Suppose $|r| < 1$. If $r = 0$, then the geometric series is simply a. In this case, we find that the series sums to $\dfrac{a}{1-r}$. So, assume that $r \neq 0$. We would like to see that the difference between s_n and $\dfrac{a}{1-r}$ can be made as small as we please by choosing n large. Write $s = \left|\dfrac{a}{1-r}\right|$. Now,

$$\left|s_n - \frac{a}{1-r}\right| = \left|\frac{a}{1-r}\right| \cdot |r|^n = s|r|^n.$$

Since $|r| < 1$, $\dfrac{1}{|r|} > 1 = 1 + h$ for some $h > 0$. By the Binomial theorem,

$$\frac{1}{|r|^n} = (1+h)^n = 1 + \binom{n}{1}h + \binom{n}{2}h^2 + \cdots + \binom{n}{n}h^n > 1 + nh \quad \text{for } n > 1.$$

Then $|r|^n < \dfrac{1}{1+nh} < \dfrac{1}{nh}$ for $n > 1$.

Let $\epsilon > 0$. Choose a natural number $m > \dfrac{s}{\epsilon h}$. Let $n > m$, then $n > 1$ and $n > \dfrac{s}{\epsilon h}$. This gives

$$nh > \frac{s}{\epsilon} \Rightarrow \frac{1}{nh} < \frac{\epsilon}{s} \Rightarrow |r|^n < \frac{\epsilon}{s} \Rightarrow s|r|^n < \epsilon.$$

Using the earlier inequality, we see the following:

For each $\epsilon > 0$, there exists $m \in \mathbb{N}$ such that for each $n > m$, we have

$$\left| s_n - \frac{a}{1-r} \right| < \epsilon.$$

Therefore, $a + ar + ar^2 + ar^3 + \cdots = \dfrac{a}{1-r}$ for $|r| < 1$. □

(3.2) ***Example***

The series $1 + \dfrac{1}{2} + \dfrac{1}{3} + \dfrac{1}{4} + \cdots$ is called the **harmonic series**. We show that the harmonic series diverges to ∞.

For this, suppose $r > 0$. choose a natural number $k \geq 2(r-1)$ and take $m = 2^k$, and let $n > m$. Then

$$\begin{aligned}
s_n > s_m &= 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^k} \\
&= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^{k-1}+1} + \frac{1}{2^{k-1}+2} + \cdots + \frac{1}{2^k}\right) \\
&\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^k} + \frac{1}{2^k} + \cdots + \frac{1}{2^k}\right) \\
&= 1 + \frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2} = 1 + \frac{k}{2} \geq 1 + (r-1) = r.
\end{aligned}$$

That is, given any $r > 0$, there exists $m \in \mathbb{N}$ such that for each natural number $n > m$, we have $s_n > r$. Therefore, the series diverges to ∞. □

From the above example, we see that the following series also diverges to ∞:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \cdots.$$

One of the many ways of showing that a series is convergent is the following. There are many tests of convergence for series, but we will not deal with them.

3.2 Taylor series

We have seen that the geometric series converges for $|r| < 1$, and in this case, its sum is $\dfrac{1}{1-r}$. That is,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots \quad \text{for} \quad -1 < x < 1.$$

It says that for each $x \in (-1, 1)$, the sum of the series on the RHS is equal to the number $\dfrac{1}{1-x}$. Thus, we say that the function $f(x) = \dfrac{1}{1-x}$ *has the series expansion* as the series $1 + \sum_{n=1}^{\infty} x^n$ for $-1 < x < 1$.

Given a function how do we obtain its series expansion?

To answer this, we use Taylor's formula in (1.50). It says that if $f(x)$ defined on $(a-\delta, a+\delta)$ is $n+1$ times differentiable for $n \in \mathbb{N}$, then we have

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

where

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \quad \text{for some } c \text{ between } a \text{ and } x.$$

The above formula suggests a series for a given function $f(x)$. The series

$$f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f(n)(a)}{n!}(x-a)^n + \cdots$$

is called the **Taylor series** of the function $f(x)$ at $x = a$. In particular, when $a = 0$, the Taylor series is

$$f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f(n)(0)}{n!}x^n + \cdots .$$

It is called the **Maclaurin series** of $f(x)$.

Notice that the Taylor series at $x = a$ may or may not converge; and even if it converges, it may or may not sum to $f(x)$. From Taylor's formula, we see that the Taylor series sums to $f(x)$ provided that

for each $\epsilon > 0$, there exists $m \in \mathbb{N}$ such that for each $n > m$, $|R_n(x)| < \epsilon$.

This condition is abbreviated to $\boldsymbol{|R_n(x)| \to 0}$ and read as **$|R_n(x)|$ converges to 0**.

In the expression for $|R_n(x)|$, the constant c is a fixed number between a and x. This constant depends on the function and the point a, and as such is unknown to us. Thus, when needed, we prove that $|R_n(x)| \to 0$ for all c between a and x while $a - \delta < x < a + \delta$ for a suitable $\delta > 0$. We then obtain the following:

If $f(x)$ has derivatives of all orders, and $|R_n(x)| \to 0$ for all $x \in (a-\delta, a+\delta)$ and for all c between a and x, then the Taylor series of $f(x)$ sums to $f(x)$ for $a-\delta < x < a+\delta$. That is,

$$f(x) = f(a) + \sum_{n=1}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \quad \text{for } a - \delta < x < a + \delta.$$

When the Taylor series of $f(x)$ sums to $f(x)$ for each $x \in (a-\delta, a+\delta)$, where δ is some positive real number, we also say that the function **$f(x)$ has the Taylor series expansion at $x = a$** as that series.

Notice that the condition "$|R_n(x)| \to 0$ for all $x \in (a-\delta, a+\delta)$ and for all c between a and x" is a sufficient condition to show that $f(x)$ has the Taylor series expansion at $x = a$. This condition is not a necessary condition. If we can prove that the Taylor series sums to $f(x)$ while $a - \delta < x < a + \delta$ by some other means, then also it follows that the Taylor series so obtained is the Taylor series expansion of $f(x)$. Sometimes this method produces a larger interval around $x = a$, where the Taylor series sums to the function.

(3.3) ***Example***

Find the Taylor series expansion of the function $f(x) = 1/x$ at $x = 2$. In which interval around $x = 2$, the series converges?

We see that

$$f(x) = \frac{1}{x},\ f(2) = \frac{1}{2}; \quad \cdots; f^{(n)}(x) = \frac{(-1)^n n!}{x^{n+1}},\ f^{(n)}(2) = \frac{(-1)^n n!}{2^{n+1}}.$$

Hence the Taylor series for $f(x) = 1/x$ at $x = 2$ is

$$\frac{1}{2} - \frac{x-2}{2^2} + \frac{(x-2)^2}{2^3} - \cdots + (-1)^n \frac{(x-2)^n}{2^{n+1}} + \cdots \tag{3.2.1}$$

We now require the remainder term in the Taylor series expansion. The absolute value of the remainder term is

$$|R_n(x)| = \left|\frac{f^{(n+1)}(c)}{(n+1)!}(x-2)^{n+1}\right| = \left|\frac{(x-2)^{n+1}}{c^{n+2}}\right|$$

Here, $2 - \delta < x < 2 + \delta$ for some $\delta > 0$, and c lies between 2 and x. Let us take $\delta = 1$. Then, $1 < x < 3$ and c is between 2 and x. Thus, $|c| > 1$ and $|x-2| < 1 < c$. Then, $|(x-2)/c| < 1$. It follows that $|R_n(x)| \to 0$ as $n \to \infty$ for $1 < x < 3$. That is, $f(x) = 1/x$ has the Taylor series expansion (3.2.1) for $1 < x < 3$.

It looks we cannot take δ to be larger than 1 if we follow this method to estimate $|R_n(x)|$. However, the series here is a geometric series with ratio $r = -(x-2)/2$. Hence it converges whenever

$$|r| < 1, \quad \text{i.e.,} \quad |x-2| < 2 \quad \text{i.e.,} \quad 0 < x < 4.$$

Thus the $f(x) = 1/x$ has the Taylor series expansion (3.2.1) for $0 < x < 4$. That is,

$$\frac{1}{x} = \frac{1}{2} - \frac{x-2}{2^2} + \frac{(x-2)^2}{2^3} - \cdots + (-1)^n \frac{(x-2)^n}{2^{n+1}} + \cdots \quad \text{for } 0 < x < 4. \qquad \square$$

(3.4) ***Example***

Find the Maclaurin series expansion of e^x.

Let $f(x) = e^x$. For its Maclaurin series (Taylor series at $x = 0$), we find that $f(0) = 1,\ f'(0) = 1, \cdots, f^{(n)}(0) = 1, \cdots$.

Hence the Maclaurin series for e^x is $1 + x + \dfrac{x^2}{2!} + \cdots + \dfrac{x^n}{n!} + \cdots$.

Does this series converge? Does it sum to e^x? If so, in which interval around $x = 0$? To answer these questions, we consider the remainder term. We see that

$$|R_n(x)| = \left|\frac{f^{(n+1)}(c)}{(n+1)!}x^{n+1}\right| = \left|\frac{e^c x^{n+1}}{(n+1)!}\right| \to 0 \text{ as } n \to \infty$$

for any $x \in \mathbb{R}$ and any c between 0 to x. Why?

Let $\epsilon > 0$. Choose $N > 2|x|$. Then $\dfrac{|x|}{k} < \dfrac{1}{2}$ for all $k > N$. Since $\dfrac{1}{2^n} \to 0$ as $n \to \infty$, there exists $N' > N$ such that $\frac{1}{2^{N'-N}}\dfrac{e^c|x|^N}{N} < \epsilon$. Then, for all $n > N'$,

$$\left|\frac{e^c x^{n+1}}{(n+1)!}\right| = \frac{|x|^{n+1-N}}{(n+1)n\cdots(N+1)}\frac{e^c|x|^N}{N!} \le \frac{1}{2^{n+1-N}}\frac{e^c|x|^N}{N} < \frac{1}{2^{N'-N}}\frac{e^c|x|^N}{N} < \epsilon.$$

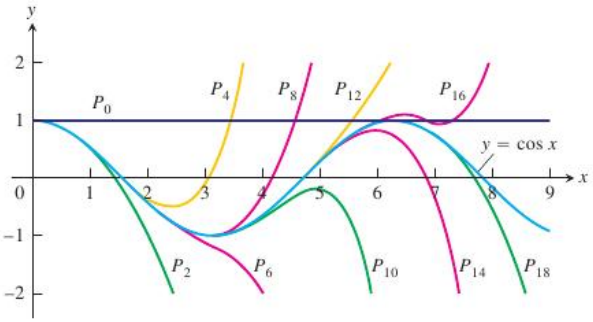
Hence,

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots \qquad \text{for each } x \in \mathbb{R}.$$

That is, this series is the Maclaurin series expansion of e^x for $x \in \mathbb{R}$. □

(3.5) ***Example***

You can show that $\cos x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$. The Taylor polynomials approximating $\cos x$ at $x = 0$ are $P_{2n}(x) = \displaystyle\sum_{k=0}^{n} \frac{(-1)^k x^{2k}}{(2k)!}$. The following picture shows how these polynomials $P_{2n}(x)$ approximate $\cos x$ for $0 \le n \le 9$.



For the above Maclaurin series expansion of $\cos x$, we see that

$$|R_{2n}(x)| = \frac{|x|^{2n+1}}{(2n+1)!} \to 0 \text{ as } n \to \infty$$

for each $x \in \mathbb{R}$. Hence the series represents $\cos x$ for all $x \in \mathbb{R}$. □

Similarly, you can show that $\sin x = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n-1}}{(2n-1)!}$ for all $x \in \mathbb{R}$.

(3.6) ***Example*** (Binomial expansion)

Let $m \in \mathbb{R}$. For any $n \in \mathbb{N}$, write $\binom{m}{n} = \frac{m(m-1)\cdots(m-n+1)}{n!}$. Then,

$$(1+x)^m = 1 + \sum_{n=1}^{\infty} \binom{m}{n} x^n \quad \text{for } -1 < x < 1.$$

For this, we find the derivatives of the given function:

$$f(x) = (1+x)^m, \quad f^{(n)}(x) = m(m-1)\cdots(m-n+1)x^{m-n}.$$

Then the Maclaurin series for $f(x)$ is the given series. It can be shown that the Maclaurin series for $(1+x)^m$ sums to $(1+x)^m$ for $-1 < x < 1$. (It requires tools we have not yet discussed; so we skip its proof.)

The series so obtained is called a **binomial series** expansion of $(1+x)^m$. Substituting values of m, we get series for different functions. Notice that when $m \in \mathbb{N}$, the binomial series terminates to give

$$(1+x)^m = 1 + mx + \binom{m}{2}x^2 + \cdots + \binom{m}{n}x^n + \cdots + mx^{m-1} + x^m \quad \text{for } x \in \mathbb{R}.$$

This is the Binomial theorem as you have studied earlier. □

There are three facts that help us in obtaining Taylor series expansions of some functions from those of known ones.

Fact 1: Taylor series expansions for two functions can be added term by term, and multiplied like polynomials. It means the following.

Suppose $f(x)$ and $g(x)$ have the following Taylor series expansions at $x = a$:

$$f(x) = f(a) + \sum_{n=1}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \quad \text{for } a-\delta < x < a+\delta \tag{3.2.2}$$

$$g(x) = g(a) + \sum_{n=1}^{\infty} \frac{g^{(n)}(a)}{n!}(x-a)^n \quad \text{for } a-\delta < x < a+\delta$$

for some $\delta > 0$. Then, the following are true for $a - \delta < x < a + \delta$:

$$f(x) + g(x) = \big(f(a) + g(a)\big) + \sum_{n=1}^{\infty} \frac{f^{(n)}(a) + g^{(n)}(a)}{n!}(x-a)^n.$$

$$f(x) \cdot g(x) = \big(f(a) \cdot g(a)\big) + \sum_{n=1}^{\infty} a_n (x-a)^n$$

$$\text{where } a_n = f(a)\frac{g^{(n)}(a)}{n!} + \sum_{k=1}^{n-1} \frac{f^{(k)}(a) \cdot g^{(n-k)}(a)}{k!(n-k)!} + \frac{f^{(n)}(a)}{n!} g(a).$$

In general, if $f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n$ for $a-\delta < x < a+\delta$, and $g(x) = \sum_{n=0}^{\infty} b_n (x-a)^n$ for $a - \delta < x < a + \delta$, then $f(x) \cdot g(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$ for $a - \delta < x < a + \delta$, where $c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_{n-1} b_1 + a_n b_0$.

(3.7) ***Example***

Find the Maclaurin series expansion of $\dfrac{e^x}{1-x}$.

We know that

$$e^x = 1 + \sum_{n=1}^{\infty} \frac{x^n}{n!} \quad \text{for } x \in \mathbb{R}, \qquad \frac{1}{1-x} = 1 + \sum_{n=1}^{\infty} x^n \quad \text{for } -1 < x < 1.$$

So, the coefficient c_n of x^n in the Maclaurin series expansion of $\dfrac{e^x}{1-x} = \sum_{n=0}^{\infty} c_n x^n$ is

$$c_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}.$$

That is, $\dfrac{e^x}{1-x} = 1 + \sum_{n=1}^{\infty} \left(1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}\right) x^n$ for $-1 < x < 1$. □

Fact 2: The Taylor series expansion of a function can be differentiated and integrated term by term. It means the following:

Suppose the Taylor series expansion of $f(x)$ at $x = a$ is as given in (3.2.2). Then, the following are true for $a - \delta < x < a + \delta$:

$$f'(x) = \sum_{n=1}^{\infty} \frac{f^{(n)}(a)}{(n-1)!}(x-a)^{n-1}.$$

$$\int_a^x f(t)\,dt = f(a)(x-a) + \sum_{n=1}^{\infty} \frac{f^{(n)}(a)}{(n+1)!}(x-a)^{n+1}.$$

(3.8) ***Example***

Find the Maclaurin series expansion of $\tan^{-1} x$ for $-1 < x < 1$.

We know that $\dfrac{1}{1+t^2} = 1 + \sum_{n=1}^{\infty} (-1)^n t^{2n}$ for $|-t^2| < 1$; that is, for $-1 < t < 1$.

Integrating from 0 to x term by term, we obtain

$$\tan^{-1} x = x + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} \quad \text{for } -1 < x < 1. \qquad \square$$

Fact 3: If the Taylor series expansion of $f(x)$ in (3.2.2) converges at $x = \pm\delta$, then those will sum to the respective limits of $f(x)$ at these points. It means the following:

If $f(a) + \sum_{n=1}^{\infty} \dfrac{f^{(n)}(a)}{n!} (\delta - a)^n$ converges, then its sum is equal to $f(\delta-)$.

If $f(a) + \sum_{n=1}^{\infty} \dfrac{f^{(n)}(a)}{n!} (-\delta - a)^n$ converges, then its sum is equal to $f(-\delta+)$.

Here, $f(\delta-) = \lim_{x \to \delta-} f(x)$ and $f(-\delta+) = \lim_{x \to -\delta+} f(x)$.

(3.9) ***Example***

Using a series expansion of $\tan^{-1} x$ give an approximation scheme for π.

From (3.8), we have

$$\tan^{-1} x = x + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} \quad \text{for } -1 < x < 1.$$

It can be shown that the series on the RHS converges for $x = 1$. Then,

$$1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n+1} = \lim_{x \to 1-} \tan^{-1} x = \tan^{-1} 1 = \frac{\pi}{4}.$$

Therefore, $\pi = 4 - \dfrac{4}{3} + \dfrac{4}{5} - \dfrac{4}{7} + \cdots$. Truncating the series up to n terms, we obtain approximations of π. $\square$

(3.10) ***Example***

For the function $f(x) = \sqrt{1+x}$, the binomial series expansion gives (with $m = 1/2$)

$$\sqrt{1+x} = 1 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(2n)!}{2^{2n}(n!)^2(2n-1)} x^n = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \cdots \quad \text{for } -1 < x < 1.$$

It can be shown that the series converges for $x = 1$. Thus,

$$1 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(2n)!}{2^{2n}(n!)^2(2n-1)} = \lim_{x \to 1-} f(x) = \sqrt{2}.$$

Similarly, convergence at $x = -1$ gives

$$1 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(2n)!}{2^{2n}(n!)^2(2n-1)}(-1)^n = \lim_{x \to -1+} f(x) = 0. \qquad \square$$

3.3 Fourier series

In the power series for $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \cdots$, the periodicity of $\cos x$ is not obvious. Recall that $f(x)$ is called **2ℓ-periodic** for $\ell > 0$ iff $f(x + 2\ell) = f(x)$ for all $x \in \mathbb{R}$. For the time being, we restrict to 2π-periodic functions. We will see that a 2π-periodic function can be expanded in a series involving sines and cosines instead of powers of x.

A **trigonometric series** is a series in the form $\dfrac{1}{2}a_0 + \displaystyle\sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$.

Since both cosine and sine functions are 2π-periodic, if a trigonometric series converges to a function $f(x)$, then necessarily $f(x)$ is also 2π-periodic. Thus,

$$f(0) = f(2\pi) = f(4\pi) = f(6\pi) = \cdots \quad \text{and} \quad f(-\pi) = f(\pi), \quad \text{etc.}$$

Moreover, if $f(x) = \dfrac{a_0}{2} + \displaystyle\sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$ for all $x \in [-\pi, \pi]$, then the coefficients can be determined from $f(x)$. Towards this, multiply $f(t)$ by $\cos mt$ and integrate to obtain:

$$\int_{-\pi}^{\pi} f(t) \cos mt \, dt = \frac{a_0}{2}\int_{-\pi}^{\pi} \cos mt \, dt + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi} \cos nt \cos mt \, dt + \sum_{n=1}^{\infty} b_n \int_{-\pi}^{\pi} \sin nt \cos mt \, dt.$$

For $m, n = 0, 1, 2, 3, \ldots,$

$$\int_{-\pi}^{\pi} \cos nt \cos mt \, dt = \begin{cases} 0 & \text{if } n \neq m \\ \pi & \text{if } n = m > 0 \\ 2\pi & \text{if } n = m = 0 \end{cases} \quad \text{and} \quad \int_{-\pi}^{\pi} \sin nt \cos mt \, dt = 0.$$

Thus, we obtain

$$\int_{-\pi}^{\pi} f(t) \cos mt \, dt = \pi a_m, \quad \text{for } m = 0, 1, 2, 3, \cdots$$

Similarly, by multiplying $f(t)$ by $\sin mt$ and integrating, and using the fact that

$$\int_{-\pi}^{\pi} \sin nt \sin mt \, dt = \begin{cases} 0 & \text{if } n \neq m \\ \pi & \text{if } n = m > 0 \\ 0 & \text{if } n = m = 0 \end{cases}$$

we obtain

$$\int_{-\pi}^{\pi} f(t) \sin mt \, dt = \pi b_m, \quad \text{for } m = 1, 2, 3, \cdots$$

We thus define our series of interest as follows.

Let $f : \mathbb{R} \to \mathbb{R}$ be a 2π-periodic function integrable on $[-\pi, \pi]$. Write

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt \, dt, \quad \text{for } n = 0, 1, 2, 3, \ldots$$
$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt \, dt \quad \text{for } n = 1, 2, 3, \ldots.$$

The trigonometric series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

is called the **Fourier series** of $f(x)$. The numbers a_n and b_n are called the **Fourier coefficients** of $f(x)$.

Analogous to Taylor series, the Fourier series of a function may or may not converge; and even if it converges, it need not sum to $f(x)$. For convergence of a Fourier series, we introduce the notion of a piecewise differentiable function.

Let $f(x)$ be defined on $[a, b]$. We say that $f(x)$ is **piecewise differentiable** iff $f(x)$ has a right derivative at $x = a$, $f(x)$ has a left derivative at $x = b$, and $f(x)$ is differentiable at every point of (a, b) except possibly at a finite number of points in (a, b) where $f(x)$ has both a left derivative and a right derivative.

Recall that $f(c-) = \lim_{h \to 0+} f(c-h)$ and $f(c+) = \lim_{h \to 0+} f(c+h)$ are the left and right side limits of $f(x)$ at $x = c$, respectively.

(3.11) ***Theorem*** (Convergence of Fourier Series)

Let $f(x)$ be a 2π-periodic function defined on $\mathbb{R}$. Suppose $f(x)$ is a piecewise differentiable function on $[-\pi, \pi]$. Let $c \in \mathbb{R}$. Then the Fourier series of $f(x)$ converges at $x = c$. Further, the Fourier series of $f(x)$ at $x = c$ sums to

(a) $f(c)$ *if* $f(x)$ *is continuous at* $x = c$*;*

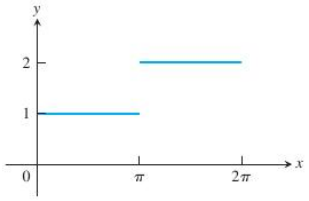
(b) $\frac{1}{2}[f(c-) + f(c+)]$ *if* $f(x)$ *is not continuous at* c*.*

We remark that Fourier series of *piecewise smooth functions* and *functions of bounded variation* converge. But the defining conditions of these more general types of functions are difficult to verify. Also, most functions that are met in applications are piecewise differentiable. So, we restrict our study to piecewise differentiable functions.

(3.12) ***Example***

Find the Fourier series of the function $f(x)$ given by the following which is extended to $\mathbb{R}$ with periodicity 2π:

$$f(x) = \begin{cases} 1 & \text{if } 0 \le x < \pi \\ 2 & \text{if } \pi \le x < 2\pi. \end{cases}$$



Due to periodic extension, we can rewrite the function $f(x)$ on $[-\pi, \pi)$ as

$$f(x) = \begin{cases} 2 & \text{if } -\pi \le x < 0 \\ 1 & \text{if } 0 \le x < \pi. \end{cases}$$

Observe that this function is piecewise differentiable. The coefficients of the Fourier series are computed as follows:

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{0} f(t)\,dt + \frac{1}{\pi}\int_{0}^{\pi} f(t)\,dt = 3.$$
$$a_n = \frac{1}{\pi}\int_{-\pi}^{0} 2\cos nt\,dt + \frac{1}{\pi}\int_{0}^{\pi} \cos nt\,dt = 0.$$
$$b_n = \frac{1}{\pi}\int_{-\pi}^{0} 2\sin nt\,dt + \frac{1}{\pi}\int_{0}^{\pi} \sin nt\,dt = \frac{(-1)^n - 1}{n\pi}.$$

Notice that $b_1 = -\frac{2}{\pi}$, $b_2 = 0$, $b_3 = -\frac{2}{3\pi}$, $b_4 = 0, \ldots$. Therefore,

$$f(x) = \frac{3}{2} - \frac{2}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right).$$

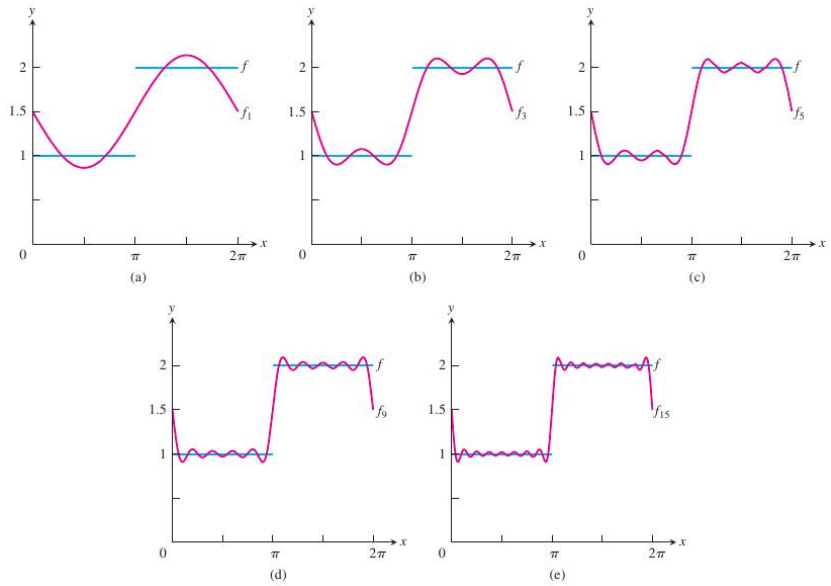
Here, the last expression for $f(x)$ holds for all $x \in \mathbb{R}$, where ever $f(x)$ is continuous; in particular, for $x \in [-\pi, \pi)$ except at $x = 0$. At $x = 0$, the Fourier series sums to $[f(0-) + f(0+)]/2 = 3/2$. □

Once we have a series representation of a function, we should see how the **partial sums** of the series approximate the function, to satisfy our curiosity. In the above

example, let us write

$$f_m(x) = \frac{1}{2}a_0 + \sum_{n=1}^{m}(a_n \cos nx + b_n \sin nx).$$

The approximations $f_1(x)$, $f_3(x)$, $f_5(x)$, $f_9(x)$ and $f_{15}(x)$ to $f(x)$ are shown in the figure below.



Remark 3.13 In the above example, we found that

$$f(x) = \frac{3}{2} - \frac{2}{\pi}\Big(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\Big).$$

In particular, at $x = \pi$, the function $f(x)$ is not differentiable. However, each term on the right hand side is differentiable at $x = \pi$. It means that each term of a Fourier series can be differentiable at a point, but the sum of the series is not differentiable at that point. Indeed, if we differentiate each term of the series and sum the terms after evaluation at π, we get the divergent series $\frac{2}{\pi}\big(-1-1-1-\cdots\big)$.

(3.14) ***Example***

Show that the Fourier series for $f(x) = x^2$ defined on $[0, 2\pi)$ is given by

$$\frac{4\pi^2}{6} + \sum_{n=1}^{\infty}\Big(\frac{4}{n^2}\cos nx - \frac{4\pi}{n}\sin nx\Big).$$

Extend $f(x)$ to $\mathbb{R}$ by periodicity 2π. We thus have $f(2\pi) = f(0) = 0$. Notice that the function when extended to $\mathbb{R}$ by periodicity has left and right derivatives at each point, and it is piecewise continuous.

If $x \in [-\pi, 0)$, then $x + 2\pi \in [\pi, 2\pi)$ so that $f(x) = f(x + 2\pi) = (x + 2\pi)^2$. For instance, $f(-\pi) = f(-\pi + 2\pi) = f(\pi) = \pi^2 = (-\pi + 2\pi)^2$ and $f(-\pi/2) = f(-\pi/2 + 2\pi) = f(3\pi/2) = (3\pi/2)^2 = (-\pi/2 + 2\pi)^2$. Thus the function $f(x)$ on $[-\pi, \pi)$ is given by

$$f(x) = \begin{cases} (x + 2\pi)^2 & \text{if } -\pi \le x < 0 \\ x^2 & \text{if } 0 \le x < \pi. \end{cases}$$

The coefficients of the Fourier series for $f(x)$ are

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt = \frac{1}{\pi}\int_0^{2\pi} t^2\,dt = \frac{8\pi^2}{3}.$$
$$a_n = \frac{1}{\pi}\int_0^{2\pi} t^2 \cos nt\,dt = \frac{4}{n^2} \quad \text{for } n = 1, 2, 3, \ldots.$$
$$b_n = \frac{1}{\pi}\int_0^{2\pi} t^2 \sin nt\,dt = -\frac{4\pi}{n} \quad \text{for } n = 1, 2, 3, \ldots.$$

Hence the Fourier series for $f(x)$ is as claimed.

As per the extension of $f(x)$ to $\mathbb{R}$, we see that in the interval $\big(2k\pi, 2(k+1)\pi\big)$, the function is given by $f(x) = (x - 2k\pi)^2$. Thus it has discontinuities at the points $x = 0,\ \pm 2\pi,\ \pm 4\pi,\ \ldots$ At such a point $x = 2k\pi$, the series converges to the average value of the left and right side limits, i.e., the series when evaluated at $2k\pi$ yields the value

$$\frac{1}{2}\Big[\lim_{x\to 2k\pi-} f(x) + \lim_{x\to 2k\pi+} f(x)\Big] = \frac{1}{2}\Big[\lim_{x\to 2k\pi-} (x-2k\pi)^2 + \lim_{x\to 2k\pi+} (x-2(k+1)\pi)^2\Big] = 2\pi^2.$$

Thus, $x^2 = \dfrac{4\pi^2}{6} + \displaystyle\sum_{n=1}^{\infty}\left(\frac{4}{n^2}\cos nx - \frac{4\pi}{n}\sin nx\right)$ for $0 < x < 2\pi$

and at $x = 0$, the Fourier series sums to $2\pi^2$. □

3.4 Odd and even functions

If $f(x)$ is an odd function, that is, if $f(-x) = -f(x)$, then the Fourier coefficients for $f(x)$ are

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos nt\,dt = 0 \quad \text{for } n = 0, 1, 2, \ldots.$$
$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin nt\,dt = \frac{2}{\pi}\int_0^{\pi} f(t)\sin nt\,dt \quad \text{for } n = 1, 2, 3, \ldots.$$

In this case, the Fourier series for $f(x)$ is given by

$$\sum_{n=1}^{\infty} b_n \sin nx, \quad b_n = \frac{2}{\pi}\int_0^{\pi} f(t)\sin nt\,dt \quad \text{for } n = 1, 2, 3, \ldots.$$

Similarly, if $f(x)$ is an even function, that is, $f(-x) = f(x)$, then its Fourier series is given by

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx, \quad a_n = \frac{2}{\pi}\int_0^{\pi} f(t)\cos nt\,dt \quad \text{for } n = 0, 1, 2, \ldots.$$

(3.15) ***Example***

Show that $x^2 = \dfrac{\pi^2}{3} + 4\displaystyle\sum_{n=1}^{\infty}(-1)^n \frac{\cos nx}{n^2}$ for all $x \in [-\pi, \pi]$.

Let $f(x)$ be the periodic extension of the function $x \mapsto x^2$ on $[-\pi, \pi)$ to $\mathbb{R}$. Since $\pi^2 = (-\pi)^2$, such an extension with period 2π exists, and it is continuous. The extension of $f(x) = x^2$ to $\mathbb{R}$ is not the function x^2. For instance, in the interval $[\pi, 3\pi]$, its extension looks like $f(x) = (x - 2\pi)^2$. With this understanding, we go for the Fourier series expansion of $f(x) = x^2$ in the interval $[-\pi, \pi]$. We also see that $f(x)$ is an even function. Its Fourier series is a cosine series. The coefficients of the series are as follows:

$$a_0 = \frac{2}{\pi}\int_0^{\pi} t^2\,dt = \frac{2}{3}\pi^2.$$
$$a_n = \frac{2}{\pi}\int_0^{\pi} t^2\cos nt\,dt = \frac{4}{n^2}(-1)^n \quad \text{for } n = 1, 2, 3, \ldots.$$

Therefore,

$$f(x) = x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}(-1)^n \frac{\cos nx}{n^2} \quad \text{for all } x \in [-\pi, \pi].$$

In particular, by taking $x = 0$ and $x = \pi$, we have

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Due to the periodic extension of $f(x)$ to $\mathbb{R}$, we see that

$$(x - 2\pi)^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}(-1)^n \frac{\cos nx}{n^2} \quad \text{for all } x \in [\pi, 3\pi].$$

It also follows that the series sums to $(x-4\pi)^2$ for $x \in [3\pi, 5\pi]$, etc. □

(3.16) ***Example***

Show that for $0 < x < 2\pi$, $\displaystyle\sum_{n=1}^{\infty} \frac{\sin nx}{n} = \frac{1}{2}(\pi - x)$.

Let $f(x) = \pi - x$ for $0 < x < 2\pi$. We consider $f(x)$ as the corresponding function extended to $\mathbb{R}$ by taking periodicity as 2π.

For $x \in (-\pi, 0)$, we have $x + 2\pi \in (\pi, 2\pi)$. Then,

$$f(x) = f(x+2\pi) = \pi - (x+2\pi) = -\pi - x \quad \text{for } x \in (-\pi, 0).$$

If $x \in (0, \pi)$, then $f(x) = \pi - x$ and $-x \in (-\pi, 0)$. Then,

$$f(-x) = -\pi - (-x) = -(\pi - x) = -f(x) \quad \text{for } x \in (0, \pi).$$

You should now draw a graph of $f(x)$ for $x \in (-\pi, 0) \cup (0, \pi)$.

Since $f(x)$ has period 2π, it follows that $f(-x) = -f(x)$ for all $x \in \mathbb{R}$ excluding $x = 0$ and thereby excluding all multiples of 2π. We then define $f(2k\pi) = 0$ for all integers k. That is, we consider the 2π-periodic function

$$f(x) = \begin{cases} -\pi - x & \text{if } x \in (-\pi, 0) \\ \pi - x & \text{if } x \in (0, \pi) \\ 0 & \text{if } x = -\pi, 0, \pi. \end{cases}$$

Now, the function $f(x)$ is an odd function with periodicty 2π. So, its Fourier series is a sine series. We need only the coefficients b_n. They are given by

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(t)\sin nt\, dt = \frac{2}{\pi}\int_0^{\pi} (\pi - t)\sin nt\, dt = \frac{2}{n}.$$

By the convergence theorem,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx = 2\sum_{n=1}^{\infty} \frac{\sin nx}{n} \quad \text{for } x \in \mathbb{R}.$$

Notice that at $x = 0$, $f(x)$ is not continuous. Thus, at $x = 0$, the series sums to

$$\frac{1}{2}[f(0-) + f(0+)] = \frac{1}{2}\Big[\lim_{x\to 0-}(-\pi - x) + \lim_{x\to 0+}(\pi - x)\Big] = 0 = f(0).$$

Since $f(x) = \pi - x$ for $0 < x < 2\pi$, we obtain

$$\pi - x = 2\sum_{n=1}^{\infty} \frac{\sin nx}{n} \quad \text{for } 0 < x < 2\pi.$$ □

3.5 Half range Fourier series

Suppose a piecewise differentiable function $f : (0, \pi) \to \mathbb{R}$ is given. To find its Fourier series, we need to extend it to $\mathbb{R}$ so that the extended function is 2π-periodic. Such an extension can be done in at least two ways. We may extend the function in such a way that it is 2π-periodic and it is an even function; or we may extend it to a 2π-periodic function which is an odd function. The Fourier series so obtained for each case will be different. However, on $(0, \pi)$ the sum of the Fourier series obtained both ways will agree.

1. Even Extension:

First, extend $f(x)$ to $(-\pi, \pi)$ by requiring that $f(x)$ is an even function. This requirement forces $f(-x) = f(x)$ for each $x \in (-\pi, \pi)$. Next, we extend this $f(x)$ which has now been defined on $(-\pi, \pi)$ to $\mathbb{R}$ with periodicity 2π. Notice that since $f(0)$, $f(-\pi)$ and $f(\pi)$ are not given, we may have to define them. However, $f(x)$ is required to be even and 2π-periodic; so we must take $f(\pi) = f(-\pi)$, $f(0) = f(2\pi)$, etc. If possible, we define these values in such a way that the extended function is continuous at these points, so that the Fourier series will converge to these values at these points. Since the extended function is even and of period 2π, its Fourier series is a cosine series given by

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx, \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(t) \cos nt \, dt, \quad n = 0, 1, 2, 3, \ldots, \quad x \in \mathbb{R}.$$

In this case, we say that the Fourier series is a **half-range cosine series** for $f(x)$.

In case, domain of f is $[0, \pi)$, $(0, \pi]$ or $[0, \pi]$, the values $f(0)$ and/or $f(\pi)$ may be already available, and we do not need to define those.

2. Odd Extension:

First, extend $f(x)$ from $(0, \pi)$ to $(-\pi, \pi)$ by requiring that $f(x)$ is an odd function. This requirement forces $f(-x) = -f(x)$ for each $x \in (-\pi, \pi)$. Since the extended function is also required to be odd and 2π-periodic, we must take $f(-\pi) = f(\pi) = f(0) = 0$. Thus, the odd extension of $f(x)$ with periodicity 2π will satisfy $f(k\pi) = 0$ for all integers k. By the convergence theorem, the Fourier series of the extended odd function $f(x)$ will be equal to $f(x)$ at each $x \neq k\pi$.

In addition, if f is already defined on $[0, \pi]$ with $f(0) = f(\pi) = 0$, then the Fourier series of the odd extension of $f(x)$ with period 2π will represent the function $f(x)$ at all points $x \in \mathbb{R}$.

The Fourier series expansion of this extended $f(x)$ is a sine series given by

$$\sum_{n=1}^{\infty} b_n \sin nx, \quad b_n = \frac{2}{\pi}\int_0^{\pi} f(t)\sin nt\,dt, \quad n = 1, 2, 3, \ldots, \quad x \in \mathbb{R}.$$

In this case, we say that the Fourier series is a **half-range sine series** for $f(x)$.

(3.17) ***Example***

Find half-range cosine and sine series for $f(x) = \begin{cases} x & \text{if } 0 \le x \le \pi/2 \\ \pi - x & \text{if } \pi/2 < x \le \pi. \end{cases}$

Here, $f(0) = 0 = f(\pi)$. So, its extension to $\mathbb{R}$ with period 2π will have $f(k\pi) = 0$ for all integers k.

1. With an even extension, the Fourier coefficients a_n are given by $a_0 = \pi/2$ and for $n = 1, 2, 3, \ldots$,

$$\begin{aligned} a_n &= \frac{2}{\pi}\int_0^{\pi} f(t)\cos nt\,dt = \frac{2}{\pi}\int_0^{\pi/2} t\cos nt\,dt + \frac{2}{\pi}\int_{\pi/2}^{\pi}(\pi - t)\cos nt\,dt \\ &= \frac{2}{\pi n^2}\Big(2\cos\frac{n\pi}{2} - 1 - \cos n\pi\Big). \end{aligned}$$

Notice that $a_n = 0$ when n is odd, and $a_n = 0$ when $n = 4k$ for any integer k. Thus

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi}\Big(\frac{\cos 2x}{1^2} + \frac{\cos 6x}{3^2} + \frac{\cos 10x}{5^2} + \cdots\Big) \quad \text{for } x \in [0, \pi].$$

2. With an odd extension, the Fourier coefficients b_n for $n = 1, 2, 3, \ldots$ are

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^{\pi} f(t)\sin nt\,dt = \frac{2}{\pi}\int_0^{\pi/2} t\sin nt\,dt + \frac{2}{\pi}\int_{\pi/2}^{\pi}(\pi - t)\sin nt\,dt \\ &= \frac{2}{\pi}\Big[-\frac{t}{n}\cos nt + \frac{1}{n^2}\sin nt\Big]_0^{\pi/2} + \frac{2}{\pi}\Big[\frac{t-\pi}{n}\cos nt - \frac{1}{n^2}\sin nt\Big]_{\pi/2}^{\pi} \\ &= \frac{2}{\pi}\Big(-\frac{\pi}{2n}\cos\frac{n\pi}{2} + \frac{1}{n^2}\sin\frac{n\pi}{2}\Big) + \frac{2}{\pi}\Big(\frac{\pi}{2n}\cos\frac{2\pi}{n} + \frac{1}{n^2}\sin\frac{n\pi}{2}\Big) \\ &= \frac{4}{\pi n^2}\sin\frac{n\pi}{2} = \begin{cases} \frac{4}{\pi n^2}(-1)^{(n-1)/2} & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even.} \end{cases} \end{aligned}$$

Thus $f(x) = \frac{4}{\pi}\Big(\frac{\sin x}{1^2} - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \cdots\Big)$ for $x \in [0, \pi]$.

Justify why do we write the equalities $f(x) = \cdots$ in both the cases. □

(3.18) ***Example***

Find the sine series expansion of $\cos x$ in $[0, \pi]$.

We work with the odd extension of $\cos x$ with period 2π to $\mathbb{R}$. Observe that the odd extension $f(x)$ of $\cos x$ has the following values in $[-\pi, \pi]$:

$$f(x) = \begin{cases} -\cos x & \text{if } -\pi < x < 0 \\ \cos x & \text{if } 0 < x < \pi \\ 0 & \text{if } x = -\pi, 0, \pi. \end{cases}$$

The Fourier coefficients b_n for $n = 1, 2, 3, \ldots$ are given by

$$b_n = \frac{2}{\pi} \int_0^{\pi} \cos t \sin nt \, dt = \begin{cases} 0 & \text{for } n \text{ odd} \\ \dfrac{4n}{\pi(n^2 - 1)} & \text{for } n \text{ even.} \end{cases}$$

Therefore, the Fourier sine expansion of $\cos x$ is $\dfrac{8}{\pi} \displaystyle\sum_{n=1}^{\infty} \frac{n \, \sin(2nx)}{4n^2 - 1}$ in $[0, \pi]$.
At $x = 0$ and $x = \pi$, the Fourier series sums to $0 = f(0) = f(\pi)$, but different from $\cos 0$ and $\cos \pi$. At other points, the series represents $\cos x$. That is,

$$\cos x = \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{n \, \sin(2nx)}{4n^2 - 1} \quad \text{for } x \in (0, \pi).$$

Similarly, you can verify that $\sin x = \dfrac{2}{\pi} - \dfrac{4}{\pi} \displaystyle\sum_{n=1}^{\infty} \frac{\cos(2nx)}{4n^2 - 1}$ for $x \in (0, \pi)$. □

What are the cosine expansion of $\cos x$ and sine expansion of $\sin x$ for $0 < x < \pi$?

There is another approach to construct a Fourier series for a function with domain as $(0, \pi)$, which we discuss in the last section.

3.6 Functions defined on $(-\ell, \ell)$

Suppose $f : (-\ell, \ell) \to \mathbb{R}$ is given. We first extend $f(x)$ to a 2ℓ-periodic function on $\mathbb{R}$. While constructing such an extension, we may have to define $f(-\ell)$. Especially, we take $f(-\ell) = \lim\limits_{x \to -\ell+} f(x)$ so that the extended function is continuous at $x = -\ell$. Of course, if originally f is a function given on $[-\ell, \ell)$, then this definition of $f(-\ell)$ is not necessary.

Next, we change the independent variable by taking $x = \ell y/\pi$, equivalently, $y = \pi x/\ell$. That is, we define the function $g(y) = f(\ell y/\pi)$. Since $f(x)$ is a 2ℓ-periodic function, $g(y)$ is a 2π-periodic function.

Then, we construct the Fourier series for $g(y)$ and in this Fourier series substitute $y = \pi x/\ell$ for obtaining the Fourier series for $f(x)$.

The Fourier series for $g(y)$ is given by

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \big(a_n \cos ny + b_n \sin ny\big),$$

where the Fourier coefficients are (In a_n, $n = 0, 1, 2, 3, \ldots$; and in b_n, $n = 1, 2, 3, \ldots$.)

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f\Big(\frac{\ell s}{\pi}\Big)\cos ns\, ds, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f\Big(\frac{\ell s}{\pi}\Big)\cos ns\, ds.$$

Substituting $t = \dfrac{\ell}{\pi}s, \quad ds = \dfrac{\pi}{\ell}dt$, we have

$$a_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\cos\Big(\frac{n\pi t}{\ell}\Big)\,dt, \quad b_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\sin\Big(\frac{n\pi t}{\ell}\Big)\,dt.$$

Substituting $y = \pi x/\ell$ in the above Fourier series, we obtain the Fourier series for $f(x)$, which is

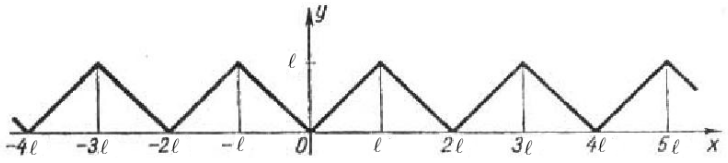
$$\frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi x}{\ell}\right) + b_n \sin\left(\frac{n\pi x}{\ell}\right)\right].$$

In general, such a Fourier series need neither be a sine series nor a cosine series. Sometimes, this series is called a **full range** Fourier series.

(3.19) *Example*

Construct the Fourier series for $f(x) = |x|$ for $x \in [-\ell, \ell)$ for a given $\ell > 0$.

We extend the given function to $f : \mathbb{R} \to \mathbb{R}$ with period 2ℓ. It is shown in the following figure:



Notice that the function $f : \mathbb{R} \to \mathbb{R}$ is not $|x|$; it is $|x|$ on $[-\ell, \ell)$. Due to its period as 2ℓ, it is $|x - 2\ell|$ on $[\ell, 3\ell)$ etc. However, it is an even function; so its Fourier series is a cosine function. The Fourier coefficients are

$$b_n = 0, \quad a_0 = \frac{2}{\ell}\int_0^{\ell} |t|\,dt = \frac{2}{\ell}\int_0^{\ell} t\,dt = \ell,$$

$$a_n = \frac{2}{\ell}\int_0^{\ell} t\cos\Big(\frac{n\pi t}{\ell}\Big)\,dt = \begin{cases} 0 & \text{for } n \text{ even} \\ -\dfrac{4\ell}{n^2\pi^2} & \text{for } n \text{ odd} \end{cases}$$

Therefore the (full range) Fourier series for $f(x)$ shows that in $[-\ell, \ell]$,

$$|x| = \frac{\ell}{2} - \frac{4\ell}{\pi^2}\left[\frac{\cos(\pi x/\ell)}{1^2} + \frac{\cos(3\pi x/\ell)}{3^2} + \cdots + \frac{\cos((2n+1)\pi x/\ell)}{(2n+1)^2} + \cdots\right].$$

Why does the equality holds for $x = \pm\ell$? □

3.7 Functions defined on $(0, \ell)$

Suppose that $f : (0, \ell) \to \mathbb{R}$ is given for which we require a Fourier series. As in the case of functions defined on $(0, \pi)$, we have essentially two approaches to the problem. One, by an even extension, and the other, by an odd extension.

1. Even extension: In this approach, we extend $f(x)$ to $(-\ell, \ell)$ so that the extended function is an even function. Then, we find its Fourier series following the idea of the last section. That is, we first impose $f(-x) = f(x)$ for $x \in (0, \ell)$ and define $f(0)$ (anyway we like) so that f on $(-\ell, \ell)$ is an even function. usually, to satisfy the continuity conditions, we define $f(0) = \lim_{x\to 0-} f(x)$. Of course, if $f : [0, \ell) \to \mathbb{R}$, then we do not have to define $f(0)$. Next, we extend this $f(x)$ to $\mathbb{R}$ by making it periodic with period 2ℓ. Then, we find the Fourier series for the function $g(y) = f(\ell y/\pi)$, and then substitute $y = \pi x/\ell$ in this Fourier series to obtain the Fourier series for $f(x)$. Since $f(x)$ has been first extended to an even function of period 2ℓ, the function $g(y)$ is even and it has period 2π so that the final Fourier series is a cosine series. In this case, the Fourier series for $g(y)$ is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty}\Big(a_n \cos ny\Big), \quad a_n = \frac{2}{\pi}\int_0^{\pi} f\Big(\frac{\ell s}{\pi}\Big)\cos ns\, ds, \quad n = 0, 1, 2, 3, \ldots.$$

Substituting $t = \dfrac{\ell s}{\pi}$, $ds = \dfrac{\pi}{\ell}dt$, and $y = \dfrac{\pi x}{\ell}$ in the above Fourier series, we obtain the Fourier series for $f(x)$. It is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty}\Big(a_n \cos\frac{n\pi x}{\ell}\Big), \quad a_n = \frac{2}{\ell}\int_0^{\ell} f(t)\cos\Big(\frac{n\pi t}{\ell}\Big)\, dt, \quad n = 0, 1, 2, 3, \ldots.$$

The Fourier series sums to the function $f : \mathbb{R} \to \mathbb{R}$, where $f(x)$ is an even function of period 2ℓ and it agrees with the given function $f(x)$ on $(0, \ell)$. In particular, if $f(x)$ is piecewise continuous on $(0, \ell)$, then at $x = 0$, the Fourier series sums to $\lim_{x\to 0+} f(x)$ and at $x = \ell$, the Fourier series sums to $\lim_{x\to \ell-} f(x)$.

The Fourier series so obtained is called the **half-range cosine series** as earlier.

(3.20) ***Example***

Find the half-range cosine series for $f(x) = 2x - 1$ for $0 < x < 1$ and show that

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}.$$

Here, the half-range cosine series is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{1}\right),$$

where $a_0 = \dfrac{2}{1}\displaystyle\int_0^1 (2t-1)dt = 0$ and

$$a_n = \frac{2}{1}\int_0^1 (2t-1)\cos\left(\frac{n\pi t}{1}\right)dt = \frac{4}{n^2\pi^2}\left[(-1)^n - 1\right] = \begin{cases} 0 & \text{for } n \text{ even} \\ \dfrac{-8}{n^2\pi^2} & \text{for } n \text{ odd.} \end{cases}$$

Notice that $f(x)$ is continuous on $0 < x < 1$. Hence,

$$2x - 1 = -\frac{8}{\pi^2}\left[\frac{\cos \pi x}{1^2} + \frac{\cos 3\pi x}{3^2} + \frac{\cos 5\pi x}{5^2} + \cdots\right] \quad \text{for } 0 < x < 1.$$

At $x = 0$, the Fourier series sums to $\lim\limits_{x\to 0+}(2x-1) = -1$. Hence,

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}.$$

□

2. Odd extension: In this approach, we extend $f(x)$ to $(-\ell, \ell)$ so that the extended function is an odd function. Then, we find its Fourier series following the idea of the last section. Notice that our goal is to extend $f(x)$ to $\mathbb{R}$ which will be 2ℓ-periodic and an odd function. Thus, $f(-\ell) = f(\ell)$ and $f(-\ell) = -f(\ell)$. It implies $f(-\ell) = f(\ell) = 0$. Similarly, $f(0) = 0$. So, we take $f(-\ell) = 0 = f(0)$ and impose $f(-x) = -f(x)$ for $x \in (0, \ell)$ so that f on $[-\ell, \ell)$ is an odd function. Next, we extend this $f(x)$ to $\mathbb{R}$ by making it periodic with period 2ℓ. Then, we find the Fourier series for the function $g(y) = f(\ell y/\pi)$. Finally, we substitute $y = \pi x/\ell$ in this Fourier series to obtain the Fourier series for $f(x)$. Since $f(x)$ has been first extended to an odd function of period 2ℓ, the function $g(y)$ is odd and it has period 2π so that its Fourier series is a sine series. The Fourier series for $g(y)$ is

$$\sum_{n=1}^{\infty}\Big(b_n \sin ny\Big), \quad b_n = \frac{2}{\pi}\int_0^{\pi} f\Big(\frac{\ell}{\pi}s\Big)\sin ns\, ds, \quad n = 1, 2, 3, \ldots.$$

Substituting $t = \dfrac{\ell}{\pi}s$, $ds = \dfrac{\pi}{\ell}dt$, and $y = \dfrac{\pi x}{\ell}$ in the Fourier series, we obtain the Fourier series for $f(x)$. It is

$$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{\ell}\right), \quad b_n = \frac{2}{\ell}\int_0^{\ell} f(t)\sin\left(\frac{n\pi t}{\ell}\right)dt, \quad n = 1, 2, 3, \ldots.$$

The Fourier series sums to the function $f : \mathbb{R} \to \mathbb{R}$, where $f(x)$ is an odd function of period 2ℓ and it agrees with the given function $f(x)$ on $(0, \ell)$. In particular, at $x = 0$ and at $x = \pm\ell$, the Fourier series sums to 0.

The series so obtained is called a **half-range sine series** as earlier.

(3.21) ***Example***

Find the half-range sine series for $f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } 1 \le x < 2. \end{cases}$

Here, $\ell = 2$. Thus, the Fourier sine series is $\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$, where

$$b_n = \frac{2}{2}\int_0^2 f(t)\sin\left(\frac{n\pi t}{2}\right)dt = \int_0^1 \sin\left(\frac{n\pi t}{2}\right)dt = \frac{2}{n\pi}\left[1 - \cos\left(\frac{n\pi}{2}\right)\right]. \qquad \square$$

3.8 Functions defined on (a, b)

We now consider determining Fourier series for functions $f : (a, b) \to \mathbb{R}$. In applications, we do not usually encounter such a case. We go ahead for theoretical interest. Since we know how to construct Fourier series for a function with domain as $(0, \pi)$ or as $(-\pi, \pi)$, we have the following three approaches to the problem.

1. In the *first approach*, we define the continuous bijection $g : (0, \pi) \to (a, b)$ given by

$$g(y) = a + \frac{b-a}{\pi}\, y \quad \text{for } y \in (0, \pi).$$

Now, the composition $f \circ g$ is a function from $(0, \pi)$ to $\mathbb{R}$. Next, we take an even extension of $f \circ g$ with periodicity 2π; and call this extended function as h. We then construct the Fourier series for $h(y) = f(g(y))$. Finally, we substitute

$$y = g^{-1}(x) = \frac{\pi(x-a)}{b-a} \quad \text{for } x \in (a, b)$$

in the Fourier series for $h(y)$. This gives the **half-range cosine series** for $f(x)$ on (a, b).

2. In the *second approach*, we use the same continuous bijection $g : (0, \pi) \to (a, b)$ given by

$$g(y) = a + \frac{b-a}{\pi}\, y \quad \text{for } y \in (0, \pi).$$

The composition map $f \circ g$ is a function from $(0, \pi)$ to $\mathbb{R}$. Next, we take an odd extension of $f \circ g$ with periodicity 2π; and call this extended function as h. We then construct the Fourier series for $h(y)$. Finally, we substitute

$$y = g^{-1}(x) = \frac{\pi(x-a)}{b-a} \quad \text{for } x \in (a, b)$$

in the Fourier series for $h(y)$ to get the **half-range sine series** for $f(x)$ on (a, b).

3. In the *third approach*, we define the continuous bijection $g : (-\pi, \pi) \to (a, b)$ given by

$$g(y) = \frac{a+b}{2} + \frac{b-a}{2\pi} y \quad \text{for } y \in (-\pi, \pi).$$

The composition $f \circ g$ is a function from $(-\pi, \pi)$ to $\mathbb{R}$. Next, we extend $f \circ g$ with periodicity 2π; and call this extended function as h. We then construct the Fourier series for $h(y)$. Finally, in the Fourier series for $h(y)$, we substitute

$$y = g^{-1}(x) = \frac{\pi}{b-a}(2x - a - b)$$

to obtain the Fourier series for $f(x)$ on (a, b).

In general, such a Fourier series need neither be a sine series nor a cosine series. Sometimes, this series is called a *full range* Fourier series.

Observe that if a function f has domain $(0, \pi)$, then we can scale it to length 2π by using the third approach. That is, we take the function $g : (-\pi, \pi) \to (0, \pi)$ as

$$g(y) = \frac{\pi + y}{2} \quad \text{for } y \in (0, \pi).$$

Then, we find the Fourier series for $h(y) = f(g(y))$. Finally, in the Fourier series for $h(y)$, we substitute $y = g^{-1}(x) = 2x - \pi$ to obtain the Fourier series for $f(x)$. In general, such a Fourier series may involve both sine and cosine terms.

(3.22) *Example*

We consider Example 3.17 once again to illustrate the third approach when the function is initially defined on $[0, \pi]$. There, we had

$$f(x) = \begin{cases} x & \text{if } 0 \le x \le \pi/2 \\ \pi - x & \text{if } \pi/2 < x \le \pi. \end{cases}$$

Here, $f(x)$ is also given at the end-points. We use the same formula for $g(x)$ at the end-points also. That is, we define $g : [-\pi, \pi] \to [0, \pi]$ by

$$x = g(y) = \frac{\pi + y}{2}, \quad h(y) = f\Big(\frac{\pi + y}{2}\Big) = \begin{cases} (\pi + y)/2 & \text{if } -\pi \le y \le 0 \\ (\pi - y)/2 & \text{if } 0 < y \le \pi. \end{cases}$$

Notice that the function $h(y)$ happens to be an even function with $h(-\pi) = h(\pi) = 0$. Then the Fourier coefficients are given by $b_n = 0$ and

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{0} \frac{\pi + t}{2}\, dt + \frac{1}{\pi}\int_{0}^{\pi} \frac{\pi - t}{2}\, dt = \frac{\pi}{2}.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{0} \frac{\pi + t}{2}\cos nt\, dt + \frac{1}{\pi}\int_{0}^{\pi} \frac{\pi - t}{2}\cos nt\, dt = \begin{cases} \frac{2}{\pi n^2} & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even.} \end{cases}$$

The Fourier series for $h(y)$ is given by

$$\frac{\pi}{4} + \sum_{n \text{ odd}} \frac{2}{\pi n^2}\cos ny.$$

Using $y = g^{-1}(x) = 2x - \pi$, we have the Fourier series for $f(x)$. Also $f(x)$ is continuous at $x = 0$ and $x = \pi$. Therefore,

$$f(x) = \frac{\pi}{4} + \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{1}{(2n+1)^2}\cos\big((2n+1)(2x - \pi)\big) \quad \text{for } x \in [0, \pi].$$

Notice that it is the same series we obtained earlier in Example 3.17(1) by even extension. This is so because $h(y)$ happens to be an even function. □

Remark 3.23 Notice that if a function $f(x)$ is defined on $(-\ell, \ell)$, we have considered scaling this interval to $(-\pi, \pi)$ and obtain the Fourier series; and again scale the series back to the interval $(-\ell, \ell)$. This is the full range expansion. The methods of this section tells that we can also scale $(-\ell, \ell)$ to $(0, \pi)$ and have half range expansions. Similarly, the interval $(0, \ell)$ could be scaled to $(-\pi, \pi)$ to have full range expansions. However, these methods are not followed in practice.

3.9 A Fun Problem

Show that the nth partial sum of the Fourier series for $f(x)$ can be written as the following integral:

$$s_n(x) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x + t)\frac{\sin(2n+1)t/2}{2\sin t/2}\, dt.$$

We know that $s_n(x) = \dfrac{a_0}{2} + \displaystyle\sum_{k=1}^{n}(a_k \cos kx + b_k \sin kx)$, where

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos kt\, dt, \quad b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin kt\, dt.$$

Substituting these values in the expression for $s_n(x)$, we have

$$\begin{aligned} s_n(x) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\,dt + \frac{1}{\pi}\sum_{k=1}^{n}\Big[\int_{-\pi}^{\pi} f(t)\cos kx\,\cos kt\,dt + \int_{-\pi}^{\pi} f(t)\sin kx\,\sin kt\,dt\Big] \\ &= \frac{1}{\pi}\int_{-\pi}^{\pi}\Big[\frac{f(t)}{2} + \sum_{k=1}^{n}\{f(t)\cos kx\,\cos kt + f(t)\sin kx\,\sin kt\}\Big]\,dt \\ &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\Big[\frac{1}{2} + \sum_{k=1}^{n}\cos k(t-x)\Big]\,dt := \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sigma_n(t-x)\,dt. \end{aligned}$$

The expression $\sigma_n(z)$ for $z = t - x$ can be re-written as follows:

$$\sigma_n(z) = \frac{1}{2} + \cos z + \cos 2z + \cdots + \cos nz.$$

Thus

$$\begin{aligned} &2\sigma_n(z)\cos z \\ &= \cos z + 2\cos z\cos z + 2\cos z\cos 2z + \cdots + 2\cos z\cos nz \\ &= \cos z + [1+\cos 2z] + [\cos z + \cos 3z] + \cdots + [\cos(n-1)z + \cos(n+1)z] \\ &= 1 + 2\cos z + 2\cos 2z + \cdots + 2\cos(n-1)z + 2\cos nz + 2\cos(n+1)z \\ &= 2\sigma_n(z) - \cos nz + \cos(n+1)z. \end{aligned}$$

This gives

$$\sigma_n(z) = \frac{\cos nz - \cos(n+1)z}{2(1-\cos z)} = \frac{\sin(2n+1)z/2}{2\sin z/2}.$$

Substituting $\sigma_n(z)$ with $z = t - x$, we have

$$s_n(x) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\frac{\sin(2n+1)(t-x)/2}{2\sin(t-x)/2}\,dt.$$

Since the integrand is 2π-periodic, the value of the integral remains same on any interval of length 2π. Thus

$$s_n(x) = \frac{1}{\pi}\int_{x-\pi}^{x+\pi} f(t)\frac{\sin(2n+1)(t-x)/2}{2\sin(t-x)/2}\,dt.$$

Introduce a new variable $y = t - x$, i.e., $t = x + y$. And then write the integral in terms of t instead of y to obtain

$$s_n(x) = \int_{-\pi}^{\pi} f(x+y)\frac{\sin(2n+1)y/2}{2\sin y/2}\,dy = \int_{-\pi}^{\pi} f(x+t)\frac{\sin(2n+1)t/2}{2\sin t/2}\,dt.$$

This integral is called the **Dirichlet Integral**. In particular, taking $f(x) = 1$, we see that $a_0 = 2$, $a_k = 0$ and $b_k = 0$ for $k \in \mathbb{N}$; and then we get the identity

$$\frac{1}{\pi}\int_{-\pi}^{\pi}\frac{\sin\big((2n+1)t/2\big)}{2\sin\big(t/2\big)}\,dt = 1 \quad \text{for each } n \in \mathbb{N}.$$

3.10 Exercises for Chapter 3

1. Give examples for the following:
   (a) $\sum a_n$ and $\sum b_n$ both diverge, but $\sum(a_n+b_n)$ converges to a nonzero number.
   (b) $\sum a_n$ and $\sum b_n$ both diverge, and $\sum(a_n + b_n)$ diverges to ∞.
   (c) $\sum a_n$ and $\sum b_n$ both diverge, and $\sum(a_n + b_n)$ diverges to $-\infty$.
2. Determine Maclaurin series expansion of (a) $\ln(1+x)$ (b) $\dfrac{\ln(1+x)}{1-x}$.
3. The function $\dfrac{1}{1-x}$ has Taylor series expansion $\displaystyle\sum_{n=0}^{\infty} x^n$ for $-1 < x < 1$. Prove that the function has Taylor series expansion at any $a \neq 1$.
4. Assuming that the alternating harmonic series $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{n+1}$ is convergent find its sum.
5. Give an approximation scheme for $\displaystyle\int_0^a \frac{\sin x}{x}\,dx$ where $a > 0$.
6. Show that $\sin^{-1} x = x + \dfrac{1}{2}\cdot\dfrac{1}{3}x^3 + \dfrac{1\cdot 3}{2\cdot 4}\cdot\dfrac{1}{5}x^5 + \dfrac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\cdot\dfrac{1}{7}x^7 + \cdots$
   for $-1 < x < 1$. Assuming that the series converges at $x = 1$, find its sum.
7. Find the Fourier series of $f(x) = \begin{cases} 0 & \text{if } -\pi \le x < 0 \\ 1 & \text{if } 0 \le x < \pi. \end{cases}$
   Then, give a series that sums to $\pi/4$.
8. Considering the Fourier series for $|x|$, deduce that $\dfrac{\pi^2}{8} = \displaystyle\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$.
9. Considering the Fourier series for x, deduce that $\dfrac{\pi}{4} = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$.
10. Considering the Fourier series for $f(x) = \begin{cases} -1 & \text{if } -\pi \le x < 0 \\ 1 & \text{if } 0 \le x \le \pi \end{cases}$
    deduce that $\dfrac{\pi}{4} = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$.
11. Find the cosine series of the function $f(x) = 1 - |x|$ for $-1 \le x \le 1$.
12. Find the half range Fourier cosine series expansion of $f(x) = x^2$ for $0 \le x < 1$.
13. Find the half range Fourier sine series expansion of $f(x) = x$ for $0 \le x \le 4$.

14. Considering $f(x) = x^2$, show that for each $x \in [0, \pi)$,

$$\frac{\pi^2}{6} + 2\sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n^2} = \sum_{n=1}^{\infty} \frac{n^2\pi^2(-1)^{n+1} + 2(-1)^n - 2}{n^3\pi} \sin nx.$$

Answers:
4. $\ln 2$ 5. $a + \sum_{n=1}^{\infty}(-a)^{2n+1}/[(2n+1)^2(2n)!]$
7. $(2/\pi)\sum_{n=0}^{\infty}(2n+1)^{-1}\sin[(2n+1)x]$
11. $1/2 + (4/\pi^2)\sum_{n=0}^{\infty}(2n+1)^{-2}\cos[(2n+1)\pi x]$
12. $x^2 = \frac{1}{3} + \sum_{n=1}^{\infty}\frac{4}{n^2\pi^2}(-1)^n\cos(n\pi x)$. 13. $x = \frac{8}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin(n\pi x/4)$.

A

One Variable Summary

This appendix is devoted to summarizing some results and formulas from calculus of functions of one real variable that we may use in the course.

A.1 Graphs of Functions

The **absolute value** of $x \in \mathbb{R}$ is defined as $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

Thus $|x| = \sqrt{x^2}$. And $|-a| = a$ or $a \geq 0$; $|x - y|$ is the distance between real numbers x and y. Moreover, if $a, b \in \mathbb{R}$, then

$$|-a| = |a|,\ |ab| = |a|\,|b|,\ \left|\frac{a}{b}\right| = \frac{|a|}{|b|} \text{ if } b \neq 0,\ |a+b| \leq |a| + |b|,\ \big||a| - |b|\big| \leq |a-b|.$$

Let $x \in \mathbb{R}$ and let $a > 0$. The following are true:

1. $|x| = a$ iff $x = \pm a$.
2. $|x| < a$ iff $-a < x < a$ iff $x \in (-a, a)$.
3. $|x| \leq a$ iff $-a \leq x \leq a$ iff $x \in [-a, a]$.
4. $|x| > a$ iff $-a < x$ or $x > a$ iff $x \in (-\infty, -a) \cup (a, \infty)$ iff $x \in \mathbb{R} - [-a, a]$.
5. $|x| \geq a$ iff $-a \leq x$ or $x \geq a$ iff $x \in (-\infty, -a] \cup [a, \infty)$ iff $x \in \mathbb{R} - (-a, a)$.

Therefore, for $a \in \mathbb{R}, \delta > 0$, $\quad |x - a| < \delta$ iff $a - \delta < x < a + \delta$.

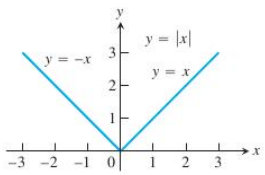
The following statements are useful in proving equalities from inequalities:

Let $a, b \in \mathbb{R}$.

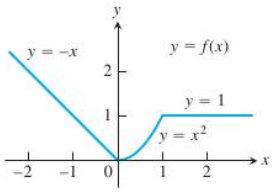
1. If for each $\epsilon > 0,\ |a| < \epsilon$, then $a = 0$.
2. If for each $\epsilon > 0,\ a < b + \epsilon$, then $a \leq b$.

Graphs of some known functions including $|\cdot|$, are as follows:

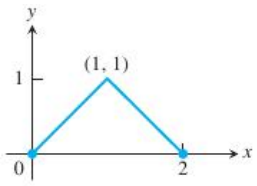
1. $y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$



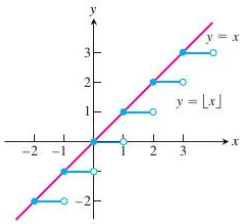
2. $y = \begin{cases} -x & \text{if } x < 0 \\ x^2 & \text{if } 0 \le x \le 1 \\ 1 & \text{if } x > 1 \end{cases}$



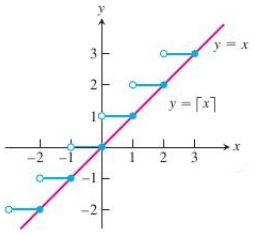
3. $y = f(x) = \begin{cases} x & \text{if } 0 \le x \le 1 \\ 2 - x & \text{if } 1 < x \le 2 \end{cases}$



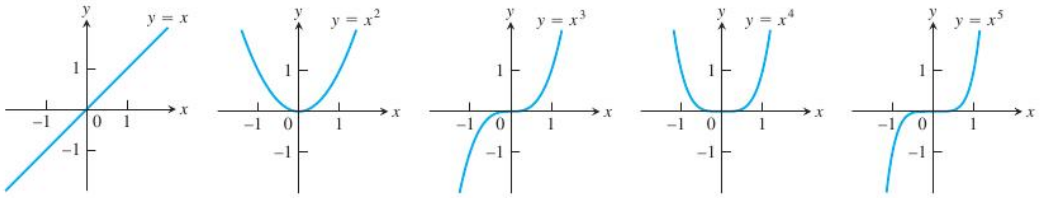
4. $y = \lfloor x \rfloor = n$ if $n \le x < n + 1$ for $n \in \mathbb{N}$. It is the largest integer less than or equal to x.
The largest integer function or the **floor** function. Sometimes we write $\lfloor\ \rfloor$ as $[\ ]$.



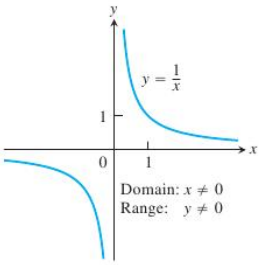
5. $y = \lfloor x \rfloor = n + 1$ if $n < x \le n + 1$ for $n \in \mathbb{N}$. It is the smallest integer greater than or equal to x.
The smallest integer function or the **ceiling** function.

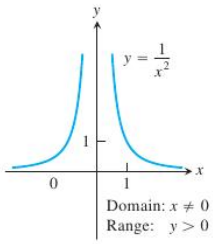


6. The **power function** $y = x^n$ for $n = 1, 2, 3, 4, 5$ look like

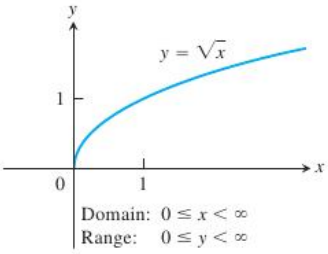


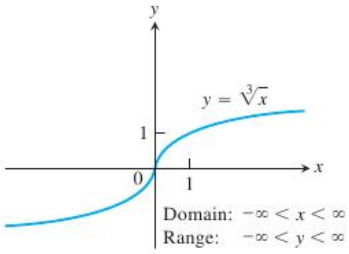
7. The power function $y = x^n$ for $n = -1$ and $n = -2$ look like

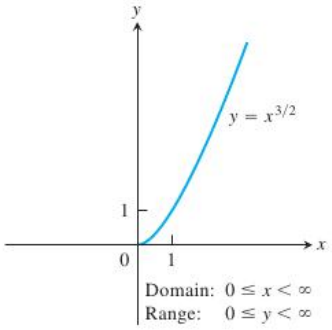


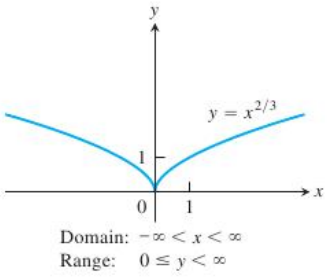


8. The graphs of the power function $y = x^a$ for $a = \frac{1}{2}, \frac{1}{3}, \frac{3}{2}$ and $a = \frac{2}{3}$ are

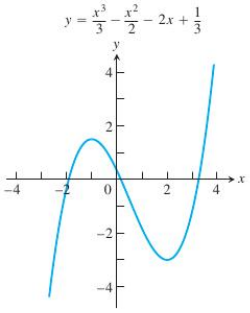


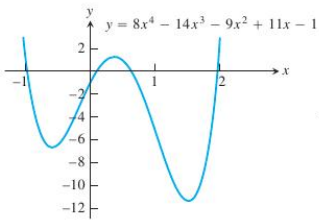


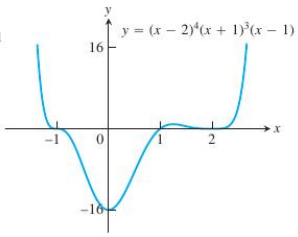




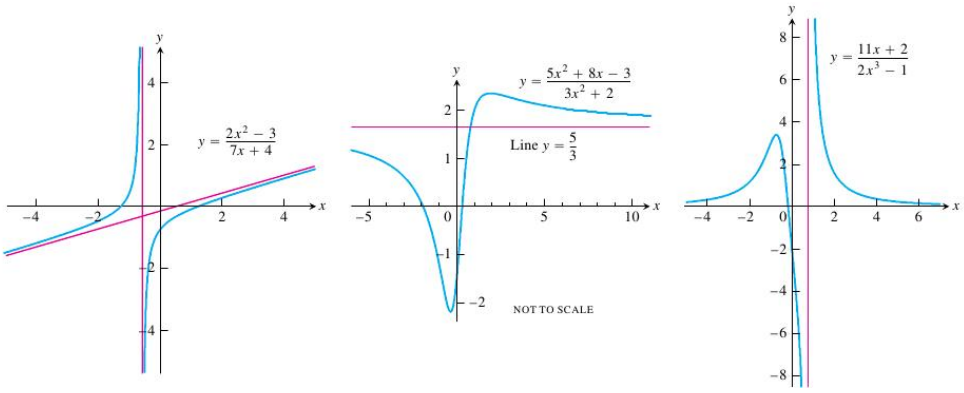
9. **Polynomial** functions are $y = f(x) = a_0 + a_1x + a_2x^2 + \cdots a_nx^n$ for some $n \in \mathbb{N} \cup \{0\}$. Here, the coefficients of powers of x are some given real numbers $a_0, \ldots, a_n$ and $a_n \neq 0$. The highest power n in the polynomial is called the **degree** of the polynomial. Graphs of some polynomial functions are as follows:



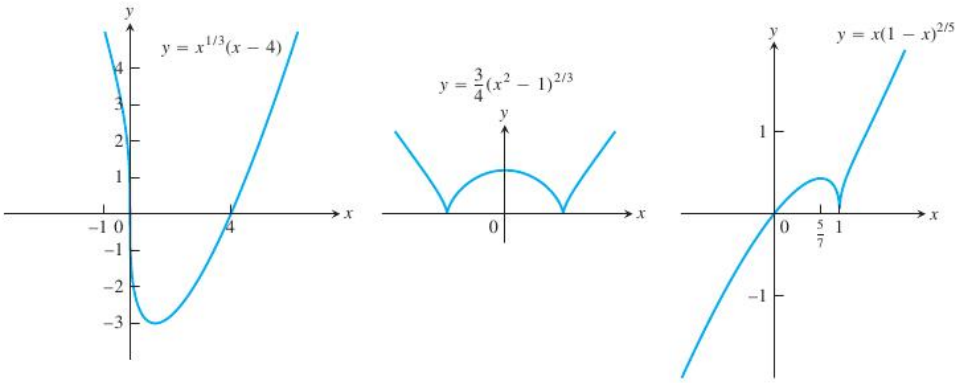




10. A **rational function** is a ratio of two polynomials; $f(x) = \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials, may or may not be of the same degree. Graphs of some rational functions are as follows:



11. **Algebraic functions** are obtained by adding subtracting, multiplying, dividing or taking roots of polynomial functions. Rational functions are special cases of algebraic functions. Some graphs of alhebraic functions:



12. **Trigonometric functions** come from the ratios of sides of a right angled triangle. The angles are measured in radian. The trigonometric functions have a period. That is, $f(x+p) = f(x)$ happens for some $p > 0$. The **period** of $f(x)$ is the minimum of such p. The period for $\sin x$ is 2π.

The functions $\cos x$ and $\sec x$ are **even functions** and all others are **odd functions**. Recall that $f(x)$ is even if $f(-x) = f(x)$ and it is odd if $f(-x) = -f(x)$ for each x in the domain of the function. Some of the useful inequalities are as follows:

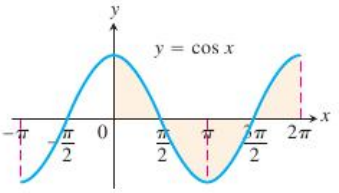
For each $x \in \mathbb{R}$,

$$-|x| \le \sin x \le |x|, \quad -1 \le \sin x,\ \cos x \le 1, \quad 0 \le 1 - \cos x \le |x|.$$

$$\sin x \le x \le \tan x \quad \text{for } x \in (0, \pi/2).$$

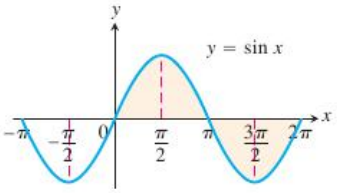
$$\sin x < |x| \quad \text{for } x \ne 0.$$

Graphs of the trigonomaetric functions are as follows:



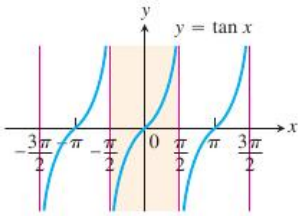
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(a)



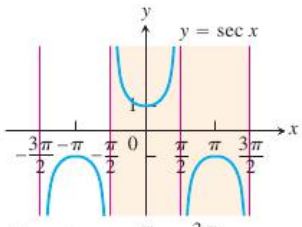
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(b)



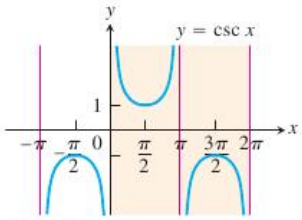
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $-\infty < y < \infty$
Period: π

(c)



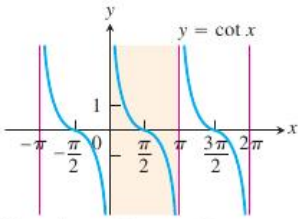
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $y \le -1$ and $y \ge 1$
Period: 2π

(d)



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $y \le -1$ and $y \ge 1$
Period: 2π

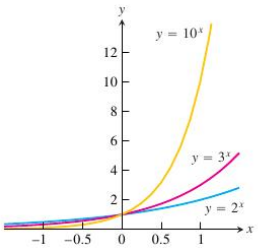
(e)

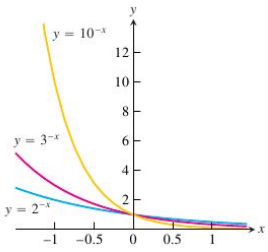


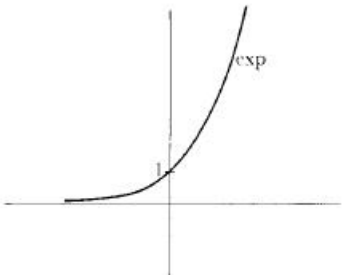
Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $-\infty < y < \infty$
Period: π

(f)

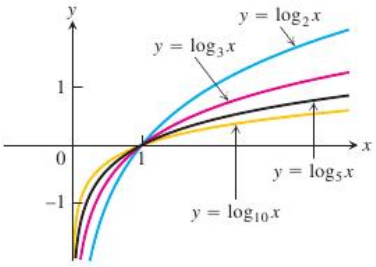
13. **Exponential functions** are in the form $y = a^x$ for some $a > 0$ and $a \neq 1$. All exponential functions have domain $(-\infty, \infty)$ and co-domain $(0, \infty)$. They never assume the value 0. Graphs of some exponential functions:

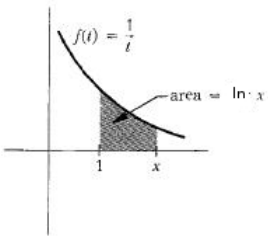


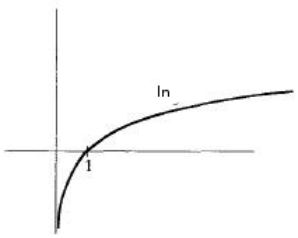




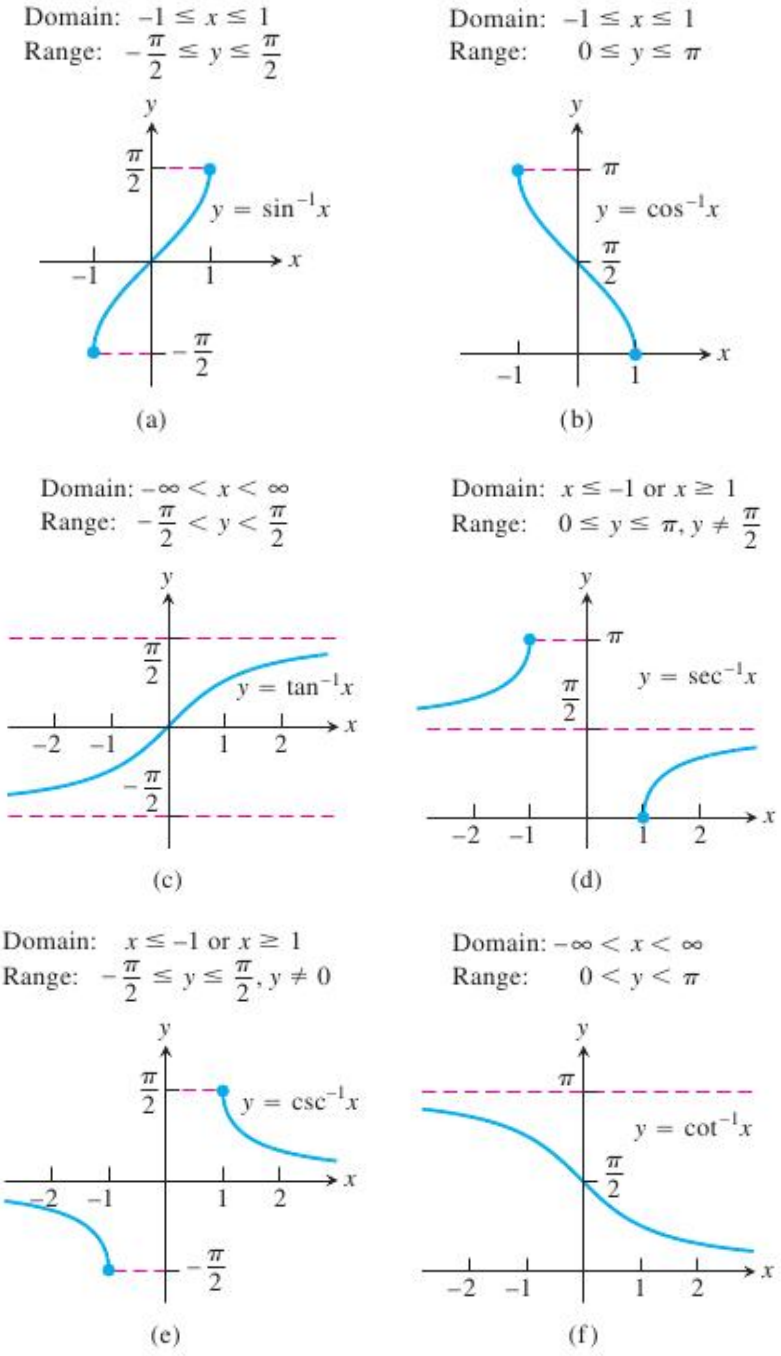
14. **Logarithmic functions** are inverse of exponential functions. That is, $a^{\log_a x} = \log_a(a^x) = x$. Some examples:







15. **Inverse trigonometric functions**:



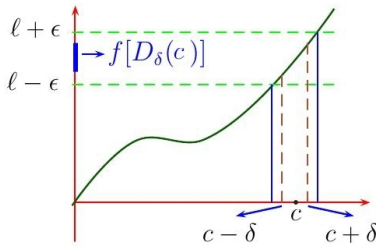
Functions that are not algebraic are called transcendental functions. Trigonometric functions, exponential functions, logarithmic functions and inverse trigonometric functions are examples of transcendental functions.

A.2 Concepts and Facts

Let $a < c < b$. Let $f : D \to R$ be a function whose domain D contains the union $(a, c) \cup (c, b)$. Let $\ell \in \mathbb{R}$. We say that **the limit of** $f(x)$ **as** x **approaches** c **is** ℓ and write it as

$$\lim_{x \to c} f(x) = \ell$$

iff for each $\epsilon > 0$, there exists a $\delta > 0$ such that for each $x \in (a, c) \cup (c, b)$ with $0 < |x - c| < \delta$, we have $|f(x) - \ell| < \epsilon$.



Limit Properties: Let k be a constant; or a constant function.

1. $\lim_{x \to c} k = k$ and $\lim_{x \to c} x = c$.
2. $\lim_{x \to c}(f(x) \pm g(x)) = \lim_{x \to c} f(x) \pm \lim_{x \to c} g(x)$.
3. $\lim_{x \to c} kf(x) = k \lim_{x \to c} f(x)$.
4. $\lim_{x \to c}[f(x)g(x)] = \lim_{x \to c} f(x) \lim_{x \to c} g(x)$.
5. $\lim_{x \to c}[f(x)/g(x)] = \left[\lim_{x \to c} f(x)\right] / \left[\lim_{x \to c} g(x)\right]$ if $\lim_{x \to c} g(x) \neq 0$.
6. $\lim_{x \to c}(f(x))^r = (\lim_{x \to c} f(x))^r$ if taking powers are meaningful.
7. $\lim_{x \to c} f(x)$ is a unique real number if it exists.
8. If $\lim_{x \to c} g(x) = 0$, and $\lim_{x \to c}[f(x)/g(x)]$ exists, then $\lim_{x \to c} f(x) = 0$.
9. (**Sandwich**) Let f, g, h be functions whose domain include $(a, c) \cup (c, b)$ for $a < c < b$. Suppose that $g(x) \leq f(x) \leq h(x)$ for all $x \in (a, c) \cup (c, b)$. If $\lim_{x \to c} g(x) = \ell = \lim_{x \to c} h(x)$, then $\lim_{x \to c} f(x) = \ell$.
10. (**Domination**) Let f, g be functions whose domains include $(a, c) \cup (c, b)$ for $a < c < b$. Suppose that both $\lim_{x \to c} f(x)$ and $\lim_{x \to c} g(x)$ exist. If $f(x) \leq g(x)$ for all $x \in (a, c) \cup (c, b)$, then $\lim_{x \to c} f(x) \leq \lim_{x \to c} g(x)$.

Let I be (a, ∞) or $[a, \infty)$ for some $a \in \mathbb{R}$. Let $f : I \to \mathbb{R}$. Let $\ell \in \mathbb{R}$. We say that $\lim_{x \to \infty} f(x) = \ell$ if for each $\epsilon > 0$, there exists an $m > 0$ such that if x is any real number greater then m, then $|f(x) - \ell| < \epsilon$.

Let $f(x)$ have a domain containing (a, c). Then $\lim_{x\to c-} f(x) = \infty$ iff for each $m > 0$, there exists a $\delta > 0$ such that for every x with $c - \delta < x < c$, we have $f(x) > m$.

That is, $\lim_{x\to c-} f(x) = \infty$ iff, "as x increases to c, $f(x)$ increases without bound".

Let $f : D \to \mathbb{R}$ be a function. Let c be an interior point of D. We say that $f(x)$ is **continuous at** c if $\lim_{x\to c} f(x) = f(a)$.

If $D = [a, b)$ or $D = [a, b]$, then $f(x)$ is called continuous at the left end-point a if $\lim_{x\to a+} f(x) = f(a)$.

If $D = (a, b]$ or $D = [a, b]$, then $f(x)$ is called continuous at the right end-point b if $\lim_{x\to b-} f(x) = f(a)$.

$f(x)$ is called **continuous** if it is continuous at each point of its domain D.

The sum, multiplication by a constant, and product of continuous functions is continuous. In addition, the following are some properties of continuous functions:

1. Let $f(x)$ be continuous at $x = c$, where the domain of $f(x)$ includes a neighborhood of c. If $f(c) > 0$, then there exists a neighborhood $(c - \delta, c + \delta)$ such that $f(x) > 0$ for each point $x \in (c - \delta, c + \delta)$.
2. Let $f(x)$ be a continuous function, whose domain contains $[a, b]$ for $a < b$. Then there exist $\alpha,\ \beta \in \mathbb{R}$ such that $\{f(x) : x \in [a, b]\} = [\alpha, \beta]$.
3. **(Extreme Value Theorem)** Let $f(x)$ be continuous on a closed bounded interval $[a, b]$. Then there exist numbers $c, d \in [a, b]$ such that $f(c) \leq f(x) \leq f(d)$ for each $x \in [a, b]$.
4. **(Intermediate Value Theorem)** Let $f(x)$ be continuous on a closed bounded interval $[a, b]$. Let d be a number between $f(a)$ and $f(b)$. Then there exists $c \in [a, b]$ such that $f(c) = d$.

Let $f(x)$ be a function whose domain includes an open interval (a, b). Let $c \in (a, b)$. If the limit

$$\lim_{h\to 0} \frac{f(c + h) - f(c)}{h}$$

exists, we say that $f(x)$ is **differentiable at** $x = c$; and we write the limit as $f'(c)$ and call it the derivative of $f(x)$ at $x = c$. If $f'(c)$ exists for each $c \in (a, b)$, then we write $f'(x)$ as $\frac{df}{dx}$.

Also, derivative of f defined on a closed interval $[a, b]$ at the end-point a is taken as the left hand derivative, where in the defining limit of the derivative we take $h \to a-$. Similarly, derivative at b is taken as the limit of that ratio for $h \to 0+$.

Let $f(x)$ be a function defined on an interval I.

We say that $f(x)$ is **increasing on** I if for all $s < t \in I,\ f(s) < f(t)$.

Similarly, we say that $f(x)$ is **decreasing on** I if for all $s < t \in I,\ f(s) > f(t)$.

A **monotonic** function on I is one which either increases on I or decreases on I.

The sum, multiplication by a constant, and product of differentiable functions is differentiable. In addition, the following are some properties of differentiable functions:

1. Each function differentiable at $x = c$ is continuous at $x = c$.
2. Derivatives of Sum, product etc. are respectives equal to sum, product etc of derivatives.
3. **(Chain Rule)** $\frac{dg(f(x))}{dx} = \frac{dg(f(x))}{df(x)} \cdot \frac{df(x)}{dx}$.
4. **(Rolle's Theorem)** Suppose that $f : [a, b] \to \mathbb{R}$ is continuous, $f(x)$ is differentiable on (a, b), and $f(a) = f(b)$. Then $f'(c) = 0$ for some $c \in (a, b)$.
5. **(Mean value Theorem)** Suppose that $f : [a, b] \to \mathbb{R}$ is continuous and $f(x)$ is differentiable on (a, b). Then there exists $c \in (a, b)$ such that $f(b) - f(a) = f'(c)(b - a)$.
6. Let I be an interval containing at least two points. Let $f : I \to \mathbb{R}$ be differentiable. If $f'(x) = 0$ for each $x \in I$, iff $f(x)$ is a constant function.
7. **(Cauchy Mean Value Theorem)** Let $f(x)$ and $g(x)$ be continuous on $[a, b]$ and differentiable on (a, b). If $g'(x) \neq 0$ on (a, b), then there exists $c \in (a, b)$ such that $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$.
8. **(L'Hospital's Rule)** Let $f(x)$ and $g(x)$ be differentiable on a neighborhood of a point $x = a$. Suppose $f(a) = g(a) = 0$ but $g(x) \neq 0$, $g'(x) \neq 0$ in the deleted neighborhood of $x = a$. If $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists, then $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$.
9. Let $f(x)$ be continuous on $[a, b]$ and differentiable on (a, b).
   (a) If $f'(x) > 0$ on (a, b), then $f(x)$ is increasing on $[a, b]$.
   (b) If $f'(x) < 0$ on (a, b), then $f(x)$ is decreasing on $[a, b]$.

Let a function $f(x)$ have domain D. The function $f(x)$ has a **local maximum** at a point $d \in D$ if $f(x) \leq f(d)$ for every x in some neighborhood of d contained in D. in such a case, we also say that the point $x = d$ is a **point of local maximum** of the function $f(x)$.

Similarly, $f(x)$ has an **local minimum** at $b \in D$ if $f(b) \leq f(x)$ for every x in some neighborhood of b contained in D. In this case, we say that the point $x = b$ is a **point of local minimum** of the function $f(x)$.

The points of local maximum and local minimum are commonly referred to as **local extremum points**; and the function is said to have **local extrema** at those points.

Let $f(x)$ have domain D. A point $c \in D$ is called a **critical point** of $f(x)$ if c is not an interior point of D, or if $f(x)$ is not differentiable at $x = c$, or if $f'(c) = 0$.

If $f(x)$ has an extremum at $x = c$, then c is a critical point of $f(x)$.

Test for Local Extrema:
Let c be an interior point of the domain of $f(x)$ with $f'(c) = 0$.

1. $f'(x)$ changes sign from + to − at $x = c$ iff $x = c$ is a point of local maximum of $f(x)$.
2. If $f''(c) < 0$, then $x = c$ is a point of local maximum of $f(x)$.
3. $f'(x)$ changes sign from − to + at $x = c$ iff $x = c$ is a point of local minimum of $f(x)$.
4. If $f''(c) > 0$, then $x = c$ is a point of local minimum of $f(x)$.

Let $x = c$ be a left end-point of the domain of $f(x)$.

1. $f'(x) < 0$ on the immediate right of $x = c$ iff $x = c$ is a point of local maximum of $f(x)$.
2. $f'(x) > 0$ on the immediate right of $x = c$ iff $x = c$ is a point of local minimum of $f(x)$.

Let $x = c$ be a right end-point of the domain of $f(x)$.

1. $f'(x) > 0$ on the immediate left of $x = c$ iff $x = c$ is a point of local maximum of $f(x)$.
2. $f'(x) < 0$ on the left of $x = c$ iff $x = c$ is a point of local minimum of $f(x)$.

The graph of a function $y = f(x)$ is **concave up** on an open interval I if $f'(x)$ is increasing on I. The graph of $y = f(x)$ is **concave down** on an open interval I if $f'(x)$ is decreasing on I.
A **point of inflection** of $y = f(x)$ is a point where the concavity of $y = f(x)$ changes.

Second derivative test for concavity:
Let $y = f(x)$ be twice differentiable on an interval I.

1. If $f''(x) > 0$ on I, then the graph of $y = f(x)$ is concave up over I.
2. If $f''(x) < 0$ on I, then the graph of $y = f(x)$ is concave down over I.
3. If $f''(x)$ is positive on one side of $x = c$ and negative on the other side, then the point $(c, f(c))$ on the graph of $y = f(x)$ is a point of inflection.

Let $f : [a, b] \to \mathbb{R}$. Divide $[a, b]$ into smaller sub-intervals by choosing the break points as

$$a = x_0 < x_1 < \ldots < x_n = b.$$

The set $P = \{x_0, x_1, \ldots, x_n\}$ is called a **partition of** $[a, b]$.

Now P divides $[a, b]$ into n sub-intervals: $[x_0, x_1], \cdots, [x_{n-1}, x_n]$. Here, the kth sub-interval is $[x_{k-1}, x_k]$. The area under the curve $y = f(x)$ raised over the kth

sub-interval is approximated by $f(c_k)(x_k - x_{k-1})$ for some choice of the point $c_k \in [x_{k-1}, x_k]$.

Write the choice points (also called **sample points**) as a set $C = \{c_1, \ldots, c_n\}$.

Then the **Riemann sum**

$$S(f, P, C) = \sum_{k=1}^{n} f(c_k)(x_k - x_{k-1})$$

is an approximation to the whole area raised over $[a, b]$ and lying between the curve $y = f(x)$ and the x-axis. By taking the **norm** of the partition as $\|P\| = \max_k(x_k - x_{k-1})$, we would say that when the norm of the partition approaches 0, the Riemann sum would approach the required area. Thus, we define the area of the region bounded by the lines $x = a,\ x = b,\ y = 0$, and $y = f(x)$ as

$$\lim_{\|P\| \to 0} \sum_{k=1}^{n} f(c_k)(x_k - x_{k-1})$$

provided that this limit exists. We define this limit (which is the mentioned area here) as the definite integral of f on the interval $[a, b]$. That is,

$$\int_a^b f(x)\, dx = \lim_{\|P\| \to 0} \sum_{k=1}^{n} f(c_k)(x_k - x_{k-1}).$$

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then $\int_a^b f(x)\, dx$ exists.

The definite integral has the following properties:

(Properties of Definite Integral)

1. Let $f(x)$ have domain $[a, b]$. Let $c \in (a, b)$. Then $f(x)$ is integrable on $[a, b]$ iff $f(x)$ is integrable on both $[a, c]$ and $[c, b]$. In this case,

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

2. Let $f(x)$ and $g(x)$ be integrable on $[a, b]$. Then $(f + g)(x)$ is integrable on $[a, b]$ and

$$\int_a^b (f + g)(x)dx = \int_a^b (f(x) + g(x))\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx.$$

3. Let $f(x)$ be integrable on $[a, b]$. Let $c \in \mathbb{R}$. Then $(cf)(x)$ is integrable on $[a.b]$ and

$$\int_a^b (cf)(x)\, dx = \int_a^b cf(x)\, dx = c \int_a^b f(x)\, dx.$$

4. Let $f(x)$ and $g(x)$ be integrable on $[a, b]$. If for each $x \in [a, b]$, $f(x) \leq g(x)$, then
$$\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx.$$

5. Let $f(x)$ be integrable on $[a, b]$. If $m \leq f(x) \leq M$ for all $x \in [a, b]$, then
$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a).$$

6. (*Average Value Theorem*) Let $f(x)$ be continuous on $[a, b]$. Then there exists $c \in [a, b]$ such that
$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

7. Let $f(x)$ be continuous on $[a, b]$. If $f(x)$ has the same sign on $[a, b]$ and $\int_a^b f(x)\,dx = 0$, then $f(x)$ is the zero function, i.e., $f(x) = 0$ for each $x \in [a, b]$.

We extend the integral even when $a \not< b$ by the following:

1. If $a = b$, then we take $\int_a^b f(x)\,dx = 0$.
2. If $a > b$, then we take $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$.

Also, for any real number c; even when c is outside the interval (a, b) we have
$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

In all these extensions, we assume that the definite integrals exist.

The main result that shows that differentiation and integration are reverse processes is the following:

(Fundamental Theorem of Calculus) Let $f(x)$ be continuous on $[a, b]$.

1. If $F(x)$ is an antiderivative of $f(x)$, then $\int_a^b f(x)\,dx = F(b) - F(a)$.
2. The function $g(x) = \int_a^x f(t)\,dt$ is continuous on $[a, b]$ and differentiable on (a, b). Moreover, $g'(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x)$.

The chain rule for differentiation is translated to integration as follows. We call it the method of *Substitution*:

1. Let $u = g(x)$ be a differentiable function whose range is an interval I. Let $f(x)$ be continuous on I. Then
$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$
2. Let $u = g(x)$ be a continuously differentiable function on $[a, b]$ whose range is an interval I. Let $f(x)$ be continuous on I. Then
$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

The rpoduct rule for differntiation gives the *integration by parts* formula.

$$\int f(x)h(x)\,dx = f(x)\int h(x)\,dx - \int \left[f'(x)\int h(x)\,dx\right]\,dx + C.$$

We remember it as follows (Read F as first and S as second):

Integral of $F \times S = F\times$ integral of $S -$ integral of (derivative of $F \times$ integral of S).

The **natural logarithm** $\ln x$ is defined as follows:

$$\ln x = \int_1^x \frac{1}{t}\,dt \quad \text{for } x > 0.$$

The **exponential function** is the inverse of the natural logarithm. That is,

$$\exp : R \to (0, \infty); \quad y = \exp(x) \quad \text{iff} \quad x = \ln y.$$

Since $\exp(x)\ \exp(y) = \exp(x + y)$ and $\exp(0) = 1$, we write

$$\exp(x) = e^x, \quad \text{where} \quad e = \exp(1).$$

Then hyperbolic functions are defined by

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1}), \quad \cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}), \quad \tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right).$$

Notice that $\cosh^{-1}$ has domain as $x \geq 1$ and $\tanh^{-1}$ has domain as $-1 < x < 1$.

Let C be a curve given parametrically by $x = f(t), y = g(t),\ a \leq t \leq b$. Assume that both $f(t)$ and $g(t)$ are continuously differentiable.

$$\text{Length of the curve} = L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\,dt.$$

If the curve is given as a function $y = f(x),\ a \le x \le b$, then take $x = t$ and $y = f(t)$ as its parameterization. We then have the length as

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx = \int_a^b \sqrt{1 + (y')^2}\, dx.$$

Notice that this formula is applicable when $f'(x)$ is continuous on $[a, b]$.

We write $L = \int_a^b ds$ with limits a and b for the variable of integration, which may be $x,\ y$ or t. Here,

$$ds = \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy.$$

Suppose that a curve is given in polar coordinates by $r = f(\theta)$ for a continuous function $f(\theta)$, where $\alpha \le \theta \le \beta$. Then the area of the sector and the arc length of the curve are

$$\text{Area } = \int_\alpha^\beta r^2\, d\theta, \quad \text{Length } = \int_\alpha^\beta \sqrt{r^2 + (r')^2}\, d\theta.$$

A.3 Formulas

Here are some formulas for the exponential and the logarithm functions:

$$\lim_{t\to\infty} \frac{t^p}{a^t} = 0 \text{ for } p \in \mathbb{N} \text{ and } a > 1.$$

$$\ln e = 1 = e^0, \quad e^{\ln x} = x, \quad \ln(e^x) = x, \quad a^x = e^{x \ln a},$$

$$\lim_{x\to\infty} \ln x = \infty, \quad \lim_{x\to 0+} \ln x = -\infty, \quad \lim_{x\to\infty} e^x = \infty, \quad \lim_{x\to-\infty} e^x = 0,$$

$$(\ln x)' = \frac{1}{x}, \quad (e^x)' = e^x, \quad (a^x)' = (\ln a)a^x, \quad \int_1^e \frac{1}{t}\, dt = 1, \quad \int e^x\, dx = e^x.$$

$$\lim_{h\to 0} \frac{\ln(1+xh)}{xh} = 1 \text{ for } x \ne 0, \quad \lim_{h\to 0} \frac{e^h - 1}{h} = 1, \quad \lim_{h\to 0} (1+xh)^{1/h} = e^x,$$

Below are given some integrals, which are obtained from the derivatives by following the simple rule that if $g'(x) = f(x)$, then $\int f(x)\, dx = g(x) + C$.

1. $\int u\, dv = uv - \int v\, du$
2. $\int a^u\, du = \dfrac{a^u}{\ln a} + C, \quad a \ne 1,\ a > 0$

3. $\int \cos u \, du = \sin u + C$
4. $\int \sin u \, du = -\cos u + C$
5. $\int (ax+b)^n \, dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C, \quad n \neq 1$
6. $\int (ax+b)^{-1} \, dx = \dfrac{1}{a} \ln |ax+b| + C$
7. $\int x(ax+b)^n \, dx = \dfrac{(ax+b)^{n+1}}{a^2} \left[\dfrac{ax+b}{n+2} - \dfrac{b}{n+1} \right] + C, \quad n \neq -1, \ -2$
8. $\int x(ax+b)^{-1} \, dx = \dfrac{x}{a} - \dfrac{b}{a^2} \ln |ax+b| + C$
9. $\int x(ax+b)^{-2} \, dx = \dfrac{1}{a^2} \left[\ln |ax+b| + \dfrac{b}{ax+b} \right] + C$
10. $\int \dfrac{dx}{x(ax+b)} = \dfrac{1}{b} \ln \left| \dfrac{x}{ax+b} \right| + C$
11. $\int \left(\sqrt{ax+b} \right)^n \, dx = \dfrac{2}{a} \dfrac{\left(\sqrt{ax+b} \right)^{n+2}}{n+2} + C, \quad n \neq -2$
12. $\int \dfrac{\sqrt{ax+b}}{x} \, dx = 2\sqrt{ax+b} + b \int \dfrac{dx}{x\sqrt{ax+b}}$
13. $\int \dfrac{dx}{x\sqrt{ax-b}} = \dfrac{2}{\sqrt{b}} \tan^{-1} \sqrt{\dfrac{ax-b}{b}} + C$
14. $\int \dfrac{dx}{x\sqrt{ax+b}} = \dfrac{1}{\sqrt{b}} \ln \left| \dfrac{\sqrt{ax+b} - \sqrt{b}}{\sqrt{ax+b} + \sqrt{b}} \right| + C$
15. $\int \dfrac{\sqrt{ax+b}}{x^2} \, dx = -\dfrac{\sqrt{ax+b}}{x} + \dfrac{a}{2} \int \dfrac{dx}{x\sqrt{ax+b}} + C$
16. $\int \dfrac{dx}{x^2\sqrt{ax+b}} = -\dfrac{\sqrt{ax+b}}{bx} - \dfrac{b}{2a} \int \dfrac{dx}{x\sqrt{ax+b}} + C$
17. $\int \dfrac{dx}{a^2+x^2} = \dfrac{1}{a} \tan^{-1} \dfrac{x}{a} + C$
18. $\int \dfrac{dx}{(a^2+x^2)^2} = \dfrac{x}{2a^2(a^2+x^2)} + \dfrac{1}{2a^3} \tan^{-1} \dfrac{x}{a} + C$
19. $\int \dfrac{dx}{a^2-x^2} = \dfrac{1}{2a} \ln \left| \dfrac{x+a}{x-a} \right| + C$
20. $\int \dfrac{dx}{(a^2-x^2)^2} = \dfrac{x}{2a^2(a^2-x^2)} + \dfrac{1}{4a^3} \ln \left| \dfrac{x+a}{x-a} \right| + C$
21. $\int \dfrac{dx}{\sqrt{a^2+x^2}} = \sinh^{-1} \dfrac{x}{a} + C = \ln \left(x + \sqrt{a^2+x^2} \right) + C$

22. $\int \sqrt{a^2+x^2}\,dx = \frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2}\ln\left(x+\sqrt{a^2+x^2}\right)+C$

23. $\int x^2\sqrt{a^2+x^2}\,dx = \frac{x}{8}(a^2+2x^2)\sqrt{a^2+x^2} - \frac{a^4}{8}\ln\left(x+\sqrt{a^2+x^2}\right)+C$

24. $\int \frac{\sqrt{a^2+x^2}}{x}\,dx = \sqrt{a^2+x^2} - a\ln\left|\frac{a+\sqrt{a^2+x^2}}{x}\right|+C$

25. $\int \frac{\sqrt{a^2+x^2}}{x^2}\,dx = \ln\left(x+\sqrt{a^2+x^2}\right) - \frac{\sqrt{a^2+x^2}}{x}+C$

26. $\int \frac{x^2}{\sqrt{a^2+x^2}}\,dx = -\frac{a^2}{2}\ln\left(x+\sqrt{a^2+x^2}\right) + \frac{x\sqrt{a^2+x^2}}{2}+C$

27. $\int \frac{dx}{x\sqrt{a^2+x^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{a^2+x^2}}{x}\right|+C$

28. $\int \frac{dx}{x^2\sqrt{a^2+x^2}} = -\frac{\sqrt{a^2+x^2}}{a^2x}+C$

29. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}+C$

30. $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}+C$

31. $\int x^2\sqrt{a^2-x^2}\,dx = \frac{a^4}{8}\sin^{-1}\frac{x}{a} - \frac{x}{8}\sqrt{a^2-x^2}\,(a^2-2x^2)+C$

32. $\int \frac{\sqrt{a^2-x^2}}{x}\,dx = \sqrt{a^2-x^2} - a\ln\left|\frac{a+\sqrt{a^2-x^2}}{x}\right|+C$

33. $\int \frac{\sqrt{a^2-x^2}}{x^2}\,dx = -\sin^{-1}\frac{x}{a} - \frac{\sqrt{a^2-x^2}}{x}+C$

34. $\int \frac{x^2}{\sqrt{a^2-x^2}}\,dx = \frac{a^2}{2}\sin^{-1}\frac{x}{a} - \frac{x}{2}\sqrt{a^2-x^2}+C$

35. $\int \frac{dx}{x\sqrt{a^2-x^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{a^2-x^2}}{x}\right|+C$

36. $\int \frac{dx}{x^2\sqrt{a^2-x^2}} = -\frac{\sqrt{a^2-x^2}}{a^2x}+C$

37. $\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1}\frac{x}{a}+C = \ln\left|x+\sqrt{x^2-a^2}\right|+C$

38. $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right|+C$

39. $\int \left(\sqrt{x^2-a^2}\right)^n dx = \frac{x\left(\sqrt{x^2-a^2}\right)^n}{n+1} - \frac{na^2}{n+1}\int \left(\sqrt{x^2-a^2}\right)^{n-2} dx, \quad n \neq 1$

40. $\displaystyle\int \frac{dx}{\left(\sqrt{x^2-a^2}\,\right)^n} = \frac{x\left(\sqrt{x^2-a^2}\,\right)^{2-n}}{(2-n)a^2} - \frac{n-3}{(n-2)a^2}\int \frac{dx}{\left(\sqrt{x^2-a^2}\,\right)^{n-2}}, \quad n \neq 2$

41. $\displaystyle\int x^2\left(\sqrt{x^2-a^2}\,\right)^n dx = \frac{\left(\sqrt{x^2-a^2}\,\right)^{n+2}}{n+2} + C, \quad n \neq -2$

42. $\displaystyle\int x^2\sqrt{x^2-a^2}\,dx = \frac{x}{8}\left(2x^2-a^2\right)\sqrt{x^2-a^2} - \frac{a^4}{8}\ln\left|x+\sqrt{x^2-a^2}\right| + C$

43. $\displaystyle\int \frac{\sqrt{x^2-a^2}}{x}\,dx = \sqrt{x^2-a^2} - a\sec^{-1}\left|\frac{x}{a}\right| + C$

44. $\displaystyle\int \frac{\sqrt{x^2-a^2}}{x^2}\,dx = \ln\left|x+\sqrt{x^2-a^2}\right| - \frac{\sqrt{x^2-a^2}}{x} + C$

45. $\displaystyle\int \frac{x^2}{\sqrt{x^2-a^2}}\,dx = \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + \frac{x}{2}\sqrt{x^2-a^2} + C$

46. $\displaystyle\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C = \frac{1}{a}\cos^{-1}\left|\frac{a}{x}\right| + C$

47. $\displaystyle\int \frac{dx}{x^2\sqrt{x^2-a^2}} = \frac{\sqrt{x^2-a^2}}{a^2x} + C$

48. $\displaystyle\int \frac{dx}{\sqrt{2ax-x^2}} = \sin^{-1}\left(\frac{x-a}{a}\right) + C$

49. $\displaystyle\int \sqrt{2ax-x^2}\,dx = \frac{x-a}{2}\sqrt{2ax-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x-a}{a}\right) + C$

50. $\displaystyle\int \left(\sqrt{2ax-x^2}\,\right)^n dx = \frac{(x-a)\left(\sqrt{2ax-x^2}\right)^n}{n+1} + \frac{na^2}{n+1}\int \left(\sqrt{2ax-x^2}\right)^{n-2} dx$

51. $\displaystyle\int \frac{dx}{\left(\sqrt{2ax-x^2}\right)^n} = \frac{(x-a)\left(\sqrt{2ax-x^2}\right)^{2-n}}{(n-2)a^2} + \frac{n-3}{(n-2)a^2}\int \frac{dx}{\left(\sqrt{2ax-x^2}\right)^{n-2}}$

52. $\displaystyle\int x\sqrt{2ax-x^2}\,dx = \frac{(x+a)(2x-3a)\sqrt{2ax-x^2}}{6} + \frac{a^3}{2}\sin^{-1}\left(\frac{x-a}{a}\right) + C$

53. $\displaystyle\int \frac{\sqrt{2ax-x^2}}{x}\,dx = \sqrt{2ax-x^2} + a\sin^{-1}\left(\frac{x-a}{a}\right) + C$

54. $\displaystyle\int \frac{\sqrt{2ax-x^2}}{x^2}\,dx = -2\sqrt{\frac{2a-x}{x}} - \sin^{-1}\left(\frac{x-a}{a}\right) + C$

55. $\displaystyle\int \frac{x\,dx}{\sqrt{2ax-x^2}} = a\sin^{-1}\left(\frac{x-a}{a}\right) - \sqrt{2ax-x^2} + C$

56. $\displaystyle\int \frac{dx}{x\sqrt{2ax-x^2}} = -\frac{1}{a}\sqrt{\frac{2a-x}{x}} + C$

57. $\displaystyle\int \sin(ax)\,dx = -\frac{1}{a}\cos(ax) + C$

58. $\int \cos(ax)\,dx = \frac{1}{a}\sin(ax) + C$

59. $\int \sin^2(ax)\,dx = \frac{x}{2} - \frac{\sin(2ax)}{4a} + C$

60. $\int \cos^2(ax)\,dx = \frac{x}{2} + \frac{\sin(2ax)}{4a} + C$

61. $\int \sin^n(ax)\,dx = -\frac{\sin^{n-1}(ax)\cos(ax)}{na} + \frac{n-1}{n}\int \sin^{n-2}(ax)\,dx$

62. $\int \cos^n(ax)\,dx = \frac{\cos^{n-1}(ax)\sin(ax)}{na} + \frac{n-1}{n}\int \cos^{n-2}(ax)\,dx$

63. $\int \sin(ax)\,\cos(bx)\,dx = -\frac{\cos[(a-b)x]}{2(a-b)} - \frac{\cos[(a+b)x]}{2(a+b)} + C, \quad a^2 \neq b^2$

64. $\int \sin(ax)\,\sin(bx)\,dx = \frac{\sin[(a-b)x]}{2(a-b)} - \frac{\sin[(a+b)x]}{2(a+b)} + C, \quad a^2 \neq b^2$

65. $\int \cos(ax)\,\cos(bx)\,dx = \frac{\sin[(a-b)x]}{2(a-b)} + \frac{\sin[(a+b)x]}{2(a+b)} + C, \quad a^2 \neq b^2$

66. $\int \sin(ax)\,\cos(ax)\,dx = -\frac{\cos(2ax)}{4a} + C$

67. $\int \sin^n(ax)\,\cos(ax)\,dx = \frac{\sin^{n+1}(ax)}{(n+1)a} + C, \quad n \neq -1$

68. $\int \frac{\cos(ax)}{\sin(ax)}\,dx = \frac{1}{a}\ln|\sin(ax)| + C$

69. $\int \cos^n(ax)\,\sin(ax)\,dx = -\frac{\cos^{n+1}(ax)}{(n+1)a} + C, \quad n \neq -1$

70. $\int \frac{\sin(ax)}{\cos(ax)}\,dx = -\frac{1}{a}\ln|\cos(ax)| + C$

71. $\int \cos^m(ax)\,\sin^n(ax)\,dx = -\frac{\cos^{m+1}(ax)\sin^{n-1}(ax)}{a(m+n)} + \frac{n-1}{m+n}\int \cos^m(ax)\sin^{n-2}(ax)\,dx, \; n \neq -m$

72. $\int \cos^m(ax)\,\sin^n(ax)\,dx = -\frac{\cos^{m-1}(ax)\sin^{n+1}(ax)}{a(m+n)} + \frac{m-1}{m+n}\int \cos^{m-2}(ax)\sin^n(ax)\,dx, \; n \neq -m$

73. $\int \frac{dx}{b + c\sin(ax)} = \frac{-2}{a\sqrt{b^2-c^2}}\tan^{-1}\left[\sqrt{\frac{b-c}{b+c}}\,\tan\left(\frac{\pi}{4} - \frac{ax}{2}\right)\right] + C, \quad b^2 > c^2$

74. $\int \frac{dx}{b + c\sin(ax)} = \frac{-1}{a\sqrt{c^2-b^2}}\ln\left|\frac{c + b\sin(ax) + \sqrt{c^2-b^2}\cos(ax)}{b + c\sin(ax)}\right| + C, \quad b^2 < c^2$

75. $\int \frac{dx}{1 + \sin(ax)} = -\frac{1}{a}\tan\left(\frac{\pi}{4} - \frac{ax}{2}\right) + C$

76. $\int \frac{dx}{1 - \sin(ax)} = \frac{1}{a}\tan\left(\frac{\pi}{4} + \frac{ax}{2}\right) + C$

77. $\displaystyle\int \frac{dx}{b+c\cos(ax)} = \frac{1}{a\sqrt{c^2-b^2}} \ln\left|\frac{c+b\cos(ax)+\sqrt{c^2-b^2}\sin(ax)}{b+c\cos(ax)}\right| + C, \quad b^2 < c^2$

78. $\displaystyle\int \frac{dx}{1+\cos(ax)} = \frac{1}{a}\tan\frac{ax}{2} + C$

79. $\displaystyle\int \frac{dx}{1-\cos(ax)} = -\frac{1}{a}\cot\frac{ax}{2} + C$

80. $\displaystyle\int x\sin(ax)\,dx = \frac{1}{a^2}\sin(ax) - \frac{x}{a}\cos(ax) + C$

81. $\displaystyle\int x\cos(ax)\,dx = \frac{1}{a^2}\cos(ax) + \frac{x}{a}\sin(ax) + C$

82. $\displaystyle\int x^n\sin(ax)\,dx = -\frac{x^n}{a}\cos(ax) + \frac{n}{a}\int x^{n-1}\cos(ax)\,dx$

83. $\displaystyle\int x^n\cos(ax)\,dx = \frac{x^n}{a}\sin(ax) - \frac{n}{a}\int x^{n-1}\sin(ax)\,dx$

84. $\displaystyle\int \tan(ax)\,dx = \frac{1}{a}\ln|\sec(ax)| + C$

85. $\displaystyle\int \cot(ax)\,dx = \frac{1}{a}\ln|\sin(ax)| + C$

86. $\displaystyle\int \tan^2(ax)\,dx = \frac{1}{a}\tan(ax) - x + C$

87. $\displaystyle\int \cot^2(ax)\,dx = -\frac{1}{a}\cot(ax) - x + C$

88. $\displaystyle\int \tan^n(ax)\,dx = \frac{\tan^{n-1}(ax)}{a(n-1)} - \int \tan^{n-2}(ax)\,dx, \quad n \neq 1$

89. $\displaystyle\int \cot^n(ax)\,dx = -\frac{\cot^{n-1}(ax)}{a(n-1)} - \int \cot^{n-2}(ax)\,dx, \quad n \neq 1$

90. $\displaystyle\int \sec(ax)\,dx = \frac{1}{a}\ln|\sec(ax)+\tan(ax)| + C$

91. $\displaystyle\int \text{cosec}\,(ax)\,dx = -\frac{1}{a}\ln|\text{cosec}\,(ax)+\cot(ax)| + C$

92. $\displaystyle\int \sec^2(ax)\,dx = \frac{1}{a}\tan(ax) + C$

93. $\displaystyle\int \text{cosec}^{\,2}(ax)\,dx = -\frac{1}{a}\cot(ax) + C$

94. $\displaystyle\int \sec^n(ax)\,dx = \frac{\sec^{n-2}(ax)\tan(ax)}{a(n-1)} + \frac{n-2}{n-1}\int \sec^{n-2}(ax)\,dx, \quad n \neq 1$

95. $\displaystyle\int \text{cosec}^{\,n}(ax)\,dx = -\frac{\text{cosec}^{\,n-2}(ax)\cot(ax)}{a(n-1)} + \frac{n-2}{n-1}\int \text{cosec}^{\,n-2}(ax)\,dx, \quad n \neq 1$

96. $\displaystyle\int \sec^n(ax)\,\tan(ax)\,dx = \frac{\sec^n(ax)}{na} + C, \quad n \neq 0$

97. $\int \operatorname{cosec}^n(ax)\ \cot(ax)\,dx = -\dfrac{\operatorname{cosec}^n(ax)}{na} + C, \quad n \neq 0$

98. $\int \sin^{-1}(ax)\,dx = x\sin^{-1}(ax) + \dfrac{1}{a}\sqrt{1-a^2x^2}\ + C$

99. $\int \cos^{-1}(ax)\,dx = x\cos^{-1}(ax) - \dfrac{1}{a}\sqrt{1-a^2x^2}\ + C$

100. $\int \tan^{-1}(ax)\,dx = x\tan^{-1}(ax) - \dfrac{1}{2a}\ln(1+a^2x^2) + C$

101. $\int x^n \sin^{-1}(ax)\,dx = \dfrac{x^{n+1}}{n+1}\sin^{-1}(ax) - \dfrac{a}{n+1}\int \dfrac{x^{n+1}}{\sqrt{1-a^2x^2}}\,dx, \quad n \neq -1$

102. $\int x^n \cos^{-1}(ax)\,dx = \dfrac{x^{n+1}}{n+1}\cos^{-1}(ax) + \dfrac{a}{n+1}\int \dfrac{x^{n+1}}{\sqrt{1-a^2x^2}}\,dx, \quad n \neq -1$

103. $\int x^n \tan^{-1}(ax)\,dx = \dfrac{x^{n+1}}{n+1}\tan^{-1}(ax) - \dfrac{a}{n+1}\int \dfrac{x^{n+1}}{\sqrt{1+a^2x^2}}\,dx, \quad n \neq -1$

104. $\int e^{ax}\,dx = \dfrac{1}{a}e^{ax} + C$

105. $\int b^{ax}\,dx = \dfrac{b^{ax}}{a\ln b} + C, \quad b > 0,\ b \neq 1$

106. $\int x\,e^{ax}\,dx = \dfrac{e^{ax}}{a^2}(ax-1) + C$

107. $\int x^n\,e^{ax}\,dx = \dfrac{1}{a}x^n\,e^{ax} - \dfrac{n}{a}\int x^{n-1}\,e^{ax}\,dx$

108. $\int x^n\,b^{ax}\,dx = \dfrac{x^n\,b^{ax}}{a\ln b} - \dfrac{n}{a\ln b}\int x^{n-1}\,b^{ax}\,dx, \quad b > 0,\ b \neq 1$

109. $\int e^{ax}\sin(bx)\,dx = \dfrac{e^{ax}}{a^2+b^2}\big(a\,\sin(bx) - b\,\cos(bx)\big) + C$

110. $\int e^{ax}\cos(bx)\,dx = \dfrac{e^{ax}}{a^2+b^2}\big(a\,\cos(bx) + b\,\sin(bx)\big) + C$

111. $\int \ln(ax)\,dx = x\ln(ax) - x + C$

112. $\int x^m\,\big(\ln(ax)\big)^n\,dx = \dfrac{x^{m+1}\big(\ln(ax)\big)^n}{m+1} - \dfrac{n}{m+1}\int x^m\,\big(\ln(ax)\big)^{n-1}\,dx, \quad m \neq -1$

113. $\int x^{-1}\big(\ln(ax)\big)^n\,dx = \dfrac{\big(\ln(ax)\big)^{n+1}}{n+1} + C, \quad n \neq -1$

114. $\int \dfrac{dx}{x\ \ln(ax)} = \ln\big|\ln(ax)\big| + C$

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